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Scalar Curvature Comparison Geometry and Higher Index Theory



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Summary

A celebrated result in scalar curvature comparison geometry is Llarull's rigidity theorem. It states that every area non-increasing smooth non-zero degree map $f: M \rightarrow S^n$ from a closed connected spin manifold M of dimension $n \geq 3$ and $\text{scal}_M \geq n(n-1)$ onto the round sphere is already an isometry. Goette and Semmelmann generalized this theorem to smooth area non-increasing spin maps $f: M \rightarrow N$ of non-zero $\hat{A}$ -degree. Here the manifolds M and N do not necessarily have the same dimension and one requires that the target manifold N has non-zero Euler characteristic, non-negative curvature operator and $\text{scal}_N > 2 \text{Ric}_N > 0$. The theorem states that if $\text{scal}_M \geq \text{scal}_N \circ f$, then f is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$.

In this thesis we give two independent generalizations of the Goette–Semmelmann rigidity theorem. In the first one we replace the condition on the $\hat{A}$ -degree by a less restrictive condition involving the so-called higher mapping degree. This yields $S^{2n} \times F \rightarrow S^{2n}$ and $\text{pr}_1: \mathbb{RP}^{2n} \times \Sigma \rightarrow \mathbb{RP}^{2n}$ as new examples. Here F is a closed connected spin manifold of rationally non-trivial Rosenberg index, e.g. $F = T^k$ or more generally F enlargeable, and Σ is an exotic sphere of non-trivial Hitchin index. The main difficulty in the proof is that a non-vanishing higher index of an $\mathcal{A}$ -linear Dirac operator does not necessarily give rise to a non-trivial harmonic section of the corresponding bundle. We develop a new method which extracts from such a non-vanishing higher index a geometrically useful family of almost harmonic sections. In the second generalization, the author together with Oskar Riedler strengthens the rigidity conclusion. We show that the comparison map $f: M \rightarrow N$ is not only a Riemannian submersion but that M splits locally as a Riemannian product of N with some Ricci-flat manifold.

There are many scalar curvature rigidity theorems in the spirit of Goette and Semmelmann for manifolds with boundary as for instance the band-rigidity theorem by Cecchini and Zeidler as well as the rigidity theorem by Lott. The proofs of these theorems rely on the existence of a non-trivial twisted parallel spinor satisfying a suitable boundary condition, which requires an appropriate theory of boundary value problems. In this thesis we generalize the theory of boundary value problems by Bär and Ballmann so that it applies to Dirac operators acting on smooth sections of bundles of finitely generated projective Hilbert C^* -modules. Combined with the method of almost harmonic spinors developed in this thesis, this is expected to yield new rigidity theorems for manifolds with boundary.

Finally we consider a question related to the existence of non-trivial parallel spinors. In general a non-vanishing Rosenberg index does not give rise to a non-trivial harmonic spinor. In this thesis we show that every closed connected Ricci-flat spin manifold with non-vanishing Rosenberg index admits a non-trivial parallel spinor on a finite Riemannian covering. We obtain as a corollary that those manifolds have special holonomy.

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Contents

1	Introduction and main results	1
1.1	Almost parallel sections and the higher index	5
1.2	Product rigidity and the higher mapping degree	8
1.3	Boundary value problems and higher index theory	11
2	Bibliography	17
3	Appendix	21
	A — Scalar curvature rigidity and the higher mapping degree	23
	B — Ricci-flat manifolds, parallel spinors and the Rosenberg index	67
	C — Scalar rigid submersions are Riemannian products	77
	D — Boundary value problems for first-order differential operators on bundles of Hilbert C^* -modules	121

1 Introduction and main results

The scalar curvature of a Riemannian manifold M is the trace of the Ricci curvature of M . It is a smooth function $\text{scal}: M \rightarrow \mathbb{R}$ that measures at each point how the volume of a small geodesic ball deviates from the volume of a Euclidean ball of the same radius. Although scalar curvature is, in many respects, a soft and flexible geometric quantity, it still encodes subtle global information. This motivates the following question:

How much information does the scalar curvature of a Riemannian manifold carry?

Closely related is the classical problem of determining whether a given manifold admits a Riemannian metric of positive scalar curvature. In recent research on scalar curvature comparison geometry, a central theme has been the study of the space of Riemannian metrics satisfying a lower scalar curvature bound and its interaction with the global geometry and topology of the manifold. For an overview of recent topics concerning the scalar curvature of a Riemannian manifold, see the *Four Lectures on Scalar Curvature* by Gromov [Gro23]. In the next two subsections we give a brief introduction to this topic, beginning with the classical existence problem.

Obstructions to the existence of positive scalar curvature metrics

There is a deep connection between the existence of a positive scalar curvature metric on some manifold and its topology [Lic63; AS63]. Let M be a closed connected spin manifold. The spin structure of the manifold M gives rise to a vector bundle $\mathcal{S}M \rightarrow M$, the so-called *spinor bundle*, together with an elliptic formally self-adjoint first-order differential operator $\mathcal{D}$ acting on smooth section of $\mathcal{S}M$. This differential operator $\mathcal{D}$ is called the *spin Dirac operator* of M . If the manifold M is even dimensional, the spinor bundle splits into an even and odd part, and the spin Dirac operator is odd with respect to this grading and satisfies

- $\text{ind}(\mathcal{D}) := \dim \ker \mathcal{D}^+ - \dim \ker \mathcal{D}^- = \hat{A}(M) \in \mathbb{Z}$ and (Atiyah-Singer index theorem)
- $\mathcal{D}^2 = \nabla^* \nabla + \frac{1}{4} \text{scal}_M$. (Schrödinger-Lichnerowicz formula)

Here $\mathcal{D}^\pm$ denotes the splitting of $\mathcal{D}$ with respect to the grading on $\mathcal{S}M$, and $\hat{A}(M)$ -denotes the $\hat{A}$ -genus of the manifold M . The above formula show that the existence of a positive scalar curvature metric is obstructed by the $\hat{A}$ -genus. This is a celebrated theorem of Lichnerowicz:

Theorem 1 ([Lic63]). *Let M be a closed connected spin manifold. If $\hat{A}(M)$ is non-zero, then M does not admit a Riemannian metric of positive scalar curvature.*

The *Hitchin invariant* [Hit74] and the notion of *enlargeability* [GL80] yield further obstructions to the existence of positive scalar curvature metrics on closed spin manifolds. The Hitchin index is a refinement of the $\hat{A}$ -obstruction and enlargeability is a geometric concept related to the existence of certain non-zero degree maps from coverings of the original manifold onto arbitrarily large spheres. The best currently known obstruction to the existence of a positive scalar curvature metric on closed connected spin manifolds based on index theory of Dirac operators is given by the *Rosenberg index* [Ros83]. It dominates classical obstructions such as the $\hat{A}$ -genus, the Hitchin invariant as well as the enlargeability obstruction [HS06; HS07]. As further examples we mention that the Rosenberg index is non-vanishing for closed spin manifolds admitting a metric of non-positive sectional curvature, as well as closed aspherical spin manifolds whose fundamental group satisfies the strong Novikov conjecture.

We now give a definition of the Rosenberg index. Let M be a closed connected spin manifold of dimension n . The Cl_n -linear spinor bundle is the vector bundle associated to the $\text{Spin}(n)$ -principal bundle with respect to the representation of the spin group on the Clifford algebra Cl_n given by left multiplication. It is equipped with the structure of a bundle of graded Real finitely generated projective Hilbert Cl_n -modules. The Mishchenko–Fomenko bundle

$$\mathcal{L} := \widetilde{M} \times_{\pi_1(M)} \text{C}^*\pi_1(M)$$

carries the structure of a bundle of finitely generated projective Hilbert $\text{C}^*\pi_1(M)$ -modules. Here $\widetilde{M}$ is the universal covering of M and $\text{C}^*\pi_1(M)$ the maximal group C^* -algebra of the fundamental group of M . We equip $\mathcal{L}$ with the flat connection. The induced twisted Dirac operator $\mathcal{D}_{\mathcal{L}}$ gives rise to a well-defined index class in the n -th real K-theory group of $\text{C}^*\pi_1(M)$.

Definition 2 ([Ros83]). The *Rosenberg index* of a closed connected spin manifold M of dimension n is defined as

$$\alpha(M) := \text{ind}(\mathcal{D}_{\mathcal{L}}) \in \text{KO}_n(\text{C}^*\pi_1(M)).$$

See Appendix A Section 2.3 for more details concerning the definition of the Rosenberg index. Since the Mishchenko–Fomenko bundle is flat, the Schrödinger–Lichnerowicz formula holds for the twisted spin Dirac operator, and we obtain the Rosenberg obstruction to positive scalar curvature metrics on such manifolds:

Theorem 3. *Let M be a closed connected spin manifold. Then the following holds:*

- (1) [Ros83] *If the Rosenberg index of M does not vanish, then M does not admit a Riemannian metric of positive scalar curvature.*
- (2) [Sto92] *Suppose M is simply-connected and $n \geq 5$. Then M admits a Riemannian metric of positive scalar curvature if and only if its Hitchin index vanishes.*

Note that for a simply-connected manifold the Rosenberg index equals the Hitchin index. Schick [Sch05] constructed an example of a closed connected spin manifold which does not admit a Riemannian metric of positive scalar curvature but with vanishing Rosenberg index, hence part (2) in Theorem 3 fails in general for non-simply-connected manifolds.

The space of Riemannian metrics satisfying a lower scalar curvature bound

The space of Riemannian metrics satisfying a certain lower bound on the scalar curvature is object of recent research in the field of scalar curvature comparison geometry. A first theorem which shows that a suitable lower bound on the scalar curvature restricts the geometry dramatically by the following celebrated theorem of Llarull:

Theorem 4 ([Lla98]). *Let M be a closed connected Riemannian spin manifold of dimension $n \geq 3$ and $f: M \rightarrow S^n$ a non-zero degree map onto the round sphere. If $g_M \geq f^*g_N$ on 2-vectors and $\text{scal}_M \geq n(n-1)$, then f is an isometry.*

Llarull [Lla96] generalized Theorem 4 to maps of non-zero $\hat{A}$ -degree onto the round sphere S^n where the dimension of the source manifold is bigger or equal to n . Then the rigidity conclusion is that the map is a Riemannian submersion and the scalar curvature of M is equal to the scalar curvature of the sphere. Goette and Semmelmann [GS02] generalized Llarull's rigidity theorem to a larger class of target manifolds. Namely, they proved the following rigidity theorem.

Theorem 5 ([GS02, Theorem 2.4]). *Let $f: M \rightarrow N$ be a smooth spin map of non-zero $\hat{A}$ -degree where both manifolds are closed connected and Riemannian, and the manifold N has $\text{scal}_N > 2\text{Ric}_N > 0$, non-negative curvature operator and non-zero Euler characteristic. If $g_M \geq f^*g_N$ on 2-vectors and $\text{scal}_M \geq \text{scal}_N \circ f$, then the map f is a Riemannian submersion and $\text{scal}_M = \text{scal}_N \circ f$.*

Here the $\hat{A}$ -degree of the map f is defined as the $\hat{A}$ -genus of the preimage of a regular value of the map f . Note that this definition is independent of the point p . By Theorem 1, the condition on the $\hat{A}$ -degree in Theorem 5 yields that the preimage of a regular value of the map f does not admit a Riemannian metric of positive scalar curvature. We consider the following example:

Example 6. Let N be as in Theorem 5 and F some closed connected spin manifold with non-zero $\hat{A}$ -genus. Then the map $\text{pr}_1: N \times F \rightarrow N$ satisfies the topological assumptions of Theorem 5 and the rigidity statement reads as follows: Every Riemannian metric g on $N \times F$ which satisfies $g \geq \text{pr}_1^*g_N$ and $\text{scal}_g \geq \text{scal}_N \circ \text{pr}_1$ satisfies $\text{scal}_g = \text{scal}_N \circ \text{pr}_1$ and the projection map $\text{pr}_1: N \times F \rightarrow N$ is a Riemannian submersion. The only known examples of such a Riemannian metric g on $N \times F$ are product metrics $g_N \oplus g_F$ for some Ricci-flat metric g_F on F .

Let N be a manifold as in Theorem 5 and F a closed connected spin manifold. The previous example gives rise to the following two independent questions:

Question 7. *Is the projection map $\text{pr}_1: N \times F \rightarrow N$ scalar-rigid in the sense of Theorem 5 if F has non-vanishing Rosenberg index?*

Question 8. *Are all Riemannian metrics on $N \times F$ satisfying $g \geq \text{pr}_1^* g_N$ and $\text{scal}_g \geq \text{scal}_N \circ \text{pr}_1$ Riemannian products as in Example 6?*

In this thesis we answer both questions to the positive. See Theorem D in Section 1.2 as well as [Ton25b, Theorem A] and [RT26, Theorem A] in Appendix A and Appendix C, respectively. The proof of Theorem 5 is based on the fact, that the index-theoretical condition on the $\hat{A}$ -degree together with the non-vanishing Euler characteristic of N gives rise to a non-trivial parallel section of a suitable vector bundle over the manifold M . The main difficulty in answering Question 7 is that, in general, a non-vanishing higher index, such as the Rosenberg index, does not give rise to a non-trivial kernel of the corresponding Dirac operator. Therefore the author developed a new method which extracts from a non-vanishing higher index a family of almost harmonic spinors which is in extreme geometric situations even almost parallel. See Lemma A in Section 1.1 and [Ton25b, Section 3] in Appendix A.

Let F be a closed manifold of dimension $n - 1 \neq 4$ which does not admit a Riemannian metric of positive scalar curvature. The *band width conjecture* by Gromov [Gro18] states that the width of the band $V = F \times [0, 1]$ is bounded by $2\pi/n$ for every Riemannian metric g on V satisfying $\text{scal}_g \geq n(n - 1)$. The width of a Riemannian band (V, g) , denoted by $\text{width}(V, g)$, is defined as the minimal distance between the two boundary components $\partial V_{\pm}$ with respect to the metric g . Zeidler [Zei22] proved this conjecture under the additional assumption that F is spin and has non-trivial Rosenberg index:

Theorem 9 ([Zei22; Zei20]). *Let F be a closed connected spin manifold of dimension $n - 1$ with non-trivial Rosenberg index. If g is a Riemannian metric on $F \times [0, 1]$ with $\text{scal}_g \geq n(n - 1)$ then the width of $F \times [0, 1]$ with respect to g is less than $2\pi/n$.*

Cecchini and Zeidler [CZ24] proved the corresponding rigidity statement under the stronger assumption that the $\hat{A}$ -genus of F is non-zero.

Theorem 10 ([CZ24, Corollary 9.2]). *Let F be a closed connected spin manifold of dimension $n - 1$ with non-zero $\hat{A}$ -genus. Suppose there exists some $d < \pi/n$ and a Riemannian metric g on the band $V = F \times [-d, d]$ with*

$$\text{scal}_g \geq n(n - 1), \quad \text{width}(V, g) \geq 2d, \quad \text{and} \quad H_g \geq -\tan(nd/2).$$

Then there exists a Ricci-flat metric g_F on F such that g is equal to the warped product $(\cos(nx/2)^{2/n} g_F) \oplus (dx \otimes dx)$.

Here H_g denotes the mean curvature at the boundary of V with respect to the Riemannian metric g . The sign convention is chosen such that the unit ball has positive mean curvature.

The band rigidity theorem in Theorem 10 implies immediately the band width estimates in Theorem 9 for the $\hat{A}$ -genus. Therefore it is a natural question to ask whether the band rigidity theorem also holds under the weaker assumption that the Rosenberg index of F does not vanish. The proof of Theorem 10 is based on the existence of a non-trivial twisted spinor which solves a suitable differential equation and a suitable boundary condition adapted to the geometric situation. In order to answer this question, it is necessary to develop a theory of boundary value problems as in [BB12] which applies to first-order differential operators acting on smooth sections of bundles of finitely generated projective Hilbert C^* -modules. This theory is explained in Section 1.3 and worked out in Appendix D. It is expected that this theory together with the method of almost harmonic sections developed in [Ton25b, Section 3] in Appendix A (see also Section 1.1) yields a generalization of the band rigidity theorem. A further application of the new theory of boundary value problems would be a generalization of the rigidity theorem by Lott [Lot21] involving higher index theory as in [Ton25b] in Appendix A.

1.1 Almost parallel sections and the higher index

Let $\mathcal{S}$ be a graded Dirac bundle over a closed Riemannian manifold M . Then the induced Dirac operator $\mathcal{D}$ is an odd formally self-adjoint first-order elliptic differential operator and the spectrum of its L^2 -closure is discrete, i.e. every spectral point is an eigenvalue of finite multiplicity. It follows that the *index* of $\mathcal{D}$ is well-defined

$$\text{ind}(\mathcal{D}) := \dim \ker(\mathcal{D}^+) - \dim \ker(\mathcal{D}^-) \in \mathbb{Z}.$$

Here $\mathcal{D}^\pm$ are the positive and negative chiral parts of $\mathcal{D}$. A non-zero index gives rise to a non-trivial $\mathcal{D}$ -harmonic section, this is a smooth non-trivial section u of the bundle $\mathcal{S}$ satisfying $\mathcal{D}u = 0$. The existence of such a section plays a crucial role in the proof of many scalar curvature rigidity results such as in [Lla96; GS02; Zei22; Lot21; Cec20]. For an $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$, where $\mathcal{A}$ is a graded Real unital C^* -algebra, such a $\mathcal{D}$ -harmonic section does not necessarily exist if the higher index of $\mathcal{D}$ does not vanish. The reason is that the fibers of the corresponding $\mathcal{A}$ -linear Dirac bundle maybe infinite dimensional as vector spaces.

Let $\mathcal{A}$ be a graded Real unital C^* -algebra and $\mathcal{S}$ a graded Real Dirac bundle. This is a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules equipped with a connection, a Clifford multiplication, a grading and a Real structure together with all compatibility conditions as for classical Dirac bundles (see [Ton25b, Definition 2.1] in Appendix A). As for a classical Dirac operator, the L^2 -closure of such an $\mathcal{A}$ -linear Dirac operator is self-adjoint and regular as an unbounded operator on the Hilbert $\mathcal{A}$ -module $L^2(M, \mathcal{S})$, hence gives rise to a functional calculus

$$\Phi_{\mathcal{D}}: C(\overline{\mathbb{R}}) \rightarrow \text{Lin}_{\mathcal{A}}(L^2(M, \mathcal{S})).$$

Here $\text{Lin}_{\mathcal{A}}(L^2(M, \mathcal{S}))$ denotes the adjointable operators on the Hilbert $\mathcal{A}$ -module $L^2(M, \mathcal{S})$.

Since the manifold M is closed, the functional calculus restricted to the graded Real C^* -algebra of smooth functions on $\mathbb{R}$ vanishing at infinity takes values in the compact operators of $L^2(M, \mathcal{S})$. Therefore the $\mathcal{A}$ -linear Dirac operator gives rise to a well-defined index class in the spectral picture of the 0-th real K-theory group of the C^* -algebra $\mathcal{A}$. This is called the *(higher) index* of $\mathcal{D}$

$$\mathrm{ind}_{\mathrm{hi}}(\mathcal{D}) := [\Phi_{\mathcal{D}}|_{C_0(\mathbb{R})}] \in \mathrm{KO}_0(\mathcal{A}).$$

See [Ton25b, Section 2.1] in Appendix A for details concerning the construction of the higher index. As in the classical case, non-vanishing of the higher index implies that the unbounded operator $\mathcal{D}: H^1(M, \mathcal{S}) \rightarrow L^2(M, \mathcal{S})$ is not invertible, hence zero is in its spectrum. But the spectrum is not necessarily discrete, and we do not obtain that the kernel of $\mathcal{D}$ is non-trivial. Instead, we can use the functional calculus for $\mathcal{D}$ and a suitable family of smooth functions f_ε supported in $(-\varepsilon, \varepsilon)$ to obtain a family of almost $\mathcal{D}$ -harmonic sections:

Lemma 11 ([cf. Ton25b, Lemma 3.2] in Appendix A). *Let $\mathcal{S}$ be a graded Real $\mathcal{A}$ -linear Dirac bundle over a closed Riemannian manifold. If the higher index of the induced $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$ does not vanish in $\mathrm{KO}(\mathcal{A})$, then there exists a family of almost $\mathcal{D}$ -harmonic sections, i.e. a family $(u_\varepsilon)_{\varepsilon>0}$ of smooth sections of $\mathcal{S}$ such that*

$$\|u_\varepsilon\|_{L^2} = 1 \quad \text{and} \quad \|\mathcal{D}^j u_\varepsilon\|_{L^2} < \varepsilon^j \quad \forall \varepsilon > 0, \forall j \geq 1.$$

This result is standard in the literature and yields that the Rosenberg index is an obstruction to a positive scalar curvature metric on closed connected spin manifolds: Suppose (M, g) is a closed connected Riemannian spin manifold of non-vanishing Rosenberg index with $\mathrm{scal}_g \geq 0$. Let (u_ε) be a family of almost $\mathcal{D}$ -harmonic sections as in Lemma 11 for the spin Dirac operator $\mathcal{D}$ twisted with the Mishchenko–Fomenko bundle $\mathcal{L}$. Then we obtain by the Schrödinger–Lichnerowicz formula

$$\varepsilon^2 > \|\mathcal{D}_{\mathcal{L}} u_\varepsilon\|_{L^2}^2 = \|\nabla u_\varepsilon\|_{L^2}^2 + \langle \mathrm{scal} u_\varepsilon, u_\varepsilon \rangle_{L^2} \|u_\varepsilon\|_{\mathcal{A}} \geq \min_{p \in M} \mathrm{scal}_g(p),$$

hence there exists a point $p \in M$ such that $\mathrm{scal}(p) = 0$. In such a situation an even stronger result holds. Schick [Sch21] proved by purely spinorial methods that the scalar curvature vanishes identically. It is a classical result by Bourguignon [cf. KW75] that even the Ricci curvature of M vanishes identically. In order to give a purely spinorial proof that the Ricci curvature of M vanishes in such a situation, the L^2 -estimates in Lemma 11 need to be upgraded to L^∞ -estimates (see [Ton25b, Section 3] in Appendix A):

Lemma A ([Ton25b, Lemma 3.5] in Appendix A and [RT26, Proposition 4.5] in Appendix C). *Let $\mathcal{S}$ be a graded Real $\mathcal{A}$ -linear Dirac bundle over a closed Riemannian manifold. Suppose the higher index of the induced $\mathcal{A}$ -linear Dirac operator does not vanish in $\mathrm{KO}(\mathcal{A})$ and let (u_ε) be a family of almost $\mathcal{D}$ -harmonic sections as in Lemma 11. If $\|\nabla u_\varepsilon\|_{L^2} < \varepsilon$ for all $\varepsilon > 0$, then*

there exist for every $j \geq 1$ constants C_1, C_2, r such that

$$\|\mathcal{D}^j u_\varepsilon\|_\infty < C_1 \varepsilon^r \quad \text{and} \quad \|\nabla^j u_\varepsilon\|_\infty < C_2 \varepsilon^r \quad \forall \varepsilon \in (0, 1).$$

The previous lemma shows that the family u_ε given in Lemma 11 is much more powerful than the L^2 -estimates in Lemma 11 suggests. Its proof is based on Moser iteration as in [AS07]. Namely, we show that for every $\mathcal{A}$ -linear differential operator $\mathcal{P}: C^\infty(M, \mathcal{S}) \rightarrow C^\infty(M, \mathcal{S})$ and for every state $\rho: \mathcal{A} \rightarrow \mathbb{C}$ of the C^* -algebra $\mathcal{A}$ the function

$$f_{\varepsilon, \rho}: M \longrightarrow \mathbb{R}, \quad p \longmapsto \rho(|\mathcal{P}u_\varepsilon|_p^2)^{1/2}$$

satisfies $\Delta f_{\varepsilon, \rho} \leq \alpha$ for some $\alpha > 0$ independent of ε and the state ρ . It follows by Moser iteration (cf. [Ton25b, Lemma 3.4] in Appendix A or [Pet16, Theorem 9.27] and [PS99; PS03]) that there exist constants C, r such that $\|f_{\varepsilon, \rho}\|_\infty \leq C \|f_{\varepsilon, \rho}\|_{L^2}^r$. The constants C, r just depend on α and the dimension, the diameter and the Ricci curvature of the manifold M .

The L^∞ -esimates in Lemma A yield a direct proof—using purely spinorial methods—of the following classical statement:

Theorem 12 ([cf. Ton25b, Corollary 4.2] in Appendix A). *Every closed connected Riemannian spin manifold with non-vanishing Rosenberg index and non-negative scalar curvature is Ricci-flat.*

Although a higher index of a Dirac operator $\mathcal{D}$ does not give rise to a non-trivial kernel in general, a detection principle by Schick and Wraith [SW21, Proposition 3.6] yields that if the fundamental group π of the underlying manifold M admits $\mathbb{Z}^q$ as a subgroup of finite index then a non-vanishing higher index can be detected by a twist with a unitary representation of π on some finite dimensional vector space [see also RWY13]. In combination with the structure theorem for Ricci-flat manifolds [FW75, Theorem 4.5] we obtain the following statement:

Theorem B ([Ton25a, Theorem A] in Appendix B). *A closed connected Riemannian spin manifold with non-vanishing Rosenberg index is Ricci-flat if and only if it admits a non-trivial parallel spinor on a finite Riemannian covering.*

The previous theorem is well-known if the $\hat{A}$ -genus of the manifold M does not vanish, and it is conjectured that all Ricci-flat spin manifolds admit a non-trivial parallel spinor on a finite Riemannian covering. A complete simply-connected irreducible Riemannian spin manifold admits a non-trivial parallel spinor if and only if it has special holonomy. Therefore the existence of non-trivial parallel spinors is related to the following question:

Question 13. *Does every closed connected Ricci-flat manifold of dimension $n \geq 2$ have special holonomy, i.e. its reduced holonomy group is a proper subgroup of $\mathrm{SO}(n)$?*

By Theorem B we can answer Question 13 in the special case of spin manifolds with non-vanishing Rosenberg index:

Corollary C ([Ton25a, Corollary B] in Appendix B). *Every closed connected Ricci-flat spin manifold of dimension ≥ 2 with non-vanishing Rosenberg index has special holonomy.*

1.2 Product rigidity and the higher mapping degree

In this section we explain two independent generalizations of the rigidity theorem by Goette and Semmelmann [GS02, Theorem 2.4] (see also Theorem 5). In the first generalization the author proved the rigidity statement by Goette and Semmelmann where the assumption on the $\hat{A}$ -degree is replaced by a weaker assumption involving the so-called *higher mapping degree* (see [Ton25b] in Appendix A). The new theorem also applies to manifolds which are not necessarily orientable as long as the map is orientable in a suitable sense. In the second generalization the author strengthened together with Oskar Riedler the rigidity conclusion of Goette and Semmelmann by showing that the map f is not just a Riemannian submersion rather than the source manifold splits locally as a Riemannian product and the map is given by the projection onto the first factor (see [RT26] in Appendix C). This generalization is independent of the first one but also holds in combination with the higher mapping degree assumption. In total, we obtain the following statement:

Theorem D ([Ton25b, Theorem A] in Appendix A and [RT26, Theorem A] in Appendix C). *Let $f: M \rightarrow N$ be a smooth spin map between two closed connected Riemannian manifolds M and N . Suppose N has $\text{Ric}_N > 0$ and non-negative curvature operator, and*

$$\chi(N) \cdot \deg_{\text{hi}}(f) \neq 0 \in \text{KO}_*(C^*\pi_1(M)), \quad (1.2.1)$$

*$\text{scal}_M \geq \text{scal}_N \circ f$ and $g_M \geq f^*g_N$ hold. Then there exists a closed connected Ricci-flat manifold (F, g_F) such that*

$$(M, g_M) \cong (N, g_N) \times (F, g_F) \quad (\text{locally isometric}) \quad (1.2.1)$$

*and f is given by the projection onto the first factor. Moreover, if $\text{scal}_N > 2\text{Ric}_N > 0$, then the rigidity also holds if we assume $g_M \geq f^*g_N$ only on $\Lambda^2 TM$.*

Here two manifolds are called *locally isometric* if their universal coverings endowed with the lifted Riemannian metrics are isometric. A map f as in Theorem D is called *spin* if the first and second Stiefel–Whitney classes satisfy $w_1(TM) = f^*w_1(TN)$ and $w_2(TM) = f^*w_2(TN)$. The *higher mapping degree* (see [Ton25b, Definition 5.2] in Appendix A) of a spin map f is defined as the higher index of the Dirac operator induced by the $\mathbb{C}\ell$ -linear spin Dirac bundle of the spin manifold $f^{-1}(p)$ twisted with the Mishchenko–Fomenko bundle of the manifold M

$$\deg_{\text{hi}}(f) := \text{ind}_{\text{hi}}(\mathcal{D}_{Sf^{-1}(p) \otimes \mathcal{L}(M)}) \in \text{KO}(C^*\pi_1(M)).$$

Here $p \in N$ is a regular value of the map f . Note that the higher mapping degree is, by bordism invariance of the higher index, independent of the point p .

If the Euler characteristic of N and the $\hat{A}$ -degree of a map f as in Theorem D do not vanish, then the index-theoretical condition on the higher mapping degree in equation (1.2.1) is satisfied, hence Theorem D generalizes the classical rigidity statement by Goette and Semmelmann in Theorem 5. Moreover, the classical statement by Goette and Semmelmann [GS02] is not applicable to the maps $\text{pr}_1: S^{2n} \times T^k \rightarrow S^{2n}$ and $\text{pr}_1: \mathbb{RP}^{2n} \times \Sigma \rightarrow \mathbb{RP}^{2n}$ — where Σ is an exotic sphere of non-zero Hitchin invariant [Hit74] — but the generalization in Theorem D applies to these two maps. More generally, Theorem D applies to $\text{pr}_1: N \times F \rightarrow F$ where N is as in Theorem D and F some closed connected spin manifold of non-vanishing Rosenberg index or some closed enlargeable spin manifold. This answers Question 7 and Question 8. See [Ton25b, Section 5] in Appendix A and [RT26, Section 1] in Appendix C for more examples where the higher mapping degree condition is satisfied. We obtain the following corollary:

Corollary E ([RT26, Example 1.7 (4)] in Appendix C). *Let Σ be an exotic sphere of non-vanishing Hitchin invariant and $\mathbb{RP}^{2n}$ be equipped with the standard metric g_0 . Then there exists no Riemannian metric g on $\mathbb{RP}^{2n} \times \Sigma$ such that $\text{scal}_g \geq 2n(2n - 1)$ and $g \geq g_0$.*

The product rigidity in Theorem D yields a new Llarull-type theorem for non-zero degree maps onto manifolds of the form $N \times F$ for certain F including enlargeable manifolds (see [RT26, Corollary B] in Appendix C). This is in the spirit of [HSS24, Theorem 1.1] — where non-zero degree maps onto $S^3 \times T^{n-3}$ with $4 \leq n \leq 7$ are studied — and generalizes a recent rigidity theorem by Chow [Cho25, Theorem A] to all even dimensions. While the approach by Chow is based on minimal slicing the approach here is purely based on index theory for the Dirac operator, hence it works also for large dimensions.

Corollary F ([RT26, Corollary B] in Appendix C). *Let $f: M \rightarrow N \times F$ be a smooth map of non-zero degree, where*

- *M and N are closed connected Riemannian spin manifolds,*
- *N has non-zero Euler characteristic, non-negative curvature operator and $\text{Ric}_N > 0$, and*
- *F is a closed connected oriented manifold which is either (area)-enlargeable, or rationally essential and whose fundamental group satisfies the strong Novikov conjecture.*

If $g_M \geq (\text{pr}_1 \circ f)^ g_N$ and $\text{scal}_M \geq \text{scal}_N \circ \text{pr}_1 \circ f$, then M is locally isometric to the Riemannian product $N \times T^k$ for a flat torus T^k and $\text{pr}_1 \circ f$ is given by the projection onto the first factor.*

Here *rationally essential* means that the fundamental class of F maps non-trivially under the map $H_*(F; \mathbb{Q}) \rightarrow H_*(B\pi_1(F))$ induced by the classifying map $\mu: F \rightarrow B\pi_1(F)$ of the universal cover. We say that its fundamental group satisfies the *strong Novikov conjecture* if the Novikov assembly map is rationally injective [cf. Zei20, Section 2].

In the remaining section we give a brief sketch of the proof of Theorem D. See [Ton25b, Section 1] in Appendix A and [RT26, Section 2] in Appendix C for detailed proof sketches.

Let $f: M \rightarrow N$ be a spin map as in Theorem D and denote by m and n the dimensions of the manifold M and N respectively. Since f is spin, the vector bundle $TM \oplus f^*TN$ equipped with the indefinite metric $g_M \oplus (-f^*g_N)$ admits an indefinite spin structure [Bau81]. It is crucial that we change the sign of the metric on the second summand since the vector bundle $TM \oplus f^*TN$ does not necessarily admit a topological spin structure. We obtain a graded Real $\text{Cl}_{m,n}$ -linear Dirac bundle $\mathcal{S}$ together with Clifford multiplications

$$c: TM \longrightarrow \text{End}(\mathcal{S}) \quad \text{and} \quad \bar{c}: f^*TN \longrightarrow \text{End}(\mathcal{S})$$

induced by the map $(\mathbb{R}^m \oplus \mathbb{R}^n) \times \text{Cl}_{m,n} \rightarrow \text{Cl}_{m,n}$ given by left multiplication. This Dirac bundle is basically the same then in the classical proof by Goette and Semmelmann [GS02] with the difference that it is equipped with the structure of a Hilbert $\text{Cl}_{m,n}$ -module. Here it is crucial that we flipped the sign of the metric of f^*TN in order to obtain the following index theorem for the induced $\text{Cl}_{m,n}$ -linear Dirac operator twisted with the Mishchenko–Fomenko bundle of the manifold M :

$$\text{ind}_{\text{hi}}(\mathcal{D}_{\mathcal{L}(M)}) = \chi(N) \deg_{\text{hi}}(f) \in \text{KO}_{m-n}(\text{C}^*\pi_1(M)). \quad (1.2.2)$$

This is a generalization of the index theorem by Goette and Semmelmann [GS02] where the same formula holds for the $\hat{A}$ -degree instead of the higher mapping degree. The proof of equation (1.2.2) is based on the Poincaré–Hopf lemma for the Euler characteristic of N and a cut-and-paste principle for the higher index.

Similarly as in the proof by Goette and Semmelmann, we obtain the following Schrödinger–Lichnerowicz type formula

$$\mathcal{D}_{\mathcal{L}(M)}^2 \geq \nabla^* \nabla + \frac{1}{4}(\text{scal}_M - \text{scal}_N \circ f). \quad (1.2.3)$$

If the Euler characteristic of N and the classical $\hat{A}$ -degree do not vanish, the corresponding index formula in equation (1.2.2) gives rise to a non-zero $\mathcal{D}$ -harmonic section of $\mathcal{S}$ which is parallel by equation (1.2.3), hence constant. It follows from equation (1.2.3) that the map f is a Riemannian submersion and $\text{scal}_M = \text{scal}_N \circ f$. Under the weaker assumption on the higher mapping degree, we obtain by the new method in Lemma A a family of almost parallel spinors in the L^∞ -sense, and it turns out that this is enough to conclude the rigidity as in Theorem 5. This proves the first generalizations concerning the higher mapping degree condition.

The second generalization, that is the product rigidity case, follows by analyzing the Schrödinger–Lichnerowicz type formula in equation (1.2.3). In the rigidity case, we obtain a relation between the Clifford multiplications c and $\bar{c}$ on two-vectors on N which are in the

image of the curvature operator of N . This relation yields that the Ricci curvature of M and N must be equal and the mean curvature of the fibers vanishes. This implies that the A and T tensors of the Riemannian submersion f vanishes identically, hence M splits locally as a Riemannian product. The main difficulty here is that the image of the curvature operator is in general a proper subset of all two-vectors on N . It turns out that since the Ricci curvature of N is strictly positive, the image is always large enough to obtain enough algebraic equations to conclude the rigidity.

1.3 Boundary value problems and higher index theory

Let $\mathcal{A}$ be a unital C^* -algebra, E and F two bundles of finitely generated projective Hilbert $\mathcal{A}$ -modules over a Riemannian manifold M with compact boundary ∂M , and $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ a first-order $\mathcal{A}$ -linear differential operator. The *minimal extension* of $\mathcal{D}$ is the closure of the densely defined unbounded operator $\mathcal{D}_{cc}: C_{cc}^\infty(M, E) \rightarrow L^2(M, F)$, and the *maximal extension* is the adjoint of the densely defined unbounded operator $\mathcal{D}_{cc}^\dagger: C_{cc}^\infty(M, F) \rightarrow L^2(M, E)$. Here $C_{cc}^\infty(M, E)$ denotes the smooth sections compactly supported in the interior of the manifold M , and $\mathcal{D}^\dagger$ denotes the formal adjoint of the differential operator $\mathcal{D}$ (see Section 1 in Appendix D). In general $\text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\mathcal{D}_{\max})$ with equality if and only if the boundary of M is empty. We define boundary conditions such that they give rise to closed extensions between the minimal and the maximal extension

$$\text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\mathcal{D}_B) \subseteq \text{dom}(\mathcal{D}_{\max}).$$

Every closed extension between $\mathcal{D}_{\min}$ and $\mathcal{D}_{\max}$ is of this form. In order to describe boundary conditions purely by data on the boundary, we use the restriction map $\mathcal{R}$. This is a bounded extension of the map $\mathcal{R}_0: C_c^\infty(M, E) \rightarrow C^\infty(\partial M, E)$, $u \rightarrow u|_{\partial M}$, to $\text{dom}(\mathcal{D}_{\max})$ and a suitable target Sobolev space $\check{H}_A(\partial M, E)$ (see Definition G and Proposition H). We make the following two assumptions on the Dirac operator $\mathcal{D}$ (cf. [BB12, Definition 1.4] and [Ebe16, Definiton 1.13]):

- (1) $\mathcal{D}$ is *boundary symmetric* with respect to some vector field T of M along ∂M pointing into the interior of M , i.e. for all $x \in \partial M$ the principal symbol $\sigma_{\mathcal{D}}(\eta)|_x$ is invertible for all co-vector fields η of M , and

$$\sigma_{\mathcal{D}}(\tau)|_x^{-1} \circ \sigma_{\mathcal{D}}(\xi)|_x \in \text{End}(E_x, E_x) \quad \text{and} \quad \sigma_{\mathcal{D}}(\xi)|_x \circ \sigma_{\mathcal{D}}(\tau)|_x^{-1} \in \text{End}(F_x, F_x)$$

are skew-Hermitian for all co-vector fields ξ of M satisfying $\xi(T) = 0$. Here $\sigma_{\mathcal{D}}$ denotes the principal symbol of $\mathcal{D}$, and τ the co-vector field associated to T (see Definition 2.2 in Appendix D).

- (2) M is *complete for $\mathcal{D}$* , i.e. there exists a proper smooth function $f: M \rightarrow \mathbb{R}$ which is

bounded from below such that the commutator $[\mathcal{D}, f]$ extends to a bounded operator from $L^2(M, E)$ to $L^2(M, F)$ (see Definition 1.11 (3) in Appendix D).

Part (1) yields that, in a suitable collar neighborhood of the boundary, $\mathcal{D}$ takes the form $\mathcal{D} = \sigma_{\mathcal{D}}(dt)(\partial_t + A + R_t)$ for a suitable rest term R_t and a formally self-adjoint elliptic first-order $\mathcal{A}$ -linear differential operator $A: C^\infty(\partial M, E) \rightarrow C^\infty(\partial M, E)$ (see Proposition 2.6 in Appendix D). Such an operator A is called an *adapted boundary operator*. This normal form is important in the analytic calculations in the proof that the restriction map $\mathcal{R}_0$ extends to a bounded operator between $\text{dom}(\mathcal{D}_{\max})$ and $\check{H}_A(\partial M, E)$ (see Proposition H). Part (2) controls the differential operator $\mathcal{D}$ at infinity, and it implies that every section in the maximal domain of $\mathcal{D}$ can be approximated in the graph norm of $\mathcal{D}$ by compactly supported sections in the maximal domain of $\mathcal{D}$ (see Lemma 1.17 in Appendix D).

Example 14. Let M be a complete Riemannian manifold and $\mathcal{S}$ some $\mathcal{A}$ -linear Dirac bundle. Then the induced $\mathcal{A}$ -linear Dirac operator is boundary symmetric with respect to the interior unit normal field of ∂M and M is complete for $\mathcal{D}$. More generally, this also holds for Dirac-type operators (see Example 2.8 in Appendix D and [BB12, Example 1.6]).

For $k \in \mathbb{N}_0$ we define the hybrid Sobolev spaces as a mix of the fractional Sobolev spaces $H^{k-1/2}(\partial M, E)$ and $H^{k+1/2}(\partial M, E)$. Since the spectrum of A is not necessarily discrete it is important to avoid projections onto certain parts of the spectrum in contrast to the definition by Bär and Ballmann [BB12, Section 5].

Definition G (see Definition 2.14 in Appendix D). Let $k \in \mathbb{N}_0$. We define the *hybrid Sobolev space* $\check{H}_A^k(\partial M, E)$ as the closure of the smooth sections on ∂M with respect to the $\mathcal{A}$ -valued inner product

$$\langle \cdot, \cdot \rangle_{\check{H}_A} = \langle (1 + A^2)^k \check{f}(A) \cdot, \cdot \rangle_{L^2}$$

where $\check{f}: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ is some smooth function which is equal to $|x|$ for $x \leq -1$ and equal to $1/x$ for $x \geq 1$. Similarly we define the $\hat{H}_A^k$ -spaces with respect to the function $\hat{f}(x) := \check{f}(-x)$ instead of $\check{f}$. We write $\check{H}_A$ and $\hat{H}_A$ instead of $\check{H}_A^0$ and $\hat{H}_A^0$, respectively.

The unbounded operator $A: H^1(\partial M, E) \rightarrow L^2(\partial M, E)$ is self-adjoint and regular, hence it admits a functional calculus as in [Ebe16, Theorem 1.19] and $\check{f}(A)u$ is well-defined for $u \in C^\infty(\partial M, E)$ (cf. Proposition 1.4 and Theorem 1.26 in Appendix D). The fractional Sobolev space satisfies for all $k \in \mathbb{N}_0$

$$H^{k-1/2}(\partial M, E) \supseteq \check{H}_A^k(\partial M, E) \supseteq H^{k+1/2}(\partial M, E).$$

We obtain the following statement:

Proposition H (Theorem 2.3 in Appendix D). *The restriction map $\mathcal{R}_0: C_c^\infty(M, E) \rightarrow$*

$C^\infty(\partial M, E)$, $u \mapsto u|_{\partial M}$, extends to a surjective bounded operator

$$\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \longrightarrow \check{H}_A(\partial M, E)$$

with kernel given by $\text{dom}(\mathcal{D}_{\min})$.

The proof of Proposition H splits into three steps. The first step is to define a suitable extension map $\mathcal{E}: C^\infty(\partial M, E) \rightarrow \text{dom}(\mathcal{D}_{\max})$ and show that it extends to a bounded operator with domain $\check{H}_A(\partial M, E)$ (see Definition 2.14 and Proposition 2.15 in Appendix D). Here it is crucial to use the normal form of the differential operator $\mathcal{D}$ in a collar of the boundary. The second step is to show that the L^2 -scalar product extends to a perfect pairing $\langle\langle \cdot, \cdot \rangle\rangle: \hat{H}_A(\partial M, E) \times \check{H}_A(\partial M, E) \rightarrow \mathcal{A}$ which yields together with the extension map that the restriction map extends to a bounded operator from the closure of $C_c^\infty(M, E)$ with respect to the graph norm of $\mathcal{D}$ to $\check{H}_A(\partial M, E)$ (see Definition 2.12, Lemma 2.13 and Lemma 2.16 in Appendix D). The last step is to show that the compactly supported smooth sections are dense in $\text{dom}(\mathcal{D}_{\max})$ with respect to the graph norm of $\mathcal{D}$ (see Proposition 2.15 in Appendix D). This is a consequence of the fact that M is complete for $\mathcal{D}$ and the existence of local Friedrich mollifiers (see Lemma 1.22 in Appendix D).

We use the hybrid Sobolev spaces in Definition G and the restriction map $\mathcal{R}$ in Proposition H to define (right-elliptic complementable self-adjoint) boundary conditions:

Definition I. (1) An (abstract) boundary condition for $\mathcal{D}$ is an $\mathcal{A}$ -submodule $B \subseteq \check{H}_A(\partial M, E)$ which is closed with respect to the $\check{H}_A$ -norm (see Definition 3.2 in Appendix D).

(2) A boundary condition B is called *right-elliptic* if $C^\infty(\partial M, E) \cap B \subseteq \check{H}_A^k(\partial M, E) \cap B$ is dense with respect to the $\check{H}_A^k$ -norm for every $k \in \mathbb{N}_0$ and there exists for every $k \in \mathbb{N}_0$ a constant C such that

$$\|\varphi\|_{H^{k+1/2}(\partial M)} \leq C \|\varphi\|_{\check{H}_A^k(\partial M)}$$

for all $\varphi \in C^\infty(\partial M, E) \cap B$ (see Definition 3.4 in Appendix D).

(3) Suppose $\mathcal{D}$ is formally self-adjoint (in particular $E = F$). A boundary condition B is called *complementable and self-adjoint* if

$$B = B^{\text{ad}} := \{\psi \in \check{H}_A(\partial M, E) \mid \langle\langle \sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle\rangle = 0 \text{ for all } \varphi \in B\} \quad \text{and} \quad (B_\rho)^{\text{ad}} = (B^{\text{ad}})_\rho$$

for every state ρ of the C^* -algebra $\mathcal{A}$. Here $\langle\langle \cdot, \cdot \rangle\rangle: \hat{H}_A(\partial M, E) \times \check{H}_A(\partial M, E) \rightarrow \mathcal{A}$ is the perfect pairing used in the proof of Proposition H and B_ρ and $(B^{\text{ad}})_\rho$ are the localized boundary conditions with respect to the state ρ (see Definition 3.13 in Appendix D).

In Theorem J we state that a boundary condition B gives rise to a closed extension of $\mathcal{D}_{\min}$ which is self-adjoint and regular if $\mathcal{D}$ is formally self-adjoint and the boundary condition is

complementable and self-adjoint. In Theorem K we state the elliptic estimates for right-elliptic boundary conditions as well as the higher elliptic regularity principle. In Theorem L we give the main theorem which states that a right-elliptic complementable self-adjoint boundary condition gives rise to an unbounded Fredholm operator, if $\mathcal{D}$ is formally self-adjoint and elliptic. This is the crucial property for applications in scalar curvature comparison geometry.

Theorem J (Theorem 3.3 and Theorem 3.15 in Appendix D). *Every boundary condition $B \subseteq \check{H}_A(\partial M, E)$ for $\mathcal{D}$ gives rise to a closed extension*

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \longrightarrow L^2(M, E)$$

with $\text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\mathcal{D}_B) \subseteq \text{dom}(\mathcal{D}_{\max})$, and every extension between $\mathcal{D}_{\min}$ and $\mathcal{D}_{\max}$ is of this form. If $\mathcal{D}$ is formally self-adjoint and the boundary condition is complementable and self-adjoint, then $\mathcal{D}_B$ is self-adjoint and regular as an unbounded operator on the Hilbert $\mathcal{A}$ -module $L^2(M, E)$.

Theorem K (Theorem 3.6 and 3.10 in Appendix D). *Suppose the differential operator $\mathcal{D}$ is elliptic and let $B \subseteq \check{H}_A(M, E)$ be a right-elliptic boundary condition for $\mathcal{D}$. Then there exists for every $k \in \mathbb{N}_0$ and every compact $K \subseteq M$ a constant C such that*

$$\|u\|_{H^{k+1}} \leq C(\|\mathcal{D}u\|_{L^2} + \|u\|_{L^2}) \quad (\text{elliptic estimate})$$

for all $u \in C_c^\infty(M, E)$ with $u|_{\partial M} \in B$ and $\text{supp}(u) \subseteq K$. Moreover, if $u \in \text{dom}(\mathcal{D}_B) \cap H_{\text{loc}}^k(M, E)$ with $\mathcal{D}_B u \in H_{\text{loc}}^k(M, F)$, then $u \in H_{\text{loc}}^{k+1}(M, E)$.

Proof (sketch). We first show the elliptic estimates for smooth sections which vanish at the boundary and are supported in a collar of the boundary (see Proposition 2.19 in Appendix D). This follows from the interior elliptic estimates in combination with a suitable mirror map which extends sections over the boundary as in [GT77, Lemma 6.37] (see Proposition 1.24, Definition 2.21 and Lemma 2.22 in Appendix D). In order to show the elliptic estimates for general right-elliptic boundary conditions, we fix some $k \in \mathbb{N}_0$ and some compact subset $K \subseteq M$. We obtain

$$\|u\|_{H^{k+1}} \leq \|(1 - \chi)u\|_{H^{k+1}} + \|(1 - \mathcal{E}\mathcal{R})\chi u\|_{H^{k+1}} + \|\mathcal{E}\mathcal{R}\chi u\|_{H^{k+1}}$$

for every smooth section u satisfying $u|_{\partial M} \in B$ and $\text{supp}(u) \subseteq K$. Here χ is a smooth cutoff function which is identically one close to the boundary and vanishes outside a suitable collar. We estimate the first term by the interior elliptic estimates, the second term by the elliptic estimates for smooth sections vanishing at the boundary, and the last term by the properties of the extension map $\mathcal{E}$ and the restriction map $\mathcal{R}$ together with the ellipticity of B . Here it is crucial that the extension and restriction map satisfy the estimates

$$\|\mathcal{D}\mathcal{E}\varphi\|_{H^k} + \|\mathcal{E}\varphi\|_{H^k} \leq c\|\varphi\|_{\check{H}_A^k} \quad \text{and} \quad \|\mathcal{R}v\|_{\check{H}_A} \leq C(\|\mathcal{D}v\|_{H^k} + \|v\|_{H^k})$$

for all $\varphi \in C^\infty(\partial M, E)$, all $v \in C_c^\infty(M, E)$ and suitable constants c, C (see Proposition 2.15 and Lemma 2.16 in Appendix D). The elliptic regularity principle follows from the elliptic estimates by a suitable smooth approximation satisfying the boundary condition (see Lemma 3.8 in Appendix D). $\square$

Suppose the C^* -algebra $\mathcal{A}$ and the bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules is equipped with a Real structure and a grading which are compatible to each other. We obtain the following main theorem:

Theorem L (Theorem 4.5 in Appendix D). *Suppose the differential operator $\mathcal{D}$ is formally self-adjoint and elliptic, and let B be a right-elliptic complementable self-adjoint boundary condition which is compatible with the grading and the Real structure of E and $\mathcal{A}$. Then the induced unbounded operator*

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \longrightarrow L^2(M, E)$$

is Fredholm, i.e. $\mathcal{D}_B(1 + \mathcal{D}_B^2)^{-1/2}$ is invertible modulo a compact operator, hence there exists a well-defined index class $\text{ind}(\mathcal{D}_B) \in \text{KO}_0(\mathcal{A})$.

Proof (sketch). Since the unbounded operator $\mathcal{D}_B$ is self-adjoint and regular by Theorem J there exists a functional calculus for $\mathcal{D}_B$

$$\Phi_{\mathcal{D}_B}: C(\mathbb{R}) \longrightarrow \text{Lin}_{\mathcal{A}}(L^2(M, E)).$$

By the elliptic estimates in Theorem K and Rellich's theorem [MF80], we obtain that $gf(\mathcal{D}_B)$ and $f(\mathcal{D}_B)g$ are both compact for all $f \in C_0(\mathbb{R})$ and all $g \in C_c^\infty(M, \mathbb{R})$. It follows that $\mathcal{D}_B$ is Fredholm as in [Ebe16]. $\square$

We conclude the introduction with two examples of right-elliptic complementable self-adjoint boundary conditions, which both arise in the context of scalar curvature comparison geometry [cf. Lot21; CZ24].

Example 15 (Atiyah-Patody-Singer boundary conditions—see Section 5.1 in Appendix D). Suppose $\mathcal{D}$ is formally self-adjoint and there exists an adapted boundary operator A which anti-commutes with $\sigma_{\mathcal{D}}(\tau)$ such that zero is not in the spectrum of A . This is for example the case for Dirac operators. Let $\varepsilon > 0$ such that $(-\varepsilon, \varepsilon) \cap \text{spec}(A) = \emptyset$, and $f_\varepsilon \in C^\infty(\mathbb{R}, [0, 1])$ be some smooth function such that $f \equiv 1$ on $(-\infty, -\varepsilon)$ and $f \equiv 0$ on (ε, ∞) . Then

$$B_{\text{APS}} := \overline{f_\varepsilon(A) C^\infty(\partial M, E)}^{\|\cdot\|_{H^{1/2}}} \subseteq \check{H}_A(\partial M, E)$$

defines a right-elliptic complementable self-adjoint boundary condition for $\mathcal{D}$. Note that B_{APS} is independent of ε and the function f_ε .

Example 16 (Chiral boundary conditions — see Section 5.2 in Appendix D). Let χ be a formally self-adjoint involution of the bundle $E|_{\partial M}$ such that

- $\sigma_{\mathcal{D}}(\tau)$ anti-commutes with χ and
- $\sigma_{\mathcal{D}}(\xi)$ commutes with χ for all $\xi \in T\partial M \subseteq TM$.

Then $B_{\chi} := H^{1/2}(\partial M, E_1)$ defines a right-elliptic complementable self-adjoint local boundary condition for $\mathcal{D}$. Here $E_1 \subseteq E|_{\partial M}$ denotes the eigenspace corresponding to the eigenvalue one of the involution χ . This boundary condition plays a crucial role in the proof of the band rigidity theorem by Cecchini and Zeidler [CZ24] and the scalar curvature rigidity theorem by Lott [Lot21]. It is illustrated in Example 5.7 in Appendix D how the new developed theory of boundary value problems could be useful in future work to generalize [Lot21] in the spirit of Theorem D.

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3 Appendix

Appendix A

Scalar curvature rigidity and the higher mapping degree

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Scalar curvature rigidity and the higher mapping degree



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ABSTRACT

A closed connected oriented Riemannian manifold N with non-vanishing Euler characteristic, non-negative curvature operator and $0 < 2\text{Ric}_N < \text{scal}_N$ is area-rigid in the sense that any area non-increasing spin map $f: M \rightarrow N$ with non-vanishing $\hat{A}$ -degree and $\text{scal}_M \geq \text{scal}_N \circ f$ is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$. This is due to Goette and Semmelmann and generalizes a result by Llarull. In this article, we show area-rigidity for not necessarily orientable manifolds with respect to a larger class of maps $f: M \rightarrow N$ by replacing the topological condition on the $\hat{A}$ -degree by a less restrictive condition involving the so-called higher mapping degree. This includes fiber bundles over even dimensional spheres with enlargeable fibers, e.g. $\text{pr}_1: S^{2n} \times T^k \rightarrow S^{2n}$. We develop a technique to extract from a non-vanishing higher index a geometrically useful family of almost $\mathcal{D}$ -harmonic sections. This also leads to a new proof of the fact that any closed connected spin manifold with non-negative scalar curvature and non-trivial Rosenberg index is Ricci flat.

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1. Introduction and main results

A classical result in scalar curvature comparison geometry by Llarull [27] states that a smooth area non-increasing map $f: M \rightarrow S^n$ from an n -dimensional closed connected Riemannian spin manifold of non-zero degree onto the round sphere, $n \geq 2$, with $\text{scal}_M \geq n(n-1) = \text{scal}_{S^n}$, is necessarily an isometry. An important consequence is that the round sphere is area-extremal in the sense that one cannot increase the metric and the scalar curvature simultaneously. Goette and Semmelmann [15] generalized this result to area non-increasing spin maps $f: M \rightarrow N$ of non-zero $\hat{A}$ -degree onto a closed connected oriented Riemannian manifold of non-vanishing Euler characteristic and non-negative curvature operator. Further generalizations are by Lott [28] to manifolds with boundary, by Cecchini, Hanke, and Schick [10] to Lipschitz maps with lower regularity, and by Bettiol and Goodman [5], who weakened the condition on the curvature operator of the target manifold in dimension four.

In this article, we generalize the extremality and rigidity statement of Goette and Semmelmann [15, Theorem 2.4] to spin maps between non-orientable manifolds and replace the topological condition on the $\hat{A}$ -degree by a less restrictive index-theoretical condition involving the so-called higher mapping degree (see Definition 5.2). Here spin map means that the map is compatible with the first and second Stiefel-Whitney classes of the involved manifolds.

Theorem A (Theorem 5.5 and Corollary 5.6). *Let $f: (M, g) \rightarrow (N, \bar{g})$ be an area non-increasing spin map between two closed connected Riemannian manifolds of dimension $n+k$ and n , respectively. Suppose that the curvature operator of N is non-negative and*

$$\chi(N) \cdot \deg_{\text{hi}}(f) \neq 0 \in \text{KO}_k(C^*\pi_1(M)).$$

Then $\text{scal}_M \geq \text{scal}_N \circ f$ on M implies $\text{scal}_M = \text{scal}_N \circ f$. If, moreover, $\text{scal}_N > 2\text{Ric}_N > 0$ (or f is distance non-increasing and $\text{Ric}_N > 0$), then $\text{scal}_M \geq \text{scal}_N \circ f$ implies that f is a Riemannian submersion.

Theorem A applies to the projections $\text{pr}_1: S^{2n} \times T^k \rightarrow S^{2n}$ and $\text{pr}_1: \mathbb{RP}^{2n} \times \Sigma^{8k+j} \rightarrow \mathbb{RP}^{2n}$ for $j \in \{1, 2\}$ and Σ^{8k+j} an exotic sphere with non-vanishing Hitchin invariant (Example 5.9 and Example 5.10). For both projections the $\hat{A}$ -degree vanishes, hence Theorem A is a proper extension of [15, Theorem 2.4]. The second example only works because of the generalization to non-orientable manifolds (see Remark 5.11).

The motivation for Theorem A is the fact that the Rosenberg index of a closed spin manifold M is the most general known index-theoretical obstruction to the existence of a positive scalar curvature metric on M . Since the Rosenberg index of the n -torus does not vanish, the n -torus does not admit a positive scalar curvature metric. This information

cannot be read off the classical index of the spin Dirac operator, hence the higher index carries more information. In particular, the classical rigidity statement by Goette and Semmelmann [15, Theorem 2.4] does not apply to $\text{pr}_1: S^{2n} \times T^k \rightarrow S^{2n}$, but the higher version in Theorem A does. A further obstruction to a positive scalar curvature metric on a closed spin manifold is enlargeability [16, Theorem A]. It also yields that the n -torus does not admit a positive scalar curvature metric. In general, the enlargeability obstruction for a closed spin manifold is dominated by the Rosenberg obstruction [17,18], so Theorem A also applies to certain maps with enlargeable fibers (see Corollary B and Example 5.8). Alternatively, the methods used in the proof of Theorem A would carry over to almost flat bundles as they are used in [18].

Theorem A yields a rigidity statement for fiber bundles over 2-connected Riemannian manifolds of non-zero Euler characteristic and non-negative curvature operator. A classification of these manifolds (Lemma 5.12) yields the following rigidity statement:

Corollary B (Corollary 5.13). *Let N be one of the following classes of Riemannian manifolds:*

- *Sphere S^{2n} , $n \geq 2$, equipped with a metric of non-negative curvature operator and $\text{scal}_{S^{2n}} > 2 \text{Ric}_{S^{2n}} > 0$.*
- *A quaternionic Grassmannian or the Cayley projective plane equipped with their symmetric Riemannian metrics.*

Let $f: M \rightarrow N$ be a fiber bundle, where the total space M is a closed connected spin manifold and the typical fiber F satisfies

$$\chi(N) \cdot \alpha(F) \neq 0 \in \text{KO}_{\dim(F)}(C^*\pi_1(F)) \quad (\text{e.g. } F \text{ is enlargeable}).$$

Then the map f is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$ if f is area non-increasing and the scalar curvatures satisfy $\text{scal}_M \geq \text{scal}_N \circ f$. Here $\chi(N)$ denotes the Euler characteristic of N , and $\alpha(F)$ denotes the Rosenberg index of the fiber.

1.1. Outline of the proof of Theorem A

Let us first analyze the proof of the classical theorem of Goette and Semmelmann [15, Theorem 2.4]. The key ingredient is the twisted spinor bundle $\mathcal{S} := \mathcal{S}M \otimes f^*SN \rightarrow M$, which contains all necessary information about the map f and the manifolds M and N . The proof divides into three steps. At first, the estimate

$$\mathcal{D}^2 \geq \nabla^* \nabla + \frac{1}{4}(\text{scal}_M - \text{scal}_N \circ f)$$

holds by using the Schrödinger–Lichnerowicz formula together with the non-negativity of the curvature operator of N and the fact that f is area non-increasing. Second, the Atiyah–Singer index theorem yields the index formula

$$\operatorname{ind} \mathcal{D} = \chi(N) \cdot \deg_{\hat{A}}(f), \quad \text{with } \deg_{\hat{A}}(f) := (\hat{A}(M)f^*\omega)[M].$$

Here ω is the fundamental class of N in cohomology and $[M]$ the fundamental class of M . Third, the non-vanishing of the Euler characteristic of N and the $\hat{A}$ -degree gives, combined with the index formula, a non-trivial $\mathcal{D}$ -harmonic section u . Inserting u into the estimate for $\mathcal{D}^2$ gives that u is even parallel. Using again the estimate for $\mathcal{D}^2$ yields the extremality and rigidity statement.

Let us start the outline of the proof of Theorem A. The spin map $f: M \rightarrow N$ gives rise to a graded Real $\mathbb{Cl}_{n+k,n}$ -linear Dirac bundle $\mathcal{S}M \hat{\otimes} f^*\mathcal{S}N$ (see Section 2.1). This is the spinor bundle associated to the vector bundle $TM \oplus f^*TN$ equipped with the indefinite metric $g \oplus (-f^*\bar{g})$ (see Section 2.2). This vector bundle is isomorphic as a Dirac bundle to some copies of the bundle $\mathcal{S}$ used by Goette and Semmelmann [15], but carries additionally the $\mathbb{Cl}_{n+k,n}$ -linear structure. Motivated by the definition of the Rosenberg index, we perform an additional twist with the Mishchenko bundle of M (see Section 2.3)

$$\mathcal{L} := \widetilde{M} \times_{\pi_1(M)} C^*\pi_1(M) \rightarrow M.$$

Here $\widetilde{M}$ is the universal cover of M and $C^*\pi_1(M)$ the maximal group C^* -algebra of the fundamental group of M . The Mishchenko bundle is equipped with a canonical flat connection. This twisted bundle carries the structure of a graded Real $\mathbb{Cl}_{n+k,n} \hat{\otimes} C^*\pi_1(M)$ -linear Dirac bundle with induced Dirac operator $\mathcal{D}_{\mathcal{L}}$. The classical estimates for $\mathcal{D}^2$ generalize to

$$\mathcal{D}_{\mathcal{L}}^2 \geq \nabla^*\nabla + \frac{1}{4}(\operatorname{scal}_M - \operatorname{scal}_N \circ f).$$

A cut-and-paste principle for the higher index, combined with the Poincaré–Hopf lemma, leads to an analogous index formula for $\mathcal{D}_{\mathcal{L}}$ as for $\mathcal{D}$ in the classical proof.

Theorem C (Theorem 5.4 and Definition 5.2). *The higher degree of a spin map $f: M \rightarrow N$ is defined as the higher index of the Dirac operator of the spinor bundle of the fiber M_p over a regular value p of f twisted by $\mathcal{L}$ restricted to M_p . Then the following holds*

$$\operatorname{ind} \mathcal{D}_{\mathcal{L}} = \chi(N) \cdot \deg_{\text{hi}}(f) \in \operatorname{KO}_k(C^*\pi_1(M)).$$

A non-vanishing higher index of an $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$ does not necessarily give rise to a non-trivial element in its kernel. The functional calculus can be used (Lemma 2.4 and Lemma 3.2) to produce a family of *almost $\mathcal{D}$ -harmonic* smooth sections $\{u_\epsilon\}_{\epsilon>0}$, i.e. that for all $\epsilon > 0$ and all $j \geq 1$

$$\|u_\epsilon\|_{L^2} = 1 \quad \text{and} \quad \|\mathcal{D}^j u_\epsilon\|_{L^2} < \epsilon^j$$

hold. The previous estimates on the L^2 -norm, together with the Schrödinger–Lichnerowicz formula, directly imply that a non-vanishing Rosenberg index is an obstruction to a positive scalar curvature metric on a closed spin manifold. To prove extremality and rigidity statements, the estimates on the L^2 -norm seem too weak to be geometrically useful. The main part of the proof of Theorem A is to develop a technique (Section 3) that shows in the extremal geometric situation that the family $\{u_\epsilon\}_{\epsilon>0}$ of almost $\mathcal{D}$ -harmonic sections is even *almost constant*. This means there exist constants $C, r > 0$ and an element $a \in A^+$ such that

$$\|a - \langle u_\epsilon(p), u_\epsilon(p) \rangle_p\|_A < C\epsilon^r$$

holds for all $p \in M$ and all $\epsilon \in (0, 1)$. A summary of the technical results:

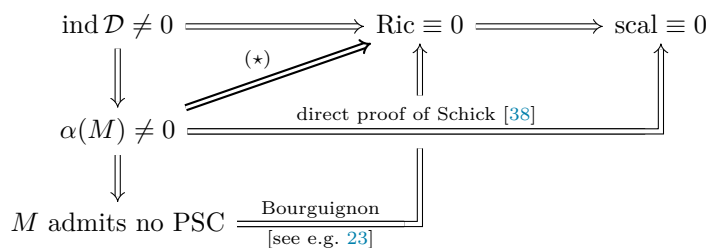
Lemma D (Lemma 2.4, Lemma 3.2, Lemma 3.3, Lemma 3.6 and Definition 3.1). *Let $\mathcal{A}$ be a graded Real unital C^* -algebra and $\mathcal{S}$ a graded Real $\mathcal{A}$ -linear Dirac bundle over a closed connected Riemannian manifold M with induced Dirac operator $\mathcal{D}$. If the higher index of $\mathcal{D}$ does not vanish in $KO_0(\mathcal{A})$, there exists a family $\{u_\epsilon\}_{\epsilon>0}$ of almost $\mathcal{D}$ -harmonic sections. If, moreover, $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ holds for all $\epsilon > 0$, the family $\{u_\epsilon\}_{\epsilon>0}$ is almost constant.*

In the proof of Lemma D, we use Moser iteration—similarly as in [2]—to deduce from $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ estimates on the L^∞ -norm (Lemma 3.4 and Lemma 3.5), and the Poincaré inequality to show that $\{u_\epsilon\}_{\epsilon>0}$ is almost constant (Lemma 3.6).

In the situation of Theorem A, Lemma D together with the index formula in Theorem C and the estimate for $\mathcal{D}_{\mathcal{L}}^2$ yields the existence of a family of almost constant sections $\{u_\epsilon\}_{\epsilon>0}$. This family replaces the $\mathcal{D}$ -harmonic section u in the classical proof of Goette and Semmelmann [15]. This property of being almost constant is sufficient to prove the extremality and rigidity statement similarly as in the classical proof.

1.2. A direct proof of a classical result about zero Ricci curvature

Let M be a closed connected spin manifold with non-negative scalar curvature. Denote the spin Dirac operator by $\mathcal{D}$. The implications in the following diagram hold [38,23]:



If the classical index of $\mathcal{D}$ does not vanish, there exists by $\text{scal} \geq 0$ a non-trivial constant section u , and a direct calculation shows that M is Ricci flat. If the Rosenberg index does not vanish, we obtain a family of almost constant sections by Lemma D and the Schrödinger–Lichnerowicz formula. The existence of such a family is enough to prove that M is Ricci flat, hence we obtain a direct proof of implication $(\star)$ in the previous diagram.

1.3. Structure

In Section 2, we introduce Dirac operators linear over C^* -algebras and their higher indices, spin structures associated to vector bundles with signature, and the Rosenberg index of a spin manifold. In Section 3, we prove Lemma D. We apply in Section 4 the new developed technique and give a direct proof of the classical result about the Rosenberg index and the Ricci curvature. In Section 5, we prove the extremality and rigidity statement (Theorem A) and the index theorem (Theorem C). There we also derive the theorem of Goette and Semmelmann [15] and treat some special versions of Theorem A involving the Rosenberg index of the fiber directly (Corollary 5.6). In the end of Section 5, we prove the rigidity statement for fiber bundles (Corollary B) over even dimensional spheres, quaternionic Grassmannians and the Cayley projective plane.

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2. Preliminaries

In this section, we summarize notations and definitions including $\mathcal{A}$ -linear Dirac operators and their higher indices, spin structures for vector bundles with indefinite metric, the Mishchenko line bundle and the Rosenberg index of a closed spin manifold. Furthermore, we give an overview of some important properties round about these concepts.

¹ Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or the European Research Council. Neither the European Union nor the granting authority can be held responsible for them.

2.1. $\mathcal{A}$ -linear Dirac operator and its higher index

Let (M, g) be an n -dimensional complete Riemannian manifold and $\mathcal{A}$ a unital graded Real C^* -algebra with norm denoted by $\|\cdot\|_{\mathcal{A}}$. At first, we generalize the definition of Dirac bundles and their induced Dirac operators [see 25, Chapter 2] to the corresponding $\mathcal{A}$ -linear structures:

Definition 2.1. An $\mathcal{A}$ -linear Dirac bundle over M is a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules $\mathcal{S} \rightarrow M$ equipped with an $\mathcal{A}$ -valued metric, a compatible $\mathcal{A}$ -linear connection and a parallel bundle map $c : TM \rightarrow \text{End}_{\mathcal{A}}(\mathcal{S})$, called the *Clifford multiplication*, such that $c(w)$ is skew-adjoint and satisfies $c(w)^2 = -g(w, w)$ for all $w \in TM$. We call an $\mathcal{A}$ -linear Dirac bundle *graded (Real)* if it is furnished with a grading (Real structure), which is compatible with the grading (Real structure) of $\mathcal{A}$, the metric, the connection and the Clifford multiplication c , i.e. $c(w)$ is odd (Real) for all $w \in TM$. If a grading and a Real structure exist, they have to be compatible with each other. The induced ($\mathcal{A}$ -linear) Dirac operator is defined as

$$\mathcal{D}_{\mathcal{S}} : C_c^{\infty}(M, \mathcal{S}) \xrightarrow{\text{connection}} C_c^{\infty}(M, T^*M \otimes \mathcal{S}) \xrightarrow{\text{metric, Clifford multiplication}} C_c^{\infty}(M, \mathcal{S}).$$

The Dirac operator induced by a (graded Real) $\mathcal{A}$ -linear Dirac bundle is, by the same considerations as in the classical case [25], an elliptic formally self-adjoint (odd Real) differential operator of first order. Details about bundles of finitely generated projective Hilbert C^* -modules are given in [29], [36] and [12], and about $\mathcal{A}$ -linear Dirac bundles in [40, §7].

Construction 2.2 (Twisted Dirac bundle). Let $\mathcal{S} \rightarrow M$ be a graded Real $\mathcal{A}$ -linear Dirac bundle, $\mathcal{B}$ a unital graded Real C^* -algebra and $E \rightarrow M$ a graded Real bundle of finitely generated projective Hilbert $\mathcal{B}$ -modules. Taking the fiberwise graded exterior tensor product of Hilbert C^* -modules with respect to the graded maximal tensor product between the C^* -algebras $\mathcal{A}$ and $\mathcal{B}$ defines a graded Real $\mathcal{A} \hat{\otimes} \mathcal{B}$ -linear Dirac bundle $\mathcal{S} \hat{\otimes} E \rightarrow M$. Here we equip $\mathcal{S} \hat{\otimes} E \rightarrow M$ with the tensor connection, the Clifford multiplication induced from the Clifford multiplication on $\mathcal{S}$, and the Real structure induced from the Real structures on $\mathcal{S}$ and E . If the C^* -algebra $\mathcal{B}$ and the Hilbert $\mathcal{B}$ -module E are not graded, we equip them in the previous construction with the trivial grading.

Notation 2.3. Let $\mathcal{S} \rightarrow M$ be an $\mathcal{A}$ -linear Dirac bundle. Denote by $\langle \cdot, \cdot \rangle_p$ the $\mathcal{A}$ -valued metric in the fiber of $\mathcal{S}$ over a point $p \in M$. Furthermore, define for compactly supported smooth sections $u, v \in C_c^{\infty}(M, \mathcal{S})$, $p \in M$ and $k, m \in \mathbb{N}_0$ the following scalar products and norms:

$$\langle u, v \rangle_p := \langle u_p, v_p \rangle_p \in \mathcal{A} \quad |u|_p := \langle u, u \rangle_p^{1/2} \in \mathcal{A}^+ \quad \|u\|_p := \|\langle u, u \rangle_p^{1/2}\|_{\mathcal{A}} \in \mathbb{R}$$

$$\begin{aligned}
\langle u, v \rangle_{L^2} &:= \int_M \langle u, v \rangle_p \, dp \in \mathcal{A} & \langle u, v \rangle_{H^k} &:= \sum_{j=0}^k \langle \nabla^j u, \nabla^j v \rangle_{L^2} \in \mathcal{A} \\
|u|_{L^2} &:= \langle u, u \rangle_{L^2}^{1/2} \in \mathcal{A}^+ & |u|_{H^k} &:= \langle u, u \rangle_{H^k}^{1/2} \in \mathcal{A}^+ \\
\|u\|_{L^2} &:= \left\| \langle u, u \rangle_{L^2}^{1/2} \right\|_{\mathcal{A}} \in \mathbb{R} & \|u\|_{H^k} &:= \left\| \langle u, u \rangle_{H^k}^{1/2} \right\|_{\mathcal{A}} \in \mathbb{R} \\
\|u\|_{\infty} &:= \max_{p \in M} \|u\|_p \in \mathbb{R} & \|u\|_{C^m} &:= \sum_{j=0}^m \|\nabla^j u\|_{\infty} \in \mathbb{R}
\end{aligned}$$

Here $\mathcal{A}^+$ denotes the set of positive elements in the C^* -algebra $\mathcal{A}$, and $\langle \nabla^j u, \nabla^j v \rangle_{L^2}$ is analogously defined as $\langle u, v \rangle_{L^2}$ with the induced $\mathcal{A}$ -valued inner product in the fibers of $T^*M^j \otimes \mathcal{S}$. Be aware of the fact that even if $|\cdot|_{H^k}$ is called a norm [see 24], it does not necessarily fulfill the triangle inequality. To prevent the risk of confusion with the $\mathcal{A}$ - and $\mathbb{R}$ -valued norms, we denote any $\mathbb{R}$ -valued norm by $\|\cdot\|$ and any $\mathcal{A}$ -valued norm by $|\cdot|$. The Hilbert $\mathcal{A}$ -module $H^k(M, \mathcal{S})$ is defined as the closure of the compactly supported smooth sections $C_c^\infty(M, \mathcal{S})$ in the norm $\|\cdot\|_{H^k}$. As classically, we write $L^2(M, \mathcal{S})$ for $H^0(M, \mathcal{S})$.

The next goal is to define the higher index of suitable $\mathcal{A}$ -linear differential operators, including certain $\mathcal{A}$ -linear Dirac operators and perturbed Dirac operators. The latter plays an important role in the proof of the index theorem (Theorem 5.4) in Section 5.

Fix a graded Real $\mathcal{A}$ -linear Dirac bundle $\mathcal{S} \rightarrow M$ together with an odd Real elliptic formally self-adjoint $\mathcal{A}$ -linear differential operator of first order $B: C_c^\infty(M, \mathcal{S}) \rightarrow C_c^\infty(M, \mathcal{S})$. The L^2 -closure, denoted by $B: \text{dom}(B) \rightarrow L^2(M, \mathcal{S})$, is self-adjoint and regular [12, Theorem 1.14] and its (possibly non-discrete) spectrum is Real [12, Proposition 1.10]. This gives rise to a functional calculus

$$\Phi_B: C(\overline{\mathbb{R}}) \longrightarrow \text{Lin}_{\mathcal{A}}(L^2(M, \mathcal{S})), \quad f \longmapsto f(B)$$

with properties as listed in [12, Theorem 1.19]. Here $C(\overline{\mathbb{R}})$ denotes the graded Real C^* -algebra of continuous functions $f: \mathbb{R} \rightarrow \mathbb{C}$ such that the limits to $\pm\infty$ exist, and $\text{Lin}_{\mathcal{A}}(L^2(M, \mathcal{S}))$ denotes the graded Real C^* -algebra of adjointable operators on $L^2(M, \mathcal{S})$.

Assume furthermore, that the unbounded operator B is *Fredholm*, i.e. that the adjointable operator $B(1 + B^2)^{-1/2}$, defined via the functional calculus of B , is invertible modulo a compact operator. This data gives directly per definition rise to a class in the picture of the Real K-theory of $\mathcal{A}$ given in [12, Definition 3.4]. We denote this class by $\text{ind } B$ and call it the *higher index* of B . Following Stolz [40, §9], we give a concrete construction of the higher index in the Fredholm picture $\text{KO}(\mathcal{A})$ of the Real K-theory of $\mathcal{A}$ [see 34, Lecture 12]. Elements of $\text{KO}_0(\mathcal{A})$ are homotopy classes of Fredholm pairs (F, E) consisting of a countably generated graded Real Hilbert $\mathcal{A}$ -module and an odd Real op-

erator $F \in \text{Lin}_{\mathcal{A}}(E)$ with $F - F^*, F^2 - 1 \in \text{Kom}_{\mathcal{A}}(E)$. Note that the Fredholm picture is equivalent to the KK-theoretic picture $\text{KK}(\mathbb{C}, \mathcal{A})$ used by Stolz [40, §9].

The unbounded operator B is Fredholm if and only if it has an *essential spectral gap*, i.e. there exists an $\epsilon > 0$ such that $f(B)$ is a compact operator for all continuous $f: \mathbb{R} \rightarrow \mathbb{R}$ with support in $(-\epsilon, \epsilon)$ [12, Lemma 2.19]. Fix such an $\epsilon > 0$. For a continuous odd function $g: \mathbb{R} \rightarrow [-1, 1]$ with $g(x) = \text{sgn}(x)$ on $\mathbb{R} \setminus (-\epsilon, \epsilon)$, the operator $g(B)^2 - 1$ is compact, hence we obtain a Fredholm pair $(g(B), L^2(M, \mathcal{S}))$. This defines a class in the Fredholm picture of the Real K-theory of $\mathcal{A}$, which is independent of ϵ and g , and matches by definition with the higher index of B

$$\text{ind } B = \left[(g(B), L^2(M, \mathcal{S})) \right] \in \text{KO}_0(\mathcal{A}) \cong \text{KK}_0(\mathbb{C}, \mathcal{A}).$$

The higher index is a cut-and-paste invariance as explained in [8, Section 2.3] and satisfies sum and product formula as mentioned in [43, Appendix B]. Furthermore, the higher index of a Dirac operator induced by a graded Real $\mathcal{A}$ -linear Dirac bundle is independent of the involved metrics by bordism invariance. The latter is a consequence of the partitioned manifold index theorem [see for instance 43, Appendix A].

For the Fredholmness of B , it is sufficient that B^2 is *uniformly positive at infinity* [12, Theorem 2.41], i.e. there exists a compact subset $K \subset M$ and a constant $c > 0$ such that

$$\langle B^2 u, u \rangle_{L^2} \geq c \langle u, u \rangle_{L^2}, \quad \forall u \in \text{dom}(B^2) \text{ with } \text{supp}(u) \cap K = \emptyset. \quad (2.1)$$

If B^2 is *uniformly positive*, i.e. equation (2.1) holds for $K = \emptyset$, the higher index vanishes:

Lemma 2.4. *Let $\mathcal{S} \rightarrow M$ be a graded Real $\mathcal{A}$ -linear Dirac bundle and $B: \text{dom}(B) \rightarrow L^2(M, \mathcal{S})$ the L^2 -closure of an odd Real elliptic formally self-adjoint differential operator of first order. Then the following holds:*

- (i) *If B^2 is uniformly positive, B is invertible.*
- (ii) *If B is invertible, B is Fredholm and its higher index vanishes in $\text{KO}_0(\mathcal{A})$.*

Proof. For the first statement see [12, Proposition 1.20 and 1.21]. Suppose B is invertible. Then zero is not in the spectrum of B , and there exists an $\epsilon > 0$ such that $(-\epsilon, \epsilon)$ does not intersect with the spectrum of B . Since $f(B)$ just depends on the values the function f takes on the spectrum of B , we obtain $f(B) = 0$ for any function f supported in $(-\epsilon, \epsilon)$, hence B has an essential spectral gap. This shows that B is Fredholm. Furthermore, $g(B)^2 - 1$ is equal to zero for a function g as in the construction of the higher index, hence $(g(B), L^2(M, \mathcal{S}))$ vanishes in $\text{KO}_0(\mathcal{A})$ by [34, Lemma 12.6]. $\square$

So far, the underlying manifold was not necessarily closed. From now on assume that the manifold M is closed. Then the induced Dirac operator $\mathcal{D}$ of an $\mathcal{A}$ -linear Dirac bundle $\mathcal{S} \rightarrow M$ with initial domain $C^\infty(M, \mathcal{S})$ extends for all $k \in \mathbb{N}_0$ to an unbounded

operator $\mathcal{D} : H^{k+1}(M, \mathcal{S}) \rightarrow H^k(M, \mathcal{S})$ [29, Lemma 3.2]. Furthermore, the L^2 -extended Dirac operator is obviously uniformly positive at infinity, hence Fredholm.

The classical elliptic regularity principle carries over to $\mathcal{A}$ -linear Dirac operators, which means precisely that the classical Sobolev embedding theorem [25, Chapter III Theorem 2.5 and Theorem 2.15] and the classical fundamental elliptic estimate [25, Chapter III Theorem 5.2] generalizes as follows:

Lemma 2.5. *Let $k, m, l \in \mathbb{N}_0$ with $k > m + \frac{n}{2}$ and $\mathcal{S} \rightarrow M$ an $\mathcal{A}$ -linear Dirac bundle with induced Dirac operator $\mathcal{D}$. There exists a constant C such that for all $u \in H^{l+1}(M, \mathcal{S})$ and all $v \in H^k(M, \mathcal{S})$ the following holds:*

- (i) $\|v\|_{C^m} \leq C\|v\|_{H^k}$ (Sobolev inequality)
- (ii) $\|u\|_{H^{l+1}} \leq C(\|\mathcal{D}u\|_{H^l} + \|u\|_{H^l})$ (fundamental elliptic estimate)

Sketch of proof. Since the manifold M is closed, it is as in the classical case enough to show the estimates for the trivial bundle $T^n \times P$ over the flat torus and a first-order elliptic differential operator. Here P denotes a finitely generated projective Hilbert $\mathcal{A}$ -module. The general case follows by considering an appropriate partition of unity and picking a good system of trivializations of the bundle $\mathcal{S}$ as explained in [25, Chapter III §2]. On the torus, we can use Fourier transform methods, which generalize from $\mathbb{C}^m$ -valued functions to functions with values in the Hilbert $\mathcal{A}$ -module P [see 29, §3]. This implies the Sobolev inequality by the same considerations as in [25, Chapter III §2]. The fundamental elliptic estimate follows by induction from the Gårding's inequality, whose classical proof in [21, Section 10.4.4] carries over without change. $\square$

2.2. Spin structure of vector bundles with indefinite metric

In this section, we define spin structures for vector bundles equipped with a non-degenerate not necessarily positive definite metric. For this purpose, we mix the definition of spin structures for vector bundles with a positive definite metric [25, Chapter II] with the notion of a spin structure for semi-Riemannian manifolds [4, 11, 1].

For $m, n \in \mathbb{N}_0$ denote by $\mathbb{Cl}_{m,n}$ the graded Real C^* -algebra generated by $\mathbb{R}^m \oplus \mathbb{R}^n$ such that $(v, w)^* = (-v, w)$ and

$$(v, w)(v', w') + (v', w')(v, w) = -2\langle v, v' \rangle_{\text{eu}} + 2\langle w, w' \rangle_{\text{eu}}$$

hold for all $(v, w), (v', w') \in \mathbb{R}^m \oplus \mathbb{R}^n$, and all elements of $\mathbb{R}^m \oplus \mathbb{R}^n$ are odd [see 12, Section 1.1]. The pin group $\text{Pin}_{m,n}$ is the subgroup of $\mathbb{Cl}_{m,n}$ generated by $(v, w) \in \mathbb{R}^n \oplus \mathbb{R}^m \subset \mathbb{Cl}_{m,n}$ with $\|v\|_{\text{eu}} - \|w\|_{\text{eu}} = \pm 1$. Furthermore, we define the subgroups $\text{Spin}_{m,n}$ and $\text{Spin}_{m,n}^+$ of $\text{Pin}_{m,n}$ as the preimage of $\text{SO}(m, n)$ and $\text{SO}^+(m, n)$, respectively, along the two-folded covering

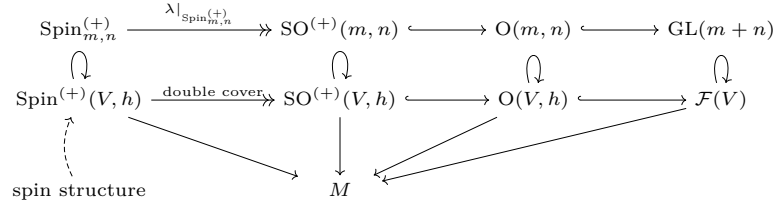


Fig. 1. Spin structure of a rank $m + n$ vector bundle V over a connected manifold M equipped with a metric h of signature (m, n) .

$$\lambda: \text{Pin}_{m,n} \longrightarrow \text{O}(m, n), \quad \lambda(q)(x) := (-1)^{\deg(q)} q x q^{-1}.$$

Here $\text{SO}^+(m, n)$ denotes the identity component of $\text{SO}(m, n)$. The spin group $\text{Spin}_{m,n}$ matches with the even part of $\text{Pin}_{m,n}$, and $\text{Spin}_{m,n}^+$ is equal to the identity component of $\text{Spin}_{m,n}$.

Let M be a connected manifold and V an orientable vector bundle of rank $m + n \geq 3$ equipped with a metric h of signature (m, n) . The existence of such a metric is equivalent to the fact that the frame bundle $\mathcal{F}(V)$ reduces to a $\text{O}(m, n)$ -principle bundle $\text{O}(V, h)$ [4, Satz 0.47]. We fix a splitting of V into two orthogonal sub-bundles V_1 and V_2 such that the induced metrics on V_1 and V_2 are positive and negative definite, respectively. The Stiefel–Whitney classes of V_1 and V_2 do not depend on that choice. If V_1 and V_2 are both orientable, the principle bundle $\text{O}(V, h)$ has four connected components and it reduces to a connected $\text{SO}^+(m, n)$ -principle bundle $\text{SO}(V, h)$. Otherwise, it has 2-connected components and reduces to a connected $\text{SO}(m, n)$ -principle bundle $\text{SO}(V, h)$ [4, Satz 0.51]. From now on, we distinguish between these two cases and write “+” in parentheses.

A *spin structure* of (V, h) is a reduction of $\text{SO}^{(+)}(V, h)$ to a $\text{Spin}_{m,n}^{(+)}$ -principle bundle $\text{Spin}(V, h)$ along the restriction of λ onto $\text{Spin}_{m,n}^{(+)}$. An overview of all reductions and lifts of the frame bundle of V is given in Fig. 1. There exists a spin structure for (V, h) if and only if $w_2(V_1) + w_2(V_2) = 0$ holds [4, Satz 2.2]. Note that if both sub-bundles V_1 and V_2 are orientable, this condition is equivalent to

$$w_2(V) = w_2(V_1) + w_2(V_2) + w_1(V_1)w_1(V_2) = 0,$$

hence it is equivalent to the existence of a *topological spin structure* on V . This is a lift of $\mathcal{F}^+(V)$ along the two-fold covering $\widetilde{\text{GL}}^+(m+n) \rightarrow \text{GL}^+(m+n)$, where $\mathcal{F}^+(V)$ denotes a reduction of $\mathcal{F}(V)$ along the inclusion $\text{GL}^+(m+n) \hookrightarrow \text{GL}(m+n)$.

In the remaining section, we will construct the $\text{Cl}_{m,n}$ -linear spinor bundle from a given spin structure on (V, h) . Let $K_{m,n}^{(\pm)} \subset \text{Spin}_{m,n}^{(\pm)}$ denote the maximal compact subgroups covering $S(\text{O}_m \times \text{O}_n) \subset \text{SO}_{m,n}$ and $\text{SO}_m \times \text{SO}_n \subset \text{SO}_{m,n}^+$, respectively. Since $K_{m,n}^{(\pm)} \subset \text{Spin}_{m,n}^{(\pm)}$ is a deformation retract, any $\text{Spin}_{m,n}^{(\pm)}$ -principle bundle reduces to a $K_{m,n}^{(\pm)}$ -principle bundle. Therefore, a fixed spin structure on (V, h) gives rise to a $K_{m,n}^{(\pm)}$ -principle bundle $P_{K_{m,n}^{(\pm)}}(V, h)$, and we obtain a vector bundle

$$\mathcal{S}(V, h) := P_{K_{m,n}^{(\pm)}}(V, h) \times_{K_{m,n}^{(\pm)}} \mathbb{C}l_{m,n},$$

where $K_{m,n}^{(\pm)} \subset \mathbb{C}l_{m,n}$ acts as left multiplication on $\mathbb{C}l_{m,n}$. The vector bundle $\mathcal{S}(V, h)$ inherits a Real structure, a grading and a right $\mathbb{C}l_{m,n}$ -action. The fiberwise canonical $\mathbb{C}l_{m,n}$ -valued metric on $\mathbb{C}l_{m,n}$ defines a global positive definite $\mathbb{C}l_{m,n}$ -valued metric on $\mathcal{S}(V, h)$. Here it is important that we used the reduction of the $\text{Spin}_{m,n}^{(\pm)}$ -principle bundle $\text{Spin}^{(+)}(V, h)$ to the subgroup $K_{m,n}^{(\pm)}$ in the construction of $\mathcal{S}(V, h)$. The map $(\mathbb{R}^m \oplus \mathbb{R}^n) \times \mathbb{C}l_{m,n} \rightarrow \mathbb{C}l_{m,n}$ given by left multiplication induces a well-defined bundle map

$$\text{cl}: V \cong \text{SO}_{m,n}^{(+)}(m, n) \times_{\text{SO}_{m,n}^{(+)}} (\mathbb{R}^m \oplus \mathbb{R}^n) \longrightarrow \text{End}_{\mathbb{C}l_{m,n}}(\mathcal{S}(V, h)),$$

which is odd, Real, satisfies $\text{cl}(v)^2 = -h(v, v)$ for all $v \in V$, and $\text{cl}(v)$ is skew-adjoint for $v \in V_1$ and self-adjoint for $v \in V_2$. If the vector bundle V is equipped with a metric connection, the vector bundle $\mathcal{S}(V, h)$ inherits a canonical connection making all structures parallel. We call $\mathcal{S}(V, h)$ the $\mathbb{C}l_{m,n}$ -linear spinor bundle induced by $\text{Spin}(V, h)$.

For an orientable Riemannian manifold (M, g) of dimension m with non-vanishing second Stiefel–Whitney class the previous construction with $(V, h) = (TM, g)$ matches with the classical spin structure in [25, Chapter II §1 and §2]. If the tangent bundle is equipped with the Levi-Civita connection, the induced spinor bundle $\mathcal{S} := \mathcal{S}(TM, g)$ is the $\mathbb{C}l_m$ -linear Dirac bundle in [25, Chapter II §8].

2.3. The Mishchenko bundle and the Rosenberg index

Let M be a closed connected manifold of dimension m with fundamental group π and universal cover $\widetilde{M}$. Denote by $C^*\pi$ the maximal group C^* -algebra of π equipped with the Real structure induced by the natural Real structure on $\mathbb{C}$. The *Mishchenko (line) bundle* of M [29] is defined by

$$\mathcal{L}(M) := \mathcal{L} := \widetilde{M} \times_{\pi} C^*\pi,$$

where π acts on $C^*\pi$ by left multiplication. The bundle $\mathcal{L}$ comes with a canonical flat connection and a $C^*\pi$ -valued product given by $\langle a, b \rangle := a^*b$, hence $\mathcal{L}$ is a flat bundle of finitely generated projective Hilbert $C^*\pi$ -modules together with a Real structure.

Suppose the manifold M is equipped with a spin structure. Twisting the $\mathbb{C}l_m$ -linear spinor bundle $\mathcal{S}$ of M by the Mishchenko bundle $\mathcal{L}$ as in Construction 2.2 gives a graded Real $\mathbb{C}l_m \widehat{\otimes} C^*\pi$ -linear Dirac bundle $\mathcal{S}_{\mathcal{L}} := \mathcal{S} \widehat{\otimes} \mathcal{L}$. The higher index of the induced $\mathbb{C}l_m \widehat{\otimes} C^*\pi$ -linear Dirac operator $\mathcal{D}_{\mathcal{L}}$, denoted by

$$\alpha(M) := \text{ind}(\mathcal{D}_{\mathcal{L}}) \in \text{KO}_0(\mathbb{C}l_m \widehat{\otimes} C^*\pi) \cong \text{KO}_m(C^*\pi), \quad (2.2)$$

is called the *Rosenberg index of M* [35]. There is a well-known connection between the Rosenberg index and the scalar curvature of M :

Proposition 2.6. *If the scalar curvature of the manifold M is positive, then the Rosenberg index of M vanishes.*

Proof. Assume $\text{scal}_M > 0$. Since the Mishchenko bundle $\mathcal{L}$ is flat, the Schrödinger–Lichnerowicz formula yields $\mathcal{D}_{\mathcal{L}}^2 = \nabla^* \nabla + \frac{\text{scal}_M}{4}$, and we obtain $\langle \mathcal{D}_{\mathcal{L}}^2 u, u \rangle_{L^2} \geq \min_{p \in M} \text{scal}_M(p) \langle u, u \rangle_{L^2}$ for all $u \in H^2(M, \mathcal{S}_{\mathcal{L}})$. Note that $\min_{p \in M} \text{scal}_M(p)$ is strictly positive by assumption on the scalar curvature. By Lemma 2.4 part (1), the Dirac operator $\mathcal{D}_{\mathcal{L}}$ is invertible and the Rosenberg index of M vanishes by part (2) of the same lemma. $\square$

There is a meta-conjecture by Schick [37, Conjecture 1.5], that any obstruction to positive scalar curvature on a closed spin manifold (of dimension at least five) based on index theory of Dirac operators can be read off the Rosenberg index. The following proposition is a well-known result in this direction and justifies why the Mishchenko bundle is the right bundle for the twist with the spinor bundle. In particular, it states that any obstruction to a positive scalar curvature metric on M , obtained from a higher index of the spin Dirac operator twisted with any Real flat bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules, can be read off the Rosenberg index.

Proposition 2.7. *Let $\mathcal{S} \rightarrow M$ be a graded Real $\mathcal{A}$ -linear Dirac bundle, $\mathcal{B}$ a unital Real C^* -algebra and $E \rightarrow M$ a Real flat bundle of finitely generated projective $\mathcal{B}$ -modules. Denote by $\mathcal{D}_E$ and $\mathcal{D}_{\mathcal{L}}$ the induced Dirac operators of $\mathcal{S}$ twisted with the bundle E , respectively the Mishchenko bundle $\mathcal{L}$. Then the following holds:*

$$\text{ind } \mathcal{D}_E \neq 0 \in \text{KO}(\mathcal{A} \widehat{\otimes} \mathcal{B}) \implies \text{ind } \mathcal{D}_{\mathcal{L}} \neq 0 \in \text{KO}(\mathcal{A} \widehat{\otimes} C^*\pi)$$

Proof. We orientate towards the proof of a similar, but slightly weaker, result by Schick and Wraith [39, Proposition 3.1 and Theorem 3.2]. Let P be the typical fiber of the flat bundle E . Then there exists a representation $\rho: \pi \rightarrow U_{\mathcal{B}}(P)$ on the unitary operators on the Real finitely generated projective Hilbert $\mathcal{B}$ -module P such that $E \cong \widetilde{M} \times_{\rho} P$. By the universal property of the maximal group C^* -algebra, this representation extends to a C^* -algebra homomorphism $\bar{\rho}: C^*\pi \rightarrow \text{Lin}_{\mathcal{B}}(P)$ that respects the grading and the Real structure. The representation

$$\psi := \text{id}_{\mathcal{A}} \widehat{\otimes} \bar{\rho}: \mathcal{A} \widehat{\otimes} C^*\pi \rightarrow \mathcal{A} \widehat{\otimes} \text{Lin}_{\mathcal{B}}(P) \cong \text{Lin}_{\mathcal{A} \widehat{\otimes} \mathcal{B}}(\mathcal{A} \widehat{\otimes} P)$$

induces as in the proof of [34, Lemma 12.17] a homomorphism on Real K-theory, which takes in the Fredholm picture the form

$$(\psi)_0: \text{KO}_0(\mathcal{A} \widehat{\otimes} C^*\pi) \longrightarrow \text{KO}_0(\mathcal{A} \widehat{\otimes} \mathcal{B}), [(F, Q)] \longmapsto [(F \otimes \text{id}_{\mathcal{A} \widehat{\otimes} P}, Q \otimes_{\psi} (\mathcal{A} \widehat{\otimes} P))].$$

Here $\otimes_{\psi}$ denotes the interior tensor product with respect to the representation ψ . Furthermore, the bundle E is related to the Mishchenko bundle via $E \cong \widetilde{M} \times_{\rho} P \cong \widetilde{M} \times_{\rho} (C^*\pi \otimes_{\bar{\rho}} P) \cong \mathcal{L} \otimes_{\bar{\rho}} P$, which implies

$$L^2(M, \mathcal{S} \widehat{\otimes} \mathcal{L}) \otimes_{\psi} (\mathcal{A} \widehat{\otimes} P) \cong L^2(M, (\mathcal{S} \widehat{\otimes} \mathcal{L}) \otimes_{\psi} (\mathcal{A} \widehat{\otimes} P)) \cong L^2(M, \mathcal{S} \widehat{\otimes} E).$$

Note that the tensor product of a bundle of Hilbert C^* -modules with a Hilbert C^* -module is always taken fiberwise. Together with $\mathcal{D}_{\mathcal{L}} \otimes \text{id}_{\mathcal{A} \widehat{\otimes} P} = \mathcal{D}_E$, we obtain

$$(\psi)_0: \text{ind } \mathcal{D}_{\mathcal{L}} = [(\mathcal{D}_{\mathcal{L}}, L^2(M, \mathcal{S} \widehat{\otimes} \mathcal{L}))] \longmapsto \text{ind } \mathcal{D}_E = [(\mathcal{D}_E, L^2(M, \mathcal{S} \widehat{\otimes} E))],$$

and the proposition is proved. $\square$

3. Almost harmonic, almost parallel and almost constant sections

Let $\mathcal{S}$ be a classical graded Dirac bundle over a closed connected Riemannian manifold M . If the index of the induced Dirac operator $\mathcal{D}$ is non-zero, there exists a non-trivial smooth section u in the kernel of $\mathcal{D}$. Furthermore, if u is additionally parallel, i.e. $\nabla u = 0$, the map $f: M \rightarrow \mathbb{R}$, $p \mapsto \|u\|_p$, is constant. In this section, we generalize this result to an $\mathcal{A}$ -linear Dirac operator induced by a graded Real $\mathcal{A}$ -linear Dirac bundle over a closed connected Riemannian manifold. It turns out that if the higher index of the Dirac operator does not vanish in $\text{KO}(\mathcal{A})$, there exists a family of almost $\mathcal{D}$ -harmonic sections (see Definition 3.1). We will furthermore study under which conditions these sections are almost constant in the sense of Definition 3.1.

At first, we make the notion of almost $\mathcal{D}$ -harmonic, almost parallel and almost constant precise. Second, we construct almost $\mathcal{D}$ -harmonic sections under the hypothesis that the $\mathcal{A}$ -linear Dirac operator is not invertible (Lemma 3.2), which is the case, if its higher index does not vanish. Last, we study the relation between almost $\mathcal{D}$ -harmonic, almost parallel and almost constant sections in general.

We fix a graded Real $\mathcal{A}$ -linear Dirac bundle $\mathcal{S}$ over a closed connected Riemannian manifold M of dimension n with induced Dirac operator $\mathcal{D}$.

Definition 3.1. A family $\{u_{\epsilon}\}_{\epsilon>0}$ of smooth sections of $\mathcal{S}$, which is L^2 -normalized, i.e. $\|u_{\epsilon}\|_{L^2} = 1$ for all $\epsilon > 0$, is called

- (1) *almost $\mathcal{D}$ -harmonic* if $\|\mathcal{D}^j u_{\epsilon}\|_{L^2} < \epsilon^j$ for all $\epsilon > 0$ and all $j \geq 1$.
- (2) *almost parallel* if it satisfies $\|\nabla u_{\epsilon}\|_{L^2} < \epsilon$ for all $\epsilon > 0$ and all covariant derivatives are uniformly bounded in $\epsilon \in (0, 1)$, i.e. for all $m \in \mathbb{N}_0$ there exists a constant C_m such that $\|\nabla^m u_{\epsilon}\|_{\infty} < C_m$ for all $\epsilon \in (0, 1)$.
- (3) *almost constant* if there exist constants $C, r > 0$ such that

$$\|\overline{u}_{\epsilon} - |u_{\epsilon}|_p^2\|_{\mathcal{A}} < C\epsilon^r, \quad \text{with} \quad \overline{u}_{\epsilon} := \frac{1}{\text{vol}M} \int_M |u_{\epsilon}|_q^2 dq,$$

for all $\epsilon \in (0, 1)$ and all $p \in M$.

In the next lemma, we give a connection between the existence of a family of almost $\mathcal{D}$ -harmonic sections and the invertibility of the Dirac operator $\mathcal{D}$. Combined with Lemma 2.4 (2), we obtain: If the index of $\mathcal{D}$ does not vanish in $\mathrm{KO}_0(\mathcal{A})$, there exists a family of almost $\mathcal{D}$ -harmonic sections.

Lemma 3.2. *If the Dirac operator $\mathcal{D}$ is not invertible, there exists a family of almost $\mathcal{D}$ -harmonic sections.*

Proof. We fix for all $\epsilon > 0$ a smooth function $f_\epsilon: \mathbb{R} \rightarrow \mathbb{R}$ with $f_\epsilon(\mathbb{R}) \subset [0, 2]$, $\mathrm{supp}(f_\epsilon) \subset (-\frac{\epsilon}{2}, \frac{\epsilon}{2})$ and $f_\epsilon(0) = 2$. By the functional calculus, we obtain adjointable operators $f_\epsilon(\mathcal{D}): L^2(M, \mathcal{S}) \rightarrow L^2(M, \mathcal{S})$ satisfying

$$\|f_\epsilon(\mathcal{D})\|_{\mathrm{op}} = \sup_{t \in \sigma(\mathcal{D})} |f_\epsilon(t)| = 2. \quad (3.1)$$

Here we used in the second step that zero is in the spectrum of $\mathcal{D}$. We conclude from equation (3.1) that there exists $\tilde{v}_\epsilon \in L^2(M, \mathcal{S})$ with $\|\tilde{v}_\epsilon\|_{L^2} = 1$ and $\|f_\epsilon(\mathcal{D})\tilde{v}_\epsilon\|_{L^2} \geq 1$. Define a family of sections via

$$u_\epsilon := f_\epsilon(\mathcal{D})v_\epsilon \in L^2(M, \mathcal{S}) \quad \text{with} \quad v_\epsilon := \frac{\tilde{v}_\epsilon}{\|f_\epsilon(\mathcal{D})\tilde{v}_\epsilon\|_{L^2}}.$$

All u_ϵ satisfy $\|u_\epsilon\|_{L^2} = 1$ by definition, hence it remains to show that all u_ϵ are smooth, and the family is indeed almost $\mathcal{D}$ -harmonic. We fix an arbitrary $j \in \mathbb{N}$ and $\epsilon > 0$. Using the properties of the functional calculus as in [12, Theorem 1.19] inductively gives that u_ϵ is in the domain of $\mathcal{D}^j$ and $\mathcal{D}^j u_\epsilon$ equals $g_\epsilon^j(\mathcal{D})v_\epsilon$ for $g_\epsilon^j(t) = t^j f_\epsilon(t)$, $t \in \mathbb{R}$. This gives

$$\|\mathcal{D}^j u_\epsilon\|_{L^2} = \|g_\epsilon^j(\mathcal{D})v_\epsilon\|_{L^2} \leq \|g_\epsilon^j(\mathcal{D})\|_{\mathrm{op}} \underbrace{\|v_\epsilon\|_{L^2}}_{\leq 1} \leq \sup_{t \in \sigma(\mathcal{D})} |t^j f_\epsilon(t)| < \epsilon^j.$$

The domain of $\mathcal{D}^j$ equals $H^j(M, \mathcal{S})$, so that u_ϵ is in $H^j(M, \mathcal{S})$ by the previous considerations. It follows, by the Sobolev inequality in Lemma 2.5 (i), that u_ϵ is smooth, and the lemma is proved. $\square$

We will see in Lemma 3.6, that any family of almost parallel sections is almost constant. This turns the existence of a family of almost parallel sections into a useful tool to deduce geometric consequence, as the extremality and rigidity statements in Section 5. The following lemma connects the existence of almost $\mathcal{D}$ -harmonic sections with the existence of almost parallel sections:

Lemma 3.3. *Any family $\{u_\epsilon\}_{\epsilon>0}$ of almost $\mathcal{D}$ -harmonic sections with $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ for all $\epsilon > 0$ is almost parallel.*

Proof. Let $\{u_\epsilon\}_{\epsilon>0}$ be a family of almost $\mathcal{D}$ -harmonic sections satisfying $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ for all $\epsilon > 0$. We fix an $m \geq 0$ and a $k > m + \frac{n}{2}$. There exist, by the Sobolev inequality in Lemma 2.5 (i) and an iterated version of the fundamental elliptic estimate in Lemma 2.5 (ii), positive constants C_1 and C_m such that

$$\|\nabla^m u_\epsilon\|_\infty \leq \|u_\epsilon\|_{C^m} \leq C_1 \|u_\epsilon\|_{H^k} \leq C_m \sum_{j=0}^k \|\mathcal{D}^j u_\epsilon\|_{L^2} < C_m(1+k)$$

holds for all $\epsilon \in (0, 1)$. Here we used in the first step the definition of the C^m -norm and in the last step that $\{u_\epsilon\}_{\epsilon>0}$ is almost $\mathcal{D}$ -harmonic. This shows that for all $m \in \mathbb{N}_0$ the m -th covariant derivative of u_ϵ is uniformly bounded in $\epsilon \in (0, 1)$, hence $\{u_\epsilon\}_{\epsilon>0}$ is almost parallel. $\square$

The remaining section is dedicated to the statement that any family of almost parallel sections is almost constant (Lemma 3.6). In the first step we use Moser iteration (Lemma 3.4) to deduce from the L^2 -estimates on the covariant derivations on u_ϵ , estimates on the L^∞ -norm. It is crucial for this step, that all covariant derivatives of u_ϵ are uniformly bounded in $\epsilon \in (0, 1)$. In the second step, we use the Poincaré inequality to show that any family of almost parallel sections is almost constant.

The version of Moser iteration—as stated in the following lemma—is a combination of [31, Theorem 9.2.7] and the corrected version of [32, Lemma 3.1] in the erratum [33, Lemma 0.1]. We use the non-negative Laplace operator defined via $\Delta f := -\text{tr}(\text{Hess } f)$.

Lemma 3.4 (Moser iteration). *Let $f: M \rightarrow \mathbb{R}^+ := \mathbb{R}_{\geq 0}$ be a continuous function that is smooth on $\{f > 0\}$ and satisfies $\Delta f \leq \alpha$ on $\{f > 0\}$ for a positive constant α . Then there exist constants $C > 0$ and $r \in (e^{-(n-2)\ln(2)/2}, 1]$ such that*

$$\|f\|_\infty \leq C \|f\|_{L^2}^r$$

holds. The constant C just depends on the constant α and the dimension, the diameter and the Ricci curvature of the manifold M .

Lemma 3.5. *Let $\{u_\epsilon\}_{\epsilon>0}$ be a family of almost parallel sections. Then there exist constants $C, r > 0$ such that $\|\nabla u_\epsilon\|_\infty < C\epsilon^r$ holds for all $\epsilon \in (0, 1)$.*

Proof. Let $\{u_\epsilon\}_{\epsilon>0}$ be a family of almost parallel sections. Note that the norm of a self-adjoint element $a \in \mathcal{A}$ is given by

$$\|a\|_{\mathcal{A}} = \max_{\rho \text{ state of } \mathcal{A}} |\rho(a)|, \quad (3.2)$$

where a state of the C^* -algebra $\mathcal{A}$ is a positive normalized functional $\rho: \mathcal{A} \rightarrow \mathbb{C}$ [30, Section 3.3]. We define for any $\epsilon \in (0, 1)$ and any state ρ of the C^* -algebra $\mathcal{A}$ a function

$$f_{\epsilon,\rho}: M \longrightarrow \mathbb{R}^+, p \longmapsto \rho(|\nabla u_\epsilon|_p^2)^{1/2}.$$

Since all u_ϵ are smooth, all functions $f_{\epsilon,\rho}$ are continuous and smooth on $\{f_{\epsilon,\rho} > 0\}$. We will show, that there exists a constant $\alpha > 0$, independent of ϵ and the state ρ , such that

$$\Delta f_{\epsilon,\rho}(p) \leq \alpha \quad (3.3)$$

holds for all $p \in M$ with $f_{\epsilon,\rho}(p) \neq 0$, all $\epsilon \in (0, 1)$ and all states ρ . Moser iteration (Lemma 3.4) would be applicable and would give for a suitable constant $C > 0$, independent of ϵ and ρ , and constants $r_{\epsilon,\rho} \in (e^{-(n-2)\ln(2)/2}, 1]$

$$\begin{aligned} \rho(|\nabla u_\epsilon|_p^2) &\leq \|f_{\epsilon,\rho}\|_\infty^2 \leq C^2 \|f_{\epsilon,\rho}\|_{L^2}^{2r_{\epsilon,\rho}} = C^2 \left(\int_M \rho(|\nabla u_\epsilon|_p^2) \, dp \right)^{r_{\epsilon,\rho}} \\ &\leq C^2 \|\nabla u_\epsilon\|_{L^2}^{2r_{\epsilon,\rho}} \\ &< C^2 \epsilon^{2r} \end{aligned} \quad (3.4)$$

for all $p \in M$, all $\epsilon \in (0, 1)$ and all states ρ . Here we used equation (3.2), the almost parallelity of u_ϵ and $r := e^{-(n-2)\ln(2)/2} < r_{\epsilon,\rho}$. This would yield immediately

$$\|\nabla u_\epsilon\|_p^2 = \max_{\rho \text{ state}} \rho(|\nabla u_\epsilon|_p^2) \stackrel{(3.4)}{<} C^2 \epsilon^{2r}$$

for all $p \in M$, and the lemma would be proved. It remains to show equation (3.3). Fix for the rest of the proof a point $p \in M$ with $f(p) \neq 0$, $\epsilon \in (0, 1)$, a state ρ and write f for $f_{\epsilon,\rho}$. We calculate $\Delta f^2(p)$ in two different ways:

$$\begin{aligned} \bullet \quad \Delta f^2(p) &= \rho\left(\Delta\langle\nabla u_\epsilon, \nabla u_\epsilon\rangle_p\right) \\ &= \rho\left(\langle\nabla^*\nabla(\nabla u_\epsilon), \nabla u_\epsilon\rangle_p + \langle\nabla u_\epsilon, \nabla^*\nabla(\nabla u_\epsilon)\rangle_p\right) - 2\rho\left(|\nabla\nabla u_\epsilon|_p^2\right) \\ \bullet \quad \Delta f^2(p) &= 2f(p)\Delta f(p) - 2\|\nabla f\|_p^2 \end{aligned} \quad (3.5)$$

Here we used that ρ commutes with the Laplacian and the connection. Note that

$$\rho(a^* + a)^2 = \left(\overline{\rho(a)} + \rho(a)\right)^2 \leq 4\overline{\rho(a)}\rho(a) = 4\rho(a^*)\rho(a) \quad \text{and} \quad (3.6)$$

$$\rho(\langle x, y \rangle)\rho(\langle y, x \rangle) \leq \rho(\langle x, x \rangle)\rho(\langle y, y \rangle) \quad (3.7)$$

hold for all $a, b \in \mathcal{A}$ and all x, y in a Hilbert $\mathcal{A}$ -module P with inner product $\langle \cdot, \cdot \rangle$. The estimate in (3.7) is the Cauchy-Schwarz inequality of the semi-inner product space P with $\rho(\langle \cdot, \cdot \rangle)$. Let $e_1, \dots, e_n$ be a local orthonormal frame of TM around p . We obtain

$$\begin{aligned}
4f^2(p)\|\nabla f\|_p^2 &= \sum_{i=1}^n (\nabla_i(f^2(p)))^2 = \sum_{i=1}^n \rho(\langle \nabla_i \nabla u_\epsilon, \nabla u_\epsilon \rangle + \langle \nabla u_\epsilon, \nabla_i \nabla u_\epsilon \rangle)^2 \\
&\stackrel{(3.6)}{\leq} 4 \sum_{i=1}^n \rho(\langle \nabla_i \nabla u_\epsilon, \nabla u_\epsilon \rangle) \rho(\langle \nabla u_\epsilon, \nabla_i \nabla u_\epsilon \rangle) \\
&\stackrel{(3.7)}{\leq} 4 \sum_{i=1}^n \rho(|\nabla_i \nabla u_\epsilon|_p^2) \rho(|\nabla u_\epsilon|_p^2) \\
&= 4f^2(p) \rho(|\nabla \nabla u_\epsilon|_p^2)
\end{aligned} \tag{3.8}$$

Dividing equation (3.8) by $2f^2(p)$ gives together with equation (3.5)

$$2f(p)\Delta f(p) \leq \rho(\langle \nabla^* \nabla(\nabla u_\epsilon), \nabla u_\epsilon \rangle_p + \langle \nabla u_\epsilon, \nabla^* \nabla(\nabla u_\epsilon) \rangle_p). \tag{3.9}$$

Since the third covariant derivative of u_ϵ is uniformly bounded in $\epsilon \in (0, 1)$ by the almost parallelity of $\{u_\epsilon\}_{\epsilon>0}$, there exist positive constants C_1 and α , independent of ϵ , such that

$$\rho(|\nabla^* \nabla(\nabla u_\epsilon)|_p^2) \leq \|\nabla^* \nabla(\nabla u_\epsilon)\|_p^2 \leq C_1 \|\nabla^3 u_\epsilon\|_\infty \leq \alpha \tag{3.10}$$

holds. Finally, we obtain with (3.9), (3.6), (3.7) and (3.10)

$$(2f(p)\Delta f(p))^2 \leq 4\rho(|\nabla^* \nabla(\nabla u_\epsilon)|_p^2) \rho(|\nabla u_\epsilon|_p^2) \leq 4\alpha f^2(p).$$

This proves equation (3.3) and the lemma is proved. $\square$

Lemma 3.6. *Any family of almost parallel sections of $\mathcal{S}$ is almost constant in the sense of Definition 3.1.*

Proof. Let $\{u_\epsilon\}_{\epsilon>0}$ be a family of almost parallel sections. There exists a constant $C_1 > 0$ such that for all $g \in W^{1,\infty}(M, \mathbb{R})$ the Poincaré inequality

$$\|\bar{g} - g\|_\infty \leq C_1 \|\nabla g\|_\infty \quad \text{with} \quad \bar{g} := \frac{1}{\text{vol}M} \int_M g \, \text{dvol}$$

holds [13, Theorem 1 Section 5.8]. Since all u_ϵ are smooth, the function

$$M \longrightarrow \mathbb{R}^+, \quad p \longmapsto \rho(|u_\epsilon|_p^2)$$

is an element of $W^{1,\infty}(M, \mathbb{R})$ for any state $\rho: \mathcal{A} \rightarrow \mathbb{C}$ and any $\epsilon \in (0, 1)$. Applying the Poincaré inequality to this function gives

$$|\rho(\bar{u}_\epsilon) - \rho(|u_\epsilon|_p^2)| \leq C_1 \|\nabla \rho(|u_\epsilon|_\bullet^2)\|_\infty \stackrel{(3.12)}{\leq} 2C_1 \|u_\epsilon\|_\infty \|\nabla u_\epsilon\|_\infty \tag{3.11}$$

for all $p \in M$, all $\epsilon \in (0, 1)$ and all states ρ . The second step follows from

$$\|\nabla \rho(|u_\epsilon|^2)\|_q^2 \leq 4\rho(|u_\epsilon|^2)\rho(|\nabla u_\epsilon|^2) \leq 4\|u_\epsilon\|_q^2 \|\nabla u_\epsilon\|_q^2, \quad (3.12)$$

which holds by the same considerations as in equation (3.8). There exist, by the almost parallelity of $\{u_\epsilon\}_{\epsilon>0}$ and Lemma 3.5, positive constants r , C_2 and C_3 such that $\|u_\epsilon\|_\infty < C_2$ and $\|\nabla u_\epsilon\|_\infty < C_3\epsilon^r$ hold for all $\epsilon \in (0, 1)$. This yields

$$\begin{aligned} \|\bar{u}_\epsilon - |u_\epsilon|_p^2\|_{\mathcal{A}} &\stackrel{(3.2)}{=} \max_{\rho \text{ state}} |\rho(\bar{u}_\epsilon) - \rho(|u_\epsilon|_p^2)| \stackrel{(3.11)}{\leq} 2C_1\|u_\epsilon\|_\infty \|\nabla u_\epsilon\|_\infty \\ &< 2C_1C_2C_3\epsilon^r \end{aligned}$$

for all $p \in M$ and all $\epsilon \in (0, 1)$, hence the lemma is proved. $\square$

4. Non-trivial Rosenberg index and the Ricci curvature

In this section, we apply the techniques from Section 3 to give a direct proof of the well-known fact that any closed connected spin manifold with non-negative scalar curvature and non-vanishing Rosenberg index is Ricci flat.

Theorem 4.1. *Let (M, g) be a closed connected spin manifold of dimension n , $\mathcal{A}$ a unital Real C^* -algebra and $E \rightarrow M$ a Real flat bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules. Let $\mathcal{D}_E$ be the induced Dirac operator of the $\mathbb{C}l_n \hat{\otimes} \mathcal{A}$ -linear Dirac bundle obtained by twisting the spinor bundle $\mathcal{S}$ by E as in Construction 2.2. If $\text{scal} \geq 0$ and the higher index of $\mathcal{D}_E$ is non-zero in $\text{KO}_n(\mathcal{A})$, the Ricci curvature vanishes.*

Applying the previous theorem to the maximal group C^* -algebra of the fundamental group of M , equipped with its natural Real structure, and the Mishchenko bundle yields:

Corollary 4.2. *Any closed connected spin manifold with $\text{scal} \geq 0$ and non-vanishing Rosenberg index is Ricci flat.*

Let us take a look at the proof of Corollary 4.2 in the case where the classical index of the spin Dirac operator $\mathcal{D}$ is non-zero: There exists a L^2 -normalized $\mathcal{D}$ -harmonic section u of the spinor bundle $\mathcal{S}$, which is by the Schrödinger–Lichnerowicz formula even parallel and satisfies $\langle u(p), u(p) \rangle_p = (\text{vol} M)^{-1}$ for all $p \in M$. For a local orthonormal frame $e_1, \dots, e_n$ of $TM|_U$ and any vector field X supported in U , we obtain

$$\|\text{Ric}(X)\|_{L^2} = \sqrt{\text{vol}(M)} \|\text{Ric}(X) \cdot u\|_{L^2} \stackrel{(4.2)}{=} 2\sqrt{\text{vol}(M)} \left\| \sum_j e_j \cdot K_{X, e_j} u \right\|_{L^2} = 0, \quad (4.1)$$

hence $\text{Ric} \equiv 0$. Here “ $\cdot$ ” denotes the Clifford multiplication by vector fields, and K denotes the curvature tensor of $\mathcal{S}$. We used in the second step the following formula [see 14, Section 3.1]

$$\operatorname{Ric}(X) \cdot u = -2 \sum_j e_j \cdot K_{X, e_j} u. \quad (4.2)$$

The situation in Theorem 4.1 gives rise to a family of almost $\mathcal{D}_E$ -harmonic section (Lemma 2.4 and Lemma 3.2). The proof of Theorem 4.1 splits into two parts. First, we show that the family of almost $\mathcal{D}_E$ -harmonic sections is almost parallel by the extreme geometric situation, hence it is almost constant by Lemma 3.6. Second, we conclude that M is Ricci flat by similar calculations as for the corresponding classical statement explained in the previous paragraph.

Lemma 4.3. *Assume a situation as in Theorem 4.1. Then there exists a family $\{u_\epsilon\}_{\epsilon>0}$ of almost constant sections. Moreover, $\{u_\epsilon\}_{\epsilon>0}$ can be chosen such that there exists a constant C such that $\|\nabla^2 u_\epsilon\|_{L^2} < C\sqrt{\epsilon}$ holds for all $\epsilon \in (0, 1)$.*

Proof. The Dirac operator $\mathcal{D}_E$ is not invertible by Lemma 2.4, hence there exists by Lemma 3.2 a family $\{u_\epsilon\}_{\epsilon>0}$ of almost $\mathcal{D}_E$ -harmonic sections. Using the almost $\mathcal{D}_E$ -harmonicity gives together with the Schrödinger–Lichnerowicz formula, the L^2 -self-adjointness of $\mathcal{D}_E$ and $\operatorname{scal} \geq 0$

$$\epsilon^2 > \|\mathcal{D}_E u_\epsilon\|_{L^2}^2 = \|\langle \mathcal{D}_E^2 u_\epsilon, u_\epsilon \rangle_{L^2}\|_{\mathcal{A}} = \left\| |\nabla u_\epsilon|_{L^2}^2 + \frac{1}{4} \int_M \operatorname{scal}_p |u_\epsilon|_p^2 \, dp \right\|_{\mathcal{A}} \geq \|\nabla u_\epsilon\|_{L^2}^2$$

for all $\epsilon > 0$. Lemma 3.3 gives that $\{u_\epsilon\}_{\epsilon>0}$ is almost parallel, hence almost constant by Lemma 3.6. For the proof of the second part, we fix an $\epsilon \in (0, 1)$. By a commutator formula for ∇ and $\nabla^* \nabla$ given in [2, Lemma A.3], we obtain

$$(\nabla^* \nabla) \nabla u_\epsilon = \nabla (\nabla^* \nabla) u_\epsilon + (\operatorname{Ric} \otimes \operatorname{id}) \nabla u_\epsilon + 2c_{12}(\operatorname{id} \otimes K) \nabla u_\epsilon + c_{12}(\nabla K) u_\epsilon. \quad (4.3)$$

Here K denotes the curvature tensor of the bundle $\mathcal{S}_E$, c_{12} the metric contraction in the first and second slot, and Ric acts on a 1-form via pre-composition. Let $\operatorname{scal}_{\max}$ be the maximum of the scalar curvature of M . We obtain, by equation (4.3) together with the almost $\mathcal{D}_E$ -harmonicity,

$$\begin{aligned} \|\nabla \nabla u_\epsilon\|_{L^2}^2 &= \|\langle (\nabla^* \nabla) \nabla u_\epsilon, \nabla u_\epsilon \rangle_{L^2}\|_{\mathcal{A}} \\ &\stackrel{(4.3)}{\leq} \underbrace{\|\nabla^* \nabla u_\epsilon\|_{L^2}^2}_{\stackrel{(*)}{\leq} (1 + \frac{1}{2} \sqrt{\operatorname{scal}_{\max}})^2 \epsilon^2} + \underbrace{\|\langle (\operatorname{Ric} \otimes \operatorname{id}) \nabla u_\epsilon, \nabla u_\epsilon \rangle_{L^2}\|_{\mathcal{A}}}_{\stackrel{(**)}{\leq} C_1 \|\nabla u_\epsilon\|_{L^2}^2} \\ &\quad + \underbrace{\|\langle 2c_{12}(\operatorname{id} \otimes K) \nabla u_\epsilon, \nabla u_\epsilon \rangle_{L^2}\|_{\mathcal{A}}}_{\stackrel{(**)}{\leq} C_1 \|\nabla u_\epsilon\|_{L^2}^2} + \underbrace{\|\langle c_{12}(\nabla K) u_\epsilon, \nabla u_\epsilon \rangle_{L^2}\|_{\mathcal{A}}}_{\stackrel{(**)}{\leq} C_1 \|\nabla u_\epsilon\|_{L^2} \|u_\epsilon\|_{L^2}} \\ &< C^2 \epsilon \end{aligned}$$

for suitable constants $C_1, C > 0$, which are both independent of ϵ . It remains to show the estimates $(*)$ and $(**)$.

($\star$) We obtain, by the Schrödinger–Lichnerowicz formula, $\text{scal} \geq 0$ and the almost $\mathcal{D}_E$ -harmonicity of the section u_ϵ ,

$$|\mathcal{D}_E u_\epsilon|_{L^2}^2 = |\nabla u_\epsilon|_{L^2}^2 + \frac{1}{4} \langle \text{scal} \cdot u_\epsilon, u_\epsilon \rangle_{L^2} \geq \frac{1}{4} \langle \text{scal} \cdot u_\epsilon, u_\epsilon \rangle_{L^2} \geq 0 \text{ and} \quad (4.4)$$

$$\|\text{scal} \cdot u_\epsilon\|_{L^2}^2 \leq \text{scal}_{\max} \underbrace{\left\| \int_M \text{scal}_p |u_\epsilon|_p^2 dp \right\|_{\mathcal{A}}}_{=\langle \text{scal} \cdot u_\epsilon, u_\epsilon \rangle_{L^2}} \stackrel{(4.4)}{\leq} 4 \text{scal}_{\max} \|\mathcal{D}_E u_\epsilon\|_{L^2}^2 < 4 \text{scal}_{\max} \epsilon^2. \quad (4.5)$$

Equation ($\star$) follows by the Schrödinger–Lichnerowicz formula, the triangle inequality, equation (4.5), the $\mathcal{D}_E$ -harmonicity of u_ϵ , and $\epsilon \in (0, 1)$:

$$\|\nabla^* \nabla u_\epsilon\|_{L^2} = \|\mathcal{D}_E^2 u_\epsilon - \frac{1}{4} \text{scal} \cdot u_\epsilon\|_{L^2} \leq \|\mathcal{D}_E^2 u_\epsilon\|_{L^2} + \frac{1}{4} \|\text{scal} \cdot u_\epsilon\|_{L^2} < \left(1 + \frac{1}{2} \sqrt{\text{scal}_{\max}}\right) \epsilon.$$

($\star\star$) The maps

$$\begin{aligned} \text{Ric} \otimes \text{id}, c_{12}(\text{id} \otimes K): C^\infty(M, T^*M \otimes \mathcal{S}_E) &\rightarrow C^\infty(M, T^*M \otimes \mathcal{S}_E) \text{ and} \\ c_{12}(\nabla K): C^\infty(M, \mathcal{S}_E) &\rightarrow C^\infty(M, T^*M \otimes \mathcal{S}_E) \end{aligned}$$

decent fiberwise to adjointable maps between the associated Hilbert $\mathcal{A}$ -modules for all $p \in M$, hence they extend to adjointable maps between the corresponding L^2 -Hilbert $\mathcal{A}$ -modules. By the Cauchy-Schwarz inequality, [24, Proposition 1.2] and the almost $\mathcal{D}_E$ -parallelity of the section u_ϵ , we obtain for a suitable constant C_1 the estimates in ($\star\star$), and the lemma is proved. $\square$

Proof (Theorem 4.1). Fix an arbitrary vector field X of the manifold M with support in a chart U . Let $\{u_\epsilon\}_{\epsilon>0}$ be a family of almost constant sections as in Lemma 4.3. There exists a constant $C > 0$ such that

$$\|\text{Ric}(X)\|_{L^2} \stackrel{(\star)}{\leq} 2\sqrt{\text{vol}M} \|\text{Ric}(X) \cdot u_\epsilon\|_{L^2} \stackrel{(\star\star)}{<} C\sqrt{\epsilon} \quad (4.6)$$

holds² for all sufficiently small $\epsilon > 0$. Equation (4.6) yields $\text{Ric}(X) = 0$ by taking $\epsilon \rightarrow 0$, hence M is Ricci flat. It remains to show ($\star$) and ($\star\star$).

($\star$) Since the family $\{u_\epsilon\}_{\epsilon>0}$ is almost constant, there exist positive constants C_1 and r such that

$$\|\overline{u}_\epsilon - |u_\epsilon|_p^2\|_{\mathcal{A}} < C_1 \epsilon^r \quad (4.7)$$

for all $p \in M$ and all $\epsilon \in (0, 1)$. Fix an $\epsilon \in (0, 1)$ with $4C_1 \text{vol}M \epsilon^r \leq 3$, which is equivalent to

² See the analogy of equation (4.6) to the classical calculation in equation (4.1).

$$\frac{1}{\text{vol}M} - C_1\epsilon^r \geq \frac{1}{4\text{vol}M}. \quad (4.8)$$

Multiplying both sides of equation (4.7) with $\|\text{Ric}(X)\|_p^2$ and integrating over M gives

$$\int_M \|\text{Ric}(X)\|_p^2 \|\bar{u}_\epsilon - |u_\epsilon|_p^2\|_{\mathcal{A}} \, dp \leq C_1\epsilon^r \|\text{Ric}(X)\|_{L^2}^2. \quad (4.9)$$

We estimate the left-hand side of equation (4.9) by taking the $\mathcal{A}$ -norm out of the integral and using the reversed triangle inequality as follows:

$$\begin{aligned} \int_M \|\text{Ric}(X)\|_p^2 \|\bar{u}_\epsilon - |u_\epsilon|_p^2\|_{\mathcal{A}} \, dp &\geq \left\| \int_M \|\text{Ric}(X)\|_p^2 \bar{u}_\epsilon \, dp - \int_M \|\text{Ric}(X)\|_p^2 |u_\epsilon|_p^2 \, dp \right\|_{\mathcal{A}} \\ &\geq \left| \left\| \bar{u}_\epsilon \int_M \|\text{Ric}(X)\|_p^2 \, dp \right\|_{\mathcal{A}} - \left\| \int_M \|\text{Ric}(X)\|_p^2 |u_\epsilon|_p^2 \, dp \right\|_{\mathcal{A}} \right| \\ &= \underbrace{\left| \frac{1}{\text{vol}M} \|\text{Ric}(X)\|_{L^2}^2 - \|\text{Ric}(X) \cdot u_\epsilon\|_{L^2}^2 \right|}_{=:\Theta} \end{aligned} \quad (4.10)$$

In the last step, we used $\|\bar{u}_\epsilon\|_{\mathcal{A}} = (\text{vol}M)^{-1}$, which holds by the L^2 -normalization of u_ϵ and the definition of $\bar{u}_\epsilon$ (Definition 3.1). Using that ϵ satisfies equation (4.8) gives, by combining equation (4.9) with equation (4.10),

$$\|\text{Ric}(X) \cdot u_\epsilon\|_{L^2}^2 \geq \begin{cases} \left(\frac{1}{\text{vol}M} - C_1\epsilon^r \right) \|\text{Ric}(X)\|_{L^2}^2 & \text{if } \Theta > 0, \\ \frac{1}{\text{vol}M} \|\text{Ric}(X)\|_{L^2}^2 & \text{if } \Theta \leq 0, \end{cases} \geq \frac{1}{4\text{vol}M} \|\text{Ric}(X)\|_{L^2}^2,$$

hence equation $(\star)$ is proved.

($\star\star$) We fix a local orthonormal frame $e_1, \dots, e_n$ of $TM|_U$, and we use the convention that indices i and j always run over $\{1, \dots, n\}$. The starting point for the proof of equation $(\star\star)$ is, as in the classical case, the formula in equation (4.2) that relates the Ricci tensor with the curvature tensor of $\mathcal{S}_E$. For $X = \sum_i a^i e_i$ and $a := \max_i \|a^i\|_\infty$, we obtain

$$\|\text{Ric}(X) \cdot u_\epsilon\|_{L^2} = 2 \left\| \sum_{i,j} a^j e_j \cdot K_{e_i, e_j} u_\epsilon \right\|_{L^2} \leq 2n^2 a \max_{i,j} \|e_j \cdot K_{e_i, e_j} u_\epsilon\|_{L^2}. \quad (4.11)$$

By Lemma 4.3, there exists a constant $C_2 > 0$ such that

$$\begin{aligned} \|e_j \cdot K_{e_i, e_j} u_\epsilon\|_{L^2} &= \|K_{e_i, e_j} u_\epsilon\|_{L^2} \\ &\leq \|\nabla_i \nabla_j u_\epsilon\|_{L^2} + \|\nabla_j \nabla_i u_\epsilon\|_{L^2} + \|\nabla_{[i,j]} u_\epsilon\|_{L^2} \\ &< C_2 \sqrt{\epsilon} \end{aligned} \quad (4.12)$$

holds for all $\epsilon \in (0, 1)$ and all i, j . Equation $(\star\star)$ follows by combining equation (4.11) with equation (4.12). This proves Theorem 4.1. $\square$

5. Extremality and rigidity for scalar curvature

Let (M, g) and $(N, \bar{g})$ be two closed connected Riemannian manifolds of dimension m and n , respectively, and $f: M \rightarrow N$ a smooth *area non-increasing spin map*, i.e. that

- $\|v \wedge w\|_p \geq \|f_*v \wedge f_*w\|_{f(p)} \quad \forall p \in M \quad \forall v, w \in T_p M \quad \text{and} \quad (\text{area non-increasing})$
- $w_1(TM) = f^*w_1(TN), \quad w_2(TM) = f^*w_2(TN) \quad (\text{spin})$

holds. Here we use the induced norm on $\bigwedge^2 TM$ and $\bigwedge^2 TN$, and denote by w_1 and w_2 the first and second Stiefel–Whitney class, respectively. We equip the vector bundle $TM \oplus f^*TN$ with the metric $h = g \oplus (-f^*\bar{g})$ of signature (m, n) . By the naturality of the Stiefel–Whitney class and the preliminaries in Section 2.2, the spin condition on f is equivalent to the existence of a spin structure on $(TM \oplus f^*TN, h)$. Fix such a spin structure. The construction in Section 2.2 gives rise to a $\mathbb{Cl}_{m,n}$ -linear spinor bundle $\mathcal{S} := SM \hat{\otimes} f^*SN \rightarrow M$. We denote the Clifford multiplications by

$$c: TM \rightarrow \text{End}(\mathcal{S}) \quad \text{and} \quad \bar{c}: f^*TN \rightarrow \text{End}(\mathcal{S}). \quad (5.1)$$

Note that $c(v)$ is skew-adjoint, $c(v)^2 = -g(v, v)$, $\bar{c}(w)$ is self-adjoint, and $\bar{c}(w)^2 = \bar{g}(w, w)$ for all $v \in TM$ and all $w \in f^*TN$. In particular, the $\mathbb{Cl}_{m,n}$ -linear spinor bundle is a graded Real $\mathbb{Cl}_{m,n}$ -linear Dirac bundle $\mathcal{S}M \hat{\otimes} f^*SN \rightarrow M$. Denote the induced Dirac operator by $\mathcal{D}$.

The curvature tensor R^N of the manifold N induces a self-adjoint curvature operator $\mathcal{R}_N$ on $\bigwedge^2 TN$ satisfying locally

$$\bar{g}(\mathcal{R}_N(\bar{e}_i \wedge \bar{e}_j), \bar{e}_k \wedge \bar{e}_l) = -\bar{g}(R_{\bar{e}_i, \bar{e}_j}^N \bar{e}_k, \bar{e}_l) \quad (5.2)$$

for all $i, j, k, l \in \{1, \dots, n\}$ and every local orthonormal frame $\bar{e}_1, \dots, \bar{e}_n$ of TN . Goette and Semmelmann [15, Lemma 1.6] showed that $\text{scal}_M \geq \text{scal}_N \circ f$ implies $\text{scal}_M = \text{scal}_N \circ f$ if the manifolds M and N are orientable, the curvature operator $\mathcal{R}_N$ is non-negative, and $\mathcal{S}M \hat{\otimes} f^*SN$ admits a non-zero $\mathcal{D}$ -harmonic section. Here non-negativity of the curvature operator means that all its eigenvalues are pointwise non-negative. Furthermore, they give a sufficient condition, involving estimates for the scalar and the Ricci curvature of M and N , that f is a Riemannian submersion.

In this section, we generalize this result to the situation, where we include the non-orientable case and replace the existence of a $\mathcal{D}$ -harmonic section on $\mathcal{S}M \hat{\otimes} f^*SN$ by the existence of a family of almost $\mathcal{D}_E$ -harmonic sections. Here $E \rightarrow M$ is a Real flat bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules for a unital Real C^* -algebra $\mathcal{A}$, and $\mathcal{D}_E$ denotes the induced Dirac operator of the $\mathbb{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundle $\mathcal{S}M \hat{\otimes} f^*SN \hat{\otimes} E$.

The main step in the proof will be that the almost $\mathcal{D}_E$ -harmonic sections are by the extreme geometric situation assumed in the theorem almost $\mathcal{D}_E$ -parallel, hence they are almost constant in the sense Definition 3.1.

Furthermore, we will introduce in Definition 5.2 the higher degree of f in analogy to the $\hat{A}$ -degree in [15, Section 2.b]. It turns out that the cut-and-paste principle for the higher index can be used to prove an index theorem (Theorem 5.4) similar to the classical one for the $\hat{A}$ -degree [see 15, Proof of Theorem 2.4]. This yields that if the product of the higher degree of f with the Euler characteristic of N is non-zero, there exists a family of almost $\mathcal{D}_E$ -harmonic sections. This gives a generalization of [15, Theorem 2.4] involving the higher degree of f instead of the $\hat{A}$ -degree.

Given manifolds M, N and a map $f: M \rightarrow N$ as in the beginning of this section, we denote the preimage of a regular value p under f by M_p , the induced graded Real $\mathbb{C}l_{m,n}$ -linear Dirac bundle $\mathcal{S}M \hat{\otimes} f^*\mathcal{S}N$ by $\mathcal{S}$, the induced Dirac operator by $\mathcal{D}$ and the Clifford multiplications given in equation (5.1) by c and $\bar{c}$. Moreover, given a unital Real C^* -algebra $\mathcal{A}$ and a Real flat bundle E of finitely generated projective Hilbert $\mathcal{A}$ -modules, we denote the graded Real $\mathbb{C}l_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundle $\mathcal{S} \hat{\otimes} E$ by $\mathcal{S}_E$ and the induced Dirac operator by $\mathcal{D}_E$.

Theorem 5.1. *Let $f: M \rightarrow N$ be a smooth area non-increasing spin map between two closed connected Riemannian manifolds (M, g) and $(N, \bar{g})$ of dimension m and n , respectively. Suppose the curvature operator of N is non-negative, and there exists a unital Real C^* -algebra $\mathcal{A}$ and a Real flat bundle E of finitely generated projective Hilbert $\mathcal{A}$ -modules together with a family of almost $\mathcal{D}_E$ -harmonic sections. Then $\text{scal}_M \geq \text{scal}_N \circ f$ on M implies $\text{scal}_M = \text{scal}_N \circ f$. If, moreover, $\text{scal}_N > 2\text{Ric}_N > 0$ (or f is distance non-increasing and $\text{Ric}_N > 0$), then $\text{scal}_M \geq \text{scal}_N \circ f$ implies that f is a Riemannian submersion.*

Proof. We adapt the proof of Goette and Semmelmann [15, Lemma 1.6]. In the first step, we recap the estimates of $\mathcal{D}_E^2$ using the Lichnerowicz formula for twisted Dirac operators, the non-negativity of the curvature operator of N and the fact that f is area non-increasing. In the second step, we show that the family of almost $\mathcal{D}_E$ -harmonic sections is almost constant, and we prove the extremality and rigidity statements.

Suppose $\text{scal}_M \geq \text{scal}_N \circ f$ on M . Fix open subsets $U \subset M$ and $\bar{U} \subset N$ trivializing TM and TN , respectively, with $f^{-1}(\bar{U}) = U$. Let $e_1, \dots, e_m$ be a g -orthonormal frame of TU and $\bar{e}_1, \dots, \bar{e}_n$ a $\bar{g}$ -orthonormal frame of $T\bar{U}$ such that there exists non-negative $\mu_1, \dots, \mu_{\min(m,n)} \in C^\infty(U, \mathbb{R})$ with

$$f_*e_i = \begin{cases} \mu_i \bar{e}_i & \text{if } i \leq \min(m, n) \\ 0 & \text{otherwise.} \end{cases}$$

There exist locally compatible spin structures of M and N such that the $\mathbb{C}l_{m,n}$ -linear Dirac bundle $\mathcal{S}$ coincides with the graded tensor product of $\mathcal{S}M$ with $f^*\mathcal{S}N$, where $\mathcal{S}N$

has typical fiber $\mathbb{C}l_{0,n}$. Using the Lichnerowicz formula for twisted spinor bundles ([25, Theorem 8.17]) and the formula for the curvature transform ([25, Theorem 4.15]) gives locally

$$\begin{aligned} \mathcal{D}_E^2 &= \nabla^* \nabla + \frac{\text{scal}_M}{4} + \underbrace{\left(\frac{1}{2} \sum_{i,j=1}^m c(e_i) c(e_j) (\text{id}_{SM} \hat{\otimes} R_{e_i, e_j}^{f^* SN}) \right)}_{=\mathfrak{R}^{f^* SN}} \hat{\otimes} \text{id}_E \\ \mathfrak{R}^{f^* SN} &= \frac{1}{8} \sum_{i,j=1}^m \sum_{k,l=1}^n \bar{g} \left(R_{f_* e_i, f_* e_j}^N \bar{e}_k, \bar{e}_l \right) c(e_i) c(e_j) \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_l) \\ &= -\frac{1}{8} \sum_{i,j=1}^m \sum_{k,l=1}^n \bar{g}(\mathcal{R}_N f_*(e_i \wedge e_j), \bar{e}_k \wedge \bar{e}_l) c(e_i) c(e_j) \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_l). \end{aligned} \quad (5.3)$$

Here $R^{f^* SN}$ denotes the curvature tensor of $f^* SN$ and $\mathcal{R}_N$ the curvature operator of N on $\bigwedge^2 TN$ as defined in equation (5.2). Note that the different behaviors of c and $\bar{c}$ do not influence the calculations in [25], since all calculations are in an orthonormal frame. By the non-negativity of the curvature operator $\mathcal{R}_N$, there exists a self-adjoint square root $L \in \text{End}(\bigwedge^2 TN)$ satisfying

$$g(L(\bar{e}_i \wedge \bar{e}_j), L(\bar{e}_k \wedge \bar{e}_l)) = g(\mathcal{R}_N(\bar{e}_i \wedge \bar{e}_j), \bar{e}_k \wedge \bar{e}_l)$$

for all $i, j, k, l \in \{1, \dots, n\}$. The same estimates as in [15, p.5-6] yield locally

$$\begin{aligned} \frac{\text{scal}_M}{4} - \frac{1}{8} \sum_{i,j=1}^m \sum_{k,l=1}^n \bar{g}(\mathcal{R}_N f_*(e_i \wedge e_j), \bar{e}_k \wedge \bar{e}_l) c(e_i) c(e_j) \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_l) \\ \geq \frac{\text{scal}_M}{4} - \frac{\text{scal}_N \circ f}{8} - \frac{1}{8} \sum_{i,j=1}^n \mu_i^2 \mu_j^2 (R_{ijji}^N \circ f) \quad =: h_1 \\ \geq \frac{\text{scal}_M - \text{scal}_N \circ f}{4} \quad =: h_2 \\ \geq 0. \end{aligned} \quad (5.4)$$

Let V be an open subset of U such that there exists a smooth cut-off function $\psi: M \rightarrow \mathbb{R}$, which is identically one on V , zero outside U and $\text{im}(\psi) \subset [0, 1]$. Then equation (5.4) still holds for h_1 and h_2 replaced by ψh_1 and ψh_2 , respectively, and we obtain together with equation (5.3) globally

$$\mathcal{D}_E^2 \geq \nabla^* \nabla + \psi h_1 \geq \nabla^* \nabla + \psi h_2 \geq \nabla^* \nabla. \quad (5.5)$$

By assumption, there exists a family $\{u_\epsilon\}_{\epsilon>0}$ of almost $\mathcal{D}_E$ -harmonic sections, which satisfy by equation (5.5)

$$|\mathcal{D}_E u_\epsilon|_{L^2}^2 \geq |\nabla u_\epsilon|_{L^2}^2 + \langle \psi h_1 \cdot u_\epsilon, u_\epsilon \rangle_{L^2} \geq |\nabla u_\epsilon|_{L^2}^2 + \langle \psi h_2 \cdot u_\epsilon, u_\epsilon \rangle_{L^2} \geq |\nabla u_\epsilon|_{L^2}^2 \quad (5.6)$$

for all $\epsilon > 0$. It follows, by the almost $\mathcal{D}_E$ -harmonicity, equation (5.6) and Lemma 3.3, that $\{u_\epsilon\}_{\epsilon>0}$ is almost parallel, hence almost constant by Lemma 3.6. By similar calculations as in the proof of Theorem 4.1 part ($\star$), we obtain

$$\|\psi h_i\|_{L^1} \leq 2\text{vol}(M) \|\langle \psi h_i u_\epsilon, u_\epsilon \rangle_{L^2}\|_{\mathcal{A}} \stackrel{(5.6)}{\leq} 2\text{vol}(M) \|\mathcal{D}_E u_\epsilon\|_{L^2}^2 < 2\text{vol}(M) \epsilon^2$$

for all sufficiently small ϵ and $i = 1, 2$. This yields in the limit $\epsilon \rightarrow 0$

$$h_1 = \frac{\text{scal}_M}{4} - \frac{\text{scal}_N \circ f}{8} - \frac{1}{8} \sum_{i,j=1}^n \mu_i^2 \mu_j^2 (R_{ijji}^N \circ f) \equiv 0 \text{ and} \quad (5.7)$$

$$h_2 = \frac{\text{scal}_M - \text{scal}_N \circ f}{4} \equiv 0 \quad (5.8)$$

on V , hence the extremality statement is shown by equation (5.8). Furthermore, equation (5.7) together with $\text{scal}_M = \text{scal}_N \circ f$ implies $\sum_{i,j=1}^n \mu_i^2 \mu_j^2 (R_{ijji}^N \circ f) = \text{scal}_N \circ f$ on V , and the rigidity statement follows as in [15, Section 1.c]. $\square$

Let $f: M \rightarrow N$ be as in Theorem 5.1, and assume additionally that both manifolds are orientable and of dimension $m = n + k$ and n , respectively. The $\hat{A}$ -degree of f is given by

$$\deg_{\hat{A}}(f) := (\hat{A}(M) f^* \omega)[M],$$

where $\omega \in H^n(N; \mathbb{Z})$ is the cohomological fundamental class of N corresponding to the orientation on N , $\hat{A}(M) \in H^*(M; \mathbb{Q})$ the $\hat{A}$ -class of M and $[M] \in H_m(M; \mathbb{Z})$ the fundamental class of M corresponding to its orientation. One sufficient condition for the existence of a non-trivial $\mathcal{D}$ -harmonic section is that the Euler characteristic of N and the $\hat{A}$ -degree of f do not vanish [15, Section 2.b]. This follows by

$$\text{ind } \mathcal{D} = \chi(N) \cdot \deg_{\hat{A}}(f), \quad (5.9)$$

and the fact that a non-zero classical index implies the existence of a $\mathcal{D}$ -harmonic section. We know, by Lemma 2.4 (2) and Lemma 3.2, that a non-vanishing higher index of $\mathcal{D}_E$ implies the existence of a family of almost $\mathcal{D}_E$ -harmonic sections. We want to define a higher version of the $\hat{A}$ -degree such that equation (5.9) still holds for the higher index of an appropriate twisted Dirac operator.

Let us take a closer look at the $\hat{A}$ -degree. Fix a regular value p of f and pick a sufficiently small neighborhood U_p of p such that there exists an open subset $V_p \subset \mathbb{R}^k$ and an isomorphism $\Phi: f^{-1}(U_p) \rightarrow M_p \times V_p$ with $f|_{f^{-1}(U_p)} = \text{pr}_1 \circ \Phi$. Let ω be an n -form on N that represents the cohomological fundamental class of N . Then we can rewrite the $\hat{A}$ -degree of f as follows:

$$\deg_{\hat{A}}(f) = \int_{M_p \times V_p} \hat{A}(M) f^* \omega = \int_{M_p} \hat{A}(M_p) = \text{ind}(\mathcal{D}_{SM_p}) \quad (5.10)$$

The last step holds by the Atiyah–Singer index theorem, where $\mathcal{D}_{SM_p}$ is the spin Dirac operator of the spin manifold M_p (see Lemma 5.3). This matches with the original notion of $\hat{A}$ -degree by Gromov and Lawson [16]. We define in analogy to equation (5.10) the higher mapping degree:

Definition 5.2. Let $f : M \rightarrow N$ be a smooth spin map between closed connected manifolds of dimension $n + k$ and n , respectively. The *higher degree* of f is defined via

$$\deg_{\text{hi}}(f) := \text{ind}(\mathcal{D}_{SM_p \hat{\otimes} \mathcal{L}(M)|_{M_p}}) \in \text{KO}_k(C^*\pi_1(M))$$

for a regular value p of f . Here SM_p denotes the spinor bundle of M_p , $\mathcal{L}(M)$ the Mishchenko bundle of M and $C^*\pi_1(M)$ the maximal group C^* -algebra of the fundamental group of M .

The higher mapping degree is well-defined by the following lemma and by the bordism invariance of the higher index.

Lemma 5.3. Let $f : M \rightarrow N$ be a spin map between two closed manifolds and p a regular value of f . Then the manifold M_p is orientable and admits a spin structure.

Proof. Since p is a regular value of f , the restriction of the tangent bundle to M_p splits $TM|_{M_p} = TM_p \oplus f^*TN|_{M_p}$. This gives

$$\begin{aligned} w_1(TM_p) + w_1(f^*TN|_{M_p}) &= w_1(TM|_{M_p}) = w_1(f^*TN|_{M_p}) \quad \text{and} \\ w_2(TM_p) + w_2(f^*TN|_{M_p}) + w_1(TM_p)w_1(f^*TN|_{M_p}) &= w_2(TM|_{M_p}) = w_2(f^*TN|_{M_p}) \end{aligned}$$

by the spin property of f together with the naturality and the Whitney sum formula for the Stiefel–Whitney class. It follows that the first and second Stiefel–Whitney classes of TM_p vanish, hence the lemma is proved. $\square$

We will show, by the Poincaré–Hopf lemma and the cut-and-paste principle in [8, Section 2.3], that the higher degree of f shows up in an analogous index formula as in equation (5.9).

Theorem 5.4 (Index theorem). Let $f : M \rightarrow N$ be a smooth spin map between two closed connected Riemannian manifolds of dimension $m = n + k$ and n , respectively, $\mathcal{A}$ a unital Real C^* -algebra and $E \rightarrow M$ a Real flat bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules. Then the higher index of the twisted Dirac operator $\mathcal{D}_E$ is given by

$$\text{ind}(\mathcal{D}_E) = \chi(N) \cdot \text{ind}(\mathcal{D}_{SM_p \hat{\otimes} E|_{M_p}}) \in \text{KO}_k(\mathcal{A})$$

for any regular value p of f . In particular, we obtain for the higher degree of f

$$\text{ind}(\mathcal{D}_{\mathcal{L}(\mathcal{M})}) = \chi(N) \cdot \deg_{\text{hi}}(f) \in \text{KO}_k(C^*\pi_1(M)).$$

Proof. The main idea of the proof is to use the cut-and-paste principle in [8, Section 2.3] to split off the higher degree of f from the index of $\mathcal{D}_E$. The remaining factor can be identified with the Euler characteristic of N via the Poincaré–Hopf lemma if we perform the cut-and-paste principle with respect to a suitable vector field on N . At first, we construct the vector field. Second, we show how this vector field gives rise to a situation in which the cut-and-paste principle is applicable. And finally, we verify the index formula.

Let ξ be a vector field on N transversal to the zero section, i.e. for any zero q of ξ and any chart ϕ around q the differential of the induced map on Euclidean space has full rank at the point $\phi(q)$. Such a vector field always exists, since transversality is a generic property. It follows that the set of zeros of ξ is discrete, hence finite by the compactness of N . We deform the vector field without changing the transversality such that all zeros $p_1, \dots, p_l$ of ξ are regular values of f , which is possible because almost all points in N are regular values of f by Sard's theorem. Fix for all $i \in \{1, \dots, l\}$ a chart

$$\phi_i: N \supset U_i \longrightarrow \phi_i(U_i) \subset \mathbb{R}^n, \quad p_i \longmapsto 0,$$

around p_i such that $\cap_{i=1}^l U_i = \emptyset$. Fix $R > \tau > r > \epsilon > 0$ such that $B_{R+\epsilon}(0) \subset \phi_i(U_i)$ for all $i \in \{1, \dots, l\}$. Since ξ is transversal to the zero section, we can deform the vector field to a transversal vector field ξ_1 , which satisfies for suitable non-zero constants $\lambda_{ij} \in \mathbb{R}$

$$\left(d_{\phi_i^{-1}(x)}\phi_i\right)\left((\xi_1 \circ \phi_i^{-1})(x)\right) = \begin{cases} \text{diag}(\lambda_{i1}, \dots, \lambda_{in})x & \text{if } x \in B_\epsilon(0), \\ \text{diag}(\lambda_{i1}, \dots, \lambda_{in})\frac{x}{\|x\|}\tau & \text{if } x \in B_R(0) \setminus B_r(0) \end{cases}$$

for all $i \in \{1, \dots, l\}$ and has the same zeros as ξ . By the Poincaré–Hopf lemma, the Euler characteristic of N is given by

$$\chi(N) = \sum_{i=1}^l \deg(\xi_1, p_i) = \sum_{i=1}^l \text{sgn}(\det \text{diag}(\lambda_{i1}, \dots, \lambda_{in})) = \sum_{i=1}^l \prod_{j=1}^n \text{sgn}(\lambda_{ij}). \quad (5.11)$$

Here $\deg(\xi_1, p_i)$ denotes the local Poincaré–Hopf index of ξ_1 at p_i [6, Section 1.1]. In the next step, we define the manifolds and bundles involved in the cut-and-paste process (see Fig. 2). Since all p_i are regular values of f , there exists for any $i \in \{1, \dots, l\}$ an isomorphism $\psi_i: (\phi_i \circ f)^{-1}(B_R(0)) \rightarrow B_R(0) \times M_{p_i}$ compatible to ϕ_i in the sense that

$$\begin{array}{ccc} (\phi_i \circ f)^{-1}(B_R(0)) & \xrightarrow{f} & \phi_i^{-1}(B_R(0)) \\ \psi_i \downarrow \cong & & \cong \downarrow \phi_i \\ B_R(0) \times M_{p_i} & \xrightarrow{\text{pr}_1} & B_R(0) \end{array}$$

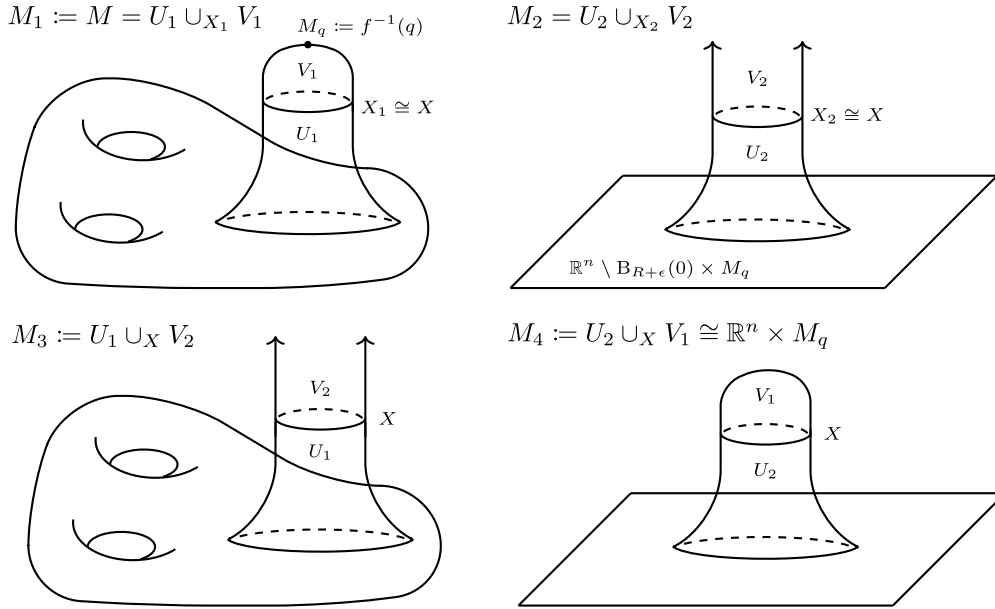


Fig. 2. An illustration of the manifolds involved in the cut-and-paste principle in the proof of the index theorem. Here the vector field ξ_1 has just one zero $q \in N$.

commutes. We define

$$\begin{aligned}
 M_1 &:= M = U_1 \cup_{X_1} V_1 \text{ with} \\
 U_1 &:= M \setminus \bigsqcup_{i=1}^l (\phi_i \circ f)^{-1}(B_r(0)^\circ), \quad X_1 := \bigsqcup_{i=1}^l \partial(\phi_i \circ f)^{-1}(B_r(0)) \text{ and} \\
 V_1 &:= \bigsqcup_{i=1}^l (\phi_i \circ f)^{-1}(B_r(0)), \\
 M_2 &:= \bigsqcup_{i=1}^l \underbrace{\left(\mathbb{R}^n \setminus B_r(0)^\circ \cup_{\partial B_r(0)} \partial B_r(0) \times [0, \infty) \right)}_{\cong S^{n-1} \times \mathbb{R} =: N_2^i} \times M_{p_i} = U_2 \cup_{X_2} V_2 \text{ with} \\
 U_2 &:= \bigsqcup_{i=1}^l \mathbb{R}^n \setminus B_r(0)^\circ \times M_{p_i}, \quad X_2 := \bigsqcup_{i=1}^l \partial B_r(0) \times M_{p_i} \text{ and} \\
 V_2 &:= \bigsqcup_{i=1}^l \left(B_r(0) \setminus B_r(0)^\circ \cup_{\partial B_r(0)} \partial B_r(0) \times [0, \infty) \right) \times M_{p_i}.
 \end{aligned}$$

Taking the union of the maps ψ_i and restrict properly gives an isomorphism between neighborhoods of X_1 and X_2

$$\Gamma: \underbrace{\bigsqcup_{i=1}^l (\phi_i \circ f)^{-1}(B_R(0)^\circ) \setminus (\phi_i \circ f)^{-1}(B_r(0))}_{=: \mathcal{U}(X_1)} \longrightarrow \underbrace{\bigsqcup_{i=1}^l B_R(0)^\circ \setminus B_r(0) \times M_{p_i}}_{=: \mathcal{U}(X_2)},$$

which restricts to a diffeomorphism from X_1 to X_2 . Equip M_1, M_2 and N with metrics turning $\mathcal{U}(X_1), \mathcal{U}(X_2)$ and $\bigsqcup_{i=1}^l \phi_i^{-1}(\mathbb{B}_R(0)^\circ \setminus \mathbb{B}_r(0))$ into tubular neighborhoods of X_1, X_2 and $\bigsqcup_{i=1}^l \partial \phi_i^{-1}(\mathbb{B}_r(0))$, respectively, turning Γ into an isometry, and carrying on the part $\bigsqcup_{i=1}^l \mathbb{R}^n \setminus \mathbb{B}_{R+\epsilon}(0) \times M_{p_i}$ the product metric of the Euclidean metric and the induced metric on M_{p_i} . We denote for $i \in \{1, \dots, l\}$ the projections of $N_2^i \times M_{p_i}$ to the first and second factor by pr_1^i and pr_2^i , respectively, and define

$$\text{pr}_1 := \bigsqcup_{i=1}^l \text{pr}_1^i : M_2 \longrightarrow N_2 := \bigsqcup_{i=1}^l N_2^i.$$

Furthermore, we obtain by radial parallel transport an isomorphism

$$\bigsqcup_{i=1}^l (\text{pr}_2^i \circ \psi_i|_{\mathcal{U}(X_1)})^* E|_{M_{p_i}} \cong E|_{\mathcal{U}(X_1)}$$

respecting the bundle metric and the connection. Consider the graded Real $\mathbb{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundles

$$\begin{aligned} W_1 &:= \mathcal{S}_E = SM \hat{\otimes} f^* SN \hat{\otimes} E \rightarrow M = M_1 \text{ and} \\ W_2 &:= SM_2 \hat{\otimes} \bigsqcup_{i=1}^l (\text{pr}_1^i)^* SN_2^i \hat{\otimes} \bigsqcup_{i=1}^l (\text{pr}_2^i)^* E|_{M_{p_i}} \rightarrow M_2 \end{aligned}$$

together with their induced Dirac operators $\mathcal{D}_1$ and $\mathcal{D}_2$, respectively. Here we consider SN_2^i as the $\mathbb{Cl}_{0,n}$ -linear spinor bundle of N_2^i . The bundle W_2 comes by construction with a self-adjoint Clifford multiplication by the pullback of tangent vectors of N_2 along pr_1 , which we denote again by $\bar{c}$. Furthermore, we define for $\lambda > 0$ perturbed Dirac operators

$$B_1 := \mathcal{D}_1 + \lambda \bar{c}(f^* \xi_1) \text{ and } B_2 := \mathcal{D}_2 + \lambda \bar{c}(\text{pr}_1^* \xi_2).$$

Here ξ_2 denotes a nowhere vanishing vector field on N_2 , which satisfies

$$\xi_2(x) = \begin{cases} \text{diag}(\lambda_{i1}, \dots, \lambda_{in}) \frac{x}{\|x\|} \tau & \text{if } x \in \mathbb{B}_R(0) \setminus \mathbb{B}_r(0) \subset N_2^i \subset N_2, \\ \text{diag}(\lambda_{i1}, \dots, \lambda_{in}) x & \text{if } x \in \mathbb{R}^n \setminus \mathbb{B}_{R+\epsilon}(0) \subset N_2^i \subset N_2 \end{cases}$$

for all $i \in \{1, \dots, l\}$. The unbounded operators B_1 and B_2 are odd Real formally self-adjoint elliptic differential operators of first order. Here it is crucial that $\bar{c}$ is self-adjoint and not skew-adjoint.³ By construction, there is a bundle isometry $\tilde{\Gamma}: W_1|_{\mathcal{U}(N_1)} \rightarrow W_2|_{\mathcal{U}(N_2)}$ covering Γ and satisfying $B_2 = \tilde{\Gamma} \circ B_1 \circ \tilde{\Gamma}^{-1}$. Fix a manifold X isomorphic to $X_1 \cong X_2$. Now we perform the cut-and-paste procedure and obtain graded Real $\mathbb{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundles

³ This justifies that we consider the spinor bundle associated to $TM \oplus f^*TN$ equipped with the indefinite metric of signature (m, n) instead of the induced positive definite metric.

$$\begin{aligned} W_3 &:= W_1|_{U_1} \cup_X W_2|_{V_2} \rightarrow M_3 := U_1 \cup_X V_2 \text{ and} \\ W_4 &:= W_2|_{U_2} \cup_X W_1|_{V_1} \rightarrow M_4 := U_2 \cup_X V_1 \end{aligned}$$

with induced Dirac operators $\mathcal{D}_3$ and $\mathcal{D}_4$, together with perturbed Dirac operators

$$B_3 := \mathcal{D}_3 + \lambda \bar{c}(\xi_3) \text{ and } B_4 := \mathcal{D}_4 + \lambda \bar{c}(\xi_4).$$

Here ξ_3 and ξ_4 are vector fields on M_3 and M_4 , respectively, obtained from $f^*\xi_1$ and $\text{pr}_1^*\xi_2$ via the cut-and-paste procedure. If B_2^2 is uniformly positive at infinity, the index formula in [8, Theorem 2.12] gives

$$\text{ind } B_1 + \text{ind } B_2 = \text{ind } B_3 + \text{ind } B_4 \in \text{KO}_k(\mathcal{A}).$$

It remains to show that for a regular value p of f and a sufficiently big $\lambda > 0$ the following holds:

- (1) $\text{ind } \mathcal{D}_E = \text{ind } B_1$
- (2) The operators B_2^2 and B_3^2 are uniformly positive and hence $\text{ind } B_2 = \text{ind } B_3 = 0$.
- (3) $\text{ind } B_4 = \chi(N) \cdot \text{ind} \left(\mathcal{D}_{SM_p \hat{\otimes} E|_{M_p}} \right)$

Ad(1) Since the graded Real $\text{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundles $W_1 \rightarrow M_1$ and $\mathcal{S}_E \rightarrow M$ coincide except the involved metrics, the indices of the induced Dirac operators match. The index of the perturbed Dirac operator B_1 matches with the index of $\mathcal{D}_E$ by [9, Proposition 4.1].

Ad(2) We show first that B_2^2 is uniformly positive. Fix a local orthonormal frame $e_1, \dots, e_m$ of TM_2 . By inserting the definition of B_2 and using the local expression of the Dirac operator, we obtain

$$\begin{aligned} B_2^2 &= \mathcal{D}_2^2 + \lambda \mathcal{D}_2 \bar{c}(\text{pr}_1^* \xi_2) + \lambda \bar{c}(\text{pr}_1^* \xi_2) \mathcal{D}_2 + \lambda^2 \bar{c}(\text{pr}_1^* \xi_2)^2 \\ &= \mathcal{D}_2^2 + \lambda \sum_{i=1}^m \left(c(e_i) \nabla_{e_i} \bar{c}(\text{pr}_1^* \xi_2) + \bar{c}(\text{pr}_1^* \xi_2) c(e_i) \nabla_{e_i} \right) + \lambda^2 \|\text{pr}_1^* \xi_2\|^2 \\ &= \mathcal{D}_2^2 + \lambda \sum_{i=1}^m c(e_i) \bar{c}(\nabla_{e_i} \text{pr}_1^* \xi_2) + \lambda^2 \|\text{pr}_1^* \xi_2\|^2. \end{aligned}$$

Here we used in the last step the compatibility of the connection with the Clifford multiplication and the anti-commutativity of c and $\bar{c}$. Since $c(e_i) \bar{c}(\nabla_{e_i} \text{pr}_1^* \xi_2)$ is a self-adjoint operator on $L^2(M_2, W_2)$ for all $i \in \{1, \dots, m\}$, we obtain with $C_1 := \max_{i \in \{1, \dots, m\}} \|c(e_i) \bar{c}(\nabla_{e_i} \text{pr}_1^* \xi_2)\|_{\text{op}}$ and $C_2 := \min_{p \in M_2} \|\text{pr}_1^* \xi_2\|_p^2$

$$\begin{aligned}
\langle B_2^2 u, u \rangle_{L^2} &= |D_2 u|_{L^2}^2 + \lambda \sum_{i=1}^m \underbrace{\langle c(e_i) \bar{c}(\nabla_{e_i} \text{pr}_1^* \xi_2) u, u \rangle_{L^2}}_{\geq -C_1 |u|_{L^2}^2} + \lambda^2 \underbrace{\langle \|\text{pr}_1^* \xi_2\|^2 u, u \rangle_{L^2}}_{\geq C_2 |u|_{L^2}^2} \\
&\geq (C_2 \lambda^2 - C_1 m \lambda) |u|_{L^2}^2
\end{aligned} \tag{5.12}$$

for all $u \in H^2(M_2, W_2)$ and all $\lambda > 0$. The vector field ξ_2 does nowhere vanish by construction, hence $C_2 > 0$. Therefore, we can choose λ sufficiently big such that by equation (5.12) the operator B_2^2 is uniformly positive. We obtain by Lemma 2.4 that the index of B_2 vanishes. Since the vector field obtained from ξ_1 and ξ_2 by following the cut-and-paste procedure vanishes nowhere, the same calculations yield $\text{ind } B_3 = 0$ for λ sufficiently big. This proves part (2).

Ad(3) Let $\bar{\text{pr}}_1^i$ and $\bar{\text{pr}}_2^i$ be the projections of $\mathbb{R}^n \times M_{p_i}$ onto the first and second factor, respectively, for all $i \in \{1, \dots, l\}$ and $\bar{\text{pr}}_1$ the union of all $\bar{\text{pr}}_1^i$. The $\text{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundle $W_4 \rightarrow M_4$ is given by

$$W_4 = \bigsqcup_{i=1}^l \mathcal{S}(\mathbb{R}^n \times M_{p_i}) \hat{\otimes} (\bar{\text{pr}}_1^i)^* \bar{\mathcal{S}}\mathbb{R}^n \hat{\otimes} (\bar{\text{pr}}_2^i)^* E|_{M_{p_i}} \rightarrow M_4 \cong \bigsqcup_{i=1}^l \mathbb{R}^n \times M_{p_i}$$

with an appropriate metric on $\mathbb{R}^n$. Here the isomorphism $M_4 \cong \bigsqcup_{i=1}^l \mathbb{R}^n \times M_{p_i}$ is constructed from ψ_i , and we write $\bar{\mathcal{S}}\mathbb{R}^n$ for the $\text{Cl}_{0,n}$ -linear spinor bundle of $T\mathbb{R}^n$. Define a graded Real $\text{Cl}_{m,n} \hat{\otimes} \mathcal{A}$ -linear Dirac bundle $\widetilde{W}_4 \rightarrow M_4$ by using the Euclidean metric on $\mathbb{R}^n$ and the product metric on $\mathbb{R}^n \times M_{p_i}$. Denote its induced Dirac operator by $\widetilde{\mathcal{D}}_4$. Then the index of B_4 matches with the index of $\widetilde{\mathcal{D}}_4 + \lambda \bar{c}(\xi_4)$. Let $\bar{\xi}$ be the vector field on $\bigsqcup_{i=1}^l \mathbb{R}^n$ defined via

$$\bar{\xi}^i: \mathbb{R}^n \longrightarrow T\mathbb{R}^n, x \longmapsto \text{diag}(\lambda_{i1}, \dots, \lambda_{in})x$$

for $i \in \{1, \dots, l\}$. Since the vector fields $\bar{\text{pr}}_1^* \bar{\xi}$ and ξ_4 match by construction outside a compact set, the operator family

$$\widetilde{B}_4^t = \widetilde{\mathcal{D}}_4 + \lambda \bar{c}(\xi_4) + t(\lambda \bar{c}(\bar{\text{pr}}_1^* \bar{\xi}) - \lambda \bar{c}(\xi_4))$$

is uniformly positive at infinity for all $t \in [0, 1]$. We obtain by [9, Proposition 4.1] that the indices of $\widetilde{B}_4^0$ and $\widetilde{B}_4^1$ matches, hence the index of B_4 matches with the index of

$$\widetilde{B}_4 = \widetilde{\mathcal{D}}_4 + \lambda \bar{c}(\bar{\text{pr}}_1^* \bar{\xi}).$$

We use $\mathcal{S}(\mathbb{R}^n \times M_{p_i}) \cong \mathcal{S}\mathbb{R}^n \hat{\boxtimes} \mathcal{S}M_{p_i} = (\bar{\text{pr}}_1^i)^* \mathcal{S}\mathbb{R}^n \hat{\otimes} (\bar{\text{pr}}_2^i)^* \mathcal{S}M_{p_i}$ and rewrite the bundle $\widetilde{W}_4$ into

$$\widetilde{W}_4 \cong \bigsqcup_{i=1}^l (\mathcal{S}\mathbb{R}^n \hat{\otimes} \bar{\mathcal{S}}\mathbb{R}^n) \hat{\boxtimes} (\mathcal{S}M_{p_i} \hat{\otimes} E|_{M_{p_i}}) \rightarrow \bigsqcup_{i=1}^l \mathbb{R}^n \times M_{p_i}.$$

We finally obtain, by sum and product formulas for the higher index,

$$\begin{aligned}
 \text{ind } B_4 &= \text{ind} \left(\widetilde{\mathcal{D}}_4 + \lambda \bar{c} (\bar{\text{pr}}_1^* \bar{\xi}) \right) \\
 &= \sum_{i=1}^l \prod_{j=1}^n \underbrace{\text{ind}(e_1 \partial_x + \lambda \lambda_{ij} x \bar{e}_1)}_{\stackrel{(*)}{=} \text{sgn}(\lambda_{ij}) \in \mathbb{Z} \cong \text{KO}_0(\mathbb{C}l_{1,1})} \cdot \underbrace{\text{ind}(\mathcal{D}_{SM_{p_i}} \widehat{\otimes}_{E|_{M_{p_i}}})}_{\stackrel{(**)}{=} \text{ind}(\mathcal{D}_{SM_p} \widehat{\otimes}_{E|_{M_p}})} \\
 &\stackrel{(5.11)}{=} \chi(N) \cdot \text{ind}(\mathcal{D}_{SM_p} \widehat{\otimes}_{E|_{M_p}}).
 \end{aligned} \tag{5.13}$$

Here the “.” denotes the K-theory product

$$\text{KO}_0(\mathbb{C}l_{n,n}) \times \text{KO}_0(\mathbb{C}l_{k,0} \widehat{\otimes} \mathcal{A}) \rightarrow \text{KO}_0(\mathbb{C}l_{n+k,n} \widehat{\otimes} \mathcal{A})$$

and the operator $e_1 \partial_x + \lambda \lambda_{ij} x \bar{e}_1$ is considered as an operator from $H^1(\mathbb{R}, \mathbb{C}l_{1,1})$ to $L^2(\mathbb{R}, \mathbb{C}l_{1,1})$, where $e_1, \bar{e}_1$ are generators of $\mathbb{C}l_{1,1}$. Equation $(**)$ holds by the bordism invariance of the index. It remains to show equation $(*)$. We have an isomorphism of graded Real C^* -algebras $\rho: \mathbb{C}l_{1,1} \rightarrow \text{Mat}_2(\mathbb{C}) \cong \text{Kom}_{\mathbb{C}}(\mathbb{C}^2)$ given by

$$1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad e_1 \mapsto \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad \bar{e}_1 \mapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad e_1 \bar{e}_1 \mapsto \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

hence $\mathbb{C}l_{1,1}$ is strongly Morita equivalent to $\mathbb{C}$. Here $\mathbb{C}$ is considered as a trivially graded C^* -algebra equipped with its natural Real structure, and $\mathbb{C}^2$ is the graded Real Hilbert space with $(1, 0)^T$ even and $(0, 1)^T$ odd. This implements as in [34, Proposition 12.17] an isomorphism on Real K-theory, which takes in the Fredholm picture the form

$$\text{KO}_0(\mathbb{C}l_{1,1}) \longrightarrow \text{KO}_0(\mathbb{C}), \quad [(F, Q)] \mapsto [(F \otimes \text{id}_{\mathbb{C}^2}, Q \otimes_{\rho} \mathbb{C}^2)].$$

Together with the isomorphism $\mathbb{C}l_{1,1} \otimes_{\rho} \mathbb{C}^2 \cong \mathbb{C}^2$ and $\text{KO}_0(\mathbb{C}) \cong \mathbb{Z}$, we obtain

$$\begin{aligned}
 \text{ind}(e_1 \partial_x + \lambda \lambda_{ij} x \bar{e}_1) &\in \text{KO}_0(\mathbb{C}l_{1,1}) \\
 &\cong \text{ind} \begin{pmatrix} 0 & -\partial_x + \lambda \lambda_{ij} x \\ \partial_x + \lambda \lambda_{ij} x & 0 \end{pmatrix} \in \text{KO}_0(\mathbb{C}) \\
 &\cong \dim \ker(\partial_x + \lambda \lambda_{ij} x) - \dim \ker(-\partial_x + \lambda \lambda_{ij} x) = \text{sgn}(\lambda_{ij}) \in \mathbb{Z}.
 \end{aligned}$$

Here we consider $\begin{pmatrix} 0 & -\partial_x + \lambda \lambda_{ij} x \\ \partial_x + \lambda \lambda_{ij} x & 0 \end{pmatrix}$ as an operator from $H^1(\mathbb{R}, \mathbb{C}^2)$ to $L^2(\mathbb{R}, \mathbb{C}^2)$, and we use in the last step

$$\ker(\partial_x + \lambda \lambda_{ij} x) = \begin{cases} \left\langle e^{-\frac{1}{2} \lambda \lambda_{ij} x^2} \right\rangle & \lambda_{ij} > 0 \\ 0 & \lambda_{ij} < 0 \end{cases}, \quad \ker(-\partial_x + \lambda \lambda_{ij} x) = \begin{cases} 0 & \lambda_{ij} > 0 \\ \left\langle e^{\frac{1}{2} \lambda \lambda_{ij} x^2} \right\rangle & \lambda_{ij} < 0. \end{cases}$$

Note that $e^{-\frac{1}{2}\lambda\lambda_{ij}x^2}$ is for $\lambda_{ij} < 0$ not in $H^1(\mathbb{R}, \mathbb{C}^2)$, hence not in the kernel of $\partial_x + \lambda\lambda_{ij}x$. This proves $(\star)$ in equation (5.13), hence part (3) is shown and the index theorem is proved. $\square$

We combine Theorem 5.1 with the index formula in Theorem 5.4 and obtain the following extremality and rigidity statement:

Theorem 5.5 (*Extremality and rigidity*). *Let $f: M \rightarrow N$ be a smooth area non-increasing spin map between two closed connected Riemannian manifolds of dimension $n+k$ and n , respectively. Suppose the curvature operator of N is non-negative and the following holds:*

(T) *There exists a unital Real C^* -algebra $\mathcal{A}$ and a Real flat bundle $E \rightarrow M$ of finitely generated projective Hilbert $\mathcal{A}$ -modules such that for a regular value p of f*

$$\chi(N) \cdot \text{ind}\left(\mathcal{D}_{\mathcal{S}M_p \hat{\otimes} E|_{M_p}}\right) \neq 0 \in \text{KO}_k(\mathcal{A}). \quad (5.14)$$

Then $\text{scal}_M \geq \text{scal}_N \circ f$ on M implies $\text{scal}_M = \text{scal}_N \circ f$. If, moreover, $\text{scal}_N > 2\text{Ric}_N > 0$ (or f is distance non-increasing and $\text{Ric} > 0$), then $\text{scal}_M \geq \text{scal}_N \circ f$ implies that f is a Riemannian submersion.

Proof. Let $\mathcal{D}_E$ be the twisted Dirac operator associated to the map f and the bundle E as explained before Theorem 5.1. Since the product of the Euler characteristic of M with the index of the twisted Dirac operator over the fiber of the point p does not vanish (equation (5.14)), we obtain by the index formula in Theorem 5.4 that the index of $\mathcal{D}_E$ does not vanish in $\text{KO}_k(\mathcal{A})$. By Lemma 2.4 (2) the Dirac operator $\mathcal{D}_E$ is not invertible, hence there exists a family of almost $\mathcal{D}_E$ -harmonic sections by Lemma 3.2. The statements in the theorem follow from Theorem 5.1. $\square$

For certain types of manifolds, e.g. symmetric spaces of compact type, the estimate on the scalar and Ricci curvature are automatically fulfilled, hence the previous theorem can be simplified significantly. See the discussion in the end of Section 1 in [15].

We want to focus on the index-theoretical condition (T) and give in the next corollary three alternative versions of Theorem 5.5. In the first version, we replace the topological condition (T) by a concrete condition involving the higher mapping degree. This seems to be less general, but Proposition 2.7 shows that the higher mapping degree is the most general invariant for the purposes in Theorem 5.5. In the second version, we replace the topological condition (T) on the higher mapping degree by a condition involving the Rosenberg index of the fiber over a regular value of f . This provides a wider range of accessible applicability of the extremality and rigidity statement, since the Rosenberg index is studied for many classes of manifolds. For example, any enlargeable or area-enlargeable spin manifold has non-vanishing rational Rosenberg index. For this simplification, we have to require the existence of a *retraction* of the fiber over a

regular value p of f to the manifold M on the level of fundamental groups. This is a homomorphism $r: \pi_1(M) \rightarrow \pi_1(M_p)$ satisfying $r \circ \iota_* = id$, where $\iota_*: \pi_1(M_p) \rightarrow \pi_1(M)$ is induced by the inclusion $\iota: M_p \hookrightarrow M$. The third version contains the $\hat{A}$ -degree of f and recovers the classical result by Goette and Semmelmann [15, Theorem 2.4].

Corollary 5.6. *Theorem 5.5 still holds if we replace the index-theoretical condition (T) by one of the following conditions:*

- (T1) *The product of the Euler characteristic of N and the higher degree of f does not vanish in $KO_k(C^*\pi_1(M))$.*
- (T2) *There exists a regular value p of f such that M_p is connected,*

$$\chi(N) \cdot \alpha(M_p) \neq 0 \in KO_k(C^*\pi_1(M_p)),$$

and there exists a retraction of M_p to the manifold M on the level of fundamental groups.

- (T3) *The $\hat{A}$ -degree of f and the Euler characteristic of N do not vanish.*

Proof. The first and third part follow by choosing in Theorem 5.5 the bundle E equal to the Mishchenko bundle of M and the trivial flat line bundle $\mathbb{C} \times M \rightarrow M$, respectively. The product of the Euler characteristic of N and the $\hat{A}$ -degree of f matches with the index in equation (5.14) by equation (5.10), the isomorphism $KO_0(\mathbb{C}) \cong \mathbb{Z}$, and the fact that twisting with the trivial flat line bundle does not change the index of the induced Dirac operator. It remains to show the second part of the theorem. Let p be as in (T2) and $r: \pi_1(M) \rightarrow \pi_1(M_p)$ a retraction. We obtain a representation

$$\rho: \pi_1(M) \xrightarrow{r} \pi_1(M_p) \longrightarrow \text{Lin}_{C^*\pi_1(M_p)}(C^*\pi_1(M_p))$$

by left multiplication, which induces a Real flat bundle of finitely generated projective Hilbert $C^*\pi_1(M_p)$ -modules

$$E := \widetilde{M} \times_\rho C^*\pi_1(M_p).$$

Since r is a retraction, ι_* is injective, hence any connected component of the preimage of M_p under the projection $\widetilde{M} \rightarrow M$ is a universal cover of M_p [19, proof of 1B.11]. We fix such a connected component and denote it by $\widetilde{M}_p$. We obtain an isomorphism

$$E|_{M_p} = \widetilde{M} \times_\rho C^*\pi_1(M_p)|_{M_p} \cong \widetilde{M}_p \times_{\pi_1(M_p)} C^*\pi_1(M_p) = \mathcal{L}(M_p).$$

This gives, by the definition of the Rosenberg index in equation (2.2),

$$0 \neq \chi(N) \cdot \alpha(M_p) = \chi(N) \cdot \text{ind}\left(\mathcal{D}_{S_{M_p} \widehat{\otimes} \mathcal{L}(M_p)}\right) = \chi(N) \cdot \text{ind}\left(\mathcal{D}_{S_{M_p} \widehat{\otimes} E|_{M_p}}\right),$$

hence the topological condition (T) in Theorem 5.5 is fulfilled, and the corollary is proved. $\square$

Remark 5.7 (*Relation between topological conditions*). By the considerations in the proof of Corollary 5.6, the index formula in Theorem 5.4 and Proposition 2.7, the three topological properties in Theorem 5.5 and Corollary 5.6 are connected as follows:

$$(T2) \implies (T) \iff (T1) \Longleftarrow (T3)$$

It follows some examples, where the topological hypothesis (T2) of Corollary 5.6 is fulfilled. The examples (5.9) and (5.10) are explicit situations in which the theorem by Goette and Semmelmann [15, Theorem 2.4] is not applicable, since the $\hat{A}$ -degree vanishes.

Example 5.8 (*product*). Let F and N be closed connected manifolds and $\text{pr}_1: N \times F \rightarrow N$ the projection to the first factor. Suppose the first and second Stiefel Whitney classes of F vanish. Then pr_1 is a spin map, and since N is a retract of $N \times F$, the topological condition (T2) in Corollary 5.6 reduces to

$$\chi(N) \cdot \alpha(F) \neq 0 \in \text{KO}_k(C^*\pi_1(F)).$$

This holds for example if the Euler characteristic of N does not vanish and the fiber F has rationally non-vanishing Rosenberg index, e.g.:

- F is enlargeable or area-enlargeable [see 18].
- F admits a metric of non-positive sectional curvature.
- F is aspherical and the strong Novikov conjecture holds for its fundamental group, i.e. the rational assembly map

$$\nu: \text{KO}_*(F) \otimes_{\mathbb{Z}} \mathbb{Q} \longrightarrow \text{KO}_*(C^*\pi_1(F)) \otimes_{\mathbb{Z}} \mathbb{Q}$$

is injective [35]. Note that for aspherical manifolds $F \cong B\pi_1(F)$.

In the third example, the non-vanishing of the Rosenberg index in $\text{KO}_0(C^*\pi_1(F)) \otimes_{\mathbb{Z}} \mathbb{Q}$ follows from the fact that the KO-fundamental class of the closed spin manifold F is rationally non-zero and get mapped under the rational assembly map ν onto $\alpha(M)$. The second example reduced to the first case, since any compact spin manifold that admits a metric of non-positive sectional curvature is enlargeable by the Cartan–Hadamard theorem [26, Theorem 1.10]. The first example can be verified by analyzing the proof of Hanke and Schick [18] that any enlargeable or area-enlargeable spin manifold has non-vanishing Rosenberg index. They construct a unital C^* -algebra Q together with a group homomorphism

$$\phi_*: \text{KO}_0(C^*\pi_1(F)) \longrightarrow K_0(Q)$$

such that the group $K_0(Q)$ splits off a summand $\prod_{i \in \mathbb{N}} \mathbb{Z} / \bigoplus_{i \in \mathbb{N}} \mathbb{Z}$ and the component of $\phi_*(\alpha(F))$ is non-zero in this summand. It follows that $\alpha(F)$ is rationally non-zero.

Example 5.9 (*n-torus*). Since the n -torus is enlargeable, the rigidity statement in Theorem 5.5 applies to

$$S^{2n} \times T^k \xrightarrow{\text{pr}_1} S^{2n} \quad \text{and} \quad \mathbb{RP}^{2n} \times T^k \xrightarrow{\text{pr}_1} \mathbb{RP}^{2n}.$$

The rigidity statement for the round sphere reads for $n \geq 2$ as follows: If $T^k \times S^{2n}$ is equipped with a metric g such that pr_1 is area non-increasing and $\text{scal}_g \geq 2n(2n-1)$, then pr_1 is a Riemannian submersion with $\text{scal}_g = 2n(2n-1)$.

Example 5.10 (*exotic sphere*). Let Σ^{8k+j} be an exotic sphere with $j \in \{1, 2\}$ and non-vanishing Hitchin invariant [see 22], and let $f: M \rightarrow N$ be a smooth area non-increasing spin map between two closed connected Riemannian manifolds. Then the condition (T2) in Corollary 5.6 holds if the Euler characteristic of N is odd and there exists a regular value p of f such that the fiber M_p is isomorphic to Σ^{8k+j} . This follows from $\pi_1(\Sigma^{8k+j}) = 0$ and $\chi(N) \cdot \alpha(\Sigma^{8k+j}) \neq 0 \in \mathbb{Z}_2$. The latter holds by Bott periodicity $\text{KO}_{8k+j}(\mathbb{C}) \cong \mathbb{Z}_2$ [3] and the fact that the Rosenberg index of a simply connected manifold matches with the Hitchin invariant. Maps of this type are for example

$$\mathbb{RP}^{2n} \times \Sigma^{8k+j} \xrightarrow{\text{pr}_1} \mathbb{RP}^{2n} \quad \text{and} \quad \mathbb{CP}^{2n} \times \Sigma^{8k+j} \xrightarrow{\text{pr}_1} \mathbb{CP}^{2n}.$$

Here the manifolds $\mathbb{RP}^{2n}$ and $\mathbb{CP}^{2n}$ are equipped with metrics of non-negative curvature operator and $\text{scal} > 2 \text{Ric} > 0$. Such metrics are given for example by the standard metric on $\mathbb{RP}^{2n}$ and the Fubini–Study metric on $\mathbb{CP}^{2n}$.

Remark 5.11 (*Rigidity for non-orientable manifolds*). We assume in Theorem 5.5 that the map f is orientable, but not that the involved manifolds are orientable. The orientability assumption in the classical statement by Goette and Semmelmann [15, Theorem 2.4] does not restrict the applicability of the classical theorem, since one can always apply it to the lift of the map to the oriented double coverings. This trick also works in our case for the map $\text{pr}_1: \mathbb{RP}^{2n} \times T^k \rightarrow \mathbb{RP}^{2n}$ considered in Example 5.8, but not for the map $\mathbb{RP}^{2n} \times \Sigma^{8k+j} \xrightarrow{\text{pr}_1} \mathbb{RP}^{2n}$ in Example 5.10. The reason is that the lift $S^{2n} \times \Sigma^{8k+j} \xrightarrow{\text{pr}_1} S^{2n}$ does not fulfill the topological condition (T1), since the Euler characteristic of S^{2n} is two and the Hitchin invariant of Σ^{8k+j} is two torsion. This shows that including non-orientable manifolds in Theorem 5.5 extends the applicability of the theorem.

In Example 5.8, we considered projections from a product of two manifold onto one of its factors. A generalization of this class of examples to possible non-trivial fiber bundles as in Example 5.10 requires more topological conditions on the target manifold N . The reason is that we have to ensure, additionally to the index-theoretical condition on the Euler characteristic and the Rosenberg index of the fiber, that the condition on the

fundamental groups in (T2) is fulfilled. It is sufficient for the existence of a retraction from M_p to M on the level of fundamental groups that the target manifold is 2-connected (Corollary 5.13).

All simply connected closed Riemannian manifolds of non-negative curvature operator are classified. See for instance [41, Theorem 1.13]. This leads, together with the classification of irreducible symmetric spaces by Cartan [7], to Lemma 5.12. In Corollary 5.13, we state the rigidity statement for fiber bundles over these manifolds.

Lemma 5.12. *Any factor of the de Rham decomposition of a closed 2-connected Riemannian manifold with non-negative curvature operator and non-vanishing Euler characteristic is isometric to a sphere S^{2n} , $n \geq 2$, equipped with a metric of non-negative curvature operator, a quaternionic Grassmannian or the Cayley projective plane.*

The *quaternionic Grassmannian* is the $4pq$ -dimensional symmetric space C II, given by $\mathrm{Sp}(p+q)/\mathrm{Sp}(p) \times \mathrm{Sp}(q)$, and the *Cayley projective plane* is the 16-dimensional symmetric space F II, given by $F_4/\mathrm{Spin}(9)$. Here we denote by $\mathrm{Sp}(k)$ the compact symplectic group and by F_4 the simply connected Lie group associated to the Lie algebra $\mathfrak{f}_4$ [see 42]. Both manifolds are equipped with their unique (up to scaling) symmetric Riemannian metrics.

Proof. Let N be a factor in the de Rham decomposition of a 2-connected closed manifold with non-zero Euler characteristic and non-negative curvature operator. We obtain by [41, Theorem 1.13] that N is isometric to one of the following Riemannian manifolds:

- Sphere S^{2n} , $n \geq 2$, with non-negative curvature operator.
- Compact irreducible 2-connected symmetric space with non-zero Euler characteristic.

Any irreducible symmetric space of compact type is listed in [20, Table V on page 518] (type I) or isometric to a compact connected simple Lie group equipped with a two-sided invariant metric (type II). Since any Lie group admits a nowhere vanishing vector field, its Euler characteristic vanishes. It remains to rule out all symmetric spaces in [20, Table V on page 518], except the quaternionic Grassmannian and the Cayley projective plane by checking whether they are 2-connected and have non-zero Euler characteristic. Therefore, we use the results about the exceptional Lie groups E_6 , E_7 , E_8 , F_4 and G_2 given in [42] and the fact that for any symmetric space G/K of compact type, with G a connected simple Lie group and K a compact connected subgroup, the following holds:

- (1) The Euler characteristic of G/K is non-zero if and only if $\mathrm{rk} G = \mathrm{rk} K$.
- (2) The symmetric space G/K is 2-connected if and only if the inclusion $\iota: K \hookrightarrow G$ induces an isomorphism $\iota_*: \pi_1(K) \rightarrow \pi_1(G)$.

The second part holds by $\pi_2(G) = 0$ and the long exact sequence of homotopy groups

$$0 \longrightarrow \pi_2(G/K) \longrightarrow \pi_1(K) \xrightarrow{i_*} \pi_1(G) \longrightarrow \pi_1(G/K) \longrightarrow 0. \quad \square$$

Corollary 5.13. *Let N be a product of Riemannian manifolds of the following form:*

- Sphere S^{2n} , $n \geq 2$, equipped with a metric satisfying $\mathcal{R}_{S^n} \geq 0$ and $\text{scal}_{S^n} > 2 \text{Ric}_{S^n} > 0$.
- Quaternionic Grassmannian
- Cayley projective plane

Let $f: M \rightarrow N$ be a fiber bundle, where the total space M is a closed connected spin manifold and the typical fiber F satisfies

$$\chi(N) \cdot \alpha(F) \neq 0 \in \text{KO}_{\dim(F)}(\mathbb{C}^* \pi_1(F)). \quad (5.15)$$

Then f is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$ if the map f is area non-increasing and $\text{scal}_M \geq \text{scal}_N \circ f$ holds. Equation (5.15) holds if the typical fiber F is area-enlargeable, or F is aspherical and the strong Novikov conjecture holds for its fundamental group.

Proof. The manifold N is 2-connected, has non-zero Euler characteristic and has non-negative curvature operator by Lemma 5.12. Furthermore, it satisfies $\text{scal}_{S^n} > 2 \text{Ric}_{S^n} > 0$ by the discussion about locally symmetric spaces of compact type in [15, end of section 1c]. Let $f: M \rightarrow N$ be a fiber bundle with typical fiber F as in the theorem. Since N is 2-connected, it is spin, hence f is a spin map. From the long exact homotopy sequence of fiber bundles

$$\dots \longrightarrow \pi_2(N) \longrightarrow \pi_1(F) \xrightarrow{\iota_*} \pi_1(M) \longrightarrow \pi_1(N) \longrightarrow \dots$$

together with the 2-connectedness of N , we obtain that $\iota_*: \pi_1(F) \rightarrow \pi_1(M)$ is an isomorphism. The condition (T2) in Corollary 5.6 is fulfilled, hence the rigidity statement holds. The second part of the corollary follows from the same considerations as in Example 5.8 about area-enlargeable and aspherical manifolds. $\square$

Data availability

No data was used for the research described in the article.

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Appendix B

Ricci-flat manifolds, parallel spinors and the Rosenberg index

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Ricci-Flat Manifolds, Parallel Spinors and the Rosenberg Index

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Abstract. Every closed connected Riemannian spin manifold of non-zero $\hat{A}$ -genus or non-zero Hitchin invariant with non-negative scalar curvature admits a parallel spinor, in particular is Ricci-flat. In this note, we generalize this result to closed connected spin manifolds of non-vanishing Rosenberg index. This provides a criterion for the existence of a parallel spinor on a finite covering and yields that every closed connected Ricci-flat spin manifold of dimension ≥ 2 with non-vanishing Rosenberg index has special holonomy.

Key words: Ricci-flat manifolds; special holonomy; parallel spinor; scalar curvature; higher index theory

2020 Mathematics Subject Classification: 53C29; 58J20; 53C21; 58B34

1 Introduction and main results

Let (M, g) be a connected Riemannian manifold of dimension n whose universal cover $\tilde{M}$ admits a spin structure. We denote by $\tilde{g}$ the pullback metric of g along the covering map $\tilde{M} \rightarrow M$. If $(\tilde{M}, \tilde{g})$ carries a parallel spinor, g is Ricci-flat, i.e., its Ricci tensor vanishes identically. The converse of this statement holds for closed connected Riemannian spin manifolds with non-zero $\hat{A}$ -genus or more general non-zero Hitchin invariant [10, Section 4.2]. This is a direct consequence of the Schrödinger–Lichnerowicz formula and the non-triviality of the kernel of the spin Dirac operator. It is an open question whether the converse of this statement holds in general.

Question 1.1. *If g is Ricci-flat, does $(\tilde{M}, \tilde{g})$ admit a non-trivial parallel spinor?*

In this note, we give a positive answer to Question 1.1 for closed connected spin manifolds with non-vanishing Rosenberg index (see Theorem A). As a consequence, in dimension ≥ 2 these manifolds have special holonomy for any Ricci-flat metric (see Corollary 1.3).

The Rosenberg index of a closed spin manifold is a class in the real K-theory of the maximal group C^* -algebra of the fundamental group of the manifold. It is defined as the higher index of the $\mathbb{C}l_n$ -linear spin Dirac operator twisted by the Mishchenko bundle (see Section 2). In general, the Rosenberg index is not only non-vanishing whenever the $\hat{A}$ -genus or the Hitchin invariant is non-zero, but also for many other classes of manifolds, as for example enlargeable manifolds [7, 8], and aspherical manifolds whose fundamental group satisfies the Novikov conjecture [17].

Theorem A. *Let (M, g) be a closed connected Riemannian spin manifold with non-vanishing Rosenberg index. Then the following statements are equivalent:*

- (1) *The Riemannian manifold (M, g) is Ricci-flat.*
- (2) *The universal cover $\tilde{M}$, equipped with the pullback metric, admits a non-trivial parallel spinor.*

- (3) *There exists a finite Riemannian covering carrying a non-trivial parallel spinor with respect to the pullback spin structure.*
- (4) *The metric g has non-negative scalar curvature.*

The main part of the proof of Theorem A is to show the implication (4) $\Rightarrow$ (3). The proof strategy is as follows. Suppose there exists a non-trivial twisted harmonic spinor. Then, by the Schrödinger–Lichnerowicz formula and the assumption on the scalar curvature, this twisted spinor is already parallel. We obtain a non-trivial parallel spinor on the universal covering of M , which yields on a suitable finite covering a non-trivial parallel spinor with respect to the pullback spin structure. In the following paragraph, we outline the construction of such a twisted harmonic spinor under the assumption (4) in Theorem A.

A non-zero classical index such as the $\hat{A}$ -genus or the Hitchin invariant gives rise to a non-trivial harmonic spinor. In general, a non-vanishing higher index such as the Rosenberg index does not give rise to a non-trivial harmonic spinor. But non-vanishing of the Rosenberg index is still an obstruction — so far the best obstruction based on the index theory of Dirac operators — to the existence of a positive scalar curvature metric on a closed connected spin manifold. A classical rigidity statement by Bourguignon states that every closed Riemannian manifold that does not admit a positive scalar curvature metric is Ricci-flat if the scalar curvature is non-negative (see, e.g., [11]). Combining these two facts yield under assumption (4) that (M, g) is Ricci-flat. By the *structure theorem for Ricci-flat manifolds* (see Theorem 3.1), the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index, and it is possible to detect the non-vanishing Rosenberg index by the spin Dirac operator twisted by some flat finite-dimensional Hermitian vector bundle (see Proposition 3.2). This detection principle we are using here is due to Schick and Wraith [18], while an alternative approach is given in [15, Section 3]. Here Ramras, Willett, and Yu [15] study the class of groups for which an associated non-vanishing higher index can be detected by a finite representation.

We now discuss the relation between Theorem A and special holonomy. A Riemannian manifold is called *irreducible* if the representation of the reduced holonomy group on the general linear group of the tangent space is irreducible. By Berger’s holonomy list [1], the reduced holonomy group of any connected irreducible Ricci-flat manifold of dimension n is given by $\mathrm{SO}(n)$ in the generic case, or one of the following subgroups: $\mathrm{SU}(n/2)$ (locally Calabi–Yau), $\mathrm{Sp}(n/4)$ (locally hyper-Kähler), G_2 or $\mathrm{Spin}(7)$. In dimension ≥ 2 , the assumption that the Riemannian manifold is irreducible excludes all flat manifolds, in particular all Ricci-flat locally symmetric spaces [2, Theorems 10.72 and 7.61]. Note that the subgroups $U(n)$ (locally Kähler) and $\mathrm{Sp}(n/4) \cdot \mathrm{Sp}(1)$ (locally quaternionic Kähler) cannot occur as holonomy groups of Ricci-flat manifolds [2, Proposition 10.29 and Theorem 14.45]. While there exist examples of closed locally Calabi–Yau, locally hyper-Kähler, G_2 , and $\mathrm{Spin}(7)$ manifolds, no closed irreducible Ricci-flat manifold with generic reduced holonomy is known. This leads to the following question.

Question 1.2. *Does every closed connected Ricci-flat manifold of dimension $n \geq 2$ have special holonomy, i.e., its reduced holonomy group is a proper subgroup of $\mathrm{SO}(n)$?*

The *de Rham splitting theorem* states that every complete simply-connected Riemannian manifold is isometric to a Riemannian product of complete non-flat irreducible manifolds and a Euclidean space [2, Theorem 10.43]. We obtain that for reducible manifolds and flat manifolds the conclusion in Question 1.2 holds. Since every Ricci-flat manifold of dimension ≤ 3 is already flat, this includes all manifolds of dimension 2 and 3. In dimension ≥ 4 , Question 1.2 is still open. It is related to spin geometry as follows. Every complete simply-connected irreducible Riemannian spin manifold $(\tilde{M}, \tilde{g})$ admits a parallel spinor if and only if it has special holonomy (see [10, Theorem 1.2] and [20]). We obtain by Theorem A the following corollary, which addresses Question 1.2 in the special case of spin manifolds with non-vanishing Rosenberg index.

Corollary 1.3. *Every closed connected Ricci-flat spin manifold of dimension ≥ 2 with non-vanishing Rosenberg index has special holonomy.*

2 The Rosenberg index

In this section, we give a brief introduction to higher index theory with the goal to define the Rosenberg index [17]. See [13] for the construction of the $(\mathbb{C}l_n\text{-linear})$ spinor bundle, [12] and [4] for some background concerning Hilbert C^* -modules and differential operators acting on Hilbert C^* -module bundles, [19, Definition 2.1] for a definition of graded real Dirac bundles carrying a Hilbert C^* -module structure, and [16] for an introduction to real K-theory of C^* -algebras and in particular for a definition of the spectral picture of real K-theory.

Let (M, g) be a closed connected Riemannian spin manifold of dimension n . This means that the manifold is oriented and equipped with a spin structure $\text{Spin}(M, g)$, which is a lift of the $\text{SO}(n)$ -principal bundle of oriented orthonormal frames along the double covering $\text{Spin}(n) \rightarrow \text{SO}(n)$. The spin structure gives rise to the following two complex vector bundles

$$\begin{aligned} \mathcal{S} &:= \text{Spin}(M, g) \times_{\rho} \Delta, & \rho: \text{Spin}(n) &\rightarrow \text{Aut}(\Delta) && \text{complex spin representation,} \\ \mathfrak{S} &:= \text{Spin}(M, g) \times_{\text{cl}} \mathbb{C}l_n, & \text{cl}: \text{Spin}(n) &\rightarrow \text{Aut}(\mathbb{C}l_n) && \text{left multiplication.} \end{aligned}$$

They are called the *irreducible spinor bundle* and the $\mathbb{C}l_n$ -linear spinor bundle, respectively. Here we denote by $\mathbb{C}l_n$ the complexification of the Clifford algebra of $\mathbb{R}^n$ which carries the structure of a graded real C^* -algebra and a graded real Hilbert $\mathbb{C}l_n$ -module. The irreducible spinor bundle $\mathcal{S}$ equipped with its natural metric, connection and Clifford multiplication has the structure of a Dirac bundle. The $\mathbb{C}l_n$ -linear spinor bundle $\mathfrak{S}$, equipped with the grading, real structure and Hilbert module structure obtained fiber-wise from $\mathbb{C}l_n$, the induced $\mathbb{C}l_n$ -linear connection, and the Clifford multiplication, carries the structure of a graded real $\mathbb{C}l_n$ -linear Dirac bundle. Note that $\mathfrak{S}$ is isomorphic as a Dirac bundle to a direct sum of copies of the irreducible spinor bundle $\mathcal{S}$.

We denote by π the fundamental group of the manifold M and by $C^*\pi$ the maximal group C^* -algebra of π . The *Mishchenko bundle* $\mathcal{L}$ is the associated bundle of the universal cover of M by the representation $l: \pi \rightarrow \text{Aut}(C^*\pi)$ given by left multiplication [14]. It carries a natural flat connection, a $C^*\pi$ -valued metric and a real structure. Tensoring the $\mathbb{C}l_n$ -linear spinor bundle by the Mishchenko bundle yields a graded real $\mathbb{C}l_n \otimes C^*\pi$ -linear Dirac bundle. The induced $\mathbb{C}l_n \otimes C^*\pi$ -linear Dirac operator $\mathfrak{D}_{\mathcal{L}}$ extends to a self-adjoint and regular unbounded operator

$$\mathfrak{D}_{\mathcal{L}}: H^1(M, \mathfrak{S} \otimes \mathcal{L}) \longrightarrow L^2(M, \mathfrak{S} \otimes \mathcal{L}).$$

Here the Sobolev spaces $H^1(M, \mathfrak{S} \otimes \mathcal{L})$ and $L^2(M, \mathfrak{S} \otimes \mathcal{L})$ are countably generated Hilbert $\mathbb{C}l_n \otimes C^*\pi$ -modules which are similarly defined as the corresponding Sobolev spaces in the classical case. Since the manifold M is compact, the generalized Rellich theorem [4, Theorem 2.33] yields that the functional calculus for unbounded regular and self-adjoint operators on Hilbert C^* -modules [4, Theorem 1.19] restricts to

$$\Phi_{\mathfrak{D}_{\mathcal{L}}}: C_0(\mathbb{R}) \longrightarrow \text{Kom}_{\mathbb{C}l_n \otimes C^*\pi}(L^2(M, \mathfrak{S} \otimes \mathcal{L})). \quad (2.1)$$

This is stated in [4, Proposition 2.34] with $g \equiv 1$ and X a point. Here $\text{Kom}_{\mathcal{A}}(E)$ denotes the compact operators on a Hilbert $\mathcal{A}$ -module E for a C^* -algebra $\mathcal{A}$. The map $\Phi_{\mathfrak{D}_{\mathcal{L}}}$ defines a class in the spectral picture of the n -th real K-theory group of $C^*\pi$ (see, e.g., [16, Lecture 13]). It is called the *Rosenberg index* of the manifold M .

3 Proof of the main theorem

In this section, we give a proof of Theorem A. It is based on the structure theorem for Ricci-flat manifolds [5, Theorem 4.5], which is a direct consequence of the Cheeger–Gromoll splitting theorem [3], and a detection principle for the spectrum of Hilbert $C^*\pi$ -modules by Schick and Wraith [18].

Theorem 3.1 (structure theorem for Ricci-flat manifolds). *Every closed connected Ricci-flat manifold M admits a finite normal Riemannian covering $N \times T^q \rightarrow M$ consisting of a closed simply-connected Ricci-flat manifold N and a flat torus T^q . In particular, the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index.*

Proposition 3.2 ([18, Proposition 3.6]). *Let π be a group containing the subgroup $\mathbb{Z}^q$ with finite index d , and let $\mathcal{B}$ be a possibly unbounded self-adjoint and regular operator on a countably generated Hilbert $C^*\pi$ -module. Suppose $T^2 - 1$ is compact in the sense of Hilbert $C^*\pi$ -modules for the bounded transform $T := \mathcal{B}(\mathcal{B}^2 + 1)^{-1/2}$. If 0 is in the spectrum of $\mathcal{B}$, then there exists a representation $\rho: \pi \rightarrow U(d)$ such that 0 is in the spectrum of $\mathcal{B} \otimes_\rho \text{id}$.*

Proof of Theorem A. Let (M, g) be a closed connected Riemannian spin manifold of dimension n with non-vanishing Rosenberg index. We denote as in Section 2 the fundamental group of M by π , the universal cover of M by $\tilde{M}$, the Mishchenko bundle by $\mathcal{L}$, the irreducible spinor bundle by $\mathcal{S}$, the $\mathbb{C}l_n$ -linear spinor bundle by $\mathfrak{S}$, and the induced Dirac operators by $\mathcal{D}$ and $\mathfrak{D}$, respectively. Moreover, we equip the universal cover of M by the pullback metric $\tilde{g}$ and denote its irreducible spinor bundle by $\tilde{\mathcal{S}}$.

Since a parallel spinor on a finite covering of M lifts to a parallel spinor on the universal covering of M and the scalar curvature is by definition the trace of the Ricci curvature, the implications (3) $\Rightarrow$ (2) and (1) $\Rightarrow$ (4) are trivial. Suppose there exists a non-trivial parallel spinor u on $\tilde{M}$. The curvature tensor K of the spinor bundle $\tilde{\mathcal{S}}$ is related to the Ricci curvature via

$$\text{Ric}_{\tilde{g}}(X) \cdot u = -2 \sum_{j=1}^n e_j \cdot K_{X, e_j} u \quad (3.1)$$

for any local orthonormal frame $e_1, \dots, e_n$ of $T\tilde{M}|_U$ and any vector field X supported in an open subset U [6, Section 3.1]. Since u is non-trivial and parallel, this yields that $\tilde{M}$ is Ricci-flat, hence M is Ricci-flat. This proves the implication (2) $\Rightarrow$ (1). It remains to show the implication (4) $\Rightarrow$ (3). Suppose the scalar curvature of the Riemannian manifold (M, g) is non-negative. We have to show that there exists a finite Riemannian covering of M together with a parallel spinor with respect to the pullback spin structure. The proof splits into the following four steps.

Step 1: The Riemannian manifold (M, g) is Ricci-flat. Since the Rosenberg index does not vanish, the manifold M does not admit a positive scalar curvature metric. Since the scalar curvature of M is non-negative, it follows by Ricci flow or a classical result by Bourguignon [11] that the manifold M is Ricci-flat. A new proof using only spinorial techniques is given by the author in [19, Sections 3 and 4] and will be briefly sketched here.

The non-vanishing Rosenberg index gives rise to a family of almost $\mathfrak{D}_{\mathcal{L}}$ -harmonic sections $\{u_\epsilon\}_{\epsilon>0}$, i.e., $u_\epsilon \in C^\infty(M, \mathfrak{S} \otimes \mathcal{L})$, $\|u_\epsilon\|_{L^2} = 1$ and $\|\mathfrak{D}_{\mathcal{L}}^j u_\epsilon\|_{L^2} < \epsilon$ for all $j > 0$ and all $\epsilon > 0$ [19, Lemma 3.2]. Since the scalar curvature of M is non-negative, we obtain by the Schrödinger–Lichnerowicz formula $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ for all $\epsilon > 0$. We can improve this estimate by Moser iteration [19, Lemma 3.4 ff.] and obtain $\|\nabla u_\epsilon\|_\infty \lesssim \epsilon^r$ for all $\epsilon \in (0, 1)$ and a suitable positive constant r which is independent of ϵ . By the Poincaré inequality, we obtain that the family is almost constant, i.e.,

$$\|\bar{u}_\epsilon - \langle u_\epsilon(p), u_\epsilon(p) \rangle\|_{\mathbb{C}l_n \otimes C^*\pi} \lesssim \epsilon^r \quad \text{with} \quad \bar{u}_\epsilon := \frac{1}{\text{vol}(M)} \int_M \langle u_\epsilon(q), u_\epsilon(q) \rangle dq,$$

for all $\epsilon \in (0, 1)$ and all $p \in M$. Furthermore, we can choose the family $\{u_\epsilon\}_{\epsilon>0}$ such that $\|\nabla^2 u_\epsilon\| \lesssim \sqrt{\epsilon}$ holds for all $\epsilon > 0$ [19, Lemma 4.3]. Together with equation (3.1), we obtain

$$\|\text{Ric}(X)\|_{L^2} \lesssim \|\text{Ric}(X) \cdot u_\epsilon\|_{L^2} \lesssim \sqrt{\epsilon} \quad (3.2)$$

for any vector field X on M and all sufficiently small $\epsilon > 0$, hence M is Ricci-flat.

Step 2: There exists a flat Hermitian bundle over M carrying a non-zero twisted parallel spinor. Since the manifold M is Ricci-flat by Step 1, we can apply Theorem 3.1 and the fundamental group of M contains the subgroup $\mathbb{Z}^q$ with finite index d for some q and d . The Rosenberg index of M does not vanish by assumption, hence the twisted Dirac operator

$$\mathcal{D}_{\mathcal{L}}: H^1(M, \mathfrak{S} \otimes \mathcal{L}) \longrightarrow L^2(M, \mathfrak{S} \otimes \mathcal{L})$$

is not invertible (see, e.g., [19, Lemma 2.4 (ii)]). Since the $\mathbb{C}l_n$ -linear spinor bundle $\mathfrak{S}$ is a direct sum of copies of the irreducible spinor bundle $\mathcal{S}$, we obtain that the twisted Dirac operator

$$\mathcal{D}_{\mathcal{L}}: H^1(M, \mathcal{S} \otimes \mathcal{L}) \longrightarrow L^2(M, \mathcal{S} \otimes \mathcal{L})$$

is also not invertible. The functional calculus associated to $\mathcal{D}_{\mathcal{L}}$ restricts as in equation 2.1, hence the operator $T^2 - 1$ is compact for the bounded transform $T := \mathcal{D}_{\mathcal{L}}(\mathcal{D}_{\mathcal{L}}^2 + 1)^{-1/2}$, and we can apply Proposition 3.2. This yields a representation $\rho: \pi \rightarrow U(d)$ such that $\mathcal{D}_{\mathcal{L}} \otimes_{\rho} \text{id}$ is not invertible. Note that the representation ρ extends by the universal property of the maximal group C^* -algebra to a representation $\bar{\rho}: C^*\pi \rightarrow GL_d(\mathbb{C})$. We define V as the flat Hermitian bundle $\tilde{M} \times_{\rho} \mathbb{C}^d$. Following the identifications

$$(\mathcal{S} \otimes \mathcal{L}) \otimes_{\bar{\rho}} \mathbb{C}^d \cong \mathcal{S} \otimes (\tilde{M} \times_{\pi} C^*\pi \otimes_{\bar{\rho}} \mathbb{C}^d) \cong \mathcal{S} \otimes V,$$

we obtain $\mathcal{D}_{\mathcal{L}} \otimes_{\bar{\rho}} \text{id} = \mathcal{D}_V$ (compare [18, Proposition 3.1]), hence the twisted Dirac operator $\mathcal{D}_V$ is not invertible. Since the manifold M is closed and the twisting bundle V has finite-dimensional fibers, the spectrum of $\mathcal{D}_V$ consists of eigenvalues only, hence the kernel of $\mathcal{D}_V$ is non-zero. Let u be a non-trivial element in the kernel of $\mathcal{D}_V$. Since the scalar curvature of M is non-negative by assumption, we obtain by the Schrödinger–Lichnerowicz formula

$$\|\nabla u\|_{L^2}^2 \leq \langle \nabla^* \nabla u, u \rangle_{L^2} + \frac{1}{4} \langle \text{scal}_g u, u \rangle_{L^2} = \|\mathcal{D}_V u\|_{L^2}^2 = 0.$$

This shows that u is parallel, and the second step is proved.

Step 3: There exists a non-trivial parallel spinor on $\tilde{M}$ equipped with the pullback metric. This step corresponds to [18, Proposition 3.3]. The proof is given below for the sake of completeness. Let V be the flat Hermitian bundle over M and u be the twisted parallel spinor constructed in Step 2. We denote by $\tilde{V}$ and $\tilde{u}$ the pullback of V and u to the universal covering of M , respectively. By construction the twisted spinor $\tilde{u}$ is parallel with respect to the pullback connection. Since the covering map $\tilde{M} \rightarrow M$ is a local isometry, hence preserves all local structures, the pullback connection on $\tilde{M}$ equals the usual spinor connection twisted with the pullback connection on $\tilde{V}$. Since V is by definition in Step 2 equal to $\tilde{M} \times_{\rho} \mathbb{C}^d$ with the representation $\rho: \pi \rightarrow U(d)$, we obtain

$$\tilde{V} = \tilde{M} \times_{\tilde{\rho}} \mathbb{C}^d \cong \tilde{M} \times \mathbb{C}^d \quad \text{with} \quad \tilde{\rho}: \{1\} = \pi_1(\tilde{M}) \longrightarrow \pi \xrightarrow{\rho} U(d).$$

This yields an identification $\tilde{\mathcal{S}} \otimes \tilde{V} \cong (\tilde{\mathcal{S}})^d$ as bundles with connection, hence the parallel twisted spinor $\tilde{u}$ can be identified with a vector of d parallel spinors on $\tilde{M}$. Since $\tilde{u}$ is non-trivial, at least one component of $\tilde{u}$ is non-trivial and we obtain a non-trivial parallel spinor on $\tilde{M}$. This proves the third step.

Step 4: There exists a finite Riemannian covering of M together with a non-trivial parallel spinor with respect to the pullback spin structure. In the first part of the argument below, we follow the approach taken in the proof of [18, Lemma 2.7]. By the structure theorem for Ricci-flat manifolds (see Theorem 3.1), there exists a finite Riemannian covering $(N, g_N) \times (T^q, g_{\text{fl}}) \rightarrow (M, g)$ where (N, g_N) is a simply-connected Ricci-flat manifold and T^q the q -torus equipped with a flat metric g_{fl} . This yields that the universal covering of M is isometric to $(N, g_N) \times (\mathbb{R}^q, \tilde{g}_{\text{fl}})$. Note that a Riemannian product admits a non-trivial parallel spinor if and only if each of its factors does. Since the universal covering of M carries a non-trivial parallel spinor by Step 3, (N, g_N) carries a non-trivial parallel spinor. It now suffices to show that a suitable double covering of the q -torus equipped with a flat metric and the spin structure pulled back from M carries a non-trivial parallel spinor.

Since the q -torus is parallelizable and equipped with a flat metric, the $\text{SO}(q)$ -principal bundle of orthonormal frames $\text{SO}(T^q)$ can be trivialized such that the trivial connection coincides with the Levi-Civita connection. Let s be the corresponding global section of $\text{SO}(T^q)$. Note that double coverings over a connected manifold X are classified by homotopy classes of homomorphisms $\pi_1(X) \rightarrow \mathbb{Z}_2$ [9, Section 1.3].

Let G be $\{0\}$, $\mathbb{Z}$ or $\mathbb{Z}_2$ for $q = 1$, $q = 2$ or $q > 2$, respectively. By the trivialization of $\text{SO}(T^q)$, we obtain an isomorphism $\pi_1(\text{SO}(T^q)) \cong \mathbb{Z}^q \times G$ and the spin structure on T^q , which is a certain double cover of $\text{SO}(T^q)$, is classified by some map

$$(f_1, f_2): \mathbb{Z}^q \times G \rightarrow \mathbb{Z}_2.$$

For $n > 1$, the condition to be a spin principal bundle yields $f_2 \neq 0$ [13, Section 1, Corollary 1.5]. The isomorphism $\pi_1(T^q) \cong \mathbb{Z}^q$ together with the map $f_1: \mathbb{Z}^q \rightarrow \mathbb{Z}_2$ gives rise to a double covering $\bar{T}^q \rightarrow T^q$. By construction, the pullback of the spin structure of T^q along the map $\bar{T}^q \rightarrow T^q$, denoted by $\text{Spin}(\bar{T}^q)$, is classified by the map $(0, f_2): \mathbb{Z}^q \times G \rightarrow \mathbb{Z}_2$. Therefore, the pullback of the section s along the double covering $\bar{T}^q \rightarrow T^q$ lifts to a section of $\text{Spin}(\bar{T}^q)$, hence yields a trivialization of the $\text{Spin}(q)$ -principal bundle of $\bar{T}^q$.

Since the trivial connection on the trivialization of $\text{SO}(\bar{T}^q)$ coincides with the Levi-Civita connection and we just lifted this trivialization, the trivial connection on the trivialization of $\text{Spin}(\bar{T}^q)$ also coincides with the Levi-Civita connection. It follows that the constant sections of the associated trivialized spinor bundle over $\bar{T}^q$ are parallel, and Theorem A is proved. ■

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Appendix C

Scalar rigid submersions are Riemannian products

The following preprint, "Scalar rigid submersions are Riemannian products", is joint work with Oskar Riedler, with equal contributions from both authors. A signed co-authorship statement was attached upon submission of the dissertation and is available at the Examination Office of the University of Münster. [arXive: 2601.14117](#)

Scalar-rigid submersions are Riemannian products

Oskar Riedler and Thomas Tony

Abstract

Scalar-rigid maps are Riemannian submersions by works of Llarull, Goette–Simmelmann, and the second named author. In this article we show that they are essentially Riemannian products of the base manifold with a Ricci-flat fiber. As an application we obtain a Llarull-type theorem for non-zero degree maps onto products of manifolds of non-negative curvature operator and positive Ricci curvature with some enlargeable manifold. The proof is based on spin geometry for Dirac operators and an analysis connecting Clifford multiplication with the representation theory of the curvature operator.

Keywords— scalar curvature • comparison geometry • Dirac operator • Riemannian submersion • curvature operator

1 Introduction and main results

A celebrated result in scalar curvature comparison geometry is the rigidity theorem of Llarull [Lla98]. It states that every area non-increasing smooth non-zero degree map $f: M \rightarrow S^n$ from a closed connected spin manifold M of dimension $n \geq 3$ with $\text{scal}_M \geq n(n-1)$ onto the round sphere is necessarily an isometry. This theorem provides insight into the general question of how a lower bound on the scalar curvature of a Riemannian manifold constrains its global geometry.

Llarull [Lla96] generalized his classical rigidity theorem to a class of comparison maps between manifolds of different dimensions. Namely, every area non-increasing smooth map $f: M \rightarrow S^n$ of non-zero $\hat{A}$ -degree, where M is a closed connected spin manifold of dimension m , with $\text{scal}_M \geq n(n-1)$ is already a Riemannian submersion with $\text{scal}_M = n(n-1)$. Goette and Semmelmann [GS02, Theorem 2.4] replaced the round sphere by some closed connected Riemannian manifold N of non-negative curvature operator, non-zero Euler characteristic and $\text{scal}_N > 2 \text{Ric} > 0$. The second named author [Ton25b, Theorem A] generalized this rigidity theorem by replacing the condition on the $\hat{A}$ -degree by a less restrictive condition involving the higher mapping degree.

The previous discussion shows that scalar-rigid maps must be Riemannian submersions. While the class of Riemannian submersions admits a wide range of geometrically rich and interesting examples, no scalar-rigid maps are known beyond those that are essentially Riemannian products. The main purpose of this article is to show that this is always the case (Theorem A). This generalizes the product rigidity result of [GS01, Theorem 0.2], where product rigidity is shown for comparison maps with target $\mathbb{CP}^n$.

Theorem A. *Let (N, g_N) be a closed connected Riemannian manifold of non-negative curvature operator with $\text{Ric}_N > 0$, and let $f: M \rightarrow N$ be a smooth spin map from a closed connected Riemannian manifold (M, g_M) satisfying*

$$\chi(N) \cdot \deg_{\text{hi}}(f) \neq 0 \in \text{KO}_*(C^*\pi_1(M)). \quad (1.1)$$

*Suppose $\text{scal}_M \geq \text{scal}_N \circ f$ and $g_M \geq f^*g_N$. Then there exists a closed connected Ricci-flat manifold (F, g_F) such that*

$$(M, g_M) \cong (N, g_N) \times (F, g_F) \quad (\text{locally isometric}) \quad (1.2)$$

*and f is given by the projection onto the first factor. Moreover, if $\text{scal}_N > 2\text{Ric}_N > 0$, then the rigidity also holds if we assume $g_M \geq f^*g_N$ just on $\Lambda^2 TM$.*

In Theorem A, two Riemannian manifolds are called *locally isometric* if their universal coverings endowed with the lifted Riemannian metrics are isometric, and the map f is called *spin*, if the first and second Stiefel–Whitney classes satisfy $w_1(TM) = f^*w_1(TN)$ and $w_2(TM) = f^*w_2(TN)$. Its *higher mapping degree* is defined as the index of the spin Dirac operator of the fiber over a regular value of the map f twisted by the Mishchenko–Fomenko bundle of the manifold M (Definition 3.16). This is a generalization of the classical $\hat{A}$ -degree.

Example 1.1 (Local Riemannian product). The conclusion that M is locally isometric to a Riemannian product means that the universal cover is a Riemannian product. If we let $M = S^{2n} \times [0, 1] / \sim$ with $(x, 0) \sim (-x, 1)$, then the map

$$f : (S^{2n} \times [0, 1] / \sim, g_{\text{round}} + dt^2) \rightarrow (\mathbb{RP}^{2n}, g_{\text{round}}), \quad [(x, t)] \mapsto [x]$$

is an example of a map satisfying the conditions of Theorem A which is only locally isometric to a Riemannian product, and not globally.

Remark 1.2 (Existence of a non-trivial parallel spinor). The Ricci-flat manifold F in Theorem A carries a non-trivial parallel spinor on a finite Riemannian covering. This is a priori a stronger statement than in Theorem A, even though it is conjectured that all Ricci-flat spin manifolds carry a non-trivial parallel spinor on a finite Riemannian covering. See the discussion in [Ton25a]. In the setting of Theorem A, we have $F = f^{-1}(p)$ for some regular value p of f . Since the higher mapping degree of f does not vanish, it follows that the Rosenberg index of F is non-trivial [cf. Ton25b, Proposition 2.7]. Together with the fact that the fiber F is Ricci-flat, we obtain the existence of such a non-trivial parallel spinor by [Ton25a, Theorem A].

Remark 1.3 (Non-negative curvature operator). The category of scalar-rigid manifolds N for which we formulate Theorem A is contained in the class of closed manifolds with non-negative curvature operator and positive Ricci curvature. By the work of many mathematicians such manifolds are classified [GM75; CT06; BW08]. They are locally isometric to Riemannian products of symmetric spaces of compact type, certain Kähler metrics on $\mathbb{CP}^n$ biholomorphic to the standard complex structure, and spheres with metrics of non-negative curvature operator.

As an application of the product rigidity in Theorem A, we establish a generalization of Llarull's rigidity theorem [Lla98] to non-zero degree maps to products of the form $N \times F$ for certain F including enlargeable manifolds (Corollary B). This generalization is in the spirit of [HSS24, Theorem 1.1] — which treats non-zero degree maps of the form $M^n \rightarrow S^3 \times T^{n-3}$ with $4 \leq n \leq 7$ — and generalizes a recent rigidity theorem by Chow [Cho25, Theorem A] to all even dimensions (Example C). While the approach in this article is purely based on the Dirac operator method, the proofs by Hao, Shi, and Sun [HSS24] and Chow [Cho25] rely on minimal slicing combined with spinorial techniques and are therefore restricted to low dimensions.

Corollary B. *Let $f: M \rightarrow N \times F$ be a smooth map of non-zero degree, where*

- *M and N are closed connected Riemannian spin manifolds,*
- *N has non-zero Euler characteristic, non-negative curvature operator and $\text{Ric}_N > 0$, and*
- *F is a closed connected oriented manifold which is either (area)-enlargeable, or rationally essential and whose fundamental group satisfies the strong Novikov conjecture.*

If $g_M \geq (\text{pr}_1 \circ f)^ g_N$ and $\text{scal}_M \geq \text{scal}_N \circ \text{pr}_1 \circ f$, then M is locally isometric to the Riemannian product $N \times T^k$ for a flat torus T^k and $\text{pr}_1 \circ f$ is given by the projection onto the first factor.*

Here *rationally essential* means that the fundamental class of F maps non-trivially under the map $H_*(F; \mathbb{Q}) \rightarrow H_*(B\pi_1(F))$ induced by the classifying map $\mu: F \rightarrow B\pi_1(F)$ of the universal cover. We say that its fundamental group satisfies the *strong Novikov conjecture* if the Novikov assembly map is rationally injective.

By the assumptions on the manifold F , the map $\text{pr}_1 \circ f: M \rightarrow N$ satisfies the condition on the higher mapping degree (see Section 6). Corollary B then follows from Theorem A applied to the map $\text{pr}_1 \circ f: M \rightarrow N$.

Example C. *Let (M, g_M) be a closed connected Riemannian spin manifold of dimension $2n+k$, $(S^{2n}, g_{\text{round}})$ the round sphere, (T^k, g_{flat}) a flat torus, and*

$$f: M \rightarrow S^{2n} \times T^k$$

an area non-increasing smooth map of non-zero degree. If $\text{scal}_M \geq 2n(2n-1)$, then $M = (S^{2n}, g_{\text{round}}) \times (T^k, \bar{g}_{\text{flat}})$ for some flat metric $\bar{g}_{\text{flat}}$ and the map f is of the form $f(x, y) = (x, h(y))$ for some 1-Lipschitz map h of non-zero degree.

Remark 1.4. If the manifold N in Corollary B satisfies $\text{scal}_N > 2\text{Ric}_N > 0$, then the rigidity statement in Corollary B remains valid even when the assumptions on the Riemannian metrics are required only on 2-vectors.

Remark 1.5. Note that the manifold F in Corollary B is not equipped with a metric, and that it need not be spin. The spin condition and the geometric restrictions only enter for the map $\text{pr}_1 \circ f$, not for the map f .

Remark 1.6. The rigidity statement in Corollary B also holds if F is closed connected and oriented, and its fundamental class maps non-trivial under the composition

$$H_*(F, \mathbb{Q}) \xrightarrow{\text{ph}^{-1}} \text{KO}_*(F) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\mu_*} \text{KO}_*(B\pi_1(F)) \otimes_{\mathbb{Z}} \mathbb{Q} \xrightarrow{\nu} \text{KO}_*(C^*\pi_1(F)) \otimes_{\mathbb{Z}} \mathbb{Q}.$$

Here the first map is the inverse of the (homological) Pontryagin character ph , the second map the induced map in K-homology of $\mu: F \rightarrow B\pi_1(F)$, and the third map the rational Novikov assembly map [cf. [Zei20](#), Section 2]. Closed connected oriented manifolds which are rationally essential and the strong Novikov conjecture holds for their fundamental group satisfy this property by definition, as well as (area)-enlargeable manifolds by the proofs of [[HS06](#), Theorem 1.4] and [[HS07](#), Theorem 1.3].

In the remaining introduction, we briefly discuss the index theoretic condition in equation (1.1) in Theorem [A](#) and provide some further examples where it is satisfied. If the condition on the higher mapping degree in equation (1.1) holds, the Rosenberg index of the spin manifold $f^{-1}(p)$ does not vanish for every regular value p of the map f . The Rosenberg index of a closed connected spin manifold F , denoted by $\alpha(F)$, is the higher index of the $\mathbb{C}l$ -linear spin Dirac operator twisted by the Mishchenko–Fomenko bundle of F [[Ros83](#)]. Its non-vanishing is the most general known index-theoretic obstruction to the existence of positive scalar curvature metrics. It is well-known for many classes of manifolds that they have non-vanishing Rosenberg index. For instance, if the manifold F is enlargeable or area-enlargeable [[HS06](#); [HS07](#)], admits a metric of non-positive sectional curvature, or if F is aspherical and the strong Novikov conjecture holds for its fundamental group [[Ros83](#)]. In all these examples the Rosenberg index is even rationally non-trivial [cf. [Ton25b](#), Example 5.8]. It is useful to detect the non-vanishing of $\chi(N) \cdot \deg_{hi}(f)$ via a condition involving the Rosenberg index, rather than the higher mapping degree. This relation is worked out in general in [[Ton25b](#), Corollary 5.6], and yields the following examples.

Example 1.7 (Index theoretic condition in Theorem [A](#)). Let M and N be Riemannian manifolds as in Theorem [A](#), and F a closed connected spin manifold. In the following examples, the index theoretic condition in equation (1.1) in Theorem [A](#) holds. See [[Ton25b](#), Corollary 5.6, Example 5.8, Corollary 5.13 and Example 5.10], respectively.

- (1) $f: M \rightarrow N$ smooth spin map with non-zero $\hat{A}$ -degree and $\chi(N) \neq 0$.
- (2) $\text{pr}_1: N \times F \rightarrow N$ with $\chi(N) \cdot \alpha(F) \neq 0 \in KO_*(C^*\pi_1(F))$. This holds for instance if $\chi(N) \neq 0$ and F has rationally non-trivial Rosenberg index, e.g. F is enlargeable.
- (3) N is two-connected and $f: M \rightarrow N$ is a fiber bundle with typical fiber F satisfying $\chi(N) \cdot \alpha(F) \neq 0 \in KO_*(C^*\pi_1(F))$. This holds for instance if $\chi(N) \neq 0$ and F has rationally non-trivial Rosenberg index.
- (4) The Euler characteristic of N is odd and $f: M \rightarrow N$ is a smooth spin map such that there exists a point $p \in N$ with $f^{-1}(p)$ isomorphic to a simply connected manifold of non-vanishing Hitchin invariant [[Hit74](#)]. Examples of such manifolds are certain exotic spheres of dimension 1 and 2 modulo 8 [[Hit74](#), Section 4.3]. Since exotic spheres do not admit Ricci-flat metrics, Theorem [A](#) shows in this case that there exists no Riemannian metric g_M on M such that $g_M \geq f^*g_N$ and $\text{scal}_M \geq \text{scal}_N \circ f$ hold.

We end our discussion by remarking that it is completely unclear, already in the setting of [[GS02](#)], to which extent an index theoretic condition is necessary to achieve rigidity. Can the index condition be replaced by a synthetic condition on the fibers of f ? We formulate a natural condition in the following question:

Question 1.8. Let (N, g_N) be a closed Riemannian manifold with non-negative curvature operator and positive Ricci curvature, and (M, g_M) a closed Riemannian manifold. Let $f: (M, g_M) \rightarrow (N, g_N)$ be a submersion so that

- (1) $\text{scal}_M \geq \text{scal}_N \circ f$ and f is 1-Lipschitz.
- (2) $f^{-1}(\{p\})$ does not admit a metric of positive scalar curvature for any p .

Do there exist examples where f is not a Riemannian submersion, or even where M is not locally isometric to a product over N ?

Remark 1.9. It is known that without some global topological condition on f the rigidity conclusion is false, see e.g. [BZ25, Theorem B]. For this reason we include the condition that f is submersive. See also [Gro23, page 139], [Gro21, page 97] for a similar question in the setting of scalar curvature extremality.

1.1 Structure of article

The article is organized as follows: In Section 2 we sketch the proof of Theorem A, highlighting the main ideas in a simpler setting. In the preliminaries in Section 3 we give the main definitions and properties concerning Riemannian submersions (Section 3.1) and higher index theory (Section 3.2). In Section 4 we set up the spinor bundles in which the proof of Theorem A takes place, and recap the proof that f is a Riemannian submersion (Section 4.1). Then we establish the pointwise inequalities on which our proof is based (Section 4.2). In Section 5 we use these pointwise inequalities. First we show $\text{Ric}_M = f^* \text{Ric}_N$ in Section 5.1. Then we derive in Section 5.2 equations for the O’Neill tensors A and T involving the curvature operator of N , and in Section 5.3 we study some properties of the image of this curvature operator. The previous two subsections are combined in Section 5.4 to show that the mean curvature of $f^{-1}(\{p\})$ vanishes for all $p \in N$. The proofs of Theorem A and Corollary B are then carried out in Section 6.

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2 Sketch of the proof of Theorem A

In this section we give a brief sketch of the main steps proving Theorem A. In order to keep the presentation simple we assume that the Euler characteristic of N and the $\hat{A}$ -degree of the map f do not vanish, rather than the more general index theoretic condition in equation (1.1) involving the higher mapping degree of the map f . The general case works with similar ideas as this case, using the methods developed in [Ton25b].

Let $f: M \rightarrow N$ be a smooth map as in Theorem A and assume that the Euler characteristic of N and the $\hat{A}$ -degree of f do not vanish. We denote by m and n the dimensions of the manifolds M and N , respectively. Since the map f is spin, the indefinite vector bundle $(TM \oplus f^*TN, g_M \oplus (-g_N))$ admits an indefinite spin structure, hence we obtain a $\text{Cl}_{m,n}$ -linear Dirac bundle $\mathcal{S}$ over the manifold M together with Clifford multiplications

$$c: TM \rightarrow \text{End}(\mathcal{S}) \quad \text{and} \quad \bar{c}: f^*TN \rightarrow \text{End}(\mathcal{S}).$$

See Example 3.15 for the construction of $\mathcal{S}$ and the properties of the Clifford multiplications c and $\bar{c}$. We extend the Clifford multiplications c and $\bar{c}$ to $\text{End}(TM)$ and $\text{End}(f^*TN)$ respectively by letting

$$c(v \otimes w^b) := c(v) \cdot c(w), \quad \bar{c}(f^* \bar{v} \otimes \bar{w}^b) := \bar{c}(f^* \bar{v}) \cdot \bar{c}(f^* \bar{w})$$

for $v, w \in TM$, $\bar{v}, \bar{w} \in TN$. In this way we obtain by the identifications $\Lambda^2 TM \cong \mathfrak{so}(TM) \subset \text{End}(TM)$ also a Clifford multiplication by 2-vectors (see Notation 4.1). As in [GS02], the Dirac operator $\mathcal{D}$ of $\mathcal{S}$ satisfies

$$\text{ind}(\mathcal{D}) = \chi(N) \cdot \deg_{\hat{A}}(f) \neq 0 \quad \text{and} \tag{2.1}$$

$$\begin{aligned} \mathcal{D}^2 = \nabla^* \nabla + \underbrace{\frac{1}{4} \text{scal}_M + \frac{1}{16} \sum_{\alpha} \left(c(L\bar{\omega}_{\alpha})^2 + \bar{c}(f^* \bar{L}\bar{\omega}_{\alpha})^2 \right)}_{\geq \frac{1}{4}(\text{scal}_M - \text{scal}_N \circ f) \geq 0} + \underbrace{\sum_{\alpha} -\left(c(L\bar{\omega}_{\alpha}) - \bar{c}(f^* \bar{L}\bar{\omega}_{\alpha}) \right)^2}_{\geq 0} \end{aligned} \tag{2.2}$$

Here $\{\bar{\omega}_{\alpha}\}$ denotes a local orthonormal frame of $\Lambda^2 TN$, $\bar{L}$ is the square root of the curvature operator of N , and $L\bar{\omega}_{\alpha}$ is defined to be $(\Lambda^2 df)^T \bar{L}\bar{\omega}_{\alpha}$.

By equation (2.1) there exists a non-trivial smooth section u of the bundle $\mathcal{S}$ satisfying $\mathcal{D}u = 0$. As in [GS02], we obtain from the first parts of equation (2.2) that the section u is parallel and that the map f is a *Riemannian submersion* with $\text{scal}_M = \text{scal}_N \circ f$. From the last summand in equation (2.2) one finds

$$(c(\omega) - \bar{c}(f^* \bar{\omega}))u = 0 \tag{2.3}$$

for all $\bar{\omega} \in \text{im } \mathcal{R}_N$ with horizontal lift ω . Equation (2.3) is not utilized in [GS02], but this equation is the starting point for our investigation.

Our goal will be to show that the Riemannian submersion f is locally a Riemannian product. This is achieved by showing that the so-called O'Neill tensors A, T of f vanish. These tensors essentially describe how much the horizontal and vertical distributions $\mathcal{H}$ and $\mathcal{V}$ of f deviate from being parallel.

We find pointwise relations involving the tensors A, T by taking the covariant derivatives of equation (2.3). For readability, we use $(c - \bar{c})(\omega)$ to denote $c(\omega) - \bar{c}(f^* \Lambda^2 df \omega)$ in what follows, here $\omega \in \Lambda^2 TM$. Since $c, \bar{c}$, and u are parallel calculating the covariant derivatives is relatively simple, one gets:

Lemma 2.1. *For every point $p \in M$ there exists a non-trivial $u \in \mathcal{S}_p \cong \mathbb{C}l_{m,n}$ and a linear map $B: T_p M \rightarrow \text{End}(\Lambda^2 \mathcal{H}_p)$ such that*

$$(1) \ c(T_U^* \cdot \omega + A_U^* \cdot \omega - \omega \cdot A_U^*)u = 0, \quad \text{and} \quad (2) \ c(2A_X \cdot \omega)u + (c - \bar{c})(B_X(\omega))u = 0$$

for all vertical $U \in T_p M$, all horizontal $X \in T_p M$, and all $\omega \in \Lambda^2 T_p M$ being the horizontal lift of an element in the image of $\mathcal{R}_N$. Here $\cdot$ denotes the product of linear maps.

Here T_U^* and A_X are linear maps $\mathcal{H}_p \rightarrow \mathcal{V}_p$, and we identify ω with an element of $\text{End}(T_p M)$. See Section 3.1 for the precise definitions and relations of A, T, A^*, T^* that we use here, in particular Definition 3.5 and Lemma 3.11. In contrast, the precise form of the linear map B will be irrelevant for all further considerations.

We now explain how to use equation (2.3) to upgrade the relations of Lemma 2.1. Let $\omega \in \Lambda^2 T_p M$ be the horizontal lift of an element in the image of the curvature operator $\mathcal{R}_N$. If $\eta \in \text{End}(T_p M)$ is such that $c(\eta)u = 0$ we obtain by equation (2.3)

$$(c - \bar{c})(\omega) c(\eta)u - c(\eta) (c - \bar{c})(\omega)u = 0. \quad (2.4)$$

A simple computation (cf. Lemma 5.7) using anti-symmetry of ω however shows that

$$(c - \bar{c})(\omega) c(\eta) - c(\eta) (c - \bar{c})(\omega) = -4 c(\omega \cdot \eta - \eta \cdot \omega), \quad (2.5)$$

so that one iteratively upgrades the equation $c(\eta)u = 0$ to

$$c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(\eta))u = 0 \quad (2.6)$$

for all $\omega_1, \dots, \omega_\ell$ being horizontal lifts of elements in $\text{im}(\mathcal{R}_N)_{f(p)}$.

Suppose for the time being that N is a round sphere, or more generally that the curvature operator of N is invertible. One then finds $(c - \bar{c})(B_X(\omega))u = 0$ from equation (2.3) and we can drop this annoying term. Recalling that $A_X: \mathcal{H}_p \rightarrow \mathcal{V}_p$ and applying equation (2.6) to part (2) of Lemma 2.1 gives

$$0 = c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(A_X \omega_{\ell+1}))u = -c(A_X \cdot (\omega_1 \cdot \dots \cdot \omega_{\ell+1})^T)u$$

for any $\omega_1, \dots, \omega_{\ell+1} \in \Lambda^2 \mathcal{H}_p \cong \mathfrak{so}(\mathcal{H}_p)$. We may however write any linear map $\mathcal{H}_p \rightarrow \mathcal{H}_p$ as a linear combination of products of elements of $\mathfrak{so}(\mathcal{H}_p)$. In particular we find for any $X, Y \in \mathcal{H}_p$ that

$$0 = c(A_X \cdot YY^T)u = c(A_X Y)c(Y)u. \quad (2.7)$$

Now if $A_X Y \neq 0$ then both $c(A_X Y)$ and $c(Y)$ are invertible and equation (2.7) clearly leads to a contradiction. We find that $A_X Y = 0$ for all horizontal X, Y and so $A = 0$. Part (1) of Lemma 2.1 then simplifies to $c(T_U^* \cdot \omega)u = 0$. We then repeat the previous steps to find $T_U^* X = 0$ for all vertical U and horizontal X , giving $T = 0$. With $A = T = 0$ we have then shown that f is locally a Riemannian product (under the assumption that the curvature operator of N is invertible).

We now consider the case of a general N . We begin by explaining how to cancel the $(c - \bar{c})(B_X(\omega))u$ term in part (2) of Lemma 2.1. A computation very similar to equations (2.4), (2.5), (2.6) extends the relation $(c - \bar{c})(\omega)u = 0$ from $\omega \in (\Lambda^2 \text{df})^T(\text{im } \mathcal{R}_N) \subseteq \Lambda^2 \mathcal{H}_p$ to the Lie-algebra generated by this image, cf. Proposition 5.5. Referring to this Lie-algebra as $\mathfrak{g}$ we may assume that $B_X(\omega)$ takes values in the $\mathfrak{g}^\perp \subseteq \Lambda^2 \mathcal{H}_p$ (as the part tangential to $\mathfrak{g}$ will evaluate to 0 in Lemma 2.1).

Now applying ad_ω -chains (as sketched in equations (2.4) – (2.6), cf. Lemma 5.9) to part (2) of Lemma 2.1 gives

$$c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(2A_X \cdot \omega))u + (c - \bar{c})((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(B_X(\omega)))u = 0 \quad (2.8)$$

for all $\omega_1, \dots, \omega_\ell \in \mathfrak{g}$ and ω a horizontal lift of an element of $\text{im } \mathcal{R}_N$. Recall that

$$(\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(A_X \cdot \omega) = (A_X \cdot \omega)(\omega_1 \cdot \dots \cdot \omega_\ell)^T,$$

i.e. the first term in equation (2.8) transforms under ad -chains as the representation of $\mathfrak{g}$ on $\mathcal{H}_p$, whereas the second transforms as the adjoint representation of $\mathfrak{g}$ on $\mathfrak{g}^\perp$. It turns out that the conditions $\mathcal{R}_N \geq 0$, $\text{Ric}_N > 0$, imply that these two representations are disjoint (cf. Proposition 5.17). It is a general fact of representation theory that one may then find a linear combination $\sum \prod \text{ad}_{\omega_i}$ of ad -chains of $\mathfrak{g}$ so that

$$\sum \prod \text{ad}_{\omega_i}(A_X \cdot \omega) = A_X \cdot \omega, \quad \sum \prod \text{ad}_{\omega_i}(B_X(\omega)) = 0,$$

i.e. we may separate equation (2.8) to find

$$0 = c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(A_X \cdot \omega))u = -c(A_X \cdot (\omega_1 \cdot \dots \cdot \omega_\ell \cdot \omega))^T u \quad (2.9)$$

for all $\omega_1, \dots, \omega_\ell \in \mathfrak{g}$ and ω a horizontal lift of an element of $\text{im } \mathcal{R}_N$.

In order to work with equation (2.9) we must understand the (associative) sub-algebra of $\text{End}(\mathcal{H}_p)$ generated by $\mathfrak{g}$. Recall that in the case of N being a round sphere this turned out to be all of $\text{End}(\mathcal{H}_p)$, which allowed us to conclude $A_X Y = 0$ for all $X, Y \in \mathcal{H}_p$. The general situation is slightly more complicated. In general $\mathcal{H}_p = \bigoplus_i V_i$ decomposes into irreducible representations of $\mathfrak{g}$. With respect to this decomposition the associative algebra generated by $\mathfrak{g}$ decomposes as $\bigoplus_i \mathcal{A}_i \subseteq \bigoplus_i \text{End}(V_i)$. Further:

Lemma 2.2. *For each $\mathcal{A}_i \subseteq \text{End}(V_i)$ one has either*

- (1) $\mathcal{A}_i = \text{End}(V_i)$.
- (2) *There is an $\mathbb{R}$ -linear isometry $\iota : V_i \rightarrow \mathbb{C}^k$ for some k so that $\mathcal{A}_i = \iota^* \text{End}_{\mathbb{C}}(\mathbb{C}^k)$, i.e. $\mathcal{A}_i$ consists of all complex linear maps on V_i . The complex unit $I \in \iota^* \text{End}_{\mathbb{C}}(\mathbb{C}^k)$ then is an element of $\mathfrak{g}$.*

The decomposition described in Lemma 2.2 is an abstract consequence of the curvature operator $\mathcal{R}_N$ having non-degenerate Ricci tensor, cf. Propositions 5.12 and 5.13.

Now if $Y \in V_i \subseteq \mathcal{H}_p$, where V_i is of the first type described in Lemma 2.2, one can show as before that $A_X Y = 0$, $T_U^* Y = 0$ for all $X \in \mathcal{H}_p$ and $U \in \mathcal{V}_p$. If however V_i is of the second type, the equations of Lemma 2.1 only imply that $g_M(Y, H_p) = 0$, where $H_p = \text{Tr}(T)$ is the mean curvature of the fiber $f^{-1}(\{f(p)\})$ at p , cf. Proposition 5.31. If we combine both cases the most we can conclude at this point is:

Corollary 2.3. *For all $p \in M$ one has $H_p = 0$, where H_p denotes the mean curvature of the fiber $f^{-1}(\{f(p)\})$.*

When dealing with a Riemannian submersion the global vanishing of the mean curvature of the fibers is a useful property. A standard curvature identity for Riemannian submersions is

$$\text{Ric}_M(X, X) - \text{Ric}_N(df X, df X) = -2\|A_X\|^2 - \|T_{(\cdot)}^* X\|^2 + g_M(\nabla_X H, X) \quad (2.10)$$

for all horizontal vectors X and appropriate norms $\|\cdot\|^2$, cf. Theorem 3.8. An inequality like $\text{Ric}_M(X, X) \geq \text{Ric}_N(df X, df X)$ for all horizontal X would allow us to conclude $A = 0$, $T = 0$ from $H = 0$. We will now sketch how equation (2.3) actually implies $\text{Ric}_M(X, X) = \text{Ric}_N(df X, df X)$ for all $X \in TM$, not necessarily horizontal:

Recall that u is parallel, so that $R_\omega^S u = 0$ for all $\omega \in \Lambda^2 TM$, here

$$R_\omega^S = c(\mathcal{R}_M(\omega)) - \bar{c}(f^* \mathcal{R}_N(\Lambda^2 df \omega)) \quad (2.11)$$

is the curvature endomorphism of the spinor bundle $\mathcal{S}$. Equations (2.3) and (2.11) immediately imply that

$$c(\mathcal{R}_M \omega - (\Lambda^2 df)^T \circ \mathcal{R}_N \circ \Lambda^2 df \omega) u = (c(\mathcal{R}_M \omega) - \bar{c}(f^* \mathcal{R}_N \Lambda^2 df \omega)) u = 0.$$

Note that $\mathcal{R}_M - (\Lambda^2 df)^T \circ \mathcal{R}_N \circ \Lambda^2 df$ is a curvature operator on $T_p M$ with Ricci curvature $\text{Ric}_M - df^T \text{Ric}_N df$, which uses f being a Riemannian submersion. If ω_α is an orthonormal basis of $\Lambda^2 T_p M$ and $X \in T_p M$ a classical contraction identity then yields:

$$0 = \sum_{\alpha} c(\omega_\alpha \cdot X) c(\mathcal{R}_M \omega_\alpha - (\Lambda^2 df)^T \circ \mathcal{R}_N \circ \Lambda^2 df \omega_\alpha) u = -c(\text{Ric}_M X - df^T \text{Ric}_N df X) u,$$

which gives

$$\text{Ric}_M X = df^T \text{Ric}_N df X \quad (2.12)$$

for $X \in T_p M$ arbitrary. Combining equation (2.12) and Corollary 2.3 with equation (2.10) allows us to conclude $A = 0$, $T = 0$ also in the general case.

We conclude this section with some remarks on what modifications must be made when we generalize the index theoretic condition from a non-vanishing $\hat{A}$ -degree to $\chi(N) \cdot \deg_{\text{hi}}(f) \neq 0$. In this case one twists the bundle $\mathcal{S}$ with the Mishchenko–Fomenko bundle of M . The index condition guarantees the existence of a family $\{u_\epsilon\}_{\epsilon>0}$ of almost harmonic sections on $\mathcal{S} \otimes \mathcal{L}$ [Ton25b], i.e. $\|u_\epsilon\|_{L^2} = 1$ and $\mathcal{D}^\ell u_\epsilon$ is $O(\epsilon)$ in an L^2 sense for all $\ell \geq 1$. In our proof we use Moser iteration as in [Ton25b] to gain pointwise (that is L^∞) estimates involving u_ϵ and its derivatives, cf. Proposition 4.5. We then proceed similarly as described in this sketch, with our pointwise equations holding only up to an $O(\epsilon^r)$ term for an appropriate constant r .

3 Preliminaries

In this section we review some preliminary notions related to Riemannian submersions as well as facts from higher index theory. The purpose of the first subsection, Section 3.1, is to fix our notational conventions for the O'Neill tensors of a Riemannian submersion, and to list the relations we require in our proof. The second subsection, Section 3.2, provides a bare-bones introduction to the higher index theory used in this paper as well as an explanation of the basic examples that are relevant to our setting.

3.1 Riemannian submersions

Definition 3.1. Let $(M, g_M), (N, g_N)$ be Riemannian manifolds. A submersive map $f: M \rightarrow N$ is called a *Riemannian submersion* if for any $p \in M$ the restriction of $d_p f$ to the *horizontal space* $\ker(d_p f)^\perp \subseteq T_p M$ is an isometry onto $T_{f(p)} N$.

Definition 3.2. Let $(M, g_M), (N, g_N)$ be Riemannian manifolds and $f: M \rightarrow N$ a Riemannian submersion. We denote the *vertical distribution* $\ker(df)$ by $\mathcal{V}$ and its orthogonal complement, the *horizontal distribution*, by $\mathcal{H}$. We define the *O'Neill tensors* of f :

- (1) The *integrability tensor* of f is the linear map $A: \mathcal{H} \times \mathcal{H} \rightarrow \mathcal{V}$ defined by

$$g_M(A_X Y, U) = \frac{1}{2} g_M([X, Y], U)$$

for X, Y horizontal fields and U a vertical field.

- (2) The *second fundamental form* of f is the linear map $T: \mathcal{V} \times \mathcal{V} \rightarrow \mathcal{H}$ defined by

$$g_M(T_U V, X) = g_M(\nabla_U V, X),$$

where X is a horizontal field, U, V are vertical fields, and ∇ the Levi-Civita connection of M .

Remark 3.3. A and T are in fact tensors, i.e. their values at a point $p \in M$ depend only on the values of the relevant fields X, Y, U, V at p .

Theorem 3.4. Let $(M, g_M), (N, g_N)$ be Riemannian manifolds and $f: M \rightarrow N$ a Riemannian submersion. The following are equivalent:

- (1) The tensors A, T both vanish identically.
- (2) The map $f: M \rightarrow N$ is locally the projection map of a Riemannian product, i.e. for $p \in N$ any point there exists a neighborhood $U_p \subseteq N$ and a Riemannian manifold (F, g_F) so that $f^{-1}(U_p) \cong (U_p \times F, g_N + g_F)$ and $f|_{f^{-1}(U_p)}$ is the projection to the first factor.
- (3) For $(\widetilde{M}, g_M), (\widetilde{N}, g_N)$ the universal covers of $(M, g_M), (N, g_N)$ one has $(\widetilde{M}, g_M) = (\widetilde{N} \times F, g_N + g_F)$ for some Riemannian manifold (F, g_F) , and the induced map $\widetilde{f}: \widetilde{M} \rightarrow \widetilde{N}$ is the projection to the first factor.

- (4) *There is a Riemannian manifold (F, g_F) and a group homomorphism $\rho: \pi_1(N) \rightarrow \text{Isom}(F)$ so that $(M, g_M) \cong ((\tilde{N} \times F)/\sim, g_N + g_F)$ where $(p, q) \sim (\gamma(p), \rho(\gamma)(q))$ for $\gamma \in \pi_1(N)$.*

Since A, T are $(1, 2)$ tensors there are many ways in which they can be contracted with other tensors. This can easily lead to notational problems. In order to accommodate such contractions we introduce the following notations for the different $(1, 1)$ tensors associated to A, T :

Definition 3.5. Let (M, g) , (N, g_N) be Riemannian manifolds, $f: M \rightarrow N$ a Riemannian submersion, and $p \in M$. For $X, U \in T_p M$ horizontal and vertical respectively, we define the tensors A_X, S_X, T_U^*, A_U^* by:

$$\begin{aligned} T_U^* : \mathcal{H}_p &\rightarrow \mathcal{V}_p, & X &\mapsto g_M(T_U(\cdot), X)^\sharp, & S_X : \mathcal{V}_p &\rightarrow \mathcal{V}_p, & U &\mapsto T_U^* X, \\ A_X : \mathcal{H}_p &\rightarrow \mathcal{V}_p, & Y &\mapsto A_X Y, & A_U^* : \mathcal{H}_p &\rightarrow \mathcal{H}_p, & X &\mapsto g_M(A_X(\cdot), U)^\sharp. \end{aligned}$$

Remark 3.6. Let $X, Y \in T_p M$ be horizontal and $U \in T_p M$ vertical, then

$$g_M(T_U^* X, T_U^* Y) = g_M\left(U, \frac{S_X S_Y + S_Y S_X}{2} U\right).$$

Remark 3.7. The tensor $S_X : \mathcal{V} \rightarrow \mathcal{V}$ is also called the *shape operator* or *Weingarten map* of the fibers in direction X . If b_i is an orthonormal basis of $\mathcal{H}$ then $\text{Tr}(S_{(\cdot)})^\sharp = \sum_i \text{Tr}(S_{b_i}) b_i$ is the mean curvature of the fibers.

Theorem 3.8 ([Bes87, Proposition 9.36]). *Let (M, g_M) , (N, g_N) be Riemannian manifolds and $f: M \rightarrow N$ a Riemannian submersion. Then*

$$\text{Ric}_M(X, X) = \text{Ric}_N(df(X), df(X)) - 2\|A_X\|_2^2 - \|S_X\|_2^2 + g_M(\nabla_X^M H, X)$$

for all horizontal vectors $X \in TM$. Here $\|\cdot\|_2$ denotes the Hilbert-Schmitt norm, and H denotes the mean curvature of the fibers.

Theorem 3.8 together with the expression of the mean curvature in Remark 3.7 yield the following corollary.

Corollary 3.9. *Let (M, g_M) , (N, g_N) be Riemannian manifolds and $f: M \rightarrow N$ a Riemannian submersion. If*

$$\text{Ric}_M(X, X) = \text{Ric}_N(df(X), df(X)) \quad \text{and} \quad \text{Tr}(S_X) = 0$$

for all horizontal vectors $X \in TM$, then A and T vanish identically.

We now remark on some relations that allow us to express the tensors A and T via derivatives. Lemma 3.11 in particular will be used in the main proof to connect Clifford multiplication with the O'Neill tensors A, T .

Lemma 3.10. *Let (M, g) , (N, h) be Riemannian manifolds and $f: M \rightarrow N$ a Riemannian submersion and $p \in M$. Let $X, U \in T_p M$ be horizontal and vertical respectively, and $\bar{Y}$ a section of TN with horizontal lift $Y = df^T \bar{Y}$. Then*

$$\nabla_U Y = T_U^* Y + A_U^* Y, \quad \nabla_X Y = A_X Y + \widehat{\nabla_{df X}^N \bar{Y}},$$

where $\widehat{\cdot}$ denotes the horizontal lift of a section of N .

Proof. These are standard calculations making use of the Koszul formula and functoriality of the Lie bracket, see e.g. [ONe66, Lemmas 1 to 3]. $\square$

Lemma 3.11. *Let $(M, g), (N, h)$ be Riemannian manifolds, $f: M \rightarrow N$ a Riemannian submersion, $p \in M$. Let $X, U \in T_p M$ be horizontal and vertical respectively and let ω be the horizontal lift of a section $\bar{\omega} \in C^\infty(N, \Lambda^2 TN)$. Then*

$$\nabla_U(\omega) = -\text{ad}_\omega(T_U^* - (T_U^*)^T + A_U^*), \quad \nabla_X\omega = -\text{ad}_\omega(A_X - A_X^T) + \widehat{\nabla_X^N \bar{\omega}},$$

where $\widehat{}$ denotes the horizontal lift of a section on N , ad_ω the commutator with an element of $\mathfrak{so}(T_p M) \cong \Lambda^2 T_p M$, and T the transpose of a linear map.

Proof. Recall that T_U^*, A_X take values in $\mathcal{V}_p$ so that

$$-\text{ad}_\omega(T_U^*) = T_U^* \cdot \omega, \quad \text{ad}_\omega((T_U^*)^T) = \omega \cdot (T_U^*)^T, \quad -\text{ad}_\omega(A_X) = A_X \cdot \omega, \quad \text{ad}_\omega(A_X^T) = \omega \cdot A_X^T$$

for any horizontal ω . On the other hand A_U^* is an anti-symmetric linear map $\mathcal{H}_p \rightarrow \mathcal{H}_p$, so that

$$-\text{ad}_\omega(A_U^*) = A_U^* \cdot \omega + \omega \cdot (A_U^*)^T.$$

Now for any sections Y, Z of TM being horizontal lifts of $\bar{X}, \bar{Y}$ one has by Lemma 3.10

$$\begin{aligned} \nabla_U(Y \otimes Z^b) &= (T_U^* Y) \otimes Z^b + Y \otimes (T_U^* Z)^b + (A_U^* Y) \otimes Z^b + Y \otimes (A_U^* Z)^b \\ &= T_U^* \cdot (Y \otimes Z^b) + (Y \otimes Z^b) \cdot (T_U^*)^T + A_U^* \cdot (Y \otimes Z^b) + (Y \otimes Z^b) \cdot (A_U^*)^T, \end{aligned}$$

which shows the first relation of the lemma. The second follows in the same way. $\square$

3.2 $\mathcal{A}$ -linear Dirac bundles and the higher mapping degree

In this section we give a brief introduction to higher index theory. We define Dirac bundles linear over a C^* -algebra (Definition 3.12), their induced Dirac operators, give a summary of their most important properties and study two relevant examples: First, the Cl_m -linear spinor bundle twisted by the Mishchenko–Fomenko bundle (Example 3.14). Second, the $\text{Cl}_{m,n}$ -linear Dirac bundle induced from a spin map $f: M \rightarrow N$ (Example 3.15). In the end of this section, we define the higher mapping degree of a spin map (Definition 3.16).

For background material concerning (graded Real) unital C^* -algebras, their Real K-theory, and Hilbert C^* -modules see [Mur90; Roe17; Lan95]. For an introduction to bundles of finitely generated projective Hilbert $\mathcal{A}$ -modules as well as first-order $\mathcal{A}$ -linear differential operators see [MF80; Ebe16; Sch05]. The definition of $\mathcal{A}$ -linear Dirac bundles [cf. CZ23, Definition 2.1] is a straight forward generalization of Cl -linear Dirac bundles as treated in [LM89, Chapter II §7]. See also [Ton25b] for more details concerning higher index theory, spin maps and the higher mapping degree.

Definition 3.12. Let $\mathcal{A}$ be a graded Real unital C^* -algebra and (M, g) a closed Riemannian manifold of dimension m . A *graded Real $\mathcal{A}$ -linear Dirac bundle* is a bundle $\mathcal{S} \rightarrow M$ of finitely generated projective graded Real Hilbert $\mathcal{A}$ -modules equipped with an $\mathcal{A}$ -valued inner product,

a metric connection, and a parallel bundle map $c: TM \rightarrow \text{End}(\mathcal{S})$ such that $c(v)$ is odd, Real, skew-adjoint, $\mathcal{A}$ -linear and $c(v)^2 = -g(v, v)$ for all $v \in TM$. The grading and the Real structure have to be compatible with all other structures. The $\mathcal{A}$ -linear Dirac operator induced by a graded Real $\mathcal{A}$ -linear Dirac bundle $\mathcal{S}$ is defined via

$$\mathcal{D} = \mathcal{D}_{\mathcal{S}}: C^\infty(M, \mathcal{S}) \longrightarrow C^\infty(M, \mathcal{S}), \quad \mathcal{D}u := \sum_{i=1}^m c(e_i) \nabla_{e_i} u.$$

Here $e_1, \dots, e_m$ denotes a local orthonormal frame of TM .

Facts and Notation 3.13. Let M be a closed Riemannian manifold, $\mathcal{A}$ a graded Real unital C^* -algebra, and $\mathcal{S}$ a graded Real $\mathcal{A}$ -linear Dirac bundle. For the induced $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$ one has the following:

- (1) $\mathcal{D}$ is a Real odd elliptic formally self-adjoint first-order $\mathcal{A}$ -linear differential operator.
- (2) The $\mathcal{A}$ -linear Dirac operator $\mathcal{D}$ extends to a self-adjoint and regular unbounded operator

$$\mathcal{D}: H^1(M, \mathcal{S}) \rightarrow L^2(M, \mathcal{S})$$

[Ebe16, Theorem 1.14]. Here $L^2(M, \mathcal{S})$ and $H^1(M, \mathcal{S})$ are graded Real Hilbert $\mathcal{A}$ -modules defined as the completion of the graded Real pre-Hilbert $\mathcal{A}$ -module $C^\infty(M, \mathcal{S})$ equipped with the $\mathcal{A}$ -valued inner products

$$\langle \cdot, \cdot \rangle_{L^2} := \int_M \langle \cdot, \cdot \rangle_p d\mu(p) \quad \text{and} \quad \langle \cdot, \cdot \rangle_{H^1} := \langle \nabla \cdot, \nabla \cdot \rangle_{L^2} + \langle \cdot, \cdot \rangle_{L^2},$$

respectively. Here we write $\langle \cdot, \cdot \rangle_p$ for the $\mathcal{A}$ -valued inner product in the fiber of $\mathcal{S}$ over a point $p \in M$. In this context, we define $|\cdot|_p^2 := \langle \cdot, \cdot \rangle_p$, $\|\cdot\|_p^2 := \|\langle \cdot, \cdot \rangle_p\|_{\mathcal{A}}$, and ($\mathcal{A}$ -valued) L^2 -norms via

$$|\cdot|_{L^2}^2 := \langle \cdot, \cdot \rangle_{L^2} \in \mathcal{A} \quad \text{and} \quad \|\cdot\|_{L^2}^2 := \|\langle \cdot, \cdot \rangle_{L^2}\|_{\mathcal{A}} \in \mathbb{R}.$$

- (3) Since the manifold M is closed, the functional calculus of the graded Real self-adjoint and regular operator $\mathcal{D}: H^1(M, \mathcal{S}) \rightarrow L^2(M, \mathcal{S})$ defines a class in the spectral picture of the 0-th Real K -theory group of the C^* -algebra $\mathcal{A}$ [Ebe16]. This is called *the higher index* of $\mathcal{D}$ and denoted by $\text{ind}(\mathcal{D}) \in \text{KO}_0(\mathcal{A})$.

Example 3.14 (Cl_m -linear spinor bundle and the Rosenberg index). Let M be a closed connected Riemannian spin manifold of dimension m . The graded Real C^* -algebra Cl_m is defined as the complex Clifford algebra Cl_m equipped with the norm given in [Ebe16, p.4], and its natural grading and Real structure. We define the following two bundles over the manifold M .

- The Cl_m -linear spinor bundle $\mathcal{S}M$ is defined as the vector bundle associated to the $\text{Spin}(m)$ -principle bundle with respect to the representation $\text{Spin}(m) \rightarrow \text{Cl}_m$ given by left multiplication. It defines a graded Real Cl_m -linear Dirac bundle.

- Let $C^*\pi_1(M)$ be the maximal group C^* -algebra of the fundamental group of the manifold M together with the trivial grading and its natural Real structure. The *Mishchenko–Fomenko* bundle $\mathcal{L}(M)$ of the manifold M is defined via

$$\mathcal{L} := \mathcal{L}(M) := \widetilde{M} \times_{\pi_1(M)} C^*\pi_1(M).$$

Here $\pi_1(M)$ acts on $C^*\pi_1(M)$ by left multiplication and $\widetilde{M}$ denotes the universal covering of M . The Mishchenko–Fomenko bundle defines a flat bundle of graded Real finitely generated projective Hilbert $C^*\pi_1(M)$ -modules [MF80].

We twist the $\mathbb{C}l_m$ -linear spinor bundle with the Mishchenko–Fomenko bundle and obtain a graded Real $\mathbb{C}l_m \otimes C^*\pi_1(M)$ -linear Dirac bundle $\mathcal{SM}_{\mathcal{L}}$. The higher index of the induced $\mathbb{C}l_m \otimes C^*\pi_1(M)$ -linear Dirac operator is called the *Rosenberg index* of the manifold M [Ros83].

Example 3.15 (Spin maps and their $\mathbb{C}l_{m,n}$ -linear spinor bundle [Ton25b, Section 2.2]). Let (M, g_M) and (N, g_N) be closed connected Riemannian manifolds of dimension m and n , respectively, and $f: M \rightarrow N$ a smooth spin map. i.e. the first and second Stiefel Whitney classes satisfy

$$\omega_1(TM) = f^*\omega_1(TN) \quad \text{and} \quad \omega_2(TM) = f^*\omega_2(TN).$$

Since f is spin, the vector bundle $TM \oplus f^*TN$ equipped with the indefinite metric $g_M \oplus (-f^*g_N)$ admits an indefinite spin structure. This is a lift of the $SO(m, n)$ -principle bundle of orthonormal frames of $TM \oplus f^*TN$ to a $\text{Spin}(m, n)$ -principle bundle along the map $\lambda: \text{Spin}(m, n) \rightarrow SO(m, n)$ [Bau81]. The $\text{Spin}(m, n)$ -principle bundle gives rise to a graded Real $\mathbb{C}l_{m,n}$ -linear Dirac bundle $\mathcal{S}$, which comes from the representation of $\text{Spin}(m, n)$ on $\mathbb{C}l_{m,n}$ given by left multiplication, together with Clifford multiplications

$$c: TM \rightarrow \text{End}(\mathcal{S}) \quad \text{and} \quad \bar{c}: f^*TN \rightarrow \text{End}(\mathcal{S})$$

[Ton25b, Section 2.2]. Here $\mathbb{C}l_{m,n}$ is the graded Real C^* -algebra generated by $\mathbb{R}^m \oplus \mathbb{R}^n$ such that $(v, w)^* = (-v, w)$ and

$$(v, w)(v', w') + (v', w')(v, w) = -2\langle v, v' \rangle_{\text{euc}} + 2\langle w, w' \rangle_{\text{euc}}$$

for all $(v, w), (v', w') \in \mathbb{R}^m \oplus \mathbb{R}^n$ [see Ebe16, p.4]. Fiberwise, the Clifford multiplications c and $\bar{c}$ are induced by the map $\mathbb{R}^m \oplus \mathbb{R}^n \rightarrow \text{End}(\mathbb{C}l_{m,n})$ given by left multiplication, hence

- $c(v)$ is skew-adjoint and $c(v)^2 = -g_M(v, v)$ for all $v \in TM$, and
- $\bar{c}(w)$ is self-adjoint and $\bar{c}(w)^2 = g_N(w, w)$ for all $w \in f^*TN$.

Locally, the bundle $\mathcal{S}$ is isomorphic to the graded tensor product of the $\mathbb{C}l_{m,0}$ -linear spinor bundle $\mathcal{SM}$ and the $\mathbb{C}l_{0,n}$ -linear spinor bundle $\mathcal{SN}$ equipped with the tensor connection.

Definition 3.16 (Higher mapping degree [Ton25b, Definition 5.2]). Let $f: M \rightarrow N$ be a spin map between two closed connected Riemannian manifold M and N . The higher mapping degree of f is defined as the higher index of the $\mathbb{C}l_k$ -linear Dirac operator of the spin manifold $M_p = f^{-1}(p)$ twisted by the Mishchenko–Fomenko bundle of the manifold M :

$$\text{deg}_{\text{hi}}(f) := \text{ind}(\mathcal{D}_{\mathcal{SM}_p \otimes \mathcal{L}(M)}|_{M_p}) \in \text{KO}_0(\mathbb{C}l_k \otimes C^*\pi_1(M)) \cong \text{KO}_k(C^*\pi_1(M)).$$

Here $p \in N$ denotes a regular value of the map f , $k := \dim(M) - \dim(N)$, and $\mathcal{SM}_p$ is the $\mathbb{C}l_k$ -linear spinor bundle of the spin manifold M_p .

4 Setup and deriving the pointwise estimates

In this section we give the setup for the proof of Theorem A. We show that a map $f: M \rightarrow N$ as in Theorem A is a Riemannian submersion (Section 4.1) and establish pointwise estimates relating the Clifford multiplications of M and N with local geometric quantities (Section 4.2).

Throughout this section, let (M, g_M) and (N, g_N) be closed connected Riemannian manifolds of dimension m and n , respectively, and let $f: M \rightarrow N$ be a smooth spin map such that $\chi(N) \cdot \deg_{\text{hi}} f \neq 0$. Moreover, we assume that the curvature operator of N is non-negative, $\text{scal}_M \geq \text{scal}_N \circ f$, and either

- $g_M \geq f^*g_N$ and $\text{Ric}_N > 0$, or
- $g_M \geq f^*g_N$ on $\Lambda^2 TM$ and $\text{scal}_N > 2\text{Ric}_N > 0$.

Here $\chi(N)$ denotes the Euler characteristic of N and $\deg_{\text{hi}}(f)$ the higher mapping degree of f , recall Definition 3.16.

4.1 Setup and recap

Denote by $\mathcal{A}$ the graded Real C^* -algebra $\mathbb{C}l_{m,n} \otimes C^*\pi_1(M)$ and its norm by $\|\cdot\|_{\mathcal{A}}$. The spin map f gives rise to a $\mathbb{C}l_{m,n}$ -linear Dirac bundles $\mathcal{S}$ as in Example 3.15. We twist the bundle $\mathcal{S}$ with the flat Mishchenko–Fomenko bundle of M and obtain an $\mathcal{A}$ -linear Dirac bundle $\mathcal{S}_{\mathcal{L}} := \mathcal{S} \otimes \mathcal{L}(M)$ together with Clifford multiplications

$$c: TM \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}}) \quad \text{and} \quad \bar{c}: f^*TN \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}}).$$

We denote the induced $\mathcal{A}$ -linear Dirac operator by $\mathcal{D}_{\mathcal{L}}$.

Let $e_1, \dots, e_m$ be a local g_M -orthonormal frame of TM and $\bar{e}_1, \dots, \bar{e}_n$ a local g_N -orthonormal frame of TN such that there exists non-negative functions $\mu_1, \dots, \mu_{\min(m,n)}$ with

$$df(e_i) = \begin{cases} \mu_i \bar{e}_i & \text{if } i \leq \min(m, n) \\ 0 & \text{otherwise.} \end{cases}$$

This is possible by considering the singular value decomposition of df . We use the following notation:

Notation 4.1. (1) Let $\{\omega_{\alpha}\} = \{e_i \wedge e_j | 1 \leq i < j \leq n\}$ and similarly $\{\bar{\omega}_{\beta}\}$ be local orthonormal frames of $\Lambda^2 TM$ and $\Lambda^2 TN$, respectively.

(2) We define Clifford multiplications

$$c: \text{End}(TM) \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}}) \quad \text{and} \quad \bar{c}: \text{End}(f^*TN) \rightarrow \text{End}(\mathcal{S}_{\mathcal{L}})$$

via

$$c(A) := \sum_{i,j} g_M(Ae_i, e_j) c(e_i) c(e_j)$$

and similarly for $\bar{c}$. By the identifications

$$\Lambda^2 TM \cong \mathfrak{so}(TM) \subseteq \text{End}(TM), e_i \wedge e_j \rightarrow e_i \otimes e_j^b - e_j \otimes e_i^b$$

and similar for f^*TN , we obtain Clifford multiplications also by 2-vectors¹. Here $\mathfrak{so}$ denotes the Lie algebra of the special orthogonal group, consisting of all anti-symmetric endomorphisms.

- (3) We denote the curvature operator of N by $\mathcal{R}_N$ and its square root by $\bar{L}$, which exists since $\mathcal{R}_N \geq 0$. Moreover, we write

$$L\bar{\omega}_\alpha := \sum_{\beta} g_N(\bar{L}\bar{\omega}_\alpha, \bar{\omega}_\beta) \omega_\beta.$$

Note that $L\bar{\omega} = (\Lambda^2 df)^T \bar{L}\bar{\omega}$ for all $\bar{\omega} \in \Lambda^2 TN$.

Since the curvature operator of N is non-negative, the Dirac operator $\mathcal{D}_{\mathcal{L}}$ satisfies as in [GS02, Section 1b] the following Schrödinger-Lichnerowicz type formula.

Lemma 4.2. *The Dirac operator $\mathcal{D}_{\mathcal{L}}$ satisfies the Schrödinger-Lichnerowicz type formula*

$$\mathcal{D}_{\mathcal{L}}^2 = \nabla^* \nabla + \frac{\text{scal}_M}{4} + \frac{1}{16} \sum_{\alpha} \left(c(L\bar{\omega}_\alpha)^2 + \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)^2 - (c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha))^2 \right) \quad (4.1)$$

with

$$\sum_{\alpha} c(L\bar{\omega}_\alpha)^2 = -2 \sum_{i,j} \mu_i^2 \mu_j^2 (R_{ijji}^{TN} \circ f) \quad \text{and} \quad \sum_{\alpha} \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)^2 = -2 \text{scal}_N \circ f. \quad (4.2)$$

Proof. Since the Mishchenko–Fomenko bundle is flat, we obtain by [LM89, Theorem 4.15 and 8.17]

$$\begin{aligned} \mathcal{D}_{\mathcal{L}}^2 &= \nabla^* \nabla + \frac{\text{scal}_M}{4} - \frac{1}{8} \sum_{i,j,k,l} g_N(R_{\bar{e}_i, \bar{e}_j}^{TN} \bar{e}_k, \bar{e}_l) c(e_i) c(e_j) \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_l) \\ &= \nabla^* \nabla + \frac{\text{scal}_M}{4} + \frac{1}{8} \sum_{\alpha, \beta} g_N(\mathcal{R}_N \bar{\omega}_\alpha, \bar{\omega}_\beta) c(\omega_\alpha) \bar{c}(f^* \bar{\omega}_\beta). \end{aligned}$$

Compared to [GS02], we have a different sign in front of $1/8$. This is because we consider the negative definite metric $-f^*g_N$ in the construction of the twisting bundle $f^*\mathcal{S}N$ (see Example 3.15). Using the existence of the square root of the curvature operator of N as in [GS02, Section 1b] yields equation (4.1) and equation (4.2). $\square$

Lemma 4.3. *There exists a family u_ϵ of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic sections on $\mathcal{S}_{\mathcal{L}}$, i.e. a family $\{u_\epsilon\} \subset C_c^\infty(M, \mathcal{S}_{\mathcal{L}})$ such that*

$$\|u_\epsilon\|_{L^2} = 1 \quad \text{and} \quad \|\mathcal{D}_{\mathcal{L}}^\ell u_\epsilon\|_{L^2} < \epsilon^\ell$$

for all $\epsilon > 0$ and for all $\ell \geq 1$.

Proof. By the index theoretic assumption on the map f in equation (1.1) and [Ton25b, Theorem 5.4], the higher index of the twisted Dirac operator $\mathcal{D}_{\mathcal{L}}$ satisfies

$$\text{ind}(\mathcal{D}_{\mathcal{L}}) = \chi(N) \cdot \deg_{\text{hi}}(f) \neq 0 \in \text{KO}_*(C^*\pi_1(M)).$$

It follows that $\mathcal{D}_{\mathcal{L}}$ is not invertible, hence [Ton25b, Lemma 3.2] provides a family of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic sections. $\square$

¹The convention here differs from the notation in [GS02] by a factor 2. We have $c(e_i \wedge e_j) = 2c(e_i)c(e_j)$.

The Schrödinger-Lichnerowicz type formula in Lemma 4.2 yields:

Lemma 4.4. *For every family u_ϵ of almost $\mathcal{D}_\mathcal{L}$ -harmonic sections on $\mathcal{S}_\mathcal{L}$ the following holds.*

- (1) $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ for all $\epsilon > 0$.
- (2) For every $\ell \geq 2$ there exists a constant C such that $\|\nabla^\ell u_\epsilon\|_{L^2} < C\sqrt{\epsilon}$ for all $\epsilon \in (0, 1)$.

Proof. Since $\text{scal}_M \geq \text{scal}_N \circ f$, $\mathcal{R}_N \geq 0$, and f is area non-increasing, we obtain by Lemma 4.2

$$\mathcal{D}_\mathcal{L}^2 \geq \nabla^* \nabla + \frac{\text{scal}_M - \text{scal}_N \circ f}{4} \geq \nabla^* \nabla.$$

Since u_ϵ is almost $\mathcal{D}_\mathcal{L}$ -harmonic, we obtain $\|\nabla u_\epsilon\|_{L^2} < \epsilon$ for all $\epsilon > 0$. For fixed $\ell \geq 2$, there exists a constant C such that

$$\|\nabla^\ell u_\epsilon\|_{L^2}^2 = \|\langle (\nabla^*)^{\ell-1} \nabla^\ell u_\epsilon, \nabla u_\epsilon \rangle_{L^2}\|_{\mathcal{A}} \leq \|(\nabla^*)^{\ell-1} \nabla^\ell u_\epsilon\|_{L^2} \|\nabla u_\epsilon\|_{L^2} < C^2 \epsilon$$

for all $\epsilon \in (0, 1)$. Here $\nabla^*: C^\infty(M, T^*M \otimes \mathcal{S}_\mathcal{L}) \rightarrow C^\infty(M, \mathcal{S}_\mathcal{L})$ denotes the L^2 -adjoint of ∇ . The second step holds by the Cauchy-Schwarz inequality, and in the last step, we used part (1) and

$$\|(\nabla^*)^{\ell-1} \nabla^\ell u_\epsilon\|_{L^2} \leq \|u_\epsilon\|_{H^{2\ell-1}} \leq C_1 \sum_{j=0}^{2\ell-1} \|\mathcal{D}_\mathcal{L}^j u_\epsilon\|_{L^2} < 2\ell C_1 =: C^2, \quad (4.3)$$

which holds for a suitable constant C_1 by the fundamental elliptic estimate [cf. Ton25b, Lemma 2.5] and almost $\mathcal{D}_\mathcal{L}$ -harmonicity of u_ϵ . $\square$

For $u \in C^\infty(M, \mathcal{S}_\mathcal{L})$ we define the L^∞ -norm via $\|u\|_\infty := \max_{p \in M} \|u\|_p$. We upgrade as in the proof of [Ton25b, Lemma 3.5] the L^2 -estimates on $\mathcal{D}_\mathcal{L}^\ell u_\epsilon$ and $\nabla^\ell u_\epsilon$ to the $\|\cdot\|_\infty$ -norm by Moser iteration [cf. Ton25b, Lemma 3.4].

Proposition 4.5. *Let u_ϵ be a family of almost $\mathcal{D}_\mathcal{L}$ -harmonic sections and $r := e^{-(m-1)\ln(2)/2}$.*

- (1) For every $\ell \in \mathbb{N}_0$, the L^∞ -norm of $\nabla^\ell u_\epsilon$ is uniformly bounded by some constant.
- (2) For every differential operator $P: C^\infty(M, \mathcal{S}_\mathcal{L}) \rightarrow C^\infty(M, \mathcal{S}_\mathcal{L})$, there exists a constant C such that $\|Pu_\epsilon\|_\infty \leq C\|Pu_\epsilon\|_{L^2}^r$ for all $\epsilon \in (0, 1)$.
- (3) For every $\ell \geq 1$ there exist constants C_1 and C_2 such that

$$\|\mathcal{D}_\mathcal{L}^\ell u_\epsilon\|_\infty < C_1 \epsilon^r \quad \text{and} \quad \|\nabla^\ell u_\epsilon\|_\infty < C_2 \epsilon^{r/2}$$

for all $\epsilon \in (0, 1)$.

- (4) There exists an $\epsilon_0 \in (0, 1)$, so that $\|u_\epsilon\|_p^2 > (2\text{vol}(M))^{-1}$ for all $p \in M$ and all $\epsilon \in (0, \epsilon_0)$.

Proof. Let u_ϵ be a family of almost $\mathcal{D}_\mathcal{L}$ -harmonic sections. The first part holds by a similar consideration as in equation (4.3) and the Sobolev embedding. Let P be some differential operator on $\mathcal{S}_\mathcal{L}$ of order k . For $\epsilon > 0$ and some state ρ of the C^* -algebra $\mathcal{A}$, we define the function

$$f_{\epsilon, \rho}: M \longrightarrow \mathbb{R}^+, \quad p \rightarrow \rho(|Pu_\epsilon|_p^2)^{1/2}.$$

We obtain for all $p \in M$, as in the proof of [Ton25b, Lemma 3.5],

$$f_{\epsilon,\rho}(p)\Delta f_{\epsilon,\rho}(p) \leq 4\rho(|\nabla^* \nabla P u_\epsilon|_p^2)\rho(|P u_\epsilon|_p^2) \leq \alpha f_{\epsilon,\rho}^2(p)$$

for some $\alpha > 0$ independent of ϵ and the state ρ . In the last step, we used that the L^∞ -norm of all covariant derivatives of u_ϵ up to order $k+2$ are uniformly bounded by part (1). Moser iteration [cf. Ton25b, Lemma 3.4] yields that there exists a constant C_3 such that

$$\|P u_\epsilon\|_\infty \leq C_3 \|P u_\epsilon\|_{L^2}^r$$

for all $\epsilon \in (0, 1)$, which proves part (2). Part (3) follows immediately from part (2), the almost $\mathcal{D}_{\mathcal{L}}$ -harmonicity of u_ϵ and Lemma 4.4.

It remains to show part (4) of the proposition. By [Ton25b, Lemma 3.6], there exists a constant C_4 such that

$$\|\bar{u}_\epsilon - |u_\epsilon|_p^2\|_{\mathcal{A}} < C_4 \epsilon^r \quad \text{with} \quad \bar{u}_\epsilon := \frac{1}{\text{vol} M} \int_M |u_\epsilon|_q^2 d\mu(q) \quad (4.4)$$

for all $\epsilon \in (0, 1)$ and all points $p \in M$. For $\epsilon_0 := \min\{1, (2C_4 \text{vol}(M))^{-1/r}\}$ we obtain the desired estimate using the reverse triangle inequality and equation (4.4). $\square$

The existence of a family of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic section u_ϵ (Lemma 4.3) together with Proposition 4.5 yields the following rigidity statement:

Theorem 4.6 ([GS02; Ton25b, Theorem 2.4 and Theorem A]). *The map f is a Riemannian submersion with $\text{scal}_M = \text{scal}_N \circ f$ and the Schrödinger-Lichnerowicz type formula in Lemma 4.2 reduces to*

$$\mathcal{D}_{\mathcal{L}}^2 = \nabla^* \nabla + \frac{1}{16} \sum_{\alpha} -(c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha))^2. \quad (4.5)$$

Proof. The classical version assuming a non-trivial $\hat{A}$ -degree of f is proved by Goette and Semmelmann [GS02], and the general version by the second named author in [Ton25b, Section 5]. The argument may be summarised as follows.

We obtain by Lemma 4.2 together with $g_M \geq f^* g_N$ and $\mathcal{R}_N \geq 0$

$$\mathcal{D}_{\mathcal{L}}^2 \geq \nabla^* \nabla + \frac{\text{scal}_M}{4} - \frac{1}{8} \sum_{i,j} \mu_i^2 \mu_j^2 (R_{ijji}^{TN} \circ f) - \frac{\text{scal}_N \circ f}{8} \quad (4.6)$$

$$\geq \nabla^* \nabla + \frac{\text{scal}_M - \text{scal}_N \circ f}{4}. \quad (4.7)$$

The existence of a family of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic sections u_ϵ (Lemma 4.3) satisfying the estimates in Proposition 4.5 yields together with equation (4.7) the extremality statement $\text{scal}_M = \text{scal}_N \circ f$. Similarly, we obtain by equation (4.6)

$$\sum_{i,j} \mu_j^2 \mu_j^2 (R_{ijji}^{TN} \circ f) = \text{scal}_N \circ f, \quad (4.8)$$

which yields as in [GS02, Section 1.c] $\mu_1 = \dots = \mu_n = 1$. In other words, f is a Riemannian submersion.

It remains to show equation (4.5). By equation (4.2), $\text{scal}_M = \text{scal}_N \circ f$, and the relation in equation (4.8), we obtain

$$\sum_{\alpha} c(L\bar{\omega}_{\alpha})^2 + \sum_{\alpha} \bar{c}(f^* \bar{L}\bar{\omega}_{\alpha})^2 = -2 \sum_{i,j} \mu_i^2 \mu_j^2 (R_{ijji}^{TN} \circ f) - 2 \text{scal}_N \circ f = -4 \text{scal}_M,$$

and the Schrödinger-Lichnerowicz type formula in equation (4.1) reduces to equation (4.5). $\square$

4.2 Establishing pointwise inequalities

Recall that by Theorem 4.6 and Lemma 4.3 the map $f: M \rightarrow N$ is a Riemannian submersion, the twisted Dirac operator $\mathcal{D}_{\mathcal{L}}$ satisfies

$$\mathcal{D}_{\mathcal{L}}^2 = \nabla^* \nabla + \frac{1}{16} \sum_{\alpha} -(c(L\bar{\omega}_{\alpha}) - \bar{c}(f^* \bar{L}\bar{\omega}_{\alpha}))^2, \quad (4.9)$$

and there exists a family u_{ϵ} of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic sections of the bundle $\mathcal{S}_{\mathcal{L}}$ satisfying the properties in Proposition 4.5. Throughout this section we fix such a family u_{ϵ} , a point $p \in M$, $r := e^{-(m-1)\ln(2)/2}$ and write $\mathcal{V}$ and $\mathcal{H}$ for the vertical and horizontal projections from $T_p M$ onto its vertical and horizontal parts, respectively.

The goal of this section is to derive the pointwise estimates stated in Lemma 4.7, 4.8, 4.9 and 4.10. Although all four statements are purely pointwise, their proofs require that the sections u_{ϵ} are locally defined and almost parallel.

Lemma 4.7. *There exists a constant C such that*

$$\|R_{X,Y}^{\mathcal{S}_{\mathcal{L}}} u_{\epsilon}\|_p < C \|X\| \|Y\| \epsilon^{r/2}$$

for all $X, Y \in T_p M$ and all $\epsilon \in (0, 1)$ where $R^{\mathcal{S}_{\mathcal{L}}}$ denotes the curvature endomorphism of $\mathcal{S}_{\mathcal{L}}$.

Proof. For $X, Y \in T_p M$ and an $\epsilon \in (0, 1)$, we obtain

$$\begin{aligned} \|R_{X,Y}^{\mathcal{S}_{\mathcal{L}}} u_{\epsilon}\|_p^2 &\leq 3m^2 \sum_{i,j} g_M(X, e_i)^2 g_M(Y, e_j)^2 (\|\nabla_{e_i} \nabla_{e_j} u_{\epsilon}\|_p^2 + \|\nabla_{e_j} \nabla_{e_i} u_{\epsilon}\|_p^2 + \|\nabla_{[e_i, e_j]} u_{\epsilon}\|_p^2) \\ &\leq C^2 \|X\|^2 \|Y\|^2 \epsilon^r. \end{aligned}$$

The second step holds for an appropriate constant C independent of X , Y and ϵ by Proposition 4.5 (3). $\square$

Lemma 4.8. *There exists a constant C such that*

$$\|(c(\omega) - \bar{c}(f^* \bar{\omega})) u_{\epsilon}\|_p < C \|\omega\| \epsilon^{r/4}$$

for all $\bar{\omega} \in \text{im}(\mathcal{R}_N)$ with horizontal lift $\omega \in \Lambda^2 T_p M$ and for all $\epsilon \in (0, 1)$.

Proof. Fix some ω , $\bar{\omega}$ and $\epsilon \in (0, 1)$ as in the lemma. Since $\text{im } \mathcal{R}_N = \text{im } \bar{L}$, there exists an $\bar{\eta} \in (\ker \bar{L})^{\perp}$ such that $\bar{\omega} = \bar{L}\bar{\eta}$. We obtain

$$\bar{\omega} = \sum_{\alpha} g_N(\bar{\eta}, \bar{\omega}_{\alpha}) \bar{L}\bar{\omega}_{\alpha} \quad \text{and} \quad \omega = \sum_{\alpha} g_N(\bar{\eta}, \bar{\omega}_{\alpha}) L\bar{\omega}_{\alpha},$$

hence

$$\begin{aligned}
|(c(\omega) - \bar{c}(f^*\bar{\omega}))u_\epsilon|_p^2 &= \left| \sum_\alpha g_N(\bar{\eta}, \bar{\omega}_\alpha) (c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u_\epsilon \right|_p^2 \\
&\leq C_1 \|\bar{\omega}\|^2 \sum_\alpha |(c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u_\epsilon|_p^2 \\
&= 16C_1 \|\omega\|^2 \left(\langle \mathcal{D}_{\mathcal{L}}^2 u_\epsilon, u_\epsilon \rangle_p - \langle \nabla^* \nabla u_\epsilon, u_\epsilon \rangle_p \right).
\end{aligned} \tag{4.10}$$

The second step holds by the estimate

$$\begin{aligned}
\left| \sum_\alpha g_N(\bar{\eta}, \bar{\omega}_\alpha) (c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u \right|_p^2 &\leq \binom{n}{2} \sum_\alpha g_N(\bar{\eta}, \bar{\omega}_\alpha)^2 |(c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u|_p^2 \\
&\leq \binom{n}{2} \|\bar{\eta}\|^2 \sum_\alpha |(c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u|_p^2 \\
&\leq C_1 \|\bar{\omega}\|^2 \sum_\alpha |(c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha))u|_p^2
\end{aligned}$$

with $C_1 = \binom{n}{2} \|\bar{L}|_{(\ker \bar{L})^\perp}^{-1}\|_{\text{op}}^2$,

and the third step by the skew-adjointness of $c(L\bar{\omega}_\alpha) - \bar{c}(f^*\bar{L}\bar{\omega}_\alpha)$ and equation (4.5). We take the norm in the C^* -algebra $\mathcal{A}$ of equation (4.10), and obtain for a suitable constant C , independent of ω and ϵ ,

$$\|(c(\omega) - \bar{c}(f^*\bar{\omega}))u_\epsilon\|_p^2 \leq 16C_1 \|\omega\|^2 \left(\|\mathcal{D}_{\mathcal{L}}^2 u_\epsilon\|_p \|u_\epsilon\|_p + \|\nabla^* \nabla u_\epsilon\|_p \|u_\epsilon\|_p \right) \leq C^2 \|\omega\|^2 \epsilon^{r/2}.$$

Here we used the triangle inequality, the Cauchy-Schwarz inequality, and Proposition 4.5 (3). $\square$

In order to get information about the O'Neill tensors A and T of the Riemannian submersion f , we take covariant derivatives of equation (4.9) in the direction of vertical and horizontal vectors. We obtain Lemma 4.9 and Lemma 4.10 respectively.

Lemma 4.9. *There exists a constant C such that*

$$\|c(\nabla_U \omega)u_\epsilon\|_p < C \|U\| \|\omega\| \epsilon^{r/8}$$

for all vertical vectors $U \in T_p M$, all $\omega \in \Lambda^2 T_p M$ being the horizontal lift of some element in the image of $\mathcal{R}_N$, and all $\epsilon \in (0, 1)$.

Proof. Fix an $\epsilon \in (0, 1)$, a vertical vector $U \in T_p M$, and some $\omega \in \Lambda^2 T_p M$ the horizontal lift of $\bar{\omega} := \bar{L}\bar{\eta}$ for some $\bar{\eta} \in (\ker \bar{L})^\perp \subset \Lambda^2 T_{f(p)} N$. We obtain, similarly as in the proof of Lemma 4.8, with

$$\omega = \sum_\alpha g_N(\bar{\eta}, \bar{\omega}_\alpha) L\bar{\omega}_\alpha, \quad C_1 := \binom{n}{2} \|\bar{L}|_{(\ker \bar{L})^\perp}^{-1}\|_{\text{op}}^2$$

at the point p and a suitable constant C_2 independent of U , X , ω and ϵ

$$\begin{aligned}
\|c(\nabla_U \omega)u_\epsilon\|_p^2 &= \left\| \sum_\alpha g_N(\bar{\eta}, \bar{\omega}_\alpha) c(\nabla_U (L\bar{\omega}_\alpha))u_\epsilon \right\|_p^2 \leq C_1 \|\omega\|^2 \left\| \sum_\alpha c(\nabla_U (L\bar{\omega}_\alpha))u_\epsilon \right\|_{\mathcal{A}}^2 \\
&\stackrel{(*)}{\leq} C_1 C_2 \|U\|^2 \|\omega\|^2 \epsilon^{r/4}.
\end{aligned} \tag{4.11}$$

In the first step, we used that the term on the left-hand side is tensorial in ω . It remains to show the estimate $(\star)$ in equation (4.11). All constants in the remaining proof are chosen independent of U , ω and ϵ . We start calculating

$$\begin{aligned}
& \nabla_U^2 [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)]^2 u_\epsilon \\
&= 2c(\nabla_U(L\bar{\omega}_\alpha))^2 u_\epsilon + [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)]^2 \nabla_U^2 u_\epsilon \\
&\quad + c(\nabla_U^2(L\bar{\omega}_\alpha)) [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] u_\epsilon + [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] c(\nabla_U^2(L\bar{\omega}_\alpha)) u_\epsilon \\
&\quad + 2[c(\nabla_U(L\bar{\omega}_\alpha)) [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] + [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] c(\nabla_U(L\bar{\omega}_\alpha))] \nabla_U u_\epsilon.
\end{aligned} \tag{4.12}$$

We insert u_ϵ into equation (4.9), take ∇_U^2 , and then the fiberwise $\mathcal{A}$ -valued inner product in $(\mathcal{S}_\mathcal{L})_p$ with u_ϵ . This yields by inserting equation (4.12), rearranging the terms, taking the norm in the C^* -algebra $\mathcal{A}$, and estimating the right-hand side with the triangle and the Cauchy-Schwarz inequality that

$$\begin{aligned}
& 2 \left\| \sum_\alpha |c(\nabla_U(L\bar{\omega}_\alpha)) u_\epsilon|_p^2 \right\|_{\mathcal{A}} \\
& \leq \sum_\alpha \left[\|\nabla_U^2 u_\epsilon\|_p \| [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)]^2 u_\epsilon \|_p + 2 \|c(\nabla_U^2(L\bar{\omega}_\alpha)) u_\epsilon\|_p \| [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] u_\epsilon \|_p \right. \\
& \quad \left. + 2 \|\nabla_U u_\epsilon\|_p \| (c(\nabla_U(L\bar{\omega}_\alpha)) [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] + [c(L\bar{\omega}_\alpha) - \bar{c}(f^* \bar{L}\bar{\omega}_\alpha)] c(\nabla_U(L\bar{\omega}_\alpha))) u_\epsilon \|_p \right] \\
& \quad + \|\nabla_U^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon\|_p \|u_\epsilon\|_p + \|\nabla_U^2 \nabla^* \nabla u_\epsilon\|_p \|u_\epsilon\|_p.
\end{aligned}$$

Here we used that c and $\bar{c}$ are skew-adjoint on two vectors. By Proposition 4.5 (1) and (2), there exist constants C_3 , C_4 and C_5 such that $\|u_\epsilon\|_p \leq C_3$,

$$\begin{aligned}
\|\nabla_U^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon\|_p^2 & \leq (m-n)^2 \|U\|^4 \|\nabla^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon\|_p^2 \leq C_4 \|U\|^4 \|\nabla^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon\|_{L^2}^{2r} \\
& \leq C_4 \|U\|^4 \|\langle \mathcal{D}_\mathcal{L}^2 u_\epsilon, (\nabla^*)^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon \rangle_{L^2}\|_{\mathcal{A}}^r \\
& \leq C_4 \|U\|^4 \|\mathcal{D}_\mathcal{L}^2 u_\epsilon\|_{L^2}^r \|(\nabla^*)^2 \mathcal{D}_\mathcal{L}^2 u_\epsilon\|_{L^2}^r \\
& < C_4 \epsilon^{r/2}
\end{aligned}$$

and similarly

$$\begin{aligned}
\|\nabla_U^2 \nabla^* \nabla u_\epsilon\|_p^2 & \leq (m-n)^2 \|U\|^4 \|\nabla^2 \nabla^* \nabla u_\epsilon\|_p^2 \leq C_5^2 \|U\|^4 \|\nabla u_\epsilon\|_{L^2}^r \|(\nabla^*)^2 \nabla^* \nabla u_\epsilon\|_{L^2}^r \\
& < C_5 \epsilon^{r/2}.
\end{aligned}$$

Together with Lemma 4.8 and Proposition 4.5 (3), we obtain for a constant C_2

$$\left\| \sum_\alpha |\nabla_U(L\bar{\omega}_\alpha) u_\epsilon|_p^2 \right\|_{\mathcal{A}} < C_2 \|U\|^2 \epsilon^{r/4},$$

hence estimate $(\star)$ in equation (4.11) is proved. $\square$

Lemma 4.10. *There exists a constant C and a linear map $B: T_p M \rightarrow \text{End}(\Lambda^2 \mathcal{H}_p)$ such that*

$$\| [c(\mathcal{V} \nabla_X \omega) + c(B_X \omega) - \bar{c}(f^* \bar{B}_X \omega)] u_\epsilon \|_p < C \|X\| \|\omega\| \epsilon^{r/8}$$

for every horizontal vector $X \in T_p M$, every $\omega \in \Lambda^2 T_p M$ being the horizontal lift of an element in the image of the curvature operator of N , and for all $\epsilon \in (0, 1)$. Here $\overline{B_X \omega}$ denotes the projection of $B_X \omega$ to $\Lambda^2 T_{f(p)} N$ by the map $\Lambda^2 df$.

Proof. Fix an $\epsilon \in (0, 1)$, some horizontal vector $X \in T_p M$ and some ω the horizontal lift of some $\bar{\omega} \in \text{im } \mathcal{R}_N$. We define $B_X \omega \in \Lambda^2 T_p M$ as the horizontal lift of

$$(\nabla_{df(X)} \bar{L}) \bar{L}^{-1} \bar{\omega} \in \Lambda^2 T_{f(p)} N,$$

where $\bar{L}^{-1}$ is the linear map given by the inverse of $\bar{L}|_{(\ker \bar{L})^\perp} : (\ker \bar{L})^\perp \rightarrow \text{im } \bar{L}$ extended by zero. Pick some $\bar{\eta} \in (\ker \bar{L})^\perp$ such that $\bar{\omega} = \bar{L} \bar{\eta}$, define $\bar{X} := df(X)$, and denote by $\widehat{}$ the horizontal lift of a 2-vector of N . We obtain as in the proof of Lemma 4.8 for a suitable constant C_1

$$\begin{aligned} & \left\| [c(\mathcal{V} \nabla_X \omega) + c(B_X \omega) - \bar{c}(f^* \overline{B_X \omega})] u_\epsilon \right\|_p^2 \\ &= \left\| \sum_\alpha g(\bar{\eta}, \bar{\omega}_\alpha) [c(\mathcal{V} \nabla_X (L \bar{\omega}_\alpha)) + c(\widehat{(\nabla_{\bar{X}} \bar{L})} \bar{\omega}_\alpha) - \bar{c}(f^* (\nabla_{\bar{X}} \bar{L}) \bar{\omega}_\alpha)] u_\epsilon \right\|_p^2 \\ &\leq C_1 \left\| \sum_\alpha |c(\mathcal{V} \nabla_X (L \bar{\omega}_\alpha)) + c(\widehat{(\nabla_{\bar{X}} \bar{L})} \bar{\omega}_\alpha) - \bar{c}(f^* (\nabla_{\bar{X}} \bar{L}) \bar{\omega}_\alpha)] u_\epsilon |_p^2 \right\|_{\mathcal{A}} \end{aligned} \quad (4.13)$$

Here we used that the left-hand side of equation (4.13) is tensorial in ω , and that $B_X \omega$ is equal to the horizontal lift of $(\nabla_{\bar{X}} \bar{L}) \bar{L}^{-1} \bar{L} \bar{\eta} = (\nabla_{\bar{X}} \bar{L}) \bar{\eta}$. Note that

$$\begin{aligned} & \nabla_X (c(L \bar{\omega}_\alpha) - \bar{c}(f^* \bar{L} \bar{\omega}_\alpha)) \\ &= c(\mathcal{V} \nabla_X (L \bar{\omega}_\alpha)) + c(\mathcal{H} \nabla_X (L \bar{\omega}_\alpha)) - \bar{c}(f^* \nabla_{\bar{X}} (\bar{L} \bar{\omega}_\alpha)) \\ &= c(\mathcal{V} \nabla_X (L \bar{\omega}_\alpha)) + c(\widehat{(\nabla_{\bar{X}} \bar{L})} \bar{\omega}_\alpha) - \bar{c}(f^* (\nabla_{\bar{X}} \bar{L}) \bar{\omega}_\alpha) + c(\widehat{\bar{L} \nabla_{\bar{X}} \bar{\omega}_\alpha}) - \bar{c}(f^* (\nabla_{\bar{X}} \bar{L}) \bar{\omega}_\alpha). \end{aligned} \quad (4.14)$$

The desired estimate follows from equation (4.13) and equation (4.14) as in the proof of Lemma 4.9, namely by taking two times the covariant derivative of equation (4.9) and using Proposition 4.5 and Lemma 4.8. $\square$

5 Pointwise computations

Let $f: M \rightarrow N$ be as in Theorem A. By Theorem 4.6, the map f is a Riemannian submersion. In this section we will begin working with the pointwise inequalities derived in Section 4.2, connecting them to local geometric quantities such as the O'Neill tensors A, T and the curvatures of M, N . Lemma 5.2 below summarises these pointwise relations and brings them into a more digestible form. On first reading one may treat Lemma 5.2 as a black box.

In Section 5.1 we prove the relation $\text{Ric}_M = f^* \text{Ric}_N$. Section 5.2 expresses our pointwise equations with the tensors A, T and upgrades these by applying chains ad_{ω_i} to them, see Proposition 5.5. In order to work with these relations we need to know something about what the image of the curvature operator $\mathcal{R}_N$ looks like. Section 5.3 is then an abstract analysis of $\text{im } \mathcal{R}_N$. Section 5.4 applies this analysis to the relations of Section 5.2, allowing us to conclude that the fibres of f have vanishing mean curvature (cf. Propositions 5.28 and 5.31).

We begin by summarising the results of Section 4 in Lemma 5.2 below. Let $\mathcal{H}$ and $\mathcal{V}$ denote the horizontal and vertical subbundles of TM , respectively. When there is no risk of confusion, the same symbols also denote the projections onto these distributions. We use the following short notation:

Definition 5.1. For $p \in M$ and $\omega \in \Lambda^2 T_p M$ the horizontal lift of some $\bar{\omega} \in \Lambda^2 T_{f(p)} N$, we define

$$(c - \bar{c})(\omega) := c(\omega) - \bar{c}(f^* \bar{\omega}).$$

Lemma 5.2. For every point $p \in M$, there exists a family $\{u_\epsilon\} \subset (\mathcal{S}_{\mathcal{L}})_p \cong \text{Cl}_{m,n} \otimes \text{C}^* \pi_1(M)$ with $\|u_\epsilon\|_p = 1$ together with constants C, r and a linear map $B: T_p M \rightarrow \text{End}(\Lambda^2 \mathcal{H}_p)$ such that

- (1) $\|R_{Y,Z}^{\mathcal{S}_{\mathcal{L}}} u_\epsilon\|_p \leq C \|Y\| \|Z\| \epsilon^r$,
- (2) $\|(c - \bar{c})(\omega) u_\epsilon\|_p \leq C \|\omega\| \epsilon^r$,
- (3) $\|c(\nabla_U \omega) u_\epsilon\|_p \leq C \|U\| \|\omega\| \epsilon^r$, and
- (4) $\|c(\nabla_X \omega) u_\epsilon + (c - \bar{c})(B_X \omega) u_\epsilon\|_p \leq C \|X\| \|\omega\| \epsilon^r$

for all $\epsilon > 0$, all $Y, Z \in T_p M$, all vertical $U \in T_p M$, all horizontal $X \in T_p M$, and all $\omega \in \Lambda^2 T_p M$ being the horizontal lift of some element in the image of the curvature operator of N .

Remark 5.3. Even though derivatives appear, parts (3) and (4) of Lemma 5.2 are tensorial in ω . Furthermore, the precise form of B will be irrelevant for our considerations.

Proof of Lemma 5.2. Fix some point $p \in M$. By Lemma 4.3 there exists a family of almost $\mathcal{D}_{\mathcal{L}}$ -harmonic section. By rescaling appropriately, which is possible by Proposition 4.5 (4), and restricting to the point p , we obtain a family u_ϵ which satisfies by Lemma 4.7, 4.8, 4.9 and 4.10 for appropriate constants C, r the estimates (1)-(4) for all $\epsilon \in (0, \epsilon_0)$ for some $\epsilon_0 \in (0, 1)$. For $\epsilon \geq \epsilon_0$, we just redefine $u_\epsilon := u_{\epsilon_0/2}$ and the lemma is proved. $\square$

5.1 The Ricci curvature in the Riemannian submersion

The scalar curvature of M is by Theorem 4.6 equal to the scalar curvature of N . In this section we show that under the assumption of Theorem A this is even true for the Ricci curvatures of the manifolds M and N , to be precise the following holds:

Proposition 5.4. The Ricci curvature of M and N are related via $\text{Ric}_M = \text{Ric}_N \circ (df \otimes df)$.

Proof. We fix a point $p \in M$ and a tangent vector $X \in T_p M$. Let $\{u_\epsilon\} \subset (\mathcal{S}_{\mathcal{L}})_p$ be a family as in Lemma 5.2, and denote $\bar{X} := df(X)$. The Ricci curvature on M and the curvature tensor R^{SM} on the spinor bundle $\mathcal{S}M$ are related via

$$c(\text{Ric}_M(X)) = -2 \sum_{\gamma=1}^m c(e_\gamma) R_{X, e_\gamma}^{SM}. \quad (5.1)$$

Since the Mishchenko–Fomenko bundle $\mathcal{L}$ is flat and the Dirac bundle $\mathcal{S}_{\mathcal{L}} = \mathcal{S} \otimes \mathcal{L}$ is locally equal to the graded tensor product of $\mathcal{S}M$ and $\mathcal{S}N$, the curvature tensor on $\mathcal{S}_{\mathcal{L}}$ splits, and we obtain

$$2 \sum_{\gamma=1}^m c(e_{\gamma}) R_{e_{\gamma}, X}^{\mathcal{S}_{\mathcal{L}}} u_{\epsilon} = 2 \sum_{\gamma=1}^m c(e_{\gamma}) R_{e_{\gamma}, X}^{\mathcal{S}M} u_{\epsilon} + 2 \sum_{\gamma=1}^n c(e_{\gamma}) R_{e_{\gamma}, \bar{X}}^{\mathcal{S}N} u_{\epsilon}. \quad (5.2)$$

The first term on the right-hand side is equal to $c(\text{Ric}_M(X))u_{\epsilon}$ by equation (5.1). We rewrite the second term using the formula for the curvature tensor of the spinor bundle [LM89, Chapter II Theorem 4.15]. Note that we have a different sign, since we consider the spinor bundle associated to TN equipped with the negative definite metric $(-g_N)$ (see Example 3.15). We obtain

$$\begin{aligned} 2 \sum_{\gamma=1}^n c(e_{\gamma}) R_{e_{\gamma}, \bar{X}}^{\mathcal{S}N} u_{\epsilon} &= -2 \sum_{\gamma=1}^n c(e_{\gamma}) \sum_{k < \ell} g_N(R_{e_{\gamma}, \bar{X}}^{TN} \bar{e}_k, \bar{e}_{\ell}) \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_{\ell}) u_{\epsilon} \\ &= -2 \sum_{\gamma=1}^n c(e_{\gamma}) \sum_{k < \ell} g_N(R_{e_{\gamma}, \bar{X}}^{TN} \bar{e}_k, \bar{e}_{\ell}) c(e_k) c(e_{\ell}) u_{\epsilon} \\ &\quad + 2 \sum_{\gamma=1}^n c(e_{\gamma}) \sum_{k < \ell} g_N(R_{e_{\gamma}, \bar{X}}^{TN} \bar{e}_k, \bar{e}_{\ell}) (c(e_k) c(e_{\ell}) - \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_{\ell})) u_{\epsilon}. \end{aligned} \quad (5.3)$$

All summand in the first term in equation (5.3) with three pairwise disjoint indices vanishes by the Bianchi identity, hence this term is equal to $c(\widehat{\text{Ric}}_N(\bar{X}))u_{\epsilon}$ by a short calculation analyzing the summands with $\gamma = k$ and $\gamma = \ell$. Here $\widehat{\text{Ric}}_N(\bar{X})$ denotes the horizontal lift of $\text{Ric}_N(\bar{X})$. Putting together equation (5.1), (5.2) and (5.3) yields

$$\begin{aligned} c(\text{Ric}_M(X) - \widehat{\text{Ric}}_N(\bar{X}))u_{\epsilon} \\ = -2 \sum_{\gamma=1}^m c(e_{\gamma}) R_{e_{\gamma}, X}^{\mathcal{S}_{\mathcal{L}}} u_{\epsilon} + 2 \sum_{\gamma=1}^n c(e_{\gamma}) \sum_{k < \ell} g_N(R_{e_{\gamma}, \bar{X}}^{TN} \bar{e}_k, \bar{e}_{\ell}) (c(e_k) c(e_{\ell}) - \bar{c}(f^* \bar{e}_k) \bar{c}(f^* \bar{e}_{\ell})) u_{\epsilon}. \end{aligned} \quad (5.4)$$

Note that

$$\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma}) = \sum_{k < \ell} g_N(\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma}), \bar{e}_k \wedge \bar{e}_{\ell}) \bar{e}_k \wedge \bar{e}_{\ell} = \sum_{k < \ell} g_N(R_{e_{\gamma}, \bar{X}}^{TN} \bar{e}_k, \bar{e}_{\ell}) \bar{e}_k \wedge \bar{e}_{\ell},$$

hence the second term on the right-hand side in equation (5.4) is equal to

$$2 \sum_{\gamma=1}^n c(e_{\gamma}) (c - \bar{c})(\widehat{\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma})}) u_{\epsilon}. \quad (5.5)$$

Here $\widehat{\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma})}$ denotes the horizontal lift of $\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma})$. We insert equation (5.5) into equation (5.4), take $\|\cdot\|_p$ of both sides, and obtain

$$\begin{aligned} \|c(\text{Ric}_M(X) - \widehat{\text{Ric}}_N(\bar{X}))u_{\epsilon}\| &\leq 2 \sum_{\gamma=1}^m \|R_{e_{\gamma}, X}^{\mathcal{S}_{\mathcal{L}}} u_{\epsilon}\| + 2 \sum_{\gamma=1}^n \|(c - \bar{c})(\widehat{\mathcal{R}_N(\bar{X} \wedge \bar{e}_{\gamma})})u_{\epsilon}\| \\ &< C \|X\| \epsilon^r \end{aligned}$$

for suitable constants C, r which are independent of ϵ . Here we used in the first step the triangle inequality, that $c(e_\gamma)$ is skew-adjoint, and $c(e_\gamma)c(e_\gamma) = -1$. The second step holds by Lemma 5.2 (1) and (2). Since $\|u_\epsilon\|_p = 1$, we obtain

$$\|\text{Ric}_M(X) - \widehat{\text{Ric}}_N(\overline{X})\| < C \|X\| \epsilon^r,$$

and $\text{Ric}_M(X) = \widehat{\text{Ric}}_N(\overline{X})$ in the limit $\epsilon \rightarrow 0$. $\square$

5.2 Pointwise equation for the O'Neill tensors A and T

In this section we derive a system of pointwise equations for the O'Neill tensors A, T of the Riemannian submersion $f : (M, g_M) \rightarrow (N, g_N)$. These relations show that certain 2-vectors fail to be invertible in the Clifford algebra. These 2-vectors are built from the tensors A, T , the elements of $\text{im}(\mathcal{R}_N)_{f(p)}$, and the tensor B from Lemma 5.2. The main result of this section is the following:

Proposition 5.5. *Let $p \in M$ and let u_ϵ, r, B be as in Lemma 5.2. Then for each $\ell \in \mathbb{N}$ there exists a constant C_ℓ so that:*

- (1) *Let $\overline{\omega}$ be an element of the Lie-algebra generated by $\text{im}(\mathcal{R}_N)_{f(p)}$ and let ω denote its horizontal lift to $\Lambda^2 T_p M$. For all $\epsilon > 0$*

$$\|(c - \overline{c})(\omega)u_\epsilon\| \leq C_1 \|\omega\| \epsilon^r. \quad (5.6)$$

- (2) *Let $U \in T_p M$ be a vertical vector, and $\omega_1, \dots, \omega_\ell \in \Lambda^2 T_p M$ horizontal lifts of elements of $\text{im}(\mathcal{R}_N)_{f(p)}$. For all $\epsilon > 0$*

$$\|c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(2T_U + A_U^*))u_\epsilon\| \leq C_\ell \|\omega_1\| \cdots \|\omega_\ell\| \|U\| \epsilon^r. \quad (5.7)$$

- (3) *Let $X \in T_p M$ be horizontal, and $\omega_1, \dots, \omega_\ell \in \Lambda^2 T_p M$ horizontal lifts of elements of $\text{im}(\mathcal{R}_N)_{f(p)}$. For all $\epsilon > 0$*

$$\begin{aligned} & \|c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(2A_X))u_\epsilon + (c - \overline{c})((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_{\ell-1}})(B_X \omega_\ell))u_\epsilon\| \\ & \leq C_\ell \|\omega_1\| \cdots \|\omega_\ell\| \|X\| \epsilon^r. \end{aligned} \quad (5.8)$$

For the convenience of the reader we recall several of the definitions and objects appearing in Proposition 5.5.

Remark 5.6. (1) For a linear map $L : T_p M \rightarrow T_p M$ we had defined

$$c(L) = \sum_{ij} g_M(L e_i, e_j) c(e_i) c(e_j),$$

where $e_1, \dots, e_m$ is some orthonormal basis of $T_p M$.

- (2) We recall the definitions of A, A^*, T^* . Let $p \in M$ a point, $U, W \in \mathcal{V}_p$ vertical and $X, Y \in \mathcal{H}_p$ horizontal vectors. We had defined the tensors $T_U^* : \mathcal{H}_p \rightarrow \mathcal{V}_p$, $A_X : \mathcal{H}_p \rightarrow \mathcal{V}_p$, $A_U^* : \mathcal{H}_p \rightarrow \mathcal{H}_p$, in Definition 3.5 via the equations

$$g_M(T_U^* X, W) = -g_M(\nabla_U X, W), \quad g_M(A_X Y, U) = \frac{1}{2} g_M(U, [X, Y]) = g_M(A_U^* X, Y).$$

These identities do not depend on the choice of extensions of W, X, Y to vector fields.

- (3) B is a linear map $T_p M \rightarrow \text{End}(\Lambda^2 \mathcal{H}_p)$. Its precise form is irrelevant.
- (4) For U, X, ω as in the proposition we extend $A_X, A_U^*, T_U^*, B_X(\omega)$ to linear maps $T_p M \rightarrow T_p M$ by letting them act as zero on the orthogonal complements of their domains of definition.
- (5) If $\omega \in \Lambda^2 T_p M$ and $L \in \text{End}(T_p M)$ we use $\text{ad}_\omega(L)$ to denote the commutator $[\omega, L] = \omega \cdot L - L \cdot \omega$, where we identify $\Lambda^2 T_p M$ with $\mathfrak{so}(T_p M) \subseteq \text{End}(T_p M)$ as in Notation 4.1 point (2).
- (6) Since T_U^* maps the horizontal space $\mathcal{H}_p$ to the vertical space $\mathcal{V}_p$ and vanishes on $\mathcal{V}_p$, one has that

$$(\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(T_U) = (-1)^\ell T_U \cdot \omega_\ell \cdots \omega_1 = T_U \cdot (\omega_1 \cdots \omega_\ell)^T,$$

where $\cdot$ denotes the product in $\text{End}(T_p M)$ and T the transpose. Similarly for A_X . On the other hand A_U^* and $B_X(\omega_\ell)$ are anti-symmetric linear maps $\mathcal{H}_p$ to $\mathcal{H}_p$ and so the terms involving their commutators do not simplify.

In order to prove Proposition 5.5 we will make use of the following elementary lemma:

Lemma 5.7. *Let $(V, \langle \cdot, \cdot \rangle)$ be a real Hilbert space, $L_1, L_2 \in \text{End}(V)$. Then*

$$\begin{aligned} [c(L_1), c(L_2)] &= -2c(L_1 \cdot L_2) + 2c(L_1 \cdot L_2^T) - 2c(L_2^T \cdot L_1) + 2c(L_2 \cdot L_1). \\ [\bar{c}(L_1), \bar{c}(L_2)] &= +2\bar{c}(L_1 \cdot L_2) - 2\bar{c}(L_1 \cdot L_2^T) + 2\bar{c}(L_2^T \cdot L_1) - 2\bar{c}(L_2 \cdot L_1). \end{aligned}$$

Here $c, \bar{c}$ are the Clifford multiplications satisfying $c(v)^2 = -\langle v, v \rangle$ and $\bar{c}(v)^2 = +\langle v, v \rangle$ for any $v \in V$. $[\cdot, \cdot]$ denotes the commutator in the corresponding Clifford algebra.

Proof. Let $e_1, \dots, e_n$ be an orthonormal basis of V . Then for $i, j, k, \ell \in \{1, \dots, n\}$ one has:

$$\begin{aligned} c(e_i)c(e_j)c(e_k)c(e_\ell) &= -2\delta_{jk}c(e_i)c(e_k) - c(e_i)c(e_k)c(e_j)c(e_\ell) \\ &= -2\delta_{jk}c(e_i)c(e_\ell) + 2\delta_{j\ell}c(e_i)c(e_k) + c(e_i)c(e_k)c(e_\ell)c(e_j) \\ &= -2\delta_{jk}c(e_i)c(e_\ell) + 2\delta_{j\ell}c(e_i)c(e_k) - 2\delta_{ik}c(e_\ell)c(e_j) - c(e_k)c(e_i)c(e_\ell)c(e_j) \\ &= -2\delta_{jk}c(e_i)c(e_\ell) + 2\delta_{j\ell}c(e_i)c(e_k) - 2\delta_{ik}c(e_\ell)c(e_j) + 2\delta_{i\ell}c(e_k)c(e_j) \\ &\quad + c(e_k)c(e_\ell)c(e_i)c(e_j). \end{aligned}$$

By noting that

$$c(L_1)c(L_2) - c(L_2)c(L_1) = \sum_{ijkl} (L_1)_{ij}(L_2)_{kl} (c(e_i)c(e_j)c(e_k)c(e_\ell) - c(e_k)c(e_\ell)c(e_i)c(e_j))$$

and dutifully performing all contractions the equation for $[c(L_1), c(L_2)]$ follows. The proof for the equation with $\bar{c}$ is identical, note that the signs of all contractions are flipped. $\square$

Remark 5.8. Suppose one of L_1, L_2 is anti-symmetric, then Lemma 5.7 states

$$[c(L_1), c(L_2)] = -4c(\text{ad}_{L_1} L_2), \quad [\bar{c}(L_1), \bar{c}(L_2)] = 4\bar{c}(\text{ad}_{L_1} L_2).$$

The key computation to prove Proposition 5.5 is then the following:

Lemma 5.9. *Let $p \in M$ and u_ϵ be as in Lemma 5.2, suppose $\eta_1 \in \text{End}(T_p M)$ and $\bar{\eta}_2 \in \text{End}(f^*(TN)_p)$ so that*

$$\|c(\eta_1)u_\epsilon + \bar{c}(\bar{\eta}_2)u_\epsilon\| \leq C (\|\eta_1\| + \|\bar{\eta}_2\|) \epsilon^r \quad (5.9)$$

for all $\epsilon > 0$. Then there exist constants C'_ℓ for $\ell \in N$ so that

$$\|c((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(\eta_1))u_\epsilon + \bar{c}((\text{ad}_{\bar{\omega}_1} \circ \dots \circ \text{ad}_{\bar{\omega}_\ell})(\bar{\eta}_2))u_\epsilon\| \leq C'_\ell \|\omega_1\| \dots \|\omega_\ell\| (\|\eta_1\| + \|\bar{\eta}_2\|) \epsilon^r$$

for all $\omega_1, \dots, \omega_\ell \in \Lambda^2 T_p M$ being horizontal lifts of $\bar{\omega}_1, \dots, \bar{\omega}_\ell \in \text{im}(\mathcal{R}_N)_{f(p)}$.

Proof. We show the claim for $\ell = 1$, the general case then follows by iteratively applying the case $\ell = 1$ to

$$\eta'_1 = (\text{ad}_{\omega_2} \circ \dots \circ \text{ad}_{\omega_\ell})(\eta_1), \quad \bar{\eta}'_2 = (\text{ad}_{\bar{\omega}_2} \circ \dots \circ \text{ad}_{\bar{\omega}_\ell})(\bar{\eta}_2).$$

To this end let ω be a horizontal lift of an element $\bar{\omega} \in \text{im}(\mathcal{R}_N)_{f(p)}$. Note the smallness of the commutator

$$\begin{aligned} \|[c - \bar{c}](\omega), c(\eta_1) + \bar{c}(\bar{\eta}_2)]u_\epsilon\| &\leq \|(c - \bar{c})(\omega)\| \|c(\eta_1) + \bar{c}(\bar{\eta}_2))u_\epsilon\| + \|c(\eta_1) + \bar{c}(\bar{\eta}_2)\| \|(c - \bar{c})(\omega)u_\epsilon\| \\ &\leq C' \|\omega\| (\|\eta_1\| + \|\bar{\eta}_2\|) \epsilon^r \end{aligned}$$

for some constant C' . Here we used point (2) of Lemma 5.2 and equation (5.9). Recalling Lemma 5.7, in particular Remark 5.8, one sees that

$$[(c - \bar{c})(\omega), c(\eta_1) + \bar{c}(f^*\bar{\eta}_2)] = -4c(\text{ad}_\omega \eta_1) - 4\bar{c}(f^* \text{ad}_{\bar{\omega}} \bar{\eta}_2),$$

where we used additionally $[c(L_1), \bar{c}(\bar{L}_2)] = 0$ for any $L_1 \in \text{End}(T_p M)$, $\bar{L}_2 \in \text{End}(f^*(TN)_p)$, this commutator being of two even elements of sub-algebras that graded commute with each other. This completes the proof. $\square$

Proof of Proposition 5.5. We first show part (1) of Proposition 5.5. Recall that

$$\|(c - \bar{c})(\omega)u_\epsilon\| \leq C \|\omega\| \epsilon^r$$

for all ω being horizontal lifts of some $\bar{\omega} \in \text{im}(\mathcal{R}_N)_{f(p)}$. We apply Lemma 5.9 with $\eta_1 = \omega$, $\bar{\eta}_2 = \bar{\omega}$ to get

$$\|(c - \bar{c})((\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_\ell})(\omega))u_\epsilon\| \leq C'_\ell \|\omega_1\| \dots \|\omega_\ell\| \|\omega\| \epsilon^r.$$

Now any element of the Lie-algebra generated by $\text{im}(\mathcal{R}_N)_{f(p)}$ is by definition a linear combination of elements $(\text{ad}_{\bar{\omega}_1} \circ \dots \circ \text{ad}_{\bar{\omega}_\ell})(\bar{\omega})$. Fix some basis $\bar{\omega}_\alpha$ of this Lie-algebra, let ω_α denote the horizontal lift of the basis elements. Choosing some way of writing this basis as a linear combination of ad-chains gives a constant C' so that

$$\|(c - \bar{c})(\omega_\alpha)u_\epsilon\| \leq C' \epsilon^r,$$

for all α . Taking linear combinations of the ω_α then gives part (1) for an appropriate constant C_1 .

For part (2) note that from Lemma 5.2 part (3) and Lemma 3.11 one has

$$\|c(\text{ad}_{\omega_\ell}(2T_U^* + A_U^*))u_\epsilon\| = \|c(\nabla_U \omega_\ell)u_\epsilon\| \leq C \|U\| \|\omega_\ell\| \epsilon^r$$

for all $U \in T_p M$ vertical and ω_ℓ being a horizontal lift of an element of $\text{im}(\mathcal{R}_N)_{f(p)}$. Part (2) of the proposition then follows immediately from Lemma 5.9. The proof for part (3) is identical: from Lemma 5.2 part (4) and Lemma 3.11 one has

$$\|c(2\text{ad}_\omega A_X)u_\epsilon + (c - \bar{c})(B_X \omega)u_\epsilon\| = \|c(\mathcal{V}\nabla_X \omega)u_\epsilon + (c - \bar{c})(B_X \omega)u_\epsilon\| \leq C \|X\| \|\omega\| \epsilon^r$$

for all $X \in T_p M$ horizontal and ω a horizontal lift of an element of $\text{im}(\mathcal{R}_N)_{f(p)}$. Now one applies Lemma 5.9. $\square$

Remark 5.10. We have then seen that $(c - \bar{c})(\omega)u_\epsilon$ is small also for horizontal lifts of elements of the Lie-algebra generated by the image of $\mathcal{R}_N$, not just for the image alone. Of course one can now go through all previous computations and verify that any conclusion we had drawn for $\text{im } \mathcal{R}_N$ then also extends to the Lie-algebra generated by this image, in particular Lemma 5.2 and Proposition 5.5. In our attempt to write a legible proof we have opted not to do this.

Note however that the statement of part (2) of Proposition 5.5 can upgrade itself to the Lie-algebra, since $[\text{ad}_{\omega_1}, \text{ad}_{\omega_2}] = \text{ad}_{[\omega_1, \omega_2]}$. The only consequence of not having part (3) formulated for the Lie-algebra will be a single extra technical complication in the proof of Lemma 5.30 below.

5.3 The image of the curvature operator

From Proposition 5.5 the O'Neill tensors A, T satisfy a system of Clifford equations involving the image of $\mathcal{R}_N$, the curvature operator of N . Recall that $\mathcal{R}_N \geq 0$ with a positive Ricci tensor. This section is a purely algebraic analysis of the consequences of these two positivity conditions, although many of the initial propositions do not use them to their full extent. The conclusions of this section will be applied to Proposition 5.5 in Section 5.4.

Definition 5.11. Let $p \in N$, we let $\bar{\mathfrak{g}}(p) \subseteq \mathfrak{so}(T_p N)$ denote the Lie algebra generated by the image of the curvature operator $\mathcal{R}_N$ under the identification $\Lambda^2 T_p N \cong \mathfrak{so}(T_p N)$.

First we decompose the action of $\bar{\mathfrak{g}}(p)$ on $T_p N$ into its irreducible summands. Since the Lie-algebra is generated by the image of a curvature operator, it will also decompose accordingly.

Proposition 5.12. *Let $p \in N$. Decomposing $T_p N = \bigoplus_i \bar{V}_i$ into irreducible summands of the $\bar{\mathfrak{g}}(p)$ action, one has*

$$\bar{\mathfrak{g}}(p) = \bigoplus_i \bar{\mathfrak{g}}_i(p),$$

where $\bar{\mathfrak{g}}_i(p)$ is the Lie-algebra generated by $\mathcal{R}_N(\Lambda^2 \bar{V}_i)$. The Lie-algebra $\bar{\mathfrak{g}}_i(p)$ acts irreducibly on $\bar{V}_i$ and trivially on $\bar{V}_j$ for $j \neq i$.

Proof. Note that $\Lambda^2 T_p N = \bigoplus_i \Lambda^2 \bar{V}_i \oplus \bigoplus_{i < j} \bar{V}_i \wedge \bar{V}_j$. We first verify that

$$\mathcal{R}_N(\Lambda^2 T_p N) = \bigoplus_i \mathcal{R}_N(\Lambda^2 V_i), \quad (5.10)$$

i.e. $\mathcal{R}_N(X \wedge Y) = 0$ for $X \in \bar{V}_i$, $Y \in \bar{V}_j$ with $i \neq j$. For any $\omega \in \Lambda^2 T_p N$ one has

$$g_N(\mathcal{R}_N(X \wedge Y), \omega) = g_N(\mathcal{R}_N(\omega), X \wedge Y) = g_N(\mathcal{R}_N(\omega)Y, X) = 0,$$

since $\bar{V}_i$ is invariant under the action of $\mathfrak{g}(p)$ and orthogonal to $\bar{V}_j$. The claim then follows.

Next we show that

$$\mathcal{R}_N(\Lambda^2 V_i) \subseteq \Lambda^2 V_i, \quad (5.11)$$

i.e. $\mathcal{R}_N(X_1 \wedge X_2)Y = 0$ for $X_1, X_2 \in V_i$, $Y \in \bar{V}_j$ with $i \neq j$. This is a consequence of the previous calculation and the Bianchi identity

$$\mathcal{R}_N(X_1 \wedge X_2)Y = -\mathcal{R}_N(X_2 \wedge Y)X_1 - \mathcal{R}_N(Y \wedge X_1)X_2 = 0.$$

The proposition then follows from equations (5.10) and (5.11). $\square$

Since $\bar{\mathfrak{g}}_i(p)$ acts irreducibly on V_i and consists of anti-symmetric endomorphisms the (associative) algebra it generates in $\text{End}(\bar{V}_i)$ is easy to understand. In principle four cases are possible by Schur's Lemma (cf. [Ros16] for Schur's Lemma for real $\ast$ -algebras). Non-degeneracy of the Ricci tensor at p provides further restrictions:

Proposition 5.13. *Let $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ be as in Proposition 5.12. Then either:*

- (1) *The (associative) sub-algebra of $\text{End}(\bar{V}_i)$ generated by $\bar{\mathfrak{g}}_i(p)$ is $\text{End}(\bar{V}_i)$.*
- (2) *There is an $\mathbb{R}$ -linear isometry $\iota: \bar{V}_i \rightarrow \mathbb{C}^k$ for some k . The (associative) algebra generated by $\bar{\mathfrak{g}}_i(p)$ is $\iota^* \text{End}_{\mathbb{C}}(\mathbb{C}^k)$. Denoting with $I: \bar{V}_i \rightarrow \bar{V}_i$ the pullback of the complex unit of $\mathbb{C}^k$ one has $I \in \bar{\mathfrak{g}}_i(p)$.*

Remark 5.14. Proposition 5.13 only requires non-degeneracy of the Ricci tensor of $\mathcal{R}_N$. Furthermore, if the image of the curvature operator is parallel then $\bar{\mathfrak{g}}(p)$ is the holonomy algebra of N by a theorem of Ambrose and Singer [AS53]. Proposition 5.13 then reproduces well-known pointwise calculations, such as Calabi-Yau manifolds being Ricci-flat.

Proof of Proposition 5.13. First note that $\bar{\mathfrak{g}}_i(p) = 0$ immediately contradicts non-degeneracy of the Ricci tensor, since if $X \in \bar{V}_i$, and e_j is some orthonormal basis of $T_p N$ then

$$\text{Ric}(X, Y) = \sum_j g_N(\mathcal{R}_N(e_j \wedge Y)X, e_j) = 0$$

for all $Y \in T_p N$.

Let $\bar{\mathfrak{g}}_i(p)'$ denote the commutant of $\bar{\mathfrak{g}}_i(p)$ in $\text{End}(\bar{V}_i)$, i.e. all endomorphisms commuting with $\bar{\mathfrak{g}}_i(p)$. Since $\bar{\mathfrak{g}}_i(p) \neq 0$ consists of anti-symmetric endomorphisms and acts irreducibly on $\bar{V}_i$, we apply Schur's Lemma to find that $\bar{\mathfrak{g}}_i(p)'$ is isomorphic to one of the real division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}$.

In the case $\mathbb{R}$ one recovers for the bi-commutant $\bar{\mathfrak{g}}_i(p)'' = \text{End}(\bar{V}_i)$. Now $\bar{\mathfrak{g}}_i(p)''$ is equal to the sub-algebra of $\text{End}(\bar{V}_i)$ generated by $\bar{\mathfrak{g}}_i(p)$ by the bi-commutant theorem (which holds also for real $*$ -algebras, see e.g. Theorem 4.3.8 of [Li03]). This case corresponds to point (1) in the proposition.

For the two remaining cases $\mathbb{C}, \mathbb{H}$ one has $\bar{\mathfrak{g}}_i(p)'' = \text{End}_{\mathbb{C}}(\bar{V}_i)$ or $\bar{\mathfrak{g}}_i(p)'' = \text{End}_{\mathbb{H}}(\bar{V}_i)$ with respect to some complex or quaternion structure on $\bar{V}_i$. In both cases $\mathbb{C}, \mathbb{H}$ there is at least one complex unit $I: \bar{V}_i \rightarrow \bar{V}_i$ with $I \in \bar{\mathfrak{g}}_i(p)'$. The cases $\mathbb{C}, \mathbb{H}$ can be distinguished by considering $\bar{\mathfrak{g}}_i(p)' \cap \bar{\mathfrak{g}}_i(p)''$, which is either $\mathbb{C}$ or $\mathbb{R}$ respectively.

We then proceed by assuming there is a complex unit $I: \bar{V}_i \rightarrow \bar{V}_i$ commuting with $\bar{\mathfrak{g}}_i(p)$. Using non-degeneracy of the Ricci tensor we will construct a non-zero anti-symmetric element of $\bar{\mathfrak{g}}_i(p)' \cap \bar{\mathfrak{g}}_i(p)''$, ruling out the case $\mathbb{H}$. In fact our construction is such that $I \in \bar{\mathfrak{g}}_i(p)$. Carrying out these steps then completes the proof of the proposition.

So let $I \in \bar{\mathfrak{g}}_i(p)'$ be a complex unit, since the elements of $\bar{\mathfrak{g}}_i(p)$ commute with I they are complex linear maps with respect to I . If $e_1, \dots, e_{2k}$ is a (real) orthonormal basis of $\bar{V}_i$, the *complex trace* of an element $\omega \in \bar{\mathfrak{g}}_i(p)$ is given by:

$$2 \text{Tr}_{\mathbb{C}}(\omega) = \text{Tr}_{\mathbb{R}}(\omega) - i \text{Tr}_{\mathbb{R}}(I\omega) = \sum_j g_N(\omega e_j, e_j) + i \sum_j g_N(\omega e_j, Ie_j) \in \mathbb{C}.$$

Note that since ω is anti-symmetric the summand $\text{Tr}_{\mathbb{R}}(\omega)$ will vanish.

Claim: There is an element $\eta \in \bar{\mathfrak{g}}_i(p)$ for which $\text{Tr}_{\mathbb{C}}(\eta) \neq 0$.

To prove the claim let $X, Y \in \bar{V}_i(p)$ then

$$\begin{aligned} 2 \text{Tr}_{\mathbb{C}}(\mathcal{R}_N(X \wedge IY)) &= i \sum_j g_N(\mathcal{R}_N(X \wedge IY)e_j, Ie_j) \\ &= -i \sum_j g_N(\mathcal{R}_N(X \wedge e_j)Ie_j, IY) - i \sum_j g_N(\mathcal{R}_N(X \wedge Ie_j)IY, e_j) \end{aligned}$$

where we applied the Bianchi identity. Note that by assumption the linear isometry I commutes with the image of $\mathcal{R}_N$, whence

$$-i \sum_j g_N(\mathcal{R}_N(X \wedge e_j)Ie_j, IY) = -i \text{Ric}(X, Y).$$

Similarly using that Ie_j is also an orthonormal basis of $\bar{V}_i$ one has

$$-i \sum_j g_N(\mathcal{R}_N(X \wedge Ie_j)IY, e_j) = i \sum_j g_N(\mathcal{R}_N(X \wedge Ie_j)Y, Ie_j) = -i \text{Ric}(X, Y),$$

and so $\text{Tr}_{\mathbb{C}}(\mathcal{R}_N(X \wedge IY)) = -i \text{Ric}(X, Y)$, whence from non-degeneracy of Ric there is an $\eta \in \bar{\mathfrak{g}}_i(p)$ with $\text{Tr}_{\mathbb{C}}(\eta) \neq 0$. This proves the claim.

From this η we now construct a non-vanishing anti-symmetric element of $\bar{\mathfrak{g}}_i(p)' \cap \bar{\mathfrak{g}}_i(p)''$. Let $G_i \subseteq U(\bar{V}_i)$ denote the Lie group generated by $\bar{\mathfrak{g}}_i(p) \subseteq \mathfrak{u}(\bar{V}_i)$. Then:

$$\text{Ad}_{G_i}(\eta) := \frac{1}{\text{Vol}(G_i)} \int_{G_i} \text{Ad}_g(\eta) \, d\mu(g)$$

has the same complex trace as η and is Ad-invariant. It follows that it is non-vanishing and commutes with every element of $\bar{\mathfrak{g}}_i(p)$. This shows that it is an element of $\bar{\mathfrak{g}}_i(p)'$. Further it is a linear combination of elements $\text{Ad}_g(\eta) \in \bar{\mathfrak{g}}_i(p)$ and hence is an element of $\bar{\mathfrak{g}}_i(p) \subseteq \bar{\mathfrak{g}}_i(p)''$. This completes the proof. $\square$

We give the two cases from Proposition 5.13 names:

Definition 5.15. Let $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ be as in Proposition 5.12. We say that

- (1) $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ is of *real type* if the (associative) algebra generated by $\bar{\mathfrak{g}}_i(p)$ is $\text{End}(\bar{V}_i)$.
- (2) $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ is of *complex type* if there is an $\mathbb{R}$ -linear isometry $\iota : \bar{V}_i \rightarrow \mathbb{C}^k$ for some k and the (associative) algebra generated by $\bar{\mathfrak{g}}_i(p)$ is $\iota^* \text{End}_{\mathbb{C}}(\mathbb{C}^k)$.

Beyond the information on the associative algebra generated by $\bar{\mathfrak{g}}_i(p)$ from Proposition 5.13 we will require one more fact on the representation theory of $\bar{\mathfrak{g}}_i(p)$. In contrast to the previous points, this fact will require both non-negativity of $\mathcal{R}_N$ as well as non-degeneracy of Ric_N .

Definition 5.16. Let $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ be as in Proposition 5.12. We denote by $\bar{\mathfrak{p}}_i(p)$ the orthogonal complement of $\bar{\mathfrak{g}}_i(p)$ in $\Lambda^2 \bar{V}_i$, which is closed under the adjoint action of $\bar{\mathfrak{g}}_i(p)$.

Proposition 5.17. Let $(\bar{\mathfrak{g}}_i(p), \bar{V}_i)$ be as in Proposition 5.12. Then $\bar{\mathfrak{p}}_i(p)$ does not contain any sub-representation of $\bar{\mathfrak{g}}_i(p)$ isomorphic to $\bar{V}_i$.

Remark 5.18. Proposition 5.17 is a combination of two results from the literature. In a paper introducing so called holonomy systems, Simons [Sim62] shows via pointwise computations that what we call $\bar{\mathfrak{g}}_i(p)$ is in fact the Lie algebra of an irreducible symmetric space. When dealing with a symmetric space, a computation of Laquer [Laq92] then gives a Casimir element taking distinct values on $\bar{V}_i$ and $\bar{\mathfrak{p}}_i(p)$, showing they must be disjoint as representations.

For convenience of the reader we will provide a self-contained proof of Proposition 5.17. The proof will be carried out in several steps and is simpler than the just cited discussion. We first define an auxiliary curvature operator:

Definition 5.19. Let $G_i = \exp(\bar{\mathfrak{g}}_i(p)) \subseteq O(\bar{V}_i)$ be the Lie group generated by $\bar{\mathfrak{g}}_i(p)$. We define

$$\tilde{R}_i : \Lambda^2 \bar{V}_i \rightarrow \Lambda^2 \bar{V}_i, \quad \omega \mapsto \frac{1}{\text{Vol}(G_i)} \int_{G_i} \text{Ad}_g(\mathcal{R}_N(\text{Ad}_{g^{-1}} \omega)) \, d\mu(g).$$

Lemma 5.20. $\tilde{R}_i$ is an algebraic curvature operator and

$$\text{im } \tilde{R}_i \subseteq \bar{\mathfrak{g}}_i(p) \subseteq \{\omega \in \Lambda^2 \bar{V}_i \mid \text{ad}_\omega \circ \tilde{R}_i = \tilde{R}_i \circ \text{ad}_\omega\}.$$

Proof. As a linear combination of curvature operators $\tilde{R}_i$ is a curvature operator. Additionally $\bar{\mathfrak{g}}_i(p)$ is invariant under the action of Ad_g for all $g \in G_i$, whence $\text{Ad}_g(\mathcal{R}_N(\text{Ad}_{g^{-1}} \omega)) \in \bar{\mathfrak{g}}_i(p)$ for all $\omega \in \Lambda^2 \bar{V}_i$. This implies the first inclusion. The second inclusion follows from noting

$$\text{Ad}_g \circ \tilde{R}_i = \tilde{R}_i \circ \text{Ad}_g$$

for all $g \in G_i$ and taking the derivative by g . $\square$

Remark 5.21. From the inclusion $\text{im } \tilde{R}_i \subseteq \{\omega \in \Lambda^2 \bar{V}_i \mid \text{ad}_\omega \circ \tilde{R}_i = \tilde{R}_i \circ \text{ad}_\omega\}$ in Lemma 5.20 one finds $\text{im } \tilde{R}_i$ is a Lie-algebra:

$$[\tilde{R}_i \omega, \tilde{R}_i \eta] = \text{ad}_{\tilde{R}_i \omega}(\tilde{R}_i \eta) = \tilde{R}_i(\text{ad}_{\tilde{R}_i \omega} \eta) \in \text{im } \tilde{R}_i.$$

Lemma 5.22. *The Ricci tensor of $\tilde{R}_i$ is non-degenerate and $\text{im } \tilde{R}_i$ acts irreducibly on $\bar{V}_i$.*

Proof. Since $\tilde{R}_i$ is G_i -equivariant so is its Ricci tensor. But any quadratic form invariant under an irreducible orthogonal representation (such as G_i on $\bar{V}_i$) must be proportional to the scalar product. We now verify this proportionality factor is non-zero: Since the Ricci tensor of $\mathcal{R}_N$ is positive the partial trace of $\mathcal{R}_N$ over $\Lambda^2 \bar{V}_i$ is strictly positive. By construction this partial trace is the trace of $\tilde{R}_i$, whence $\tilde{R}_i$ has non-vanishing (and hence non-degenerate) Ricci tensor.

We now verify that $\text{im } \tilde{R}_i$ acts irreducibly on $\bar{V}_i$. From Proposition 5.12 any two irreducible sub-representations of $\text{im } \tilde{R}_i$ are orthogonal to each other. If a sub-representation of $\text{im } \tilde{R}_i$ is not orthogonal to an irreducible summand it must then contain this summand.

Suppose $W \subset \bar{V}_i$ is a non-trivial irreducible sub-representation of $\text{im } \tilde{R}_i$, since G_i acts irreducibly there is a $g \in G_i$ so that $gW \neq W$. By connectedness of G_i we may assume gW is not orthogonal to W . We will show gW is a sub-representation of $\text{im } \tilde{R}_i$, whence it must contain W by the previous paragraph. Since W and gW have the same dimension we conclude the contradiction $W = gW$. $\bar{V}_i$ then admits no non-trivial irreducible sub-representations and so must itself be irreducible.

We now show that gW is a sub-representation of $\text{im } \tilde{R}_i$. Let $x \in W$, $\omega \in \Lambda^2 \bar{V}_i$, unpacking the relation $\text{Ad}_g(\tilde{R}_i(\omega)) = \tilde{R}_i(\text{Ad}_g \omega)$ gives:

$$\tilde{R}_i(\omega)gx = g\tilde{R}_i(\text{Ad}_{g^{-1}} \omega)x \in gW$$

and so gW is closed under the action of $\text{im } \tilde{R}_i$. □

Lemma 5.23. *One has*

$$\text{im } \tilde{R}_i = \bar{\mathfrak{g}}_i(p) = \{\omega \in \Lambda^2 \bar{V}_i \mid \text{ad}_\omega \circ \tilde{R}_i = \tilde{R}_i \circ \text{ad}_\omega\}.$$

Proof. By Lemma 5.20 it is only necessary to show $\{\omega \in \Lambda^2 \bar{V}_i \mid \text{ad}_\omega \circ \tilde{R}_i = \tilde{R}_i \circ \text{ad}_\omega\} \subseteq \text{im } \tilde{R}_i$. Let $\omega \in \text{im}(\tilde{R}_i)^\perp \cap \{\omega \in \Lambda^2 \bar{V}_i \mid \text{ad}_\omega \circ \tilde{R}_i = \tilde{R}_i \circ \text{ad}_\omega\}$. Since $\tilde{R}_i$ is self-adjoint we find $\tilde{R}_i(\omega) = 0$. Now for any $\eta \in \Lambda^2 \bar{V}_i$:

$$[\omega, \tilde{R}_i \tilde{R}_i \eta] = \text{ad}_\omega(\tilde{R}_i \tilde{R}_i \eta) = \tilde{R}_i(\text{ad}_\omega(\tilde{R}_i \eta)) = -\tilde{R}_i(\text{ad}_{\tilde{R}_i \eta} \omega) = -\text{ad}_{\tilde{R}_i}(\tilde{R}_i \omega) = 0,$$

and so ω commutes with every element of $\text{im}(\tilde{R}_i^2) = \text{im } \tilde{R}$ and so lies in the commutant $\text{im}(\tilde{R}_i)'$. By Lemma 5.22 we know that $\text{im } \tilde{R}_i$ acts irreducibly on $\bar{V}_i$ and that $\tilde{R}_i$ has non-degenerate Ricci curvature. Proposition 5.13 then applies to show that all anti-symmetric elements of the commutant $\text{im}(\tilde{R}_i)'$ already lie in $\text{im } \tilde{R}_i$. We conclude $\omega = 0$, which shows the lemma. □

From Lemma 5.23 we see that restricting $\tilde{R}_i$ to $\bar{\mathfrak{g}}_i(p)$ gives an invertible $\text{ad}(\bar{\mathfrak{g}}_i(p))$ -equivariant linear map $\bar{\mathfrak{g}}_i(p) \rightarrow \bar{\mathfrak{g}}_i(p)$. With this map we can now describe the Casimir element that distinguishes $\bar{V}_i$ from $\bar{\mathfrak{p}}_i(p)$.

Lemma 5.24. *Let $\omega_1, \dots, \omega_k$ be an orthonormal basis of $\bar{\mathfrak{g}}_i(p)$. There is a scalar $\lambda_i < 0$ so that:*

- (1) *For any $v \in \bar{V}_i$ one has $\sum_{\alpha} (\tilde{R}_i^{1/2} \omega_{\alpha})^2 v = \lambda_i v$.*
- (2) *For any $\eta \in \bar{\mathfrak{p}}_i(p)$ one has $\sum_{\alpha} (\text{ad}_{\tilde{R}_i^{1/2}} \omega_{\alpha})^2 \eta = 2\lambda_i \eta$.*

Proof. Note that $\tilde{R}_i^{1/2} \omega_1, \dots, \tilde{R}_i^{1/2} \omega_k$ is an orthonormal basis for the (positive definite) $\text{ad}(\bar{\mathfrak{g}}_i(p))$ -invariant quadratic form

$$Q : \bar{\mathfrak{g}}_i(p) \times \bar{\mathfrak{g}}_i(p) \rightarrow \mathbb{R}, \quad (\omega, \eta) \mapsto g_N(\omega, \tilde{R}_i^{-1} \eta).$$

It follows for any irreducible representation ρ of $\bar{\mathfrak{g}}_i(p)$ that $\sum_{\alpha} \rho(\tilde{R}_i^{1/2} \omega_{\alpha})^2$ acts as a negative scalar, being the image of the Casimir element associated to Q . This shows point (1).

Suppose now that $\eta \in \bar{\mathfrak{p}}_i(p)$, recall by Lemma 5.23 that this is equivalent to $\eta \in \Lambda^2 \bar{V}_i$ and $\tilde{R}_i(\eta) = 0$. Choose an orthonormal basis $e_1, \dots, e_{\ell}$ of $\bar{V}_i$ so that $\eta = \sum_{ab} \eta_{ab} e_a \wedge e_b$. Then

$$\begin{aligned} \sum_{\alpha} (\text{ad}_{\tilde{R}_i^{1/2}} \omega_{\alpha})^2 \eta &= \sum_{\alpha, ab} \eta_{ab} \left(((\tilde{R}_i^{1/2} \omega_{\alpha})^2 e_a) \wedge e_b + e_a \wedge ((\tilde{R}_i^{1/2} \omega_{\alpha})^2 e_b) - 2(\tilde{R}_i^{1/2} \omega_{\alpha} e_a) \wedge (\tilde{R}_i^{1/2} \omega_{\alpha} e_b) \right) \\ &= 2\lambda_i \eta - 2 \sum_{\alpha, ab} \eta_{ab} (\tilde{R}_i^{1/2} \omega_{\alpha} e_a) \wedge (\tilde{R}_i^{1/2} \omega_{\alpha} e_b) \\ &= 2\lambda_i \eta - 2 \sum_{\alpha, abcd} \eta_{ab} g_N(\tilde{R}_i^{1/2} \omega_{\alpha} e_a, e_c) g_N(\tilde{R}_i^{1/2} \omega_{\alpha} e_b, e_d) e_c \wedge e_d \\ &= 2\lambda_i \eta - \frac{1}{2} \sum_{\alpha, abcd} \eta_{a,b} g_N(\tilde{R}_i^{1/2} \omega_{\alpha}, e_a \wedge e_c) g_N(\tilde{R}_i^{1/2} \omega_{\alpha}, e_b \wedge e_d) e_c \wedge e_d \\ &= 2\lambda_i \eta - \frac{1}{2} \sum_{abcd} \eta_{ab} g_N(\tilde{R}_i(e_a \wedge e_c), e_b \wedge e_d) e_c \wedge e_d \end{aligned} \quad (5.12)$$

where we used point (1) in the second line, as well as self-adjointness of $\tilde{R}_i^{1/2}$ and the fact that ω_{α} is an orthonormal basis of $\text{im } \tilde{R}_i$ in the third line. Since $\tilde{R}_i$ satisfies the Bianchi identity we further recover

$$g_N(\tilde{R}_i(e_a \wedge e_c), e_b \wedge e_d) = g_N(\tilde{R}_i(e_a \wedge e_b), e_c \wedge e_d) - g_N(\tilde{R}_i(e_a \wedge e_d), e_c \wedge e_b). \quad (5.13)$$

Recall that by definition $\tilde{R}_i(\eta) = 0$, which gives

$$\sum_{abcd} \eta_{ab} g_N(\tilde{R}_i(e_a \wedge e_b), e_c \wedge e_d) e_c \wedge e_d = \tilde{R}_i\left(\sum_{ab} \eta_{ab} e_a \wedge e_b\right) = \tilde{R}_i(\eta) = 0$$

so that contracting equation (5.13) with $\sum_{abcd} \eta_{ab} e_c \wedge e_d$ will cancel the first term on the righthand side of equation (5.13). Additionally using anti-symmetry $\eta_{ab} = -\eta_{ba}$ and self-adjointness of $\tilde{R}_i$ one gets from equation (5.13)

$$\begin{aligned} \sum_{abcd} \eta_{ab} g_N(\tilde{R}_i(e_a \wedge e_c), e_b \wedge e_d) e_c \wedge e_d &= - \sum_{abcd} \eta_{ab} g_N(\tilde{R}_i(e_a \wedge e_d), e_c \wedge e_b) e_c \wedge e_d \\ &= - \sum_{abcd} (-\eta_{ba}) g_N(\tilde{R}_i((-e_b) \wedge e_c), e_a \wedge e_d) e_c \wedge e_d \end{aligned}$$

$$= - \sum_{abcd} \eta_{ab} g_N(\tilde{R}_i(e_a \wedge e_c), e_b \wedge e_d) e_c \wedge e_d.$$

The sum on the righthand side of equation (5.12) then vanishes. This completes the proof. $\square$

We can now perform the proof of Proposition 5.17.

Proof of Proposition 5.17. If $\bar{\mathfrak{p}}_i(p)$ would have an irreducible sub-representation isomorphic to $\bar{V}_i$ then there must be a non-zero $\eta \in \bar{\mathfrak{p}}_i(p)$ with $\sum_{\alpha} (\text{ad}_{\tilde{R}_i \omega_{\alpha}})^2 \eta = \lambda_i \eta$ by point (1) of Lemma 5.24. This contradicts point (2) of the same lemma. $\square$

5.4 Interaction of the curvature operator and the Clifford algebra

Our goal is to use the information on $\bar{\mathfrak{g}}(f(p))$ derived in Section 5.3 in order to conclude, using simple pointwise calculations, from equation (5.7) that the shape operator S_X is traceless for any $X \in \mathcal{H}_p$. This will be the last ingredient needed to show Theorem A, see Section 6.

Definition 5.25. For $p \in M$ we let $\mathfrak{g}(p)$ denote the horizontal lift of $\bar{\mathfrak{g}}(f(p))$ to p . Similarly we let V_i , $\mathfrak{g}_i(p)$, $\mathfrak{p}_i(p)$ denote the horizontal lifts of $\bar{V}_i$, $\bar{\mathfrak{g}}_i(f(p))$, respectively $\bar{\mathfrak{p}}_i(f(p))$.

When dealing with chains $\omega_1, \dots, \omega_{\ell} \in \mathfrak{g}(p)$ we can save a significant amount of bookkeeping by identifying them with elements of the tensor algebra generated by $\mathfrak{g}(p)$. To this end we make the following definition:

Definition 5.26. Let $p \in M$.

- (1) We let $\mathfrak{U}(p) = \bigoplus_{k=1}^{\infty} \mathfrak{g}(p)^{\otimes k}$ denote the tensor algebra generated by $\mathfrak{g}(p)$. $\mathfrak{U}_i(p) \subseteq \mathfrak{U}(p)$ denotes the tensor algebra generated by $\mathfrak{g}_i(p)$.
- (2) We let $\text{ad} : \mathfrak{U}(p) \rightarrow \text{End}(\text{End}(T_p M))$ denote the algebra-morphism defined by

$$\text{ad}_{\omega_1 \otimes \dots \otimes \omega_{\ell}}(\eta) = (\text{ad}_{\omega_1} \circ \dots \circ \text{ad}_{\omega_{\ell}})(\eta)$$

for $\omega_1, \dots, \omega_{\ell} \in \mathfrak{g}(p)$.

- (3) We let $\rho : \mathfrak{U}(p) \rightarrow \text{End}(T_p M)$ denote the morphism defined by

$$\rho(\omega_1 \otimes \dots \otimes \omega_{\ell}) = \omega_1 \cdots \omega_{\ell}$$

for $\omega_1, \dots, \omega_{\ell} \in \mathfrak{g}(p)$, where $\cdot$ is the product in $\text{End}(T_p M)$.

Remark 5.27. Recall that T_U^* and A_X are maps $\mathcal{H}_p \rightarrow \mathcal{V}_p$ so that $\text{ad}_{\omega}(T_U) = T_U \cdot \rho(\omega)^T$, $\text{ad}_{\omega}(A_X) = A_X \cdot \rho(\omega)^T$ for $\omega \in \mathfrak{U}(p)$.

We deal first with the real case.

Proposition 5.28. *For every $p \in M$ one has the following: Let $(\mathfrak{g}_i(p), V_i)$ be of real type and $h \in V_i$, then $A_X h = 0$, $T_U h = 0$ for all $X \in \mathcal{H}_p$ and $U \in \mathcal{V}_p$.*

We prove this in three steps. Recall equation (5.8) from Proposition 5.5, which gives for $\omega \in \mathfrak{U}(p)$ and η a horizontal lift of an element of $\text{im}(R_N)_{f(p)}$ the estimate¹

$$\|c(\text{ad}_\omega(2A_X\eta))u_\epsilon + (c - \bar{c})(\text{ad}_\omega B_X(\eta))u_\epsilon\| \leq C\|\omega\| \|\eta\| \|X\| \epsilon^r,$$

here B_X was some element of $\text{End}(\Lambda^2 T_{f(p)}N)$. The goal of the first two steps is to show that the B_X term can be dropped from this inequality. Here the main ingredient is Proposition 5.17. Letting $\pi_i : \mathcal{H}_p \rightarrow V_i$ denote the orthogonal projection to V_i we first separate off all “off-diagonal” terms $\pi_\ell B_X(\eta)\pi_k$ for $\ell \neq k$:

Lemma 5.29. *Let $p \in M$. For all $X \in \mathcal{H}_p$, $\omega \in \mathfrak{U}(p)$, η a horizontal lift of an element of $\text{im}(R_N)_{f(p)}$, and any two distinct $(\mathfrak{g}_\ell(p), V_\ell)$, $(\mathfrak{g}_k(p), V_k)$ one has for all $\epsilon > 0$*

$$\|(c - \bar{c})(\text{ad}_\omega(\pi_\ell B_X(\eta)\pi_k))u_\epsilon\| \leq C\|\omega\| \|\eta\| \|X\| \epsilon^r.$$

Proof. Let $\omega_\ell \in \mathfrak{U}_\ell(p)$, $\omega_k \in \mathfrak{U}_k(p)$ so that $\rho(\omega_\ell) = \pi_\ell$, $\rho(\omega_k) = \pi_k$. Then $\text{ad}_{\omega_\ell \otimes \omega_k}(2A_X\eta) = 2A_X\eta\pi_k\pi_\ell = 0$ and

$$\text{ad}_{\omega_\ell \otimes \omega_k}(B_X(\eta)) = -\pi_k B_X(\eta)\pi_\ell - \pi_\ell B_X(\eta)\pi_k$$

since all mixed terms $\rho(\omega_{\ell,i})\rho(\omega_{k,j})$ vanish. Now

$$(c - \bar{c})(\pi_k B_X(\eta)\pi_\ell) = (c - \bar{c})(\pi_\ell B_X(\eta)\pi_k)$$

by anti-symmetry of $B_X(\eta)$, so one recovers for any $\omega \in \mathfrak{U}(p)$ that

$$\begin{aligned} \|(c - \bar{c})(\text{ad}_\omega(\pi_\ell B_X(\eta)\pi_k))u_\epsilon\| &= \frac{1}{2}\|c(\text{ad}_{\omega \otimes \omega_k \otimes \omega_\ell}(2A_X\eta))u_\epsilon + (c - \bar{c})(\text{ad}_{\omega \otimes \omega_k \otimes \omega_\ell}(B_X(\eta)))u_\epsilon\| \\ &\leq \frac{1}{2}C\|\omega\| \|\eta\| \|X\| \|\omega_\ell\| \|\omega_k\| \epsilon^r. \end{aligned}$$

Absorbing the norms $\|\omega_\ell\|$, $\|\omega_k\|$ into the constant then proves the lemma. $\square$

Lemma 5.30. *Let $p \in M$. For all $X \in \mathcal{H}_p$ and $\omega \in \mathfrak{U}(p)$ one has*

$$\|c(A_X \cdot \rho(\omega)^T)u_\epsilon\| \leq C\|\omega\| \|X\| \epsilon^r. \quad (5.14)$$

Proof. By Lemma 5.29 one has

$$\|c(\text{ad}_\omega(2A_X\eta))u_\epsilon + (c - \bar{c})(\text{ad}_\omega(\sum_j \pi_j B_X(\eta)\pi_j))u_\epsilon\| \leq C\|\omega\| \|\eta\| \|U\| \epsilon^r \quad (5.15)$$

for any $\omega \in \mathfrak{U}(p)$ and η a horizontal lift of an element of $\text{im}(\mathcal{R}_N)_{f(p)}$. Note first that since A_X is a linear map $\mathcal{H}_p \rightarrow \mathcal{V}_p$, so

$$\text{ad}_\omega(2A_X\eta) = 2A_X\eta \cdot \rho(\omega)^T$$

¹Note that the constant C depends on the length of the chain ω , but this is inconsequential. We could e.g. absorb this dependence into the norm on $\mathfrak{U}(p)$ or note that we only need to work with chains up to a certain length, say $100 \dim(M)^{100}$.

by Remark 5.27. Next note that $\pi_j B_X(\eta) \pi_j$ is an element of $\Lambda^2 V_j = \mathfrak{g}_j(p) \oplus \mathfrak{p}_j(p)$, and recall that $c - \bar{c}$ is $O(\epsilon^r)$ when applied to $\mathfrak{g}$ by equation (5.6) from Proposition 5.5. Equation (5.15) then implies an equation of the form

$$\|c(2A_X \eta \cdot \rho(\omega)^T) u_\epsilon + (c - \bar{c}) (\text{ad}_\omega \mathfrak{p}) u_\epsilon\| \leq C \|\omega\| \|\eta\| \|X\| \epsilon^r$$

for some element $\mathfrak{p} \in \bigoplus_j \mathfrak{p}_j(p)$. From Proposition 5.17 we recall that the two representations of $\bigoplus_j \mathfrak{g}_j(p)$ appearing here, namely ρ on $\mathcal{H}_p = \bigoplus_j V_j$ and ad on $\bigoplus_j \mathfrak{p}_j(p)$, are disjoint. By a general theorem of representation theory there is then an element $\omega' \in \mathfrak{U}(p)$ so that $\rho(\omega') = 1$ and $\text{ad}_{\omega'}(\mathfrak{p}) = 0$. We remark that appealing to the general theorem is not really necessary, since one can also describe this element explicitly. If $\omega'(j) := \sum_\alpha (\tilde{R}_j^{1/2} \omega_\alpha)^{\otimes 2}$ is the Casimir element of Lemma 5.24, then $\omega' = \sum_j \frac{1}{3\lambda_j^4} (4\lambda_j^2 \omega'(j)^{\otimes 2} - \omega'(j)^{\otimes 4})$ does the job.

For any $\omega \in \mathfrak{U}(p)$ and η any horizontal lift of an element of $\text{im}(R_N)_{f(p)}$ we then have

$$\begin{aligned} \|c(2A_X \cdot \rho(\omega \otimes \eta)^T) u_\epsilon\| &= \|c(2A_X \eta \cdot \rho(\omega \otimes \omega')^T) u_\epsilon + (c - \bar{c}) (\text{ad}_{\omega \otimes \omega'}(\mathfrak{p})) u_\epsilon\| \\ &\leq C \|\omega\| \|\omega'\| \|\eta\| \|X\| \epsilon^r. \end{aligned}$$

Absorbing $\|\omega'\|$ into the constant C and η into ω (recall that η was arbitrary in the image of the curvature operator, which generates $\mathfrak{g}(p)$) then proves the lemma. $\square$

With equation (5.14) we can now proceed with the proof of Proposition 5.28.

Proof of Proposition 5.28. Since $(\mathfrak{g}_i(p), V_i)$ is of real type $\rho(\mathfrak{U}_i(p)) = \text{End}(V_i)$ and hence there is an element $\omega_h \in \mathfrak{U}_i(p)$ so that $\rho(\omega_h) = hh^T$. Note that

$$c(A_X \cdot hh^T) = c(A_X h) \cdot c(h),$$

whence by equation (5.14)

$$\|c(A_X h)\| \|h\| = \|c(A_X \cdot hh^T) u_\epsilon\| \leq C \|\omega_h\| \|X\| \epsilon^r$$

for all $\epsilon > 0$. In the limit $\epsilon \rightarrow 0$ we recover $A_X h = 0$.

This then implies that $A_U^* h = 0$ for all U , since $g(A_U^* h, X) = g(U, A_X h)$, recall Definition 3.5. In particular $\text{ad}_\omega(A_U^*) = 0$ for all $\omega \in \mathfrak{U}_i(p)$. For $\omega \in \mathfrak{U}_i(p)$ equation (5.7) from Proposition 5.5 then becomes

$$\|c(T_U^* \cdot \rho(\omega)^T) u_\epsilon\| = \|c(\text{ad}_\omega(T_U^*)) u_\epsilon\| \leq C \|\omega\| \|U\| \epsilon^r.$$

As before plugging in an $\omega_h \in \mathfrak{U}_i(p)$ for which $\rho(\omega_h) = hh^T$ and taking $\epsilon \rightarrow 0$ shows $T_U^* h = 0$. $\square$

Now we turn to the complex case.

Proposition 5.31. *Let $p \in M$ and $(\mathfrak{g}_i(p), V_i)$ be of complex type. Then $\text{Tr}(S_h) = 0$ for all $h \in V_i$ where S_h is the shape operator of the fiber at p of direction h .*

The initial preparation is similar to the real case, except that we can directly separate the equation for T_U^* (that is equation (5.7)) and do not need to consider the equation for A_X (that is equation (5.8)).

Lemma 5.32. *Let $p \in M$ and $(\mathfrak{g}_i(p), V_i)$ be of complex type. Then*

$$\|c(\text{ad}_\omega(T_U^*))u_\epsilon\| \leq C \|\omega\| \|U\| \epsilon^r$$

for all $\omega \in \mathfrak{U}_i(p)$ and $\epsilon > 0$.

Proof. Recall equation (5.7):

$$\|c(\text{ad}_\omega(2T_U^* + A_U))u_\epsilon\| \leq C \|\omega\| \|U\| \epsilon^r.$$

Identically to the proof of Lemma 5.29 we can separate this to get the inequality

$$\|c(\text{ad}_\omega(2T_U^* + \sum_j \pi_j A_U^* \pi_j))u_\epsilon\| \leq C \|\omega\| \|U\| \epsilon^r$$

for any $\omega \in \mathfrak{U}(p)$. Since $(\mathfrak{g}_i(p), V_i)$ is of complex type there is an $\omega_i \in \mathfrak{U}_i(p)$ with $\rho(\omega_i) = I$ a complex unit on V_i (by definition, recall Proposition 5.13). Note that $\text{ad}_{\omega_i}(\pi_j A_U^* \pi_j) = 0$ for $i \neq j$, and writing

$$L_1 := \pi_i \frac{A_U^* + I A_U^* I}{2} \pi_i, \quad L_2 := \pi_i \frac{A_U^* - I A_U^* I}{2} \pi_i$$

one has $\pi_i A_U^* \pi_i = L_1 + L_2$ as well as $(\text{ad}_{\omega_i})^2(L_1) = -4L_1$, $(\text{ad}_{\omega_i})^2(L_2) = 0$. Then

$$(\text{ad}_{\omega_i})^2(2T_U^* + \sum_j \pi_j A_U^* \pi_j) = -2T_U^* \pi_i - 4L_1, \quad (\text{ad}_{\omega_i})^4(2T_U^* + \sum_j \pi_j A_U^* \pi_j) = 2T_U^* \pi_i + 16L_1.$$

It follows that for any $\omega \in \mathfrak{U}_i(p)$

$$\|c(\text{ad}_\omega(6T_U^*))u_\epsilon\| = \left\| c\left(\text{ad}_{\omega \otimes (4\omega_i^{\otimes 2} + \omega_i^{\otimes 4})}(2T_U^* + \sum_j \pi_j A_U^* \pi_j)\right)u_\epsilon \right\| \leq C \|\omega\| \|U\| \epsilon^r,$$

where we absorbed the norm of $4\omega_i^{\otimes 2} + \omega_i^{\otimes 4}$ into the constant. This proves the lemma. $\square$

Proof of Proposition 5.31. Recall that $\text{ad}_\omega(T_U^*) = T_U^* \cdot \rho(\omega)^T$ by Remark 5.27. Since $(\mathfrak{g}_i(p), V_i)$ is of complex type one has by Definition 5.15 that the image of $\rho(\mathfrak{U}_i(p))$ is equal to $\text{End}_{\mathbb{C}}(V_i)$, all linear maps commuting with a certain complex unit I on V_i . In particular for $h \in V_i$ one has that

$$hh^T - Ih h^T I = hh^T + (Ih)(Ih)^T$$

lies in the image of ρ . One then finds

$$c(T_U^* h) c(h) u_\epsilon + c(T_U^* Ih) c(Ih) u_\epsilon = O(\epsilon^r), \tag{5.16}$$

where $\|u_\epsilon\| = 1$ for all ϵ . Our initial goal is to show that equation (5.16) implies $\|T_U^* h\| = \|T_U^* Ih\|$ and $g(T_U^* h, T_U^* Ih) = 0$. First we get from equation (5.16)

$$\|T_U^* h\| \|h\| = \|c(T_U^* h) c(h) u_\epsilon\| = \|c(T_U^* Ih) c(Ih) u_\epsilon\| + O(\epsilon^r) = \|T_U^* Ih\| \|Ih\| + O(\epsilon^r).$$

Taking the limit $\epsilon \rightarrow 0$ and using $\|h\| = \|Ih\|$ implies $\|T_U^*h\| = \|T_U^*Ih\|$. Knowing this we multiply equation (5.16) on the left by $c(T_U^*h)c(Ih) + c(T_U^*Ih)c(h)$ and recover:

$$(\|T_U^*h\|^2 c(h)c(Ih) + \|Ih\|^2 c(T_U^*h)c(T_U^*Ih) + \|h\|^2 c(T_U^*Ih)c(T_U^*h) + \|T_U^*Ih\|^2 c(Ih)c(h)) u_\epsilon = O(\epsilon^r).$$

Since $\|T_U^*h\| = \|T_U^*Ih\|$, $\|h\| = \|Ih\|$, and $c(h)c(Ih) + c(Ih)c(h) = 0$ we recover

$$-2\|h\|^2 \langle T_U^*h, T_U^*Ih \rangle u_\epsilon = O(\epsilon^r)$$

and so $T_U h \perp T_U Ih$.

Recall that $S_h U = T_U^* h$ by definition and that S_h is symmetric. So we have seen that for any $U \in \mathcal{V}_p$ and $h \in V_i$ one has

$$\begin{aligned} \|S_h U\|^2 &= \|T_U^* h\|^2 = \|T_U^* Ih\|^2 = \|S_{Ih} U\|^2, \\ g(U, (S_h S_{Ih} + S_{Ih} S_h) U) &= 2g(S_h U, S_{Ih} U) = g(T_U^* h, T_U^* Ih) = 0. \end{aligned}$$

These relations imply $\text{Tr}(S_h) = 0$, since e.g. upon complexifying $TM \otimes \mathbb{C}$ with a complex unit i one computes

$$(S_h + iS_{Ih})^2 = S_h^2 - S_{Ih}^2 + i(S_h S_{Ih} + S_{Ih} S_h) = 0.$$

Then $S_h + iS_{Ih}$ is nilpotent and so traceless, but its trace is just $\text{Tr}(S_h) + i \text{Tr}(S_{Ih})$. $\square$

6 Completing the proof of Theorem A and its applications

In this section we first describe how the results of the previous sections combine to prove the main result of this paper, Theorem A. After this we prove how Theorem A implies Corollary B.

Proof of Theorem A. Recall that $f: (M, g_M) \rightarrow (N, g_N)$ is a Riemannian submersion by Theorem 4.6. By Proposition 5.4 we additionally have that $\text{Ric}_M = f^* \text{Ric}_N$. From Corollary 3.9 we see that f is locally a Riemannian product, i.e. the O'Neill tensors A and T vanish identically, if the mean curvature of the fibers vanish. We now show how the vanishing of the mean curvature follows from the results of the previous sections.

Let $p \in M$ and $X \in T_p M$ be a horizontal vector. Decompose $X = \sum_i h_i$ for $h_i \in V_i$, where $(\mathfrak{g}_i(p), V_i)$ is the decomposition of $\mathcal{H}_p$ of Sections 5.3 and 5.4, recall Definition 5.25 and Proposition 5.12. Then $S_X = \sum_i S_{h_i}$, where S is the shape operator of the fibers, recall Definition 3.5. If $(\mathfrak{g}_i(p), V_i)$ is of real type one has $S_{h_i} = 0$ by Proposition 5.28, and if $(\mathfrak{g}_i(p), V_i)$ is of complex type one has $\text{Tr}(S_{h_i}) = 0$ by Proposition 5.31. Combining both cases gives $\text{Tr}(S_X) = \sum_i \text{Tr}(S_{h_i}) = 0$. Since X was arbitrary the mean curvature of the fibers $H = \text{Tr}(S_{(\cdot)})^\sharp$ vanishes. $\square$

We now turn to the proof of Corollary B.

Proof of Corollary B. The proof strategy is to apply Theorem A to the spin map $\text{pr}_1 \circ f: M \rightarrow N$, hence we have to verify that $\chi(N) \cdot \deg_{\text{hi}}(\text{pr}_1 \circ f) \neq 0$. Let $p \in N$ be a regular value of $\text{pr}_1 \circ f$ such that there exists a point $q \in F$ for which (p, q) is a regular value of f . We define

$$M_p := f^{-1}(\{p\} \times F) = (\text{pr}_1 \circ f)^{-1}(p) \quad \text{and} \quad f_p := (\text{pr}_2 \circ f)|_{M_p}: M_p \rightarrow F.$$

Note that $\deg(f_p) = \deg(f) \neq 0$. Let $\mathcal{L}(F)$ be the Mishchenko–Fomenko bundle of the manifold F for the reduced group C^* -algebra of $\pi_1(F)$. We define $E := (\text{pr}_2 \circ f)^* \mathcal{L}(F)$, which is a flat bundle of finitely generated projective Hilbert $C^*\pi_1(F)$ -modules over the manifold M , and claim that

$$\text{ind}(\mathcal{D}_{SM_p \otimes E}|_{M_p}) \otimes_{\mathbb{Z}} \mathbb{Q} \neq 0 \in \text{KO}_{\dim(F)}(C^*\pi_1(F)) \otimes_{\mathbb{Z}} \mathbb{Q}. \quad (6.1)$$

We argue as in the proof of [Zei20, Proposition 5.2]. For some discrete group Γ , its classifying space $B\Gamma$ and some closed spin manifold X , the *Novikov assembly map* $\nu: \text{KO}_*(B\Gamma) \rightarrow \text{KO}_*(C^*\Gamma)$ satisfies $\nu(\phi_*[X]_{\text{KO}}) = \text{ind}(\mathcal{D}_{SX \otimes \phi^* \mathcal{L}(B\Gamma)})$ for every map $\phi: X \rightarrow B\Gamma$ [cf. Zei20, Section 2]. Let $\Gamma = \pi_1(F)$, $X = M_p$, and $\phi = \mu \circ f_p$. Here $\mu: F \rightarrow B\pi_1(F)$ denotes the classifying map of the universal cover of F . As in Remark 1.6 we denote by ν the rational Novikov assembly map and by ph the Pontryagin character, and obtain

$$\begin{aligned} (\nu \circ \mu_* \circ \text{ph}^{-1} \circ (f_p)_*)([M_p] \cap \hat{A}(M_p)) &= (\nu \circ (\mu \circ f_p)_* \circ \text{ph}^{-1})([M_p] \cap \hat{A}(M_p)) \\ &= \text{ind}(\mathcal{D}_{SM_p \otimes E}|_{M_p}) \otimes_{\mathbb{Z}} \mathbb{Q}. \end{aligned} \quad (6.2)$$

Here we used $\text{ph}([M_p]_{\text{KO}}) = [M_p] \cap \hat{A}(M_p)$. Since $\deg(f_p) \neq 0$, the top degree part of $(f_p)_*([M_p] \cap \hat{A}(M_p))$ is non-trivial in $H_*(F; \mathbb{Q})$. By the assumption on the manifold F and Remark 1.6, it follows that the image of $(f_p)_*([M_p] \cap \hat{A}(M_p))$ under the map $\nu \circ \mu_* \circ \text{ph}^{-1}$ is non-trivial, hence equation (6.1) holds by equation (6.2).

By [Ton25b, Remark 5.7], the condition on the higher mapping degree in equation (1.1) is fulfilled, and Theorem A is applicable to the map $\text{pr}_1 \circ f: M \rightarrow N$. We obtain that there exists a closed connected Ricci-flat spin manifold $\bar{F}$ such that M is locally isometric to the Riemannian product $N \times \bar{F}$ and $\text{pr}_1 \circ f$ is given by the projection onto the first factor. By the structure theorem for Ricci-flat manifolds [cf. FW75, Theorem 4.5], the Ricci-flat manifold $\bar{F}$ is on a finite Riemannian covering isometric to the Riemannian product of some simply-connected Ricci-flat manifold X and some flat torus. Since F is rationally essential and the map $f_p: \bar{F} \cong M_p \rightarrow F$ has non-zero degree, we obtain that X must be a point and the corollary is proved. $\square$

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Appendix D

Boundary value problems for first-order differential operators on bundles of Hilbert C^* -modules

The following manuscript, "Boundary value problems for first-order differential operators on bundles of Hilbert C^* -modules", is currently unpublished and has not yet been submitted to arXiv.

Boundary value problems for first-order differential operators on bundles of Hilbert C^* -modules

Thomas Tony

Abstract

In this article we develop a theory of boundary value problems for first-order differential operators acting on smooth sections of bundles of finitely generated projective Hilbert C^* -modules. This extends the theory by Bär and Ballmann to infinite dimensional bundles. We give a general definition of abstract boundary conditions classifying all closed extensions between the minimal and the maximal extension of a boundary symmetric first-order differential operator. Moreover, we give a definition of right-elliptic complementable self-adjoint boundary conditions for which the elliptic regularity principle holds. As a main result, we obtain that a boundary symmetric first-order elliptic formally self-adjoint differential operator together with a right-elliptic complementable self-adjoint boundary condition gives rise to an unbounded operator which is self-adjoint, regular and Fredholm.

Keywords— Elliptic boundary conditions • regularity • Hilbert C^* -module bundles • higher index theory

Contents

1	Preliminaries	2
1.1	Functional calculus for unbounded operators and Hilbert space localizations . .	2
1.2	Hilbert C^* -module bundles, Sobolev spaces and differential operators	6
1.3	Analysis on manifolds without boundary	14
2	Analytic foundations for manifolds with boundary	15
2.1	Boundary symmetric differential operators	17
2.2	Hybrid Sobolev spaces and perfect pairings	20
2.3	The extension and the restriction map	23
2.4	Elliptic estimates for smooth sections vanishing at the boundary	29
3	Boundary value problems	32
3.1	Abstract boundary conditions	33
3.2	Elliptic boundary conditions and regularity	34
3.3	Complementable and self-adjoint boundary conditions	38
4	Index theory	42
5	Examples of right-elliptic complementable self-adjoint boundary conditions	43
5.1	The generalized Atiyah-Patodi-Singer boundary condition	44
5.2	Local boundary conditions and chirality	46
	References	50

1 Preliminaries

In this section we give some preliminaries and fix our notations. In Section 1.1 we give functional analytic definitions and summarize properties concerning (self-adjoint and regular) unbounded operators on Hilbert C^* -modules, their functional calculus and the Fredholm property. See [Mur90; Lan95; KL11; Roe17] as references and further background material. In Section 1.2 we define bundles of Hilbert C^* -modules, their (fractional) Sobolev spaces and state in Proposition 1.9 the Sobolev embedding theorem and Rellich's theorem for such bundles. We give a definition of first-order differential operators acting on smooth sections of bundles of Hilbert C^* -modules, and define the minimal and maximal extension (Definition 1.14) of such differential operators. Moreover, we prove an approximation result for the maximal domain using local Friedrichs mollifiers (Proposition 1.15 and Lemma 1.22). So far all definitions and statements hold for manifolds with or without boundary. In Section 1.3 we study first-order differential operators on bundles of Hilbert C^* -modules where the underlying manifold has no boundary. We summarize the elliptic regularity principle (Theorem 1.25) and the self-adjointness of the L^2 -closure of such differential operators (Theorem 1.26). As main references see [MF80; Ebe16; Sch05; ST01].

1.1 Functional calculus for unbounded operators and Hilbert space localizations

Let $\mathcal{A}$ be a unital C^* -algebra. This is a unital Banach $*$ -algebra $\mathcal{A}$ over the complex numbers $\mathbb{C}$ where the involution $*$ is related to the norm on $\mathcal{A}$ via

$$\|xx^*\|_{\mathcal{A}} = \|x\|_{\mathcal{A}}\|x^*\|_{\mathcal{A}} \quad \text{for all } x \in \mathcal{A}. \quad [C^*\text{-identity}]$$

We make the following elementary definitions:

- A self-adjoint element $a \in \mathcal{A}$ is called *non-negative* if there exists an element $h \in \mathcal{A}$ such that $a = h^*h$.
- A *grading* on $\mathcal{A}$ is a $\mathbb{C}$ -linear $*$ -automorphism α on $\mathcal{A}$ satisfying $\alpha^2 = 1$.
- A *Real structure* on $\mathcal{A}$ is a conjugate-linear $*$ -automorphism κ on $\mathcal{A}$ satisfying $\kappa^2 = 1$.

If A is equipped with a grading and a Real structure, they have to be compatible to each other i.e. $\kappa\alpha = \alpha\kappa$. A *pre-Hilbert $\mathcal{A}$ -module* E is a right $\mathcal{A}$ -module together with an $\mathcal{A}$ -valued inner product $\langle \cdot, \cdot \rangle_E$ which satisfies the same axioms as the inner product on a complex Hilbert space by replacing the complex conjugation with the involution $*$. A pre-Hilbert $\mathcal{A}$ -module which is complete with respect to the $\mathbb{R}$ -valued norm $\|\cdot\|^2 := \|\langle \cdot, \cdot \rangle_E\|_{\mathcal{A}}$ is called a *Hilbert $\mathcal{A}$ -module*. We call a Hilbert $\mathcal{A}$ -module *finitely generated and projective* if it is isomorphic to $\mathcal{A}^\ell$ for some $\ell \in \mathbb{N}$. Moreover, a *grading* and a *Real structure* on E are defined similarly as for C^* -algebras via suitable involutions. If these structures exist on E and the underlying C^* -algebra A , then they have to be compatible to each other.

1.1.1 Unbounded operator and their functional calculus

Let E and F be two Hilbert $\mathcal{A}$ -modules. An *unbounded operator* between E and F is an $\mathcal{A}$ -linear map $T: \text{dom}(T) \rightarrow F$ which is defined on some $\mathcal{A}$ -submodule $\text{dom}(T) \subseteq E$. If $\text{dom}(T) \subseteq E$ is dense, we call T *densely defined*. The $\mathcal{A}$ -valued inner product $\langle x, y \rangle_T := \langle x, y \rangle_E + \langle Tx, Ty \rangle_F$, for $x, y \in \text{dom}(T)$, defines the structure of a pre-Hilbert $\mathcal{A}$ -module on the domain of T . The induced $\mathbb{R}$ -valued norm is called the *graph norm* of T . Sums and products of densely defined unbounded operators are defined similarly as for unbounded operators on Hilbert spaces. However, the resulting operator may not necessarily be densely defined [Lan95, Chapter 9]. We make the following elementary definitions:

Definition 1.1. The *adjoint* of a densely defined unbounded operator T is the unbounded operator $T^{\text{ad}}: \text{dom}(T^{\text{ad}}) \rightarrow E$ with domain

$$\text{dom}(T^{\text{ad}}) := \{y \in F \mid \exists z \in E : \langle Tx, y \rangle_F = \langle x, z \rangle_E \forall x \in \text{dom}(T)\}.$$

We call a densely defined unbounded operator $T: \text{dom}(T) \rightarrow F$

- *closed* if $\text{dom}(T)$ is complete with respect to the graph norm of T .
- *closable* if there exists a closed operator $\bar{T}: \text{dom}(\bar{T}) \rightarrow F$ with $\text{dom}(T) \subseteq \text{dom}(\bar{T})$.
- *regular* if T is closed, its adjoint T^{ad} is densely defined, and $1 + TT^{\text{ad}}$ has closed range.
- *self-adjoint* if T is equal to its adjoint T^{ad} .
- *symmetric* if $E = F$ and $\langle Tx, y \rangle_E = \langle x, Ty \rangle_E$ for all $x, y \in \text{dom}(T)$.

Moreover, we define the *spectrum* of T , denoted by $\text{spec}(T)$, as the subset of $\mathbb{C}$ containing all $\lambda \in \mathbb{C}$ such that the operator $T - \lambda: \text{dom}(T) \rightarrow E$ is not invertible.

We summarize in the next lemma some basic properties [see KL11, Lemma 2.1, Lemma 2.2, Lemma 2.3 and Proposition 4.1].

Lemma 1.2. *Let $T: \text{dom}(T) \rightarrow F$ with $\text{dom}(T) \subseteq E$ be a densely defined unbounded operator. Then the following holds:*

- (1) *If the adjoint of T is densely defined, then T is closable and $T^{\text{ad}} = (\bar{T})^{\text{ad}}$.*
- (2) *If T is regular, its adjoint is regular and $T = (T^{\text{ad}})^{\text{ad}}$.*
- (3) *The spectrum of T is closed in $\mathbb{C}$, and if T is self-adjoint and regular, then $\text{spec}(T) \subseteq \mathbb{R}$.*

Notation 1.3. An $\mathcal{A}$ -linear map between E and F is called an *operator*. We denote by

- $\text{Lin}_{\mathcal{A}}(E)$ the C^* -algebra of adjointable operators from E to E ,
- $\text{Kom}_{\mathcal{A}}(E) \subseteq \text{Lin}_{\mathcal{A}}(E)$ the ideal of compact operators from E to E ,
- $\text{Reg}_{\mathcal{A}}(E)$ the set of all regular operators from E to E ,

- $C(\mathbb{R})$ the $*$ -algebra of continuous complex valued functions on $\mathbb{R}$ equipped with the even-odd grading and its natural Real structure,
- $C_b(\mathbb{R}) \subset C(\mathbb{R})$ the graded Real C^* -algebra of bounded functions on $\mathbb{R}$, and
- $C_0(\mathbb{R}) \subset C_b(\mathbb{R})$ the graded Real C^* -algebra of functions on $\mathbb{R}$ vanishing at infinity.

In the following proposition we state the existence of the continuous functional calculus for self-adjoint regular operators on a Hilbert $\mathcal{A}$ -module together with its most important properties. See [Lan95, Theorem 10.9], [Ebe16, Theorem 1.19] and [HPS15, Theorem 3.1].

Proposition 1.4 (Continuous functional calculus). *Let T be a (possibly unbounded) self-adjoint regular operator on the Hilbert $\mathcal{A}$ -module E . Then there exists a linear map*

$$\pi_T: C(\mathbb{R}) \longrightarrow \text{Reg}_{\mathcal{A}}(E), \quad f \longmapsto \pi_T(f) = f(T)$$

such that the following holds:

- (1) $\pi_T(1) = \text{id}_E$ and $\pi_T(\text{id}) = T$.
- (2) $f^*(T) = f(T)^{\text{ad}}$ for all $f \in C(\mathbb{R})$.
- (3) $(f \cdot g)(T) = \overline{f(T) \circ g(T)} = \overline{g(T) \circ f(T)}$ for all $f, g \in C(\mathbb{R})$.
- (4) If $f, g \in C(\mathbb{R})$ with $|f| \leq |g|$, then $\text{dom}(g(T)) \subseteq \text{dom}(f(T))$.
- (5) Let (f_j) be a sequence in $C(\mathbb{R})$ and $F, f \in C(\mathbb{R})$. Suppose $f_j \rightarrow f$ converges uniformly on compact subsets of $\mathbb{R}$ and $|f_j| \leq |F|$ for all $j \in \mathbb{N}$. Then $f_j(T)u \rightarrow f(T)u$ for each $u \in \text{dom}(F(T))$.
- (6) π_T restricts to a C^* -algebra homomorphism $C_b(\mathbb{R}) \rightarrow \text{Lin}_{\mathcal{A}}(E)$.
- (7) $\|f(T)\|_{op} = \sup_{t \in \text{spec}(T)} |f(t)|$ for all $f \in C_b(\mathbb{R})$.
- (8) If $f \in C_b(\mathbb{R})$ such that $g(t) = tf(t) \in C_b(\mathbb{R})$, then $\text{im}(f(T)) \subseteq \text{dom}(T)$ and $Tf(T) = f(T)T = g(T)$.
- (9) If the C^* -algebra $\mathcal{A}$ and the Hilbert $\mathcal{A}$ -module E are equipped with a grading and a Real structure, the map π_T respects these structures. Here $\text{Reg}_{\mathcal{A}}(E)$ inherits the grading and the Real structure from E .

Definition 1.5 (Fredholm). An unbounded self-adjoint regular operator T on a Hilbert $\mathcal{A}$ -module E is called *Fredholm*, if the adjointable operator $T(1 + T^2)^{-1/2}$ (defined via the functional calculus in Proposition 1.4) is invertible modulo a compact operator. This means there exists a $G \in \text{Lin}_{\mathcal{A}}(E)$ such that

$$G \circ (T(1 + T^2)^{-1/2}) - \text{id}_E, (T(1 + T^2)^{-1/2}) \circ G - \text{id}_E \in \text{Kom}_{\mathcal{A}}(E).$$

Construction 1.6 (Fredholm index). Suppose the C^* -algebra $\mathcal{A}$ and the Hilbert $\mathcal{A}$ -module E are equipped with a grading and a Real structure, and E is countably generated. Let T be an

unbounded odd Real Fredholm operator on E . We construct from these data a class in the Fredholm picture of the Real K-theory group $\mathrm{KO}_0(\mathcal{A})$ of the Real C^* -algebra $\mathcal{A}$ [see [Roe17](#), Lecture 12]. Elements of $\mathrm{KO}_0(\mathcal{A})$ are homotopy classes of Fredholm pairs (F, E) consisting of a countably generated graded Real Hilbert $\mathcal{A}$ -module E and an odd Real operator $T \in \mathrm{Lin}_{\mathcal{A}}(E)$ with $T - T^*, T^2 - 1 \in \mathrm{Kom}_{\mathcal{A}}(E)$.

By [[Ebe16](#), Lemma 2.19], the (unbounded) operator T is Fredholm if and only if the adjointable operator $T(1 + T^2)^{-1/2}$ has an *essential spectral gap*. This means there exists an $\varepsilon > 0$ such that

$$f(T(1 + T^2)^{-1/2}) \in \mathrm{Kom}_{\mathcal{A}}(E)$$

for all $f \in C(\mathbb{R})$ supported in $(-\varepsilon, \varepsilon)$. For such an $\varepsilon > 0$ and a continuous odd function $g: \mathbb{R} \rightarrow [-1, 1]$ with $g(t) = \mathrm{sgn}(t)$ for $t \in \mathbb{R} \setminus (-\varepsilon, \varepsilon)$, we obtain a class

$$\mathrm{ind}(T) := [g(T(1 + T^2)^{-1/2}), E] \in \mathrm{KO}_0(\mathcal{A})$$

in the 0-th Real K-theory group of the C^* -algebra $\mathcal{A}$. This class is independent of the choice of ε and g and is called the (*Fredholm*) *index* of the Fredholm operator T .

1.1.2 Localization of Hilbert C^* -modules

Let E be a Hilbert $\mathcal{A}$ module with inner product $\langle \cdot, \cdot \rangle$ and ρ a *state* of the C^* -algebra $\mathcal{A}$, i.e. a positive linear functional $\mathcal{A} \rightarrow \mathbb{C}$. The *Gelfand-Naimark-Segal (GNS) representation* associated to ρ [[Mur90](#), Section 3.4] is defined via

$$\pi_{\rho}: \mathcal{A} \longrightarrow \mathrm{Lin}(H_{\rho}), \quad \pi_{\rho}(a)(b + N_{\rho}) := ab + N_{\rho}.$$

Here $N_{\rho} := \{a \in \mathcal{A} \mid \rho(a^*a) = 0\}$ and H_{ρ} is the Hilbert space completion of $\mathcal{A}/N_{\rho}$ with respect to the inner product $\langle a + N_{\rho}, b + N_{\rho} \rangle_{\rho} := \rho(b^*a)$ for $a, b \in \mathcal{A}$. Moreover, there exists a unique unit vector $x_{\rho} \in H_{\rho}$ which is *cyclic*, i.e. that the map $\mathcal{A} \rightarrow H_{\rho}, a \mapsto \pi_{\rho}(a)x_{\rho}$, has dense image, and that

$$\rho(a) = \langle a + N_{\rho}, x_{\rho} \rangle_{\rho} \quad \text{and} \quad \pi_{\rho}(a)x_{\rho} = a + N_{\rho} \tag{1.1}$$

hold for all $a \in \mathcal{A}$ [[Mur90](#), Theorem 5.1.1.]. We have the following constructions:

Construction 1.7. (1) The *localization* of E with respect to the state ρ is the interior tensor product $E^{\rho} := E \otimes_{\mathcal{A}} H_{\rho}$ with respect to the GNS-representation of ρ [[Lan95](#), Chapter 4]. This is the completion of the algebraic tensor product $E \odot_{\mathcal{A}} H_{\rho}$ with respect to the inner product

$$\langle e \otimes x, f \otimes y \rangle_{\rho} := \langle x, \pi_{\rho}(\langle e, f \rangle) y \rangle_{\rho} \in \mathbb{C}.$$

(2) There exists a natural map $\iota: E \rightarrow E^{\rho}$ defined via $\rho(e) := e \otimes x_{\rho}$ which has dense image (since x_{ρ} is cyclic) and satisfies by the properties in equation (1.1)

$$\langle \iota(e), \iota(e) \rangle_{\rho} = \rho(\langle e, e \rangle) \quad \text{for all } e \in E.$$

- (3) For $T \in \text{Lin}_{\mathcal{A}}(E)$ the induced map $T^\rho: T \otimes 1: E \otimes_{\mathcal{A}} H_\rho \rightarrow E \otimes_{\mathcal{A}} H_\rho$ is bounded [Lan95, p.42], and therefore yields a representation of C^* -algebras $\tilde{\pi}_\rho: \text{Lin}_{\mathcal{A}}(E) \rightarrow \text{Lin}(E^\rho)$.
- (4) Let $T: \text{dom}(T) \rightarrow E$ be a densely defined unbounded operator on E such that its adjoint is densely defined. Then the *localization* of T with respect to ρ [KW75, Section 2.4] is the closed densely defined unbounded operator T^ρ on E^ρ defined as the closure of the unbounded operator

$$T_0^\rho := T \odot_{\mathcal{A}} 1: \text{dom}(T) \odot_{\mathcal{A}} H_\rho \longrightarrow E \odot_{\mathcal{A}} H_\rho \subseteq E \otimes_{\mathcal{A}} H_\rho = E^\rho.$$

We conclude this section with a localization criterion due to Kaad and Lesch [KL11]:

Theorem 1.8 ([KL11, Theorem 1.1]). *A closed symmetric densely defined unbounded operator T on a Hilbert $\mathcal{A}$ -module E is self-adjoint and regular if and only if the localization T^ρ is self-adjoint for any state ρ of the C^* -algebra $\mathcal{A}$.*

1.2 Hilbert C^* -module bundles, Sobolev spaces and differential operators

Throughout this section, let M be a Riemannian manifold of dimension n with or without boundary ∂M , and $\mathcal{A}$ a unital C^* -algebra.

1.2.1 Bundles of Hilbert $\mathcal{A}$ -modules

A *bundle of (finitely generated projective graded Real) Hilbert $\mathcal{A}$ -modules* is a smooth fiber bundle $E \rightarrow M$ with typical fiber a (finitely generated projective graded Real) Hilbert $\mathcal{A}$ -module P and the structure group equals the group of (graded Real) unitary automorphism of P (see [Sch05; Ebe16; MF80]). Note that a bundle of Hilbert $\mathcal{A}$ -modules is equipped with a global smooth $\mathcal{A}$ -valued inner product $\langle \cdot, \cdot \rangle$. If $\mathcal{A} = \mathbb{C}$, we call a bundle of Hilbert $\mathbb{C}$ -modules a *bundle of Hilbert spaces*. Moreover, we denote by E_p the fiber over the point $p \in M$ and by

- $C^\infty(M, E)$ the space of smooth sections of E ,
- $C_c^\infty(M, E)$ the space of compactly supported smooth sections of E ,
- $C_0^\infty(M, E)$ the space of compactly supported smooth sections of E vanishing at ∂M , and
- $C_{cc}^\infty(M, E)$ the space of smooth sections of E with support in the interior of M .

Smooth means smooth up to the boundary. A *connection* on a bundle of Hilbert $\mathcal{A}$ -modules E is an $\mathcal{A}$ -linear map

$$\nabla: C^\infty(M, E) \rightarrow C^\infty(M, TM^* \otimes E)$$

which is derivative with respect to right multiplication with functions $f \in C^\infty(M, \mathcal{A})$. A connection is called *metric* if $\nabla_X \langle u, v \rangle = \langle \nabla_X u, v \rangle + \langle u, \nabla_X v \rangle$ for all $u, v \in C^\infty(M, E)$ and all vector fields X on M .

1.2.2 Sobolev spaces

Let P be a Hilbert $\mathcal{A}$ -module with $\mathcal{A}$ -valued inner product $\langle \cdot, \cdot \rangle$ and fix some $s \in \mathbb{R}$. We define an $\mathcal{A}$ -valued inner product

$$\langle u, v \rangle_s := \int_{\mathbb{R}^n} (1 + x^2)^s \langle \hat{u}(x), \hat{v}(x) \rangle dx$$

for $u, v \in C_c^\infty(\mathbb{R}^n, \mathbb{R}^n \times P)$. Here

$$\hat{u}(x) := (2\pi)^{-2/n} \int_{\mathbb{R}^n} e^{-i\langle x, y \rangle_{\text{eu}}} u(y) dy \quad \text{and} \quad \hat{v}(x) := (2\pi)^{-2/n} \int_{\mathbb{R}^n} e^{-i\langle x, y \rangle_{\text{eu}}} v(y) dy$$

are the Fourier transforms of the smooth sections u and v , respectively [MF80, § 3]. The *fractional Sobolev space* $H^s(\mathbb{R}^n, \mathbb{R}^n \times P)$ is the completion of the pre-Hilbert $\mathcal{A}$ -module $C_c^\infty(M, \mathbb{R}^n \times P)$ equipped with the inner product $\langle \cdot, \cdot \rangle_s$.

Let $E \rightarrow M$ be a bundle of Hilbert $\mathcal{A}$ -modules with typical fiber P and equipped with a metric $\mathcal{A}$ -linear connection ∇ . If the manifold M is closed we obtain the *fractional Sobolev space* $H^s(M, E)$ for $s \in \mathbb{R}$ by a partition of unity argument and the previous construction for trivial bundles over $\mathbb{R}^n$ [see MF80]. In general, we define the following *Sobolev spaces*. Denote by $\langle \cdot, \cdot \rangle_p$ the $\mathcal{A}$ -valued inner product on the fiber over a point $p \in M$, and define $\mathcal{A}$ -valued inner products for all $k \in \mathbb{N}$ and $u, v \in C_c^\infty(M, E)$ via

$$\langle u, v \rangle_{L^2} := \int_M \langle u, v \rangle_p dp \quad \text{and} \quad \langle u, v \rangle_{H^k} := \sum_{j=0}^k \langle \nabla^j u, \nabla^j v \rangle_{L^2}. \quad (1.2)$$

Here we consider ∇^j as a map from $C^\infty(M, E)$ to $C^\infty(M, (TM^*)^j \otimes E)$ and equip $(TM^*)^j \otimes E$ with the induced structure of a bundle of Hilbert $\mathcal{A}$ -modules together with its induced $\mathcal{A}$ -linear connection. We denote the $\mathcal{A}$ -valued norm by $|\cdot|_{H^k}^2 := \langle \cdot, \cdot \rangle_{H^k}$, and define for every $k \in \mathbb{N}_0$ the *Sobolev space* $H^k(M, E)$ as the completion of the pre-Hilbert $\mathcal{A}$ -module of compactly supported smooth sections equipped with respect to the $\mathcal{A}$ -valued inner product in equation (1.2). For $k = 0$ we define $L^2(M, E) := H^0(M, E)$ which is equal to the completion of the smooth sections of E which are compactly supported in the interior of M . If the manifold M is compact, the definition of the Sobolev space $H^k(M, E)$ is independent of the connection on E . For closed manifolds, this definition coincides with the definition of fractional Sobolev spaces for $s \in \mathbb{N}_0$.

Next we define *local Sobolev spaces*. For the regularity statements in Theorem 1.25 and Theorem 3.10 it is sufficient to define the intersection of the local Sobolev spaces with $L^2(M, E)$:

$$L^2(M, E) \cap H_{\text{loc}}^k(M, E) := \{u \in L^2(M, E) \mid gu \in H^k(M, E) \text{ for all } g \in C_c^\infty(M, \mathbb{R})\}. \quad (1.3)$$

For $u \in L^2(M, E)$, we write $u \in H_{\text{loc}}^k(M, E)$ instead of $u \in L^2(M, E) \cap H_{\text{loc}}^k(M, E)$. If the boundary of M is compact, it is sufficient to consider in equation (1.3) functions g which are constant at the boundary of M . In general, we have the inclusion $H^k(M, E) \subseteq L^2(M, E) \cap H_{\text{loc}}^k(M, E)$ with equality if the manifold M is compact.

As in the setting of finite-rank vector bundles, the Sobolev embedding theorem and Rellich's

theorem still hold. The classical statements are given in [LM89, Chapter III Theorem 2.5 and Theorem 2.15] and [LM89, Chapter II Theorem 2.6], and the corresponding generalizations in Proposition 1.9 are due to Mishchenko and Fomenko [MF80], see for instance [ST01, Section 2.1].

Proposition 1.9. *The following holds for all $k, \ell \in \mathbb{N}_0$:*

- (1) *If $k > \ell + \frac{n}{2}$, then there exists a constant C such that for all $u \in C^\infty(M, E)$*

$$\|u\|_{C^\ell} \leq C \|u\|_{H^k}. \quad [\text{Sobolev embedding}]$$

- (2) *Suppose M is compact and E is a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules. If $k > \ell$, then $H^k(M, E) \hookrightarrow H^\ell(M, E)$ is a compact embedding.* [Rellich's theorem]

Here the uniform C^ℓ -norm is defined for smooth sections $u \in C_c^\infty(M, E)$ via

$$\|u\|_{C^\ell} := \sum_{j=0}^{\ell} \|\nabla^j u\|_{L^\infty} \quad \text{with} \quad \|u\|_{L^\infty} := \sup_{p \in M} \|\langle u, u \rangle_p\|_{\mathcal{A}}.$$

Suppose the boundary of the manifold M is compact. The proof of the trace theorem in [Eva10, Chapter 5 Theorem 2] carries over without change. We obtain:

Lemma 1.10 (Trace theorem). *The restriction map $C_c^\infty(M, E) \rightarrow C^\infty(\partial M, E)$, $u \mapsto u|_{\partial M}$ extends to an $\mathcal{A}$ -linear bounded operator*

$$\mathcal{R}_1: H^1(M, E) \longrightarrow H^{1/2}(\partial M, E),$$

with kernel equal to $H^1(M \setminus \partial M, E)$.

1.2.3 $\mathcal{A}$ -linear differential operators

Let E and F be bundles of Hilbert $\mathcal{A}$ -modules over the manifold M with typical fibers P and P' , respectively. An $\mathcal{A}$ -linear map $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ is called an ($\mathcal{A}$ -linear) *differential operator of first order*, if in each coordinate system $x: U \rightarrow \mathbb{R}^n$ and for every smooth local trivializations $h: E|_U \rightarrow U \times P$ and $h': F|_U \rightarrow U \times P'$ the operator $\mathcal{D}$ has locally the form

$$\mathcal{D}u(x) = b(x)u(x) + \sum_{j=1}^n a_j(x) \partial_{x_j} u(x) \quad (1.4)$$

for some smooth $b, a_1, \dots, a_n: U \rightarrow \text{Lin}_{\mathcal{A}}(P, P')$. We say the operator $\mathcal{D}$ has *order zero*, if in equation (1.4) all a_j vanishes, and has *order $\leq \ell$* if it can be written as a finite sum of compositions of at most ℓ first-order $\mathcal{A}$ -linear differential operators.

Definition 1.11. Let $\mathcal{D}$ be a first-order differential operator.

- (1) The *formal adjoint* of $\mathcal{D}$ is the unique first-order $\mathcal{A}$ -linear differential operator

$$\mathcal{D}^\dagger: C_c^\infty(M, F) \longrightarrow C_c^\infty(M, E)$$

which satisfies $\langle \mathcal{D}u, v \rangle_{L^2} = \langle u, \mathcal{D}^\dagger v \rangle_{L^2}$ for all $u \in C_{cc}^\infty(M, E)$ and all $v \in C_c^\infty(M, F)$. We call $\mathcal{D}$ *formally self-adjoint*¹, if $E = F$ and $\mathcal{D} = \mathcal{D}^\dagger$.

- (2) The *principal symbol* of $\mathcal{D}$ is the unique $\mathcal{A}$ -linear adjointable bundle map $\sigma_{\mathcal{D}}: TM^* \otimes E \rightarrow F$ such that

$$\sigma(d_x f)u(x) = [\mathcal{D}, f]u(x)$$

for all $x \in M$, $u \in C^\infty(M, E)$ and all $f \in C^\infty(M, \mathbb{R})$. We call $\mathcal{D}$ *elliptic* if the principal symbol $\sigma_{\mathcal{D}}(\xi)|_x$ is invertible for all nonzero $\xi \in T_x M$ and all $x \in M$.

- (3) We call $(M, \mathcal{D})$ *complete*, and say M is *complete for $\mathcal{D}$* , if there exists a proper smooth function $f: M \rightarrow \mathbb{R}$ which is bounded from below such that the commutator $[\mathcal{D}, f]$ extends to a bounded operator from $L^2(M, E)$ to $L^2(M, F)$.

The formal adjoint $\mathcal{D}^\dagger$ can be constructed explicitly using the local form of $\mathcal{D}$ in equation (1.4), integration by parts and the fact that the local coefficients of the differential operator $\mathcal{D}$ are adjointable. Moreover, we obtain as in the classical case of finite-rank vector bundles [cf. BB12, Lemma 2.6] the following formula:

Lemma 1.12. *The differential operators $\mathcal{D}$ and $\mathcal{D}^\dagger$ satisfy*

$$\langle \mathcal{D}u, v \rangle_{L^2(M)} - \langle u, \mathcal{D}^\dagger v \rangle_{L^2(M)} = -\langle \sigma_{\mathcal{D}}(\tau)u|_{\partial M}, v|_{\partial M} \rangle_{L^2(\partial M)} \quad [\text{Green's formula}]$$

for all $u \in C_c^\infty(M, E)$ and all $v \in C_c^\infty(M, F)$. Here τ denotes the co-vector field associated to an inward pointing vector field along the boundary of M .

We continue with some functional analytic results of (first-order) $\mathcal{A}$ -linear differential operators. Suppose E and F are equipped with $\mathcal{A}$ -linear metric connections. There exist (a priori) several closed extensions associated to a first-order $\mathcal{A}$ -linear differential operator. For instance the minimal and maximal extension in Definition 1.14. They play a crucial role in the theory of boundary value problems. If the manifold M is compact, then there exist the following bounded Sobolev extensions:

Proposition 1.13 ([MF80, Lemma 3.2]). *Let $\ell \in \mathbb{N}_0$, $\mathcal{P}: C^\infty(M, E) \rightarrow C^\infty(M, E)$ be an $\mathcal{A}$ -linear differential operator of order $\leq \ell$ and suppose the manifold M is compact. Then $\mathcal{P}$ extends for all $k \in \mathbb{N}_0$ to a bounded operator*

$$\mathcal{P}: H^{k+\ell}(M, E) \longrightarrow H^k(M, E).$$

Let $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ be a first-order $\mathcal{A}$ -linear differential operator and denote its formal adjoint by $\mathcal{D}^\dagger$. The minimal and maximal extension of $\mathcal{D}$ are defined as follows:

¹A first-order differential operator $\mathcal{D}$ is formally self-adjoint if and only if $\mathcal{D}: C_{cc}^\infty(M, E) \rightarrow L^2(M, F)$ is symmetric as an unbounded operator.

Definition 1.14 (Minimal and maximal extension). We define densely defined unbounded operators $\mathcal{D}_{cc}: C_{cc}^\infty(M, E) \rightarrow L^2(M, F)$ and $\mathcal{D}_{cc}^\dagger: C_{cc}^\infty(M, F) \rightarrow L^2(M, E)$ by restricting $\mathcal{D}$ and $\mathcal{D}^\dagger$ accordingly. The *minimal* and *maximal extension* of $\mathcal{D}$,

$$\begin{aligned}\mathcal{D}_{\min} &:= \overline{\mathcal{D}_{cc}}: \text{dom}(\mathcal{D}_{\min}) \longrightarrow L^2(M, F) \quad \text{and} \\ \mathcal{D}_{\max} &:= (\mathcal{D}_{cc}^\dagger)^{\text{ad}}: \text{dom}(\mathcal{D}_{\max}) \longrightarrow L^2(M, F),\end{aligned}$$

are defined as the closure of $\mathcal{D}_{cc}$ and the adjoint of $\mathcal{D}_{cc}^\dagger$, respectively (see Definition 1.1).

Note that $\mathcal{D}_{cc}: C_{cc}^\infty(M, E) \rightarrow L^2(M, F)$ is closable since its adjoint is densely defined by the existence of the formal adjoint $\mathcal{D}^\dagger$. If the manifold M is compact², the definitions of $\mathcal{D}_{\min}$ and $\mathcal{D}_{\max}$ yield immediately

$$\begin{aligned}\text{dom}(\mathcal{D}_{\min}) &\subseteq \text{dom}(\overline{\mathcal{D}_c}) \subseteq \text{dom}(\mathcal{D}_{\max}) \\ \text{H}^1(M \setminus \partial M, E) &\subseteq \text{H}^1(M, E).\end{aligned}\tag{1.5}$$

Moreover, every closed extension of $\mathcal{D}_{cc}: C_{cc}^\infty(M, E) \rightarrow L^2(M, F)$ is contained in $\mathcal{D}_{\max}$. The vertical inclusions in equation (1.5) hold by Proposition 1.13. We will see in Proposition 1.15 that $\overline{\mathcal{D}_c}$ coincide with the maximal extension of $\mathcal{D}$ if the manifold M is complete for $\mathcal{D}$. Moreover, if the manifold is closed and the differential operator $\mathcal{D}$ is elliptic, all Hilbert $\mathcal{A}$ -modules in equation (1.5) coincide (Theorem 1.25).

Proposition 1.15. *Suppose $(M, \mathcal{D})$ is complete. Then $C_c^\infty(M, E)$ is dense in $\text{dom}(\mathcal{D}_{\max})$ with respect to the graph norm of $\mathcal{D}$. In particular, if the boundary of M is empty then $\text{dom}(\mathcal{D}_{\min}) = \text{dom}(\mathcal{D}_{\max})$.*

We prove Proposition 1.15 in two steps. In the first step we show that every $u \in \text{dom}(\mathcal{D}_{\max})$ can be approximated in the graph norm of $\mathcal{D}$ by compactly supported sections in the maximal domain of $\mathcal{D}$ (see Lemma 1.17). In the second step we show that such sections can be approximated by compactly supported smooth sections (see Lemma 1.21). We show in the second step a stronger statement namely that every compactly supported section $u \in \text{dom}(\mathcal{D}_{\max}) \cap \text{H}^k(M, E)$ with $\mathcal{D}_{\max} u \in \text{H}^k(M, E)$ can be approximated by compactly supported smooth section in the H^k -graph norm of $\mathcal{D}$ (see Definition 1.18). This will be important for the higher elliptic regularity principle in Theorem 1.25 and Theorem 3.10. The proof of Lemma 1.21 is based on the existence of Friedrichs mollifiers on $\mathbb{R}^n$ and the half-space $\mathbb{H}^n$ (Lemma 1.22).

Lemma 1.16. *Let $u \in \text{dom}(\mathcal{D}_{\max})$ and $g \in C^\infty(M, \mathbb{R})$ be a bounded smooth function. Then $gu \in \text{dom}(\mathcal{D}_{\max})$.*

Proof. Let $u \in \text{dom}(\mathcal{D}_{\max})$ and $g \in C^\infty(M, \mathbb{R})$ be a bounded smooth function. By the definitions of the maximal domain and the adjoint operator (Definition 1.14 and Definition 1.1), there exists $w \in L^2(M, E)$ such that $\langle \mathcal{D}_{cc}^\dagger v, u \rangle_{L^2} = \langle v, w \rangle_{L^2}$ for all $v \in C_{cc}^\infty(M, F)$. We obtain by the

²The upper row in equation (1.5) also holds for non-compact manifolds.

properties of the principal symbol in Definition 1.11 (2)

$$\langle v, gw - \sigma_{\mathcal{D}^\dagger}(\mathrm{d}g)^\dagger u \rangle_{L^2} = \langle \mathcal{D}_{\mathrm{cc}}^\dagger(gv) - \sigma_{\mathcal{D}^\dagger}(\mathrm{d}g)v, u \rangle_{L^2} = \langle \mathcal{D}_{\mathrm{cc}}^\dagger v, gu \rangle_{L^2}$$

for all $v \in C_c^\infty(M, E)$, hence $gu \in \mathrm{dom}((\mathcal{D}_{\mathrm{cc}}^\dagger)^{\mathrm{ad}}) = \mathrm{dom}(\mathcal{D}_{\mathrm{max}})$. Note that $gw \in L^2(M, F)$ since g is bounded. $\square$

Lemma 1.17. *Suppose $(M, \mathcal{D})$ is complete. Then there exists for every $j \in \mathbb{N}$ a smooth function $g_j \in C_c^\infty(M, \mathbb{R})$ such that for every $u \in \mathrm{dom}(\mathcal{D}_{\mathrm{max}})$ the sequence $(g_j u) \subseteq \mathrm{dom}(\mathcal{D}_{\mathrm{max}})$ converge to u in the graph norm of $\mathcal{D}$.*

Proof. Let $f: M \rightarrow \mathbb{R}$ be a smooth function as in Definition 1.11 (3) together with a constant C such that $\|\sigma_{\mathcal{D}}(\mathrm{d}f)\|_{L^2 \rightarrow L^2} = \|[\mathcal{D}, f]\|_{L^2 \rightarrow L^2} \leq C$. We fix some smooth function $\eta: \mathbb{R} \rightarrow [0, 1]$ with $\mathrm{supp}(\eta) \subset (-\infty, 2)$ and $\eta \equiv 1$ on $(-\infty, 1]$, and define for $j \in \mathbb{N}$ a smooth function

$$g_j: M \longrightarrow \mathbb{R}, \quad p \longmapsto \eta(f(p)/j).$$

Since f is proper and bounded from below, g_j is compactly supported. We obtain

$$\|[\mathcal{D}, g_j]u\|_{L^2} = \|\sigma_{\mathcal{D}}(\mathrm{d}g_j)u\|_{L^2} = 1/j \|\sigma_{\mathcal{D}}(\mathrm{d}f)u\|_{L^2} \leq C/j \|u\|_{L^2}$$

for all smooth section $u \in C^\infty(M, E)$. In particular, the commutator $[\mathcal{D}, g_j]$ extends to a bounded operator $L^2(M, E) \rightarrow L^2(M, E)$ and $[\mathcal{D}_{\mathrm{max}}, g_j] = [\mathcal{D}, g_j]$ on $\mathrm{dom}(\mathcal{D}_{\mathrm{max}})$. Note that the commutator $[\mathcal{D}_{\mathrm{max}}, g_j]$ is well-defined by Lemma 1.16. We obtain for $u \in \mathrm{dom}(\mathcal{D}_{\mathrm{max}})$

$$\begin{aligned} 2^{-1/2} \|g_j u - u\|_{\mathcal{D}} &\leq \|[\mathcal{D}_{\mathrm{max}}, g_j]u\|_{L^2} + \|g_j \mathcal{D}_{\mathrm{max}} u - \mathcal{D}_{\mathrm{max}} u\|_{L^2} + \|g_j u - u\|_{L^2} \\ &\leq C/j \|u\|_{L^2} + \|g_j \mathcal{D}_{\mathrm{max}} u - \mathcal{D}_{\mathrm{max}} u\|_{L^2} + \|g_j u - u\|_{L^2} \xrightarrow{j \rightarrow \infty} 0, \end{aligned}$$

and the lemma is proved. $\square$

Definition 1.18. For $k \in \mathbb{N}_0$, we define a Hilbert $\mathcal{A}$ -module

$$\mathrm{dom}(\mathcal{D}_{\mathrm{max}}; k) := \{u \in \mathrm{dom}(\mathcal{D}_{\mathrm{max}}) \cap H^k(M, E) \mid \mathcal{D}u \in H^k(M, F)\}$$

with the $\mathcal{A}$ -valued inner product given by $\langle \cdot, \cdot \rangle_{\mathcal{D}_{\mathrm{max}}, H^k} := \langle \mathcal{D}_{\mathrm{max}} \cdot, \mathcal{D} \cdot \rangle_{H^k} + \langle \cdot, \cdot \rangle_{H^k}$, and denote its induced $\mathcal{A}$ and $\mathbb{R}$ -valued norms by $|\cdot|_{\mathcal{D}, H^k}$ and $\|\cdot\|_{\mathcal{D}, H^k}$, respectively.

Remark 1.19. The space $\mathrm{dom}(\mathcal{D}_{\mathrm{max}}; k)$ is closed with respect to the norm $\|\cdot\|_{\mathcal{D}, H^k}$, hence it defines indeed a Hilbert $\mathcal{A}$ -module: Fix some $k \in \mathbb{N}_0$ and let $(u_j) \subset \mathrm{dom}(\mathcal{D}_{\mathrm{max}}; k)$ be a Cauchy sequence with respect to $\|\cdot\|_{\mathcal{D}, H^k}$. Then (u_j) is Cauchy in H^k and Cauchy in $\mathrm{dom}(\mathcal{D}_{\mathrm{max}})$, hence converges to some $u \in \mathrm{dom}(\mathcal{D}_{\mathrm{max}}) \cap H^k(M, E)$ with respect to the norms $\|\cdot\|_{H^k}$ and $\|\cdot\|_{\mathcal{D}}$. Moreover, $(\mathcal{D}u_j)$ is Cauchy in H^k , hence converges to some $v \in H^k(M, E)$. Since $\mathcal{D}_{\mathrm{max}}: \mathrm{dom}(\mathcal{D}_{\mathrm{max}}) \rightarrow L^2(M, E)$ is a closed unbounded operator, we obtain $\mathcal{D}_{\mathrm{max}} u = v \in H^k(M, E)$, hence $u \in \mathrm{dom}(\mathcal{D}_{\mathrm{max}}; k)$.

Remark 1.20. Suppose there exists a smooth proper function f bounded from below such that $[\mathcal{D}, f]$ extends for some $k \in \mathbb{N}$ to a bounded operator $H^k(M, E) \rightarrow H^k(M, F)$ (cf. Definition 1.11 (3)). Then even a stronger version of Proposition 1.15 holds. Namely, $C_c^\infty(M, E)$ is dense in $\text{dom}(\mathcal{D}_{\max}; k)$ with respect to the $\|\cdot\|_{\mathcal{D}, H^k}$ -norm. The crucial point is that Lemma 1.17 generalizes to $\text{dom}(\mathcal{D}_{\max}; k)$ under the stronger assumption on the commutator $[\mathcal{D}, f]$.

Lemma 1.21. *Let $k \in \mathbb{N}$ and $u \in \text{dom}(\mathcal{D}_{\max}; k)$ compactly supported. Then there exists for every open relatively compact subset U with $\text{supp}(u) \subseteq U$ a sequence $(u_j) \subset C_c^\infty(M, E)$ supported within U converging to u in the $\|\cdot\|_{\mathcal{D}, H^k}$ -norm.*

Lemma 1.22 (Existence of (local) Friedrichs mollifiers). *For $U \subseteq M$ being either isomorphic to $\mathbb{R}^n$ or $\mathbb{H}^n := \{x \in \mathbb{R}^n \mid x_1 \geq 0\}$ there exists a family of bounded operators $\mathcal{F}_\varepsilon: L^2(U, E) \rightarrow L^2(U, F)$, $\varepsilon > 0$, satisfying for all $k \in \mathbb{N}_0$:*

- (1) $\mathcal{F}_\varepsilon u$ is smooth with support contained in an ε -neighborhood of $\text{supp}(u)$ for all $u \in L^2(U, E)$.
- (2) $\mathcal{F}_\varepsilon$ extends to a bounded operator from $H^k(U, E)$ to $H^k(U, F)$ and its operator norm is uniformly bounded in $\varepsilon \in (0, 1]$.
- (3) $\|\mathcal{F}_\varepsilon u - u\|_{H^k} \rightarrow 0$ for $\varepsilon \rightarrow 0$ for all $u \in H^k(U, E)$.
- (4) [Friedrichs lemma] Let $\mathcal{P}$ be a first-order $\mathcal{A}$ -linear differential operator. Then the commutator $[\mathcal{P}, \mathcal{F}_\varepsilon]$ extends to a bounded operator $H^k(U, E) \rightarrow H^k(U, F)$, its operator norm is uniformly bounded in $\varepsilon \in (0, 1]$, and $\|[\mathcal{P}, \mathcal{F}_\varepsilon]u\| \rightarrow 0$ for $\varepsilon \rightarrow 0$ for all $u \in H^k(U, E)$.

Proof. Let $g \in C_c^\infty(\mathbb{R}^n, \mathbb{R})$ be a smooth function with $g \geq 0$, $\text{supp}(g) \subset B_1(0)$ and $\int g(x) dx = 1$. We define for every $\varepsilon > 0$ the Friedrichs mollifier $\mathcal{F}_\varepsilon$ for a smooth function $u \in C_c^\infty(U, E)$ by the ε -translated convolution with $g_\varepsilon(x) := \varepsilon^{-n}g(x/\varepsilon)$, namely

$$(\mathcal{F}_\varepsilon u)(x) := (u \star_\varepsilon g_\varepsilon)(x) := \int_{\mathbb{R}^n} u(x + (\varepsilon, 0, \dots, 0) - y) g_\varepsilon(y) dy$$

for $x \in U$. This is well-defined since $U \cong \mathbb{R}^n$ or $U \cong \mathbb{H}^n$, the ε -shift in the convolution, and $\text{supp}(g_\varepsilon) \subset B_\varepsilon(0)$. The properties (1)-(4) follow by standard mollifier techniques as in [Hör07, Section 17.1] for the case without boundary and [BL01, Section 2] for the general case. $\square$

Remark 1.23. It is possible to construct global Friedrichs mollifiers from the local ones given in Lemma 1.22. However, they satisfy the properties (1)-(3) listed in Lemma 1.22, except that in part (4) the operator norm of the commutator $[\mathcal{P}, \mathcal{F}_\varepsilon]$ is just uniformly bounded but does in general not converge to zero [HR00, Proof of Lemma 10.2.5]. The reason is that the differential operator hits the partition of unity and produces terms which are bounded but not necessarily small. For the approximation result in Lemma 1.17 we need the local Friedrichs mollifiers with the property as state in part (4) of Lemma 1.22. In the classical setting of finite-rank vector bundles, it suffices to use the global Friedrichs mollifier for these kind of approximation results. However, this approach relies on the sequential Banach–Alaoglu and Banach–Saks theorems [see proof of Ebe16, Proposition 1.16], which are both not available in the case of infinite-rank vector bundles.

Proof of Lemma 1.21. Let $k \in \mathbb{N}_0$, $u \in \text{dom}(\mathcal{D}_{\max}; k)$ and U an open relatively compact subset U with $\text{supp}(u) \subseteq U$. We pick a finite cover of U by coordinate charts U_ℓ which are either isomorphic to $\mathbb{R}^n$ or $\mathbb{H}^n$ together with a smooth subordinated partition of unity (χ_ℓ) such that $\text{supp}(\chi_\ell) \subset U_\ell$ is compact for all ℓ . For every ℓ we pick a Friedrichs mollifier $\mathcal{F}_\varepsilon^\ell$ on U_ℓ as in Lemma 1.22. We define for $j \in \mathbb{N}$ a smooth section $u_j := \sum_\ell \mathcal{F}_{1/j}^\ell \chi_\ell u \in C_c^\infty(M, E)$. Here we extend $\mathcal{F}_{1/j}^\ell \chi_\ell u$ by zero onto $M \setminus U_\ell$. Since $\chi_\ell u \in H^k(U_\ell, E)$ and $\mathcal{F}_{1/j}^\ell \chi_\ell u$ is compactly supported in U_ℓ , u_j is well-defined and supported in U . We obtain

$$\begin{aligned} \|u_j - u\|_{\mathcal{D}, H^k} &\leq \sum_\ell \|\mathcal{F}_{1/j}^\ell \chi_\ell u - \chi_\ell u\|_{\mathcal{D}, H^k} \\ &\leq \sqrt{2} \sum_\ell \left(\|\mathcal{F}_{1/j}^\ell \mathcal{D} \chi_\ell u - \mathcal{D} \chi_\ell u\|_{H^k} + \|[\mathcal{D}, \mathcal{F}_{1/j}^\ell] \chi_\ell u\|_{H^k} + \|\mathcal{F}_{1/j}^\ell \chi_\ell u - \chi_\ell u\|_{H^k} \right) \end{aligned} \quad (1.6)$$

for all $j \in \mathbb{N}$. Since $\mathcal{D} \chi_\ell u = [\mathcal{D}, \chi_\ell] u + \chi_\ell \mathcal{D} u \in H^k(M, F)$ by assumption, all terms on the right-hand side of equation (1.6) converge to zero for $j \rightarrow \infty$ by Lemma 1.22 (3) and (4), hence the lemma and also Proposition 1.15 are proved. $\square$

1.2.4 Localization of bundles of Hilbert C^* -modules

In this section we explain the localization principle for bundles of Hilbert $\mathcal{A}$ -modules and $\mathcal{A}$ -linear differential operators following the construction in [Ebe16].

Let $E \rightarrow M$ be a bundle of Hilbert $\mathcal{A}$ -modules of typical fiber P , and ρ a state of the C^* -algebra $\mathcal{A}$. Then there exists a $U(P)$ -principal bundle $Q \rightarrow M$ such that $E \cong Q \times_{U(P)} P$, and the C^* -algebra representation in Construction 1.7 (3) restricts to a continuous group homomorphism $U(P) \rightarrow U(P^\rho)$. We obtain a bundle of Hilbert spaces

$$E^\rho := Q \times_{U(P^\rho)} P^\rho,$$

called the *localization* of E (with respect to ρ). The connection on E induces a connection on E^ρ , hence the Sobolev spaces $H^k(M, E^\rho)$ in Section 1.2.2 are well-defined.

Let $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, E)$ be a first-order differential operator on E . By composing the functions $b, a_1, \dots, a_m \in U \rightarrow \text{Lin}_A(P)$ in the local expressions of $\mathcal{D}$ in equation (1.4) with $\pi_\rho: \text{Lin}_A(P) \rightarrow \text{Lin}(P^\rho)$, we obtain a first-order differential operator

$$\mathcal{D}_\rho: C^\infty(M, E^\rho) \rightarrow C^\infty(M, E^\rho)$$

on the bundle of Hilbert spaces E^ρ . It is called the *localized* differential operator of $\mathcal{D}$ (with respect to ρ). The map $\iota: P \rightarrow P^\rho$ in Construction 1.7 (2) induces a bundle map $\iota: E \rightarrow E^\rho$ which satisfies

$$\|\iota(u)\|_{H^k(M, E^\rho)} = \rho(|u|_{H^k(M, E)}) \quad \text{and} \quad \iota(\mathcal{D}u) = \mathcal{D}_\rho(\iota(u)) \quad (1.7)$$

for all $k \in \mathbb{N}_0$ and all $u \in C^\infty(M, E)$.

1.3 Analysis on manifolds without boundary

Throughout this section let M be a manifold without boundary, $\mathcal{A}$ a unital C^* -algebra, E and F bundles of Hilbert $\mathcal{A}$ -modules over the manifold M , and $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ a first-order $\mathcal{A}$ -linear differential operator. The results summarized in this section are well-known and for instance contained in [ST01] and [Ebe16].

Proposition 1.24. *Let $k \in \mathbb{N}_0$ and suppose the differential operator $\mathcal{D}$ is elliptic. Then there exists for every compact subsets $K \subseteq M$ a constant C_K such that*

$$\|u\|_{H^{k+1}} \leq C_K(\|\mathcal{D}u\|_{H^k} + \|u\|_{H^k}) \quad [\text{interior elliptic estimate}]$$

for all $u \in C^\infty(M, E)$ with support contained in K .

Proof. The proof of the classical Gårding inequality, the elliptic estimates for $k = 0$, given in [HR00, Section 10.4.4] carries over to the setting of infinite dimensional bundles. The asserted estimate follows by an induction on k . $\square$

Theorem 1.25 (Interior elliptic regularity). *Suppose the differential operator $\mathcal{D}$ is elliptic. Then the following holds for all $k \in \mathbb{N}_0$:*

$$u \in \text{dom}(\mathcal{D}_{\max}) \cap H_{\text{loc}}^k(M, E) \quad \text{and} \quad \mathcal{D}_{\max} u \in H_{\text{loc}}^k(M, F) \quad \implies \quad u \in H_{\text{loc}}^{k+1}(M, E).$$

In particular, if the manifold M is closed then $\text{dom}(\mathcal{D}_{\max}) = \text{dom}(\mathcal{D}_{\min}) = H^1(M, E)$.

Proof. Let $u \in \text{dom}(\mathcal{D}_{\max}) \cap H_{\text{loc}}^k(M, E)$ with $\mathcal{D}_{\max} u \in H_{\text{loc}}^k(M, F)$ and fix some $g \in C_c^\infty(M, \mathbb{R})$. We have to show that $gu \in H^{k+1}(M, E)$. By Lemma 1.16 we have

$$gu \in \text{dom}(\mathcal{D}_{\max}; k) \quad \text{and} \quad \mathcal{D}_{\max}(gu) = [\mathcal{D}, g]u + g \mathcal{D}_{\max} u \in H^k(M, F).$$

Here we used that $[\mathcal{D}, g]$ extends to $H^k(M, E) \rightarrow H^k(M, F)$ since g is compactly supported and that $[\mathcal{D}_{\max}, g]u = [\mathcal{D}, g]u$. We obtain that for some fixed relatively compact open subset U containing $\text{supp}(g)$, there exists by Lemma 1.21 a sequence $(u_j) \subset C^\infty(M, E)$ supported in U such that (u_j) converges to gu in the $\|\cdot\|_{\mathcal{D}, H^k}$ -norm. By the elliptic estimate in Proposition 1.24 with $K = \overline{U}$, we obtain

$$\|u_i - u_j\|_{H^{k+1}} \leq \sqrt{2} C_K \|u_i - u_j\|_{\mathcal{D}, H^k}$$

for all $i, j \in \mathbb{N}$. It follows that (u_j) is a Cauchy sequence with respect to the H^{k+1} -norm, hence it converges to some $w \in H^{k+1}(M, E)$. We obtain $\chi u = w \in H^{k+1}(M, E)$, and the lemma is proved. $\square$

Theorem 1.26 ([Ebe16, Theorem 1.14]). *Suppose E is either a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules or a bundle of Hilbert spaces. If $(M, \mathcal{D})$ is complete and $\mathcal{D}$ is formally self-adjoint, then the L^2 -closure of $\mathcal{D}$ is self-adjoint and regular.*

Proof. If $\mathcal{A} = \mathbb{C}$ and E a finite-rank vector bundle, then Theorem 1.26 is classical and due to Chernoff [Che73] and Wolf [Wol73]. As demonstrated in [Ebe16, Proposition 1.16], the proof

given by Higson and Roe [HR00, Proposition 10.10.2] carries over to bundles of Hilbert spaces. For E being a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules, Ebert [Ebe16] proved the statement in Theorem 1.26 using the localization criterion by Kaad and Lesch [KL11] stated in Theorem 1.8 together with the localization principle for bundles of Hilbert $\mathcal{A}$ -modules in Section 1.2.4. Note that this proof takes advantage of the fact that the fibers of E are finitely generated and projective [Ebe16, Lemma 1.16]. $\square$

Remark 1.27 (Fractional Sobolev spaces and differential operators). Suppose the manifold M is closed, and E is either a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules or if $\mathcal{A} = \mathbb{C}$ a bundle of Hilbert spaces. Let $A: C^\infty(M, E) \rightarrow C^\infty(M, E)$ be an elliptic formally self-adjoint first-order $\mathcal{A}$ -linear differential operator. By Theorem 1.26 the L^2 -closure of A is self-adjoint and regular. We define for $s \in \mathbb{R}$ an $\mathcal{A}$ -valued inner product

$$\langle u, v \rangle_{H^s} := \langle (1 + A^2)^s u, v \rangle_{L^2} \quad (1.8)$$

for $u, v \in C^\infty(M, E)$. Here, $(1 + A^2)^s$ is defined via the functional calculus in Proposition 1.4. The completion of the pre-Hilbert $\mathcal{A}$ -module of smooth sections of E equipped with the $\mathcal{A}$ -valued inner product is equal to the fractional Sobolev space $H^s(M, E)$ defined in Section 1.2.2. For $s \in \mathbb{N}_0$ this follows immediately from Proposition 1.13 and Proposition 1.24.

2 Analytic foundations for manifolds with boundary

In this section we give the analytic groundwork to define abstract boundary conditions in Section 3. The main results of this section are the properties of the restriction map in Theorem 2.3 and the elliptic estimates for smooth sections vanishing at the boundary in Proposition 2.19. In Section 2.1 and Section 2.2 we study boundary symmetric differential operators (Definition 2.2) and the Hybrid Sobolev spaces (Definition 2.9), respectively. We give a proof Theorem 2.3 in Section 2.3 using the extension map in Definition 2.14. In Section 2.4 we prove the elliptic estimates for smooth sections vanishing at the boundary (Proposition 2.19).

We mainly follow the approach by Bär and Ballmann [BB12] and adapt the definitions and proofs to apply to first-order differential operators acting on smooth sections of bundles of finitely generated projective Hilbert $\mathcal{A}$ -modules. The main difficulty is that the spectrum of the adapted boundary operator is not necessarily discrete as in [BB12]. However, in our setting the adapted boundary operator still admits a continuous functional calculus as in Proposition 1.4 (see Remark 2.4) and we can use similar analytic arguments as in [BB12]. Throughout this section we assume the following setup.

Setup 2.1. Let (M, g) be a Riemannian manifold of dimension n with compact boundary ∂M , $\mathcal{A}$ a unital C^* -algebra, E and F bundles of finitely generated projective Hilbert $\mathcal{A}$ -modules or if $\mathcal{A} = \mathbb{C}$ bundles of Hilbert spaces over the manifold M which are both equipped with $\mathcal{A}$ -linear metric connections, and

$$\mathcal{D}: C^\infty(M, E) \longrightarrow C^\infty(M, F)$$

a first-order $\mathcal{A}$ -linear differential operator such that M is complete for $\mathcal{D}$.

If the boundary of M is empty, the minimal and maximal extension of $\mathcal{D}$ coincide by Proposition 1.15. For a non-empty boundary this is no longer the case. A boundary condition B for the differential operator $\mathcal{D}$ will be defined such that it yields a closed extension $\mathcal{D}_B$ of the differential operator $\mathcal{D}$ satisfying

$$\text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\mathcal{D}_B) \subseteq \text{dom}(\mathcal{D}_{\max})$$

(see Definition 3.2 and Theorem 3.3). Since boundary conditions should be defined purely by data on the boundary of M , we will define them using the restriction map $\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \rightarrow \check{H}_A(\partial M, E)$. Here $\check{H}_A(\partial M, E)$ is a suitable Hilbert $\mathcal{A}$ -module such that the restriction map $\mathcal{R}$ is bounded with respect to the graph norm of $\mathcal{D}$, surjective and its kernel is equal to $\text{dom}(\mathcal{D}_{\min})$. Then an abstract boundary condition will be defined as a closed $\mathcal{A}$ -submodule of $\check{H}_A(\partial M, E)$ (Definition 3.2). In order to obtain such a restriction map we need that the first-order $\mathcal{A}$ -linear differential operator $\mathcal{D}$ is boundary symmetric:

Definition 2.2 (Boundary symmetric [compare BB12, Definition 1.4.]). Let T be a vector field of M along the boundary ∂M pointing into the interior of M and τ its associated co-vector field. We call the differential operator $\mathcal{D}$ *boundary symmetric* (with respect to T) if for all $x \in \partial M$ the principal symbol $\sigma_{\mathcal{D}}(\eta)|_x$ is invertible for all co-vector fields η of M , and

$$\sigma_{\mathcal{D}}(\tau)|_x^{-1} \circ \sigma_{\mathcal{D}}(\xi)|_x \in \text{End}(E_x, E_x) \quad \text{and} \quad \sigma_{\mathcal{D}}(\xi)|_x \circ \sigma_{\mathcal{D}}(\tau)|_x^{-1} \in \text{End}(F_x, F_x)$$

are skew-Hermitian for all co-vector fields ξ of M satisfying $\xi(T) = 0$.

If the $\mathcal{A}$ -linear differential operator $\mathcal{D}$ is boundary symmetric, it takes in a suitable collar of the boundary the normal from

$$\mathcal{D} = \sigma_{\mathcal{D}}(dt)(\partial_t + A + R_t).$$

Here A is some elliptic self-adjoint first-order $\mathcal{A}$ -linear differential operator at the boundary, which is called *adapted to $\mathcal{D}$* , and R_t a suitable rest term (see Proposition 2.6 and Definition 2.7). It is crucial for the analytic calculation that the L^2 -closure of A admits a functional calculus as in Proposition 1.4, hence the L^2 -closure of A has to be self-adjoint and regular. This requires that the fibers of the bundle E are either finitely generated projective Hilbert $\mathcal{A}$ -modules or if $\mathcal{A} = \mathbb{C}$ bundles of Hilbert spaces (see Theorem 1.26). We obtain the following main theorem of this section:

Theorem 2.3. *Let $\mathcal{D}$ be boundary symmetric with respect to a vector field T along ∂M pointing into the interior of M , and A an adapted boundary operator as in Definition 2.7. Then the restriction map $C_c^\infty(M, E) \rightarrow C^\infty(\partial M, E)$, $u \mapsto u|_{\partial M}$, extends to a surjective bounded operator*

$$\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \longrightarrow \check{H}_A(\partial M, E)$$

with kernel equal to $\text{dom}(\mathcal{D}_{\min})$. Moreover,

$$\langle \mathcal{D}_{\max} u, v \rangle_{L^2} - \langle u, (\mathcal{D}^\dagger)_{\max} v \rangle_{L^2} = -\langle \sigma_{\mathcal{D}}(\tau) \mathcal{R}u, \mathcal{R}v \rangle$$

for all $u \in \text{dom}(\mathcal{D}_{\max})$ and all $v \in \text{dom}((\mathcal{D}^\dagger)_{\max})$.

Here $\check{H}_A(\partial M, E)$ is the Hybrid Sobolev space (Definition 2.9) which behaves like $H^{1/2}(\partial M, E)$ on one side of the spectrum of A and like $H^{-1/2}(\partial M, E)$ on the other side of the spectrum of A , and

$$\langle\langle \sigma_{\mathcal{D}}(\tau) \cdot, \cdot \rangle\rangle: \check{H}_{\tilde{A}}(\partial M, F) \longrightarrow \check{H}_{\tilde{A}}(\partial M, F)$$

is the perfect pairing induced by $\langle \cdot, \cdot \rangle_{L^2}$ (see Definition 2.12, Lemma 2.13 and Lemma 2.18 (1)). Here $\tilde{A}$ is an adapted boundary operator for the formal adjoint of $\mathcal{D}$. The proof of Theorem 2.3 is given in Section 2.1-2.3, namely in Lemma 2.17 and Lemma 2.18.

Remark 2.4. We assume in Setup 2.1 that the bundle of Hilbert $\mathcal{A}$ -modules E consists either of finitely generated projective Hilbert $\mathcal{A}$ -modules or if $\mathcal{A} = \mathbb{C}$ of Hilbert spaces. We focus on these two cases since we need that the L^2 -close of a boundary operator A adapted to $\mathcal{D}$ is self-adjoint and regular (see Theorem 1.26) in order to have the functional calculus as in Proposition 1.4 available. We are mostly interested in the case where E is a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules. However, the case $\mathcal{A} = \mathbb{C}$ and E a bundle of Hilbert spaces is crucial in the proof of Theorem 3.15. There we use, similarly as in [Ebe16], a localization principle to show that for a suitable boundary condition the induced closed unbounded operator is self-adjoint and regular.

2.1 Boundary symmetric differential operators

We fix the following setup for this sections:

Setup 2.5. Let T be a vector field of M along ∂M pointing into the interior of M , and τ the co-vector field associated to T . We fix a neighborhood U of ∂M in M together with a constant $r \in (0, 1)$ and *adapted* coordinates $\Psi = (\psi, t): U \rightarrow Z_{[0, r)} := \partial M \times [0, r)$, i.e.

$$\partial M = t^{-1}(0), \psi|_{\partial M} = \text{id}_{\partial M}, d\Psi(T) = \partial_t, \tau = dt \text{ along } \partial M, \text{ and } \psi_*g = dt^2 \oplus g|_{\partial M}$$

[see BB12, Lemma 2.4]. We use parallel transport along the $\mathcal{A}$ -linear metric connection on E , and obtain an identification as Hilbert $\mathcal{A}$ -module bundles of E restricted to $Z_{[0, r)}$ with the pullback of $E|_{\partial M}$ along the projection $\text{pr}_1: Z_{[0, r)} \rightarrow \partial M$. We make the same identifications for the bundle F .

Proposition 2.6 (Normal form). *Suppose $\mathcal{D}$ is boundary symmetric with respect to T . Then there exists an $r \in (0, 1)$, a formally self-adjoint elliptic $\mathcal{A}$ -linear differential operator A of first-order,*

$$A: C^\infty(\partial M, E) \longrightarrow C^\infty(\partial M, E),$$

and a family of $\mathcal{A}$ -linear differential operators R_t of order ≤ 1 depending smoothly on $t \in [0, r)$,

$$R_t: C^\infty(\partial M, E) \longrightarrow C^\infty(\partial M, E),$$

such that R_0 is of zero-order and the following holds:

(1) The $\mathcal{A}$ -linear differential operator $\mathcal{D}$ is elliptic on the collar $Z_{[0,r)}$ and takes the form

$$\mathcal{D} = \sigma_{\mathcal{D}}(dt)(\partial_t + A + R_t) \quad (2.1)$$

with respect to the adapted coordinates $\Psi = (\psi, t)$.

(2) For every $k \in \mathbb{N}_0$, there exist constants c, C such that

$$|R_t \varphi|_{\mathbb{H}^k(\partial M)}^2 \leq c(t^2 |A\varphi|_{\mathbb{H}^k(\partial M)}^2 + |\varphi|_{\mathbb{H}^k(\partial M)}^2) \quad \text{and} \quad (2.2)$$

$$C^{-1}|u|_{\mathcal{D}_0, \mathbb{H}^k}^2 \leq |u|_{\mathcal{D}, \mathbb{H}^k}^2 \leq C|u|_{\mathcal{D}_0, \mathbb{H}^k}^2, \quad (2.3)$$

for all $t \in [0, r)$, all $\varphi \in C^\infty(\partial M, E)$ and all $u \in C_c^\infty(Z_{[0,r)}, E)$. Here $|\cdot|_{\mathcal{D}, \mathbb{H}^k}$ and $|\cdot|_{\mathcal{D}_0, \mathbb{H}^k}$ are defined as in Definition 1.18, and $\mathcal{D}_0 := \sigma_{\mathcal{D}}(\tau)(\partial_t + A)$.

Proof. Since $\mathcal{D}$ is boundary symmetric, the principal symbol is invertible along the boundary of M and therefore also on the collar $Z_{[0,r)}$, hence $\mathcal{D}$ is elliptic on $Z_{[0,r)}$ (for r sufficiently small). We fix an elliptic first-order $\mathcal{A}$ -linear differential operator $A: C^\infty(\partial M, E) \rightarrow C^\infty(\partial M, E)$ with principal symbol given by

$$\sigma_A(\xi)|_x = \sigma_{\mathcal{D}}(\tau)|_x^{-1} \circ \sigma_{\mathcal{D}}(\xi)|_x \quad (2.4)$$

for all $\xi \in T_x^* \partial M \cong \{\eta \in T_x^* \partial M \mid \eta(T) = 0\}$ and all $x \in \partial M$. Since the principal symbol of A in equation (2.4) is skew-Hermitian (see Definition 2.2), we can choose A to be formally self-adjoint. As in the proof of [BB12, Lemma 4.1] there exists a family of $\mathcal{A}$ -linear differential operators $R_t: C^\infty(\partial M, E) \rightarrow C^\infty(\partial M, E)$ of order ≤ 1 which is smooth in $t \in [0, r)$ and R_0 is of order zero such that $\mathcal{D}$ takes the normal form given in equation (2.1).

It remains to show the estimates in equation (2.2) and (2.3). We start with equation (2.2) for $k = 0$. Fix a state $\rho: \mathcal{A} \rightarrow \mathbb{C}$ of the C^* -algebra $\mathcal{A}$ and localize the bundle E of Hilbert $\mathcal{A}$ -modules, the differential operator A and the family of differential operators R_t as in Section 1.2.4. We obtain for suitable constants C_ρ and $\tilde{C}_\rho$

$$\|R_t^\rho \varphi^\rho\|_{L^2}^2 \leq C_\rho(t^2 \|\varphi^\rho\|_{\mathbb{H}^1}^2 + \|\varphi^\rho\|_{L^2}^2) \leq \tilde{C}_\rho(t^2 \|A_\rho \varphi^\rho\|_{L^2}^2 + \|\varphi^\rho\|_{L^2}^2)$$

for all $t \in [0, r)$ and all $\varphi^\rho \in C^\infty(\partial M, E^\rho)$. Here we used in the first step that the family R_t^ρ depends smoothly on $t \in [0, r)$ and that R_0^ρ is of order zero, and in the second step the elliptic estimate in Proposition 1.24 for the elliptic differential operator A_ρ . By equation (1.7) and the fact that the space of states on $\mathcal{A}$ is compact, there exists a constant c such that

$$\rho(|R_t \varphi|_{L^2}^2) \leq c\rho(|A\varphi|_{L^2}^2 + |\varphi|_{L^2}^2)$$

for all states ρ of $\mathcal{A}$, all $\varphi \in C^\infty(\partial M, E)$ and all $t \in [0, r)$. Since the constant c is independent of the state ρ and $\|a\|_{\mathcal{A}} = \max_\rho \rho(a)$ for all self-adjoint $a \in \mathcal{A}$, we obtain the desired estimate in equation (2.2) for $k = 0$.

We proceed by an induction on k . Fix $t \in [0, r)$, $\varphi \in C^\infty(\partial M, E)$ and write $\lesssim$ for $\leq$ modulo a constant which is independent of φ . By a partition of unity argument, we can assume that φ

is supported in a local chart of ∂M . For some fixed local orthonormal frame $\{e_j\}$ of $T\partial M$, we obtain

$$|R_t \varphi|_{\mathbb{H}^{k+1}}^2 = \sum_j |\nabla_{e_j} R_t \varphi|_{\mathbb{H}^k}^2 + |R_t \varphi|_{\mathbb{H}^k}^2 \lesssim \sum_j |R_t \nabla_{e_j} \varphi|_{\mathbb{H}^k}^2 + \sum_j |[\nabla_{e_j}, R_t] \varphi|_{\mathbb{H}^k}^2 + |R_t \varphi|_{\mathbb{H}^k}^2.$$

Since $[\nabla_{e_j}, R_t]$ is a differential operator of order ≤ 1 on the compact manifold ∂M , we obtain by Proposition 1.13 together with the induction hypothesis

$$\begin{aligned} |R_t \varphi|_{\mathbb{H}^{k+1}}^2 &\lesssim \sum_j (t^2 |A \nabla_{e_j} \varphi|_{\mathbb{H}^k}^2 + |\nabla_{e_j} \varphi|_{\mathbb{H}^k}^2 + |\varphi|_{\mathbb{H}^{k+1}}^2) + t^2 |A \varphi|_{\mathbb{H}^k}^2 + |\varphi|_{\mathbb{H}^k}^2 \\ &\lesssim t^2 |A \varphi|_{\mathbb{H}^{k+1}}^2 + |\varphi|_{\mathbb{H}^{k+1}}^2. \end{aligned}$$

The second step holds by commuting A with ∇_{e_j} and using again Proposition 1.13. This finishes the proof of equation (2.2).

It remains to show the estimates in equation (2.3). We will just show the case $k = 0$. The corresponding estimates for the higher Sobolev spaces $\mathbb{H}^k(\partial M, E)$ follow by an induction on k by commuting the differential operator $\mathcal{D}_0$ and $\mathcal{D}$ with the covariant derivative ∇_{e_j} as in the proof of equation (2.2). By the normal form of $\mathcal{D}$ in equation (2.1) and the definition of $\mathcal{D}_0$, we have

$$\mathcal{D} - \sigma_{\mathcal{D}}(dt) \sigma_{\mathcal{D}}(\tau)^{-1} \mathcal{D}_0 = \sigma_{\mathcal{D}}(dt) R_t \quad (2.5)$$

We fix a smooth section $u \in C_c^\infty(Z_{[0,r)}, E)$, denote its restriction to $\partial M \times \{t\}$ by u_t , and denote by $\lesssim$ the inequality $\leq$ modulo a constant which is independent of u and r . Since $\sigma_{\mathcal{D}}(dt)$ is a zero-order differential operator, we obtain

$$|\sigma_{\mathcal{D}}(dt) R_t u|_{L^2(Z_{[0,r)})}^2 \lesssim \int_0^r |R_t u_t|_{L^2(\partial M)}^2 dt \lesssim \int_0^r |A t u_t|_{L^2(\partial M)}^2 dt + |u|_{L^2(Z_{[0,r)})}^2 \quad (2.6)$$

Here we used in the last step equation (2.2) for $k = 0$. Next, we investigate the first term in equation (2.6). Note that for every $v \in C^\infty(Z_{[0,r)}, E)$

$$\begin{aligned} |(\partial_t + A)v_t|_{L^2(\partial M)}^2 &= \langle (\partial_t + A)v_t, (\partial_t + A)v_t \rangle_{L^2(\partial M)} \\ &= |\partial_t v_t|_{L^2(\partial M)}^2 + |A v_t|_{L^2(\partial M)}^2 + \partial_t \langle A v_t, v_t \rangle_{L^2(\partial M)} \end{aligned} \quad (2.7)$$

for all $t \in [0, r)$ with v_t the restriction of v to $\partial M \times \{t\}$. Here we used $\partial_t \langle A v_t, v_t \rangle_{L^2(\partial M)} = \langle \partial_t v_t, A v_t \rangle_{L^2(\partial M)} + \langle A v_t, \partial_t v_t \rangle_{L^2(\partial M)}$ since A and ∂_t commute and $\mathcal{A}$ is formally self-adjoint. Inserting $v = tu$ into equation (2.7) and integrating over $t \in [0, r)$, yields

$$\begin{aligned} \int_0^r |A(tu_t)|_{L^2(\partial M)}^2 dt &\leq |(\partial_t + A)(tu)|_{L^2(Z_{[0,r)})}^2 \lesssim |t(\partial_t + A)u|_{L^2(Z_{[0,r)})}^2 + |u|_{L^2(Z_{[0,r)})}^2 \\ &\lesssim |t \mathcal{D}_0 u|_{L^2(Z_{[0,r)})}^2 + |tu|_{L^2(Z_{[0,r)})}^2 + |u|_{L^2(Z_{[0,r)})}^2. \end{aligned} \quad (2.8)$$

In the first step we used that tu is compactly supported in $Z_{[0,r)}$ and vanishes at the boundary. Therefore, the integral over $t \in [0, r)$ of the last term in equation (2.7) vanishes. In the last

step, we replace $\partial_t + A$ by $\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0$, use that $\sigma_{\mathcal{D}}(\tau)^{-1}$ has zero order and commutes with t .

Putting together equation (2.5), (2.6) and (2.8), we obtain

$$|\mathcal{D}u - \sigma_{\mathcal{D}}(dt)\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0u|_{L^2}^2 \lesssim |t\mathcal{D}_0u|_{L^2}^2 + |tu|_{L^2}^2 + |u|_{L^2}^2. \quad (2.9)$$

Since $r \leq 1$, equation (2.9) yields

$$|u|_{\mathcal{D}}^2 \lesssim |(\mathcal{D} - \sigma_{\mathcal{D}}(dt)\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0)u|_{L^2}^2 + |\sigma_{\mathcal{D}}(dt)\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0u|_{L^2}^2 + |u|_{L^2}^2 \lesssim |u|_{\mathcal{D}_0}^2$$

and the second estimate in equation (2.3) is proved for $k = 0$. Moreover, by equation (2.9), there exist constant C_1, C_2, C_3 , which are independent of u and r , such that

$$\begin{aligned} |\mathcal{D}_0u|_{L^2}^2 &\leq C_1 |\sigma_{\mathcal{D}}(dt)\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0u|_{L^2}^2 \leq C_2 |(\mathcal{D} - \sigma_{\mathcal{D}}(dt)\sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0)u|_{L^2}^2 + |\mathcal{D}u|_{L^2}^2 \\ &\leq C_3 (r^2 |\mathcal{D}_0u|_{L^2}^2 + |\mathcal{D}u|_{L^2}^2 + |u|_{L^2}^2). \end{aligned} \quad (2.10)$$

For $r \in (0, 1)$ such that $C_3^{-1} - r^2 > 0$, we obtain by reordering the terms in equation (2.10)

$$(C_3^{-1} - r^2) |\mathcal{D}_0u|_{L^2}^2 \leq (|\mathcal{D}u|_{L^2}^2 + |u|_{L^2}^2)$$

for all $u \in C_c^\infty(Z_{[0,r]}, E)$. This proves the second estimate in equation (2.3) for $k = 0$, and the lemma is proved. $\square$

Definition 2.7. Let $\mathcal{D}$ be boundary symmetric with respect to T . We call a differential operator A as in Proposition 2.6 *adapted* to $\mathcal{D}$ and say it is an *adapted boundary operator* for $\mathcal{D}$.

Example 2.8 (Dirac type operators). We say that the first-order differential operator $\mathcal{D}$ is of *Dirac type* [see BB12, Example 4.3 (a)] if its principal symbol satisfies the Clifford relations

$$\sigma_{\mathcal{D}}(\xi)^\dagger \sigma_{\mathcal{D}}(\eta) + \sigma_{\mathcal{D}}(\eta)^\dagger \sigma_{\mathcal{D}}(\xi) = 2\langle \eta, \xi \rangle \quad \text{and} \quad \sigma_{\mathcal{D}}(\xi) \sigma_{\mathcal{D}}(\eta)^\dagger + \sigma_{\mathcal{D}}(\eta) \sigma_{\mathcal{D}}(\xi)^\dagger = 2\langle \eta, \xi \rangle$$

for all $x \in M$ and all $\xi, \eta \in T_x^*M$. Dirac type operators are boundary symmetric with respect to the interior normal vector field T along the boundary of M . If $\mathcal{D}$ is formally self-adjoint and of Dirac type, then there exists an adapted boundary operator which anti-commutes with $\sigma_{\mathcal{D}}(\tau)$ [BB16, Lemma 3.2]. This will be useful when we study the Atiyah-Patodi-Singer boundary condition in Section 5.1. A rich class of Dirac type operators are Dirac operators induced by some $\mathcal{A}$ -linear Dirac bundle [cf. Ton25b, Definition 2.1]. For instance the $\mathbb{C}l$ -linear spin Dirac operator twisted by the Mishchenko-Fomenko bundle of a connected spin manifold [cf. Ton25a, Section 2].

2.2 Hybrid Sobolev spaces and perfect pairings

We fix the identifications in a collar neighborhood of the boundary of M as in Setup 2.5, suppose that $\mathcal{D}$ is boundary symmetric with respect to T and fix a first-order elliptic differential

operator A adapted to $\mathcal{D}$. Moreover, we fix a smooth function $\check{f}: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ such that

$$\check{f}(x) = \begin{cases} |x| & \text{for } x \leq -1 \\ \frac{1}{x} & \text{for } x \geq 1 \end{cases} \quad (2.11)$$

for all $x \in \mathbb{R}$, and define $\hat{f}: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ via $\hat{f}(x) := \check{f}(-x)$.

Definition 2.9 (Hybrid Sobolev spaces). For $k \in \mathbb{Z}$, we define the Hilbert $\mathcal{A}$ -modules $\check{H}_A^k(\partial M, E)$ and $\hat{H}_A^k(\partial M, E)$ as the completion of the pre-Hilbert $\mathcal{A}$ -module $C^\infty(\partial M, E)$ equipped with the $\mathcal{A}$ -valued inner products

$$\begin{aligned} \langle u, u \rangle_{\check{H}_A^k} &:= \langle \check{f}(A)u, v \rangle_{H^k} = \langle (1 + A^2)^k \check{f}(A)u, u \rangle_{L^2} \quad \text{and} \\ \langle u, u \rangle_{\hat{H}_A^k} &:= \langle \hat{f}(A)u, v \rangle_{H^k} = \langle (1 + A^2)^k \hat{f}(A)u, u \rangle_{L^2}, \end{aligned} \quad (2.12)$$

respectively. Moreover, we define $\check{H}_A(\partial M, E) := \check{H}_A^0(\partial M, E)$ and $\hat{H}_A(\partial M, E) := \hat{H}_A^0(\partial M, E)$.

Remark 2.10. For $k = 0$, the hybrid Sobolev spaces generalize the $\check{H}$ and $\hat{H}$ -spaces by Bär and Ballmann [BB12, Section 5]. For a fixed $k \in \mathbb{Z}$, there exist constants C_1, C_2 such that

$$C_1(1 + x^2)^{k-1/2} \leq \begin{Bmatrix} (1 + x^2)^k \check{f}(x) \\ (1 + x^2)^k \\ (1 + x^2)^k \hat{f}(x) \end{Bmatrix} \leq C_2(1 + x^2)^{k+1/2}$$

for all $x \in \mathbb{R}$. We obtain by the properties of the functional calculus in Proposition 1.4 the following continuous embeddings for all $k \in \mathbb{Z}$:

$$\begin{array}{ccccc} & & \check{H}_A^k(\partial M, E) & & \\ & \swarrow & & \searrow & \\ \dots & H^{k-1/2}(\partial M, E) & \longleftrightarrow & H^k(\partial M, E) & \longleftrightarrow & H^{k+1/2}(\partial M, E) & \dots \\ & \swarrow & & \searrow & \\ & & \hat{H}_A^k(\partial M, E) & & \end{array}$$

Lemma 2.11. *The equivalence class of the norms induced by the $\mathcal{A}$ -valued inner products $\langle \cdot, \cdot \rangle_{\check{H}_A^k}$ and $\langle \cdot, \cdot \rangle_{\hat{H}_A^k}$ in equation (2.12) is independent of the choice of the function $\check{f}$.*

Proof. Fix some $k \in \mathbb{N}_0$ and let $\check{g}: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ be a function satisfying equation (2.11). Since $\check{f} = \check{g}$ on $\mathbb{R} \setminus (-1, 1)$, and $\check{g}$ is pointwise strictly positive, there exists a constant C such that $\check{f} \leq C\check{g}$. By the properties of the functional calculus in Proposition 1.4, we obtain

$$\| \langle (1 + A^2)^k \check{f}(A)u, u \rangle_{L^2} \|_{\mathcal{A}} \leq C \| \langle (1 + A^2)^k \check{g}(A)u, u \rangle_{L^2} \|_{\mathcal{A}}$$

for all $u \in C^\infty(\partial M, E)$. By the same considerations, we obtain the corresponding estimate in the other direction and both estimates for the $\hat{H}_A^k$ -norm. $\square$

Definition 2.12. Let $\mathcal{H}_1$ and $\mathcal{H}_2$ be two Hilbert $\mathcal{A}$ -modules and $\langle\!\langle \cdot, \cdot \rangle\!\rangle: \mathcal{H}_1 \times \mathcal{H}_2 \rightarrow \mathcal{A}$ a bi-additive map such that $\langle\!\langle u, va \rangle\!\rangle = \langle\!\langle u, v \rangle\!\rangle a$ and $\langle\!\langle ua, v \rangle\!\rangle = a^* \langle\!\langle u, v \rangle\!\rangle$ for all $u \in \mathcal{H}_1$, $v \in \mathcal{H}_2$ and all $a \in \mathcal{A}$. We call $\langle\!\langle \cdot, \cdot \rangle\!\rangle$ a *perfect pairing* if there exist constants C_1, C_2 and C_3 such that

$$(1) \quad \|\langle\!\langle u, v \rangle\!\rangle\|_{\mathcal{A}} \leq C_1 \|u\|_{\mathcal{H}_1} \|v\|_{\mathcal{H}_2},$$

$$(2) \quad \|u\|_{\mathcal{H}_1} \leq C_2 \sup_{w \in \mathcal{H}_2} \frac{\|\langle\!\langle u, w \rangle\!\rangle\|_{\mathcal{A}}}{\|w\|_{\mathcal{H}_2}} \quad \text{and} \quad \|v\|_{\mathcal{H}_2} \leq C_3 \sup_{w \in \mathcal{H}_1} \frac{\|\langle\!\langle w, v \rangle\!\rangle\|_{\mathcal{A}}}{\|w\|_{\mathcal{H}_1}}$$

for all $u \in \mathcal{H}_1$ and all $v \in \mathcal{H}_2$.

Lemma 2.13. For all $k \in \mathbb{N}_0$ the $\mathcal{A}$ -valued inner product $\langle \cdot, \cdot \rangle_{\mathbb{H}^k}$ induces a perfect pairing

$$\langle\!\langle \cdot, \cdot \rangle\!\rangle_k: \check{\mathbb{H}}_A^k(\partial M, E) \times \hat{\mathbb{H}}_A^k(\partial M, E) \longrightarrow \mathcal{A}.$$

Proof. Fix $k \in \mathbb{N}_0$ and let $g_0, g: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ be compactly supported smooth functions such that $g_0|_{[-1,1]} \equiv 1$ and $\check{f}\hat{f} \equiv (1+g)^2$. Note that such a function g exists by the choice of the functions $\hat{f}$ and $\check{f}$ and moreover the support of g is contained in $[-1, 1]$. We fix arbitrary $u, v \in C^\infty(\partial M, E)$ and obtain

$$\begin{aligned} \|\langle u, v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} &= \|\langle (\check{f}^{1/2}(A)\hat{f}^{1/2}(A) - g(A))u, v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} \\ &\leq \|\langle \check{f}^{1/2}(A)u, \hat{f}^{1/2}(A)v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} + \left\| \left\langle g(A)u, g_0(A)v \right\rangle_{\mathbb{H}^k} \right\|_{\mathcal{A}} \\ &\leq \|\check{f}^{1/2}(A)u\|_{\mathbb{H}^k} \|\hat{f}^{1/2}(A)v\|_{\mathbb{H}^k} + \|g(A)u\|_{\mathbb{H}^k} \|g_0(A)v\|_{\mathbb{H}^k} \\ &= \|u\|_{\check{\mathbb{H}}_A^k} \|v\|_{\hat{\mathbb{H}}_A^k} + \|g(A)u\|_{\mathbb{H}^k} \|g_0(A)v\|_{\mathbb{H}^k} \end{aligned} \tag{2.13}$$

Here we used in the first step $\check{f}\hat{f} \equiv (1+g)^2$, in the second step equation (1.8), the triangle inequality, $g_0|_{[-1,1]}$ and $\text{supp}(g) \subseteq [-1, 1]$, in the third step the Cauchy-Schwarz inequality in the Hilbert $\mathcal{A}$ -module $\mathbb{H}^k(\partial M, E)$, and in the last step that $\check{f}^{1/2}(A)$ and $\hat{f}^{1/2}(A)$ are self-adjoint (Proposition 1.4) and commute with $(1+A^2)^k$. Since g and g_0 are compactly supported, and $\hat{f}$ and $\check{f}$ are pointwise strictly positive, there exists a constant C such that $g^2 \leq C\check{f}$ and $g_0^2 \leq C\hat{f}$. We obtain

$$\begin{aligned} \|g(A)u\|_{\mathbb{H}^k}^2 \|g_0(A)v\|_{\mathbb{H}^k}^2 &= \|\langle g^2(A)u, u \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} \|\langle g_0^2(A)v, v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} \\ &\leq C^2 \|\langle \check{f}(A)u, u \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} \|\langle \hat{f}(A)v, v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}} \\ &= C^2 \|u\|_{\check{\mathbb{H}}_A^k}^2 \|v\|_{\hat{\mathbb{H}}_A^k}^2, \end{aligned}$$

which yields together with equation (2.13)

$$\|\langle u, v \rangle_{\mathbb{H}^k}\|_{\mathcal{A}}^2 = (1 + C^2) \|u\|_{\check{\mathbb{H}}_A^k}^2 \|v\|_{\hat{\mathbb{H}}_A^k}^2.$$

Since the constant C is independent of the sections u and v , we obtain that the $\mathcal{A}$ -valued $\mathbb{H}^k$ -inner product extends to a bi-additive map

$$\langle\!\langle \cdot, \cdot \rangle\!\rangle_k: \check{\mathbb{H}}_A^k(\partial M, E) \times \hat{\mathbb{H}}_A^k(\partial M, E) \longrightarrow \mathcal{A}$$

which is $\mathcal{A}$ -antilinear in the first factor, $\mathcal{A}$ -linear in the second factor and satisfies inequality (1) in Definition 2.12. It remains to show the estimates in (2) in Definition 2.12. Since the proof of both estimates is similar, we will just show that there exists a constant C_2 such that

$$\|u\|_{\check{H}_A^k} \leq C_2 \sup \left\{ \frac{\|\langle u, w \rangle_k\|_{\mathcal{A}}}{\|w\|_{\check{H}_A^k}} \mid w \in \hat{H}_A^k(\partial M, E) \right\} \quad (2.14)$$

for all $u \in \check{H}_A^k(\partial M, E)$. We fix an arbitrary $u \in C^\infty(\partial M, E)$ and define $v := \check{f}(A)u$. By the choice of v and the definition of the $\check{H}_A^k$ -norm, we obtain

$$\|\langle u, v \rangle_{H^k}\|_{\mathcal{A}} = \|\langle u, \check{f}(A)u \rangle_{H^k}\|_{\mathcal{A}} = \|u\|_{\check{H}_A^k}^2. \quad (2.15)$$

Moreover, since g is compactly supported, there exists a constant C such that $(2g + g^2) \leq C$. We use $\check{f}\hat{f} \equiv (1 + g)^2$ and obtain

$$\begin{aligned} \|v\|_{\check{H}_A^k}^2 &= \|\langle \hat{f}(A)\check{f}(A)u, \check{f}(A)u \rangle_{H^k}\|_{\mathcal{A}} = \|u\|_{\check{H}_A^k}^2 + \|\langle (2g(A) + g^2(A))\check{f}(A)u, u \rangle_{H^k}\|_{\mathcal{A}} \\ &\leq (1 + C)\|u\|_{\check{H}_A^k}^2. \end{aligned} \quad (2.16)$$

Combining equation (2.15) with equation (2.16) yields

$$\|u\|_{\check{H}_A^k} \leq (1 + C)^{1/2} \frac{\|\langle u, v \rangle_k\|_{\mathcal{A}}}{\|v\|_{\check{H}_A^k}} \leq (1 + C)^{1/2} \sup \left\{ \frac{\|\langle u, w \rangle_k\|_{\mathcal{A}}}{\|w\|_{\check{H}_A^k}} \mid w \in \hat{H}_A^k(\partial M, E) \right\},$$

and the estimate in equation (2.14) is proved. $\square$

2.3 The extension and the restriction map

In this section we give a proof of Theorem 2.3. As a main tool we use the extension map in Definition 2.14. For the higher regularity principle of right-elliptic boundary conditions in Section 3.2 we need the hybrid Sobolev spaces for $k \geq 1$. Therefore we show in this section some results for the norms of the higher Hybrid Sobolev spaces $\check{H}_A^k(\partial M, E)$ even if this is not necessary for the proof of Theorem 2.3.

Let $\mathcal{D}$ be boundary symmetric, $Z_{[0,r)}$ a collar of the boundary together with the identifications in Setup 2.5 with $r > 0$ sufficiently small such that Proposition 2.6 holds, A an adapted boundary operator for $\mathcal{D}$ as in Definition 2.7, and $\check{f}: \mathbb{R} \rightarrow \mathbb{R}$ a smooth function as in equation (2.11). Fix some smooth cutoff function $\chi: [0, r) \rightarrow [0, 1]$ satisfying

$$\chi(t) = \begin{cases} 1 & \text{for } t \leq r/3 \\ 0 & \text{for } t \geq 2r/3. \end{cases} \quad (2.17)$$

We start with the definition of the extension map and study some of its properties.

Definition 2.14. The *extension map* is defined via

$$\mathcal{E}: C^\infty(\partial M, E) \longrightarrow C_c^\infty(M, E), \quad (\mathcal{E}\varphi)(x, t) := \chi(t)(e^{-t|A|}\varphi)(x) \in E_{(x,t)}.$$

Here $e^{-t|A|}\varphi$ is defined via the functional calculus for the self-adjoint and regular unbounded operator $A: H^1(\partial M, E) \rightarrow L^2(M, E)$ (see Theorem 1.26 and Proposition 1.4). By the choice of the cutoff function χ and the identification $\Psi: U \rightarrow Z_{[0,r)}$, we obtain that $\mathcal{E}\varphi$ defines a smooth section of E compactly supported in $Z_{[0,r)}$. Therefore, the extension map is well-defined, and satisfies the following properties:

Proposition 2.15. *Let $k \in \mathbb{N}_0$. There exist constants c and C such that*

$$(1) \quad \|\mathcal{E}\varphi\|_{H^k(M)} \leq c\|\varphi\|_{H^{k-1/2}(\partial M)} \text{ and}$$

$$(2) \quad \|\mathcal{E}\varphi\|_{\mathcal{D}, H^k} \leq C\|\varphi\|_{\check{H}_A^k}$$

for all $\varphi \in C^\infty(\partial M, E)$. In particular, the extension map in Definition 2.14 extends to bounded operators $H^{k-1/2}(\partial M, E) \rightarrow H^k(M, E)$ and

$$\mathcal{E}: \check{H}_A^k(\partial M, E) \longrightarrow \text{dom}(\mathcal{D}_{\max}; k).$$

Proof. Fix some $k \in \mathbb{N}_0$. We split the proof into three steps. In step (0), we show a formula for the norm $|u|_{H^k(Z_{[0,r)})}$ in terms of $|\cdot|_{H^k(\partial M)}$ for smooth sections u supported in $Z_{[0,r)}$. In the first and second step, we show the two estimates in the proposition.

Step (0). For some fixed $u \in C_c^\infty(Z_{[0,r)}, E)$ we have

$$|u|_{H^k(Z_{[0,r)})}^2 = \sum_{\ell=0}^k |\nabla^\ell u|_{L^2(Z_{[0,r)})}^2 = \int_{Z_{[0,r)}} \sum_{\ell=0}^k |\nabla^\ell u|_p^2 \, dp. \quad (2.18)$$

Around some point $p \in Z_{[0,r)}$ we fix some local orthonormal frame $e_1, \dots, e_n$ which is synchronous at the point p and $e_n(p) = \partial_t(p)$. We denote by $\alpha = (\alpha_1, \dots, \alpha_\ell) \in \{e_1, \dots, e_{\ell-1}, \partial_t\}^\ell$ some multi-index, and define $|\alpha| := \ell$ and $\nabla^\alpha := \nabla_{\alpha_1} \dots \nabla_{\alpha_\ell}$. There exists a constant C_p , independent of the section u such that at the point p

$$\sum_{\ell=0}^k |\nabla^\ell u|_p^2 = \sum_{|\alpha| \leq k} |\nabla^\alpha u|_p^2 \leq C_p \sum_{j=0}^k \sum_{|\alpha| \leq k-j} |\nabla^\alpha \partial_t^j u|_p^2. \quad (2.19)$$

Here we change the order of the covariant derivatives in the term ∇^α and pick up some curvature terms of the bundle E . These terms as well as their covariant derivatives can be bounded by a constant independent of the chosen local frame. Inserting equation (2.19) into equation (2.18)

yields for $C_1 := \max_{p \in Z_{[0,r]}} C_p$ and suitable constants C_2 and C_3

$$\begin{aligned} |u|_{\mathbb{H}^k(Z_{[0,r]})}^2 &\leq C_1 \int_{Z_{[0,r]}} \sum_{j=0}^k \sum_{|\alpha| \leq k-j} |\nabla^\alpha \partial_t^j u|_p^2 dp \leq C_2 \sum_{j=0}^k \int_0^r |(\partial_t^j u)_t|_{\mathbb{H}^{k-j}(\partial M)}^2 dt \\ &\leq C_3 \sum_{j=0}^k \sum_{\ell=0}^{k-j} \int_0^r |(A^\ell \partial_t^j u)_t|_{L^2(\partial M)}^2 dt. \end{aligned} \quad (2.20)$$

Here $(\cdot)_t$ denotes the restriction of a smooth section on $Z_{[0,r]}$ to the slice at $\partial M \times \{t\}$. In the second step of equation (2.20), we used the identifications on the collar $Z_{[0,r]}$ in Setup 2.5, and in the last step the elliptic estimates in Proposition 1.24 for the elliptic differential operator A .

Step (1). Fix $\varphi \in C^\infty(\partial M, E)$. We obtain by the definition of the extension map in Definition 2.14, equation (2.20) and $\partial_t^j e^{-t|A|} = \pm |A|^j e^{-t|A|}$

$$\begin{aligned} |\mathcal{E}\varphi|_{\mathbb{H}^k(M)}^2 &= |\chi(t)e^{-t|A|}\varphi|_{\mathbb{H}^k(Z_{[0,r]})}^2 \leq C_3 \sum_{j=0}^k \sum_{\ell=0}^{k-j} \int_0^r |A^\ell \partial_t^j e^{-t|A|}\varphi|_{L^2(\partial M)}^2 dt \\ &\leq k C_3 \sum_{j=0}^k \int_0^r \langle A^{2j} e^{-2t|A|}\varphi, \varphi \rangle_{L^2(\partial M)} dt \end{aligned} \quad (2.21)$$

for a suitable constant C_3 independent of φ . Here we also used that $\chi(t)$ is a bounded map from $\mathbb{H}^k(M, E) \rightarrow \mathbb{H}^k(Z_{[0,r]}, E)$ and that A is formally self-adjoint. Let C_4 be a constant such that $\frac{x^{2j}}{2|x|}(1 - e^{-2r|x|}) \leq C_4(1 + x^2)^{k-1/2}$ for all $x \in \mathbb{R}$. We obtain

$$\begin{aligned} \int_0^r \langle A^{2j} e^{-2t|A|}\varphi, \varphi \rangle_{L^2(\partial M)} dt &= \left\langle A^{2j} \int_0^r e^{-2t|A|} dt \varphi, \varphi \right\rangle_{L^2(\partial M)} \\ &= \left\langle \frac{A^{2j}}{2|A|} (1 - e^{-2r|A|}) \varphi, \varphi \right\rangle_{L^2(\partial M)} \\ &\leq C_4 \langle (1 + A^2)^{k-1/2} \varphi, \varphi \rangle_{L^2(\partial M)} \end{aligned} \quad (2.22)$$

for all $j \in \{0, \dots, k\}$. Here we used the properties of the functional calculus in Proposition 1.4. We insert equation (2.22) into equation (2.21) and obtain for $c^2 := k^2 C_3 C_4$

$$|\mathcal{E}\varphi|_{\mathbb{H}^k(M)}^2 \leq k^2 C_3 C_4 \langle (1 + A^2)^{k-1/2} \varphi, \varphi \rangle_{L^2(\partial M)} = c^2 |\varphi|_{\mathbb{H}^{k-1/2}(\partial M)}^2. \quad (2.23)$$

The first estimate in the proposition follows by taking the $\mathcal{A}$ -norm of equation (2.23) and using the C^* -identity for $\mathcal{A}$.

Step (2). Fix $\varphi \in C^\infty(\partial M, E)$ and write $\lesssim$ for $\leq$ modulo a constant which is independent of the section φ . By Proposition 2.6 (2), we obtain

$$|\mathcal{E}\varphi|_{\mathcal{D}, \mathbb{H}^k}^2 \lesssim |\mathcal{E}\varphi|_{\mathcal{D}_0, \mathbb{H}^k}^2 = |\mathcal{D}_0 \mathcal{E}\varphi|_{\mathbb{H}^k(M)}^2 + |\mathcal{E}\varphi|_{\mathbb{H}^k(M)}^2 \quad (2.24)$$

for $\mathcal{D}_0 = \sigma_{\mathcal{D}}(\tau)(\partial_t + A)$. Moreover we have by the definition of the extension map (see

Definition 2.14)

$$\begin{aligned}\mathcal{D}_0(\mathcal{E}\varphi) &= \sigma_{\mathcal{D}}(\tau)(\partial_t + A)\chi(t)e^{-t|A|}\varphi \\ &= \chi(t)\sigma_{\mathcal{D}}(\tau)(\partial_t + A)e^{-t|A|}\varphi + \chi'(t)\sigma_{\mathcal{D}}(\tau)e^{-t|A|}\varphi.\end{aligned}\tag{2.25}$$

Inserting equation (2.25) into equation (2.24) and using that $\chi(t)\sigma_{\mathcal{D}}(\tau)$ and $\chi'(t)\sigma_{\mathcal{D}}(\tau)$ are bounded as maps from $H^k(Z_{[0,r]}, E)$ to $H^k(Z_{[0,r]}, F)$ yields

$$|\mathcal{E}\varphi|_{\mathcal{D}, H^k}^2 \lesssim |(\partial_t + A)e^{-t|A|}\varphi|_{H^k(M)}^2 + |e^{-t|A|}\varphi|_{H^k(M)}^2.\tag{2.26}$$

By the estimates in the proof of step (1) and $\check{H}_A^k(\partial M, E) \subseteq H^{k-1/2}(\partial M, E)$ (see Remark 2.10), the second term in equation (2.26) can be estimated via

$$|e^{-t|A|}\varphi|_{H^k(M)}^2 \lesssim |\varphi|_{H^{k-1/2}(\partial M)}^2 \lesssim |\varphi|_{\check{H}_A^k}^2.$$

It remains to estimate the first term in equation (2.26). By the formula in equation (2.20), we obtain similarly as for the estimates in step (1)

$$\begin{aligned}|(\partial_t + A)e^{-t|A|}\varphi|_{H^k(M)}^2 &\lesssim \sum_{j=0}^k \sum_{\ell=0}^{k-j} \int_0^r |(A - |A|)A^\ell \partial_t^j e^{-t|A|}\varphi|_{L^2(\partial M)}^2 dt \\ &\leq k \sum_{j=0}^k \left\langle \frac{A^{2j}(A - |A|)^2}{2|A|} (1 - e^{-2r|A|})\varphi, \varphi \right\rangle_{L^2(\partial M)}.\end{aligned}\tag{2.27}$$

The asymptotic behavior of the function

$$f: \mathbb{R} \longrightarrow \mathbb{R}, \quad x \longmapsto k \sum_{j=0}^k \frac{x^{2j}(x - |x|)^2}{2|x|} (1 - e^{-2r|x|})$$

is dominated by the asymptotic behavior of the function $(1 + x^2)^k \check{f}(x)$. Since $(1 + x^2)^k \check{f}(x)$ is strictly positive for every $x \in \mathbb{R}$, there exists a constant C_5 such that $f(x) \leq C_5(1 + x^2)^k \check{f}(x)$ for all $x \in \mathbb{R}$. Then equation (2.27) yields

$$|(\partial_t + A)e^{-t|A|}\varphi|_{H^k(M)}^2 \lesssim \langle (1 + A^2)^k \check{f}(A)\varphi, \varphi \rangle_{L^2(M)} = |\varphi|_{\check{H}_A^k}^2,$$

hence estimate (2) in the proposition is proved. $\square$

Lemma 2.16. *For every $k \in \mathbb{N}_0$, there exists a constant C such that*

$$\|u\|_{\partial M, \check{H}_A^k} \leq C(\|\mathcal{D}u\|_{H^k} + \|u\|_{H^k})$$

for all $u \in C_c^\infty(M, E)$ and all $k \in \mathbb{N}_0$.

Proof. Fix $k \in \mathbb{N}_0$. By Lemma 2.13, there exists a constant C_1 such that

$$\|u|_{\partial M}\|_{\hat{H}_A^k} \leq C_1 \sup \left\{ \frac{\|\langle u|_{\partial M}, \varphi \rangle_k\|_{\mathcal{A}}}{\|\varphi\|_{\hat{H}_A^k}} \mid \varphi \in \hat{H}_A^k(\partial M, E) \right\}$$

for all $u \in C_c^\infty(M, E)$. Therefore, we have to show that there exists a constant C_2 such that

$$\|\langle u|_{\partial M}, \varphi \rangle_k\|_{\mathcal{A}} \leq C_2 \|u\|_{\mathcal{D}} \|\varphi\|_{\hat{H}_A^k} \quad (2.28)$$

for all $u \in C_c^\infty(M, E)$ and all $\varphi \in \hat{H}_A^k(\partial M, E)$. Since $C^\infty(\partial M, E)$ is dense in $\hat{H}_A^k(\partial M, E)$, it is sufficient to show equation (2.28) for all $u \in C_c^\infty(M, E)$, all $\varphi \in C^\infty(\partial M, E)$ and the perfect pairing $\langle \cdot, \cdot \rangle_k$ replaced by $\langle \cdot, \cdot \rangle_{H^k(\partial M)}$.

We fix $u \in C_c^\infty(M, E)$, $\varphi \in C^\infty(\partial M, E)$ and write $\lesssim$ for $\leq$ modulo a constant independent of u and φ . We apply Green's formula (Lemma 1.12) to the smooth sections $(1 + A^2)^{k/2} \chi(t)u$ and $(\sigma_{\mathcal{D}}(dt)^{-1})^\dagger \chi(t) \mathcal{E}(1 + A^2)^{k/2} \varphi$, and obtain

$$\begin{aligned} \langle u|_{\partial M}, \varphi \rangle_{H^k(\partial M)} &= \langle (1 + A^2)^{k/2} u|_{\partial M}, (1 + A^2)^{k/2} \varphi \rangle_{L^2(\partial M)} \\ &= -\langle \mathcal{D}(1 + A^2)^{k/2} \chi(t)u, (\sigma_{\mathcal{D}}(dt)^{-1})^\dagger \chi(t) \mathcal{E}(1 + A^2)^{k/2} \varphi \rangle_{L^2(M)} \\ &\quad + \langle (1 + A^2)^{k/2} \chi(t)u, \mathcal{D}^\dagger (\sigma_{\mathcal{D}}(dt)^{-1})^\dagger \chi(t) \mathcal{E}(1 + A^2)^{k/2} \varphi \rangle_{L^2(M)}. \end{aligned} \quad (2.29)$$

Note that $(1 + A^2)^{k/2}$ commutes with the restriction map. We define an $\mathcal{A}$ -linear first-order differential operator

$$\mathcal{B} := (\sigma_{\mathcal{D}}(dt)^{-1} \mathcal{D} \chi(t))^\dagger : C^\infty(M, E) \rightarrow C^\infty(M, E).$$

Taking the $\mathcal{A}$ -norm of equation (2.29) and using the triangle inequality and the Cauchy-Schwarz inequality yields

$$\|\langle u|_{\partial M}, \varphi \rangle_{H^k(\partial M)}\|_{\mathcal{A}} \leq 2 \|(1 + A)^{k/2} \chi(t)u\|_{\mathcal{D}} \|\mathcal{E}(1 + A^2)^{k/2} \varphi\|_{\mathcal{B}}. \quad (2.30)$$

We estimate the first factor in equation (2.30) as follows:

$$\begin{aligned} \|(1 + A)^{k/2} \chi(t)u\|_{\mathcal{D}}^2 &\leq \|\mathcal{D}(1 + A)^{k/2} \chi(t)u\|_{L^2}^2 + \|(1 + A)^{k/2} \chi(t)u\|_{L^2}^2 \\ &\lesssim \|(1 + A^2)^{k/2} \chi(t) \mathcal{D}u\|_{L^2}^2 + \|[\mathcal{D}, (1 + A^2)^{k/2} \chi(t)]u\|_{L^2}^2 + \|u\|_{H^k}^2 \\ &\lesssim \|\mathcal{D}u\|_{H^k}^2 + \|u\|_{H^k}^2 \\ &= \|u\|_{\mathcal{D}, H^k}^2. \end{aligned} \quad (2.31)$$

In the third step we used that $[\mathcal{D}, (1 + A^2)^{k/2}]$ is a differential operator of order $\leq k$, Proposition 1.13 and that $\chi(t)$ is supported in $Z_{[0, r]}$. Before we continue with the second term in equation (2.30), we study some properties of the $\mathcal{A}$ -linear differential operator $\mathcal{B}$. Its principal

symbol is given by

$$\sigma_{\mathcal{B}}(dt) = -(\sigma_{\sigma_{\mathcal{D}}(dt)}^{-1} \mathcal{D}\chi(t)(dt))^{\dagger} = -\chi(t)(\sigma_{\mathcal{D}}(dt)^{-1} \circ \sigma_{\mathcal{D}}(dt))^{\dagger} = -\chi(t) \quad (2.32)$$

over $Z_{[0,r)}$. In particular, $\sigma_{\mathcal{B}}(\tau)|_x$ is invertible for all $x \in \partial M$. Similar considerations yield that $\mathcal{B}$ is boundary symmetric and satisfies

$$\mathcal{B} = (\sigma_{\mathcal{D}}(dt)^{-1} \mathcal{D}\chi(t))^{\dagger} = \chi(t)(-\partial_t + A + R_t^{\dagger}) \stackrel{\text{eq. (2.32)}}{=} \sigma_{\mathcal{B}}(dt)(\partial_t - A - R_t^{\dagger})$$

on $Z_{[0,r)}$. Here we used in the second step that the differential operator $\mathcal{D}$ takes in the collar $Z_{[0,r)}$ the form $\mathcal{D} = \sigma_{\mathcal{D}}(dt)(\partial_t + A + R_t)$. In particular, $-A$ is an adapted boundary operator for $\mathcal{B}$. We apply Proposition 2.15 to $\mathcal{B}$, and obtain by similar considerations as in equation (2.30)

$$\|\mathcal{E}(1 + A^2)^{k/2} \varphi\|_{\mathcal{B}} \lesssim \|\mathcal{E}\varphi\|_{\mathcal{B}, H^k} \lesssim \|\varphi\|_{\check{H}_{\mathcal{A}}^k} = \|\varphi\|_{\hat{H}_A^k}. \quad (2.33)$$

Putting together equation (2.30), (2.31) and (2.33) yields

$$\|\langle u|_{\partial M}, \varphi \rangle_{L^2(\partial M)}\|_{\mathcal{A}} \lesssim \|u\|_{\mathcal{D}, H^k} \|\varphi\|_{\hat{H}_A^k},$$

and the estimate in Lemma 2.16 is proved. $\square$

Lemma 2.17. *The restriction map extends to a surjective bounded operator $\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \rightarrow \check{H}_A(\partial M, E)$ with kernel given by $\text{dom}(\mathcal{D}_{\min})$.*

Proof. Since $C_c^{\infty}(M, E)$ is dense in $\text{dom}(\mathcal{D}_{\max})$ by Proposition 1.15, the restriction map extends by Lemma 2.16 to a bounded operator $\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \rightarrow \check{H}_A(\partial M, E)$. By Proposition 2.15 for $k = 0$, we obtain $\mathcal{R}\mathcal{E}\varphi = \varphi$ for all $\varphi \in \check{H}_A(\partial M, E)$, hence $\mathcal{R}$ is surjective. It remains to show that the kernel of $\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \rightarrow \check{H}_A(\partial M, E)$ is contained in $\text{dom}(\mathcal{D}_{\min})$. Let $u \in \text{dom}(\mathcal{D}_{\max})$ with $\mathcal{R}u = 0$ and fix some smooth cutoff function $\chi: M \rightarrow \mathbb{R}$ induced by a map as in equation (2.17). We write $u = \chi u + (1 - \chi)u$ and analyze both summands separately.

Since $(1 - \chi)u$ is supported in $M \setminus Z_{[0, r/3]}$, we can approximate $(1 - \chi)u$ in the graph norm of $\mathcal{D}$ by smooth sections compactly supported in $M \setminus Z_{[0, r/6]}$ by Lemma 1.17 and Lemma 1.21, hence $(1 - \chi)u \in \text{dom}(\mathcal{D}_{\min})$. Since $\chi u \in \text{dom}(\mathcal{D}_{\max})$ with $\mathcal{R}(\chi u) = 0$ and $\text{supp}(\chi u) \subset Z_{[0, 2r/3]}$, there exists a sequence $(u_j) \subset C_0^{\infty}(Z_{[0, 2r/3]}, E)$ converging to χu in the graph norm of $\mathcal{D}$. Note that by Lemma 1.21, we have a sequence consisting of smooth sections supported in $Z_{[0, 2r/3)}$ which does not necessarily vanish at the boundary. But since $\mathcal{R}(\chi u) = 0$, we can use the extension and restriction map in order to construct a sequence consisting of smooth sections vanishing at the boundary (see Lemma 3.8 for this construction for general boundary conditions). By the elliptic estimates in Proposition 2.19, we obtain $\chi u \in H^1(Z_{[0, 2r/3)}, E)$ with $\mathcal{R}_1 u = 0$ for the trace map $\mathcal{R}_1: H^1(Z_{[0, r)}, E) \rightarrow H^{1/2}(\partial M, E)$ given in Lemma 1.10. It follows

$$\chi u \in \ker(\mathcal{R}_1) \stackrel{\text{Lem. 1.10}}{=} \overline{C_{cc}^{\infty}(Z_{[0, r]}, E)}^{\|\cdot\|_{H^1}} \stackrel{\text{Prop. 1.13}}{\subseteq} \overline{C_{cc}^{\infty}(Z_{[0, r]}, E)}^{\|\cdot\|_{\mathcal{D}}} \subseteq \text{dom}(\mathcal{D}_{\min}),$$

and the lemma is proved. $\square$

Lemma 2.18. *Let $\tilde{A}$ be an adapted boundary operator for $\mathcal{D}^\dagger$. Then the following holds:*

- (1) *The bundle map $\sigma_{\mathcal{D}}(\tau)$ extends to a bounded operator $\check{H}_A(\partial M, E) \rightarrow \hat{H}_{\tilde{A}}(\partial M, F)$.*
- (2) *The formula*

$$\langle \mathcal{D}_{\max} u, v \rangle_{L^2} - \langle u, (\mathcal{D}^\dagger)_{\max} v \rangle_{L^2} = -\langle \sigma_{\mathcal{D}}(\tau) \mathcal{R}u, \mathcal{R}v \rangle \quad [\text{Green's formula}]$$

holds for all $u \in \text{dom}(\mathcal{D}_{\max})$ and all $v \in \text{dom}((\mathcal{D}^\dagger)_{\max})$. Here $\langle \cdot, \cdot \rangle: \hat{H}_{\tilde{A}}(\partial M, F) \times \check{H}_{\tilde{A}}(\partial M, F) \rightarrow \mathcal{A}$ denotes the perfect pairing given in Lemma 2.13.

Proof. We start with the proof of part (1). Note that by the same considerations as in the proof of Lemma 2.16, the first-order differential operator $-\tilde{A}$ is an adapted boundary operator for $\mathcal{B} := \mathcal{D}\sigma_{\mathcal{D}}(dt)^{-1}$. We obtain by Lemma 2.16 for the operator $\mathcal{B}$, the triangle inequality in $\mathcal{A}$, and Proposition 2.15 (2) for the operator $\mathcal{D}$

$$\|\sigma_{\mathcal{D}}(\tau)\varphi\|_{\check{H}_{\tilde{A}}}^2 = \|\mathcal{R}\sigma_{\mathcal{D}}(dt)\mathcal{E}\varphi\|_{\check{H}_{-\tilde{A}}}^2 \lesssim \|\sigma_{\mathcal{D}}(dt)\mathcal{E}\varphi\|_{\mathcal{B}}^2 \lesssim \|\mathcal{E}\varphi\|_{\mathcal{D}}^2 \lesssim \|\varphi\|_{H_A}^2$$

for all $\varphi \in C^\infty(\partial M, E)$. Here we denote by $\lesssim$ the estimate $\leq$ modulo a constant independent of the section φ . This shows that the map $\sigma_{\mathcal{D}}(\tau): C^\infty(\partial M, E) \rightarrow \hat{H}_{\tilde{A}}(\partial M, F)$ extends to the Sobolev space $\check{H}_A(\partial M, E)$.

It remains to show part (2). The asserted formula holds for all $u \in C_c^\infty(M, E)$ and all $v \in C_c^\infty(M, F)$ by Lemma 1.12. Since $C_c^\infty(M, E) \subseteq \text{dom}(\mathcal{D}_{\max})$ and $C_c^\infty(M, F) \subseteq \text{dom}((\mathcal{D}^\dagger)_{\max})$ are dense with respect to the graph norms of $\mathcal{D}$ and $\mathcal{D}^\dagger$, respectively (see Proposition 1.15), Green's formula holds for all $u \in \text{dom}(\mathcal{D}_{\max})$ and all $v \in \text{dom}((\mathcal{D}^\dagger)_{\max})$. To see this we fix some $v \in C_c^\infty(M, F)$ and a sequence $(u_j) \subset C_c^\infty(M, E)$ converging in the graph norm of $\mathcal{D}$ to some $u \in \text{dom}(\mathcal{D}_{\max})$. Then Green's formula holds for the sections u_j and v for all $j \in \mathbb{N}$. Moreover, since (u_j) converges to u in the graph norm of $\mathcal{D}$ as well as part (1) of this lemma and Lemma 2.13, we have

$$\begin{aligned} \langle \mathcal{D}u_j, v \rangle_{L^2} &\longrightarrow \langle \mathcal{D}u, v \rangle_{L^2}, & \langle u_j, \mathcal{D}^\dagger v \rangle_{L^2} &\longrightarrow \langle u, \mathcal{D}^\dagger v \rangle_{L^2} \quad \text{and} \\ \langle \sigma_{\mathcal{D}}(\tau) \mathcal{R}u_j, v \rangle_{L^2(\partial M)} &\xrightarrow{j \rightarrow \infty} \langle \sigma_{\mathcal{D}}(\tau) \mathcal{R}u, \mathcal{R}v \rangle \end{aligned}$$

in the C^* -algebra $\mathcal{A}$. Repeating this process for some $v \in \text{dom}((\mathcal{D}^\dagger)_{\max})$ proves Green's formula in full generality, hence the lemma is proved. $\square$

Lemma 2.17 together with Lemma 2.18 completes the proof of Theorem 2.3.

2.4 Elliptic estimates for smooth sections vanishing at the boundary

In this section we prove the elliptic estimates for smooth sections vanishing at the boundary of M . This generalizes Proposition 1.24 to manifolds with boundary and will be the crucial step for the elliptic estimates for general right-elliptic boundary conditions in Section 3.2.

Proposition 2.19. *For every $k \in \mathbb{N}_0$ there exists a constant C such that*

$$\|u\|_{H^{k+1}} \leq C(\|\mathcal{D}u\|_{H^k} + \|u\|_{H^k}) \quad [\text{elliptic estimate}]$$

for all $u \in C_0^\infty(Z_{[0,r]}, E)$.

For the proof of the elliptic estimate for compactly supported smooth sections vanishing at the boundary we reduce the situation to the interior elliptic estimates. Therefore, we define for $k \in \mathbb{N}_0$ a mirror map $\mathcal{M}_k$ which extends smooth sections of $E \rightarrow Z_{[0,r]}$ to k -times continuously differentiable sections of some bundle $\tilde{E} \rightarrow Z_{(-r,r)}$. See [GT77, Lemma 6.37] for the construction of the mirror map.

Let $Z_{(-r,r)} := \partial M \times (-r, r) \supset Z_{[0,r]}$, and define bundles $\tilde{E}$ and $\tilde{F}$ over $Z_{(-r,r)}$ by pulling back the corresponding bundles over ∂M via the projection $\text{pr}_1: Z_{(-r,r)} = \partial M \times (-r, r) \rightarrow \partial M$. The first-order differential operator $\mathcal{D}_0 = \sigma_{\mathcal{D}}(\tau)(\partial_t + A)$ extends to a first-order $\mathcal{A}$ -linear elliptic differential operator acting on smooth sections of $\tilde{E}$ which is complete for $Z_{(-r,r)}$.

Definition 2.20. We define for $k \in \mathbb{N}_0$ the *mirror map* $\mathcal{M}_k$ via

$$\begin{aligned} \mathcal{M}_k: C_c^\infty(Z_{[0,r]}, E) &\longrightarrow C_c^k(Z_{(-r,r)}, \tilde{E}), \\ (\mathcal{M}_k u)(x, t) &:= \begin{cases} u(x, t) & \text{for } (x, t) \in \partial M \times [0, r), \\ \sum_{j=1}^{k+1} c_j u(x, -t/j) & \text{for } (x, t) \in \partial M \times (-r, 0). \end{cases} \end{aligned} \quad (2.34)$$

Here the constants $c_1, \dots, c_{k+1} \in \mathbb{R}$ are a solution of the system of equations

$$\sum_{j=1}^{k+1} (-1/j)^i c_j = 1, \quad i = 0, \dots, k.$$

Lemma 2.21. *Let $k \in \mathbb{N}_0$ and $\ell \leq k$. Then the mirror map $\mathcal{M}_k: C_c^\infty(Z_{[0,r]}, E) \rightarrow H^\ell(Z_{(-r,r)}, E)$ is well-defined and extends to a bounded map with domain $H^\ell(Z_{[0,r]}, E)$. Moreover, there exists a constant C such that*

$$\|[\mathcal{D}_0, \mathcal{M}_k]u\|_{H^\ell(Z_{(-r,r)})} \leq C(\|\mathcal{D}_0 u\|_{H^\ell(Z_{[0,r]})} + \|u\|_{H^\ell(Z_{[0,r]})}) \quad (2.35)$$

for all $u \in C_0^\infty(Z_{[0,r]}, E)$. Here $\mathcal{D}_0 = \sigma(\tau)(\partial_t + A)$ as in Proposition 2.6.

Proof. Fix some $k \in \mathbb{N}$ and $\ell \leq k$. For $u \in C_c^\infty(Z_{[0,r]}, E)$, $\mathcal{M}_k u$ is smooth on $Z_{(-r,r)} \setminus (\partial M \times \{0\})$ and compactly supported in $Z_{(-r,r)}$. Around some point $(x, 0) \in \partial M \times \{0\}$, we fix a local orthonormal frame $e_1, \dots, e_{n-1}, \partial_t$ of $TZ_{(-r,r)}$, a multi-index $\alpha = (\alpha_1, \dots, \alpha_\ell) \in \{e_1, \dots, e_{n-1}, \partial_t\}^\ell$ with $\ell \leq k$, and write $\nabla^\alpha := \nabla_{\alpha_1} \dots \nabla_{\alpha_\ell}$. For $|\alpha|_t := \#\{j \in \{1, \dots, \ell\} \mid \alpha_j = \partial_t\}$, we obtain by definition of the mirror map in equation (2.34)

$$(\nabla^\alpha \mathcal{M}_k u)(x, 0) = \sum_{j=1}^{k+1} c_j (-1/j)^{|\alpha|_t} (\nabla^\alpha u)(x, 0) = (\nabla^\alpha u)(x, 0).$$

It follows that $\mathcal{M}_k u$ is k -times continuously differentiable on $\partial M \times \{0\}$, hence $\mathcal{M}_k$ is well-defined.

Moreover, there exists a constant C_1 such that

$$\begin{aligned}\|\mathcal{M}_k u\|_{\mathbf{H}^\ell(Z_{(-r,r)})}^2 &= \left\| \sum_{i=0}^{\ell} \left(\int_{\partial M} \int_{-r}^0 \left| \sum_{j=1}^{k+1} c_j \nabla^i u(x, -t/j) \right|_{(x,t)}^2 dt dx + \int_{Z_{[0,r]}} |\nabla^i u|_p^2 dp \right) \right\|_{\mathcal{A}} \\ &\leq k \sum_{j=1}^{k+1} c_j^2 \left\| \sum_{i=0}^{\ell} \int_{\partial M} \int_{-r}^0 |\nabla^i u(x, -t/j)|_{(x,t)}^2 dt dx \right\|_{\mathcal{A}} + \|u\|_{\mathbf{H}^k(Z_{[0,r]})}^2 \\ &\leq C_1 \|u\|_{\mathbf{H}^\ell(Z_{[0,r]})}^2\end{aligned}$$

for all $u \in C_c^\infty(Z_{[0,r]}, E)$. In the first step we used the definition of the $\mathbf{H}^\ell$ -norm and the definition of the mirror map in equation (2.34), and in the last step we make a change of variables $s := -t/j$ and compare with $\|u\|_{\mathbf{H}^k(Z_{[0,r]})}^2$.

It remains to show equation (2.35). Fix some $u \in C_0^\infty(Z_{[0,r]}, E)$. We write $\lesssim$ for $\leq$ modulo a constant which is independent of the section u . The commutator of $\mathcal{D}_0$ and $\mathcal{M}_k$ vanishes on $Z_{[0,r]}$ by definition of $\mathcal{M}_k$. At a point $(x, t) \in Z_{(-r,0)}$, we have by definition of $\mathcal{D}_0$ and $\mathcal{M}_k$

$$\begin{aligned}\bullet \quad (\mathcal{D}_0 \mathcal{M}_k u)(x, t) &= \sum_{j=1}^{k+1} c_j \sigma_{\mathcal{D}}(\tau) (\partial_t + A) u(x, -t/j) \\ &= \sum_{j=1}^{k+1} c_j \sigma_{\mathcal{D}}(\tau) [-1/j (\partial_t u)(x, -t/j) + (Au)(x, -t/j)] \quad \text{and} \\ \bullet \quad (\mathcal{M}_k \mathcal{D}_0 u)(x, t) &= \sum_{j=1}^{k+1} c_j \sigma_{\mathcal{D}}(\tau) ((\partial_t u)(x, -t/j) + (Au)(x, -t/j)),\end{aligned}$$

hence the commutator is given by

$$[\mathcal{D}_0, \mathcal{M}_k]u(x, t) = (\mathcal{D}_0 \mathcal{M}_k - \mathcal{M}_k \mathcal{D}_0)u(x, t) = - \sum_{j=1}^{k+1} c_j \sigma_{\mathcal{D}}(\tau) (1/j + 1) (\partial_t u)(x, -t/j). \quad (2.36)$$

We take the $\mathbf{H}^\ell$ -norm of the commutator in equation (2.36), make a change of variables in the t -direction, and obtain

$$\|[\mathcal{D}_0, \mathcal{M}_k]u\|_{\mathbf{H}^\ell(Z_{(-r,r)})}^2 \lesssim \|\partial_t u\|_{\mathbf{H}^\ell(Z_{[0,r]})}^2 \stackrel{(\star)}{\lesssim} \|\mathcal{D}_0 u\|_{\mathbf{H}^\ell(Z_{[0,r]})}^2 + \|u\|_{\mathbf{H}^\ell(Z_{[0,r]})}^2. \quad (2.37)$$

We show $(\star)$ in equation (2.37) by an induction on ℓ . Let us start with $\ell = 0$. By the same considerations as in equation (2.7), we obtain

$$|(\partial_t + A)u_t|_{L^2(\partial M)}^2 \geq |Au_t|_{L^2(\partial M)}^2 + \partial_t \langle Au_t, u_t \rangle_{L^2(\partial M)}. \quad (2.38)$$

Here we used $\langle A \partial_t u_t, u_t \rangle_{L^2(\partial M)} = \langle \partial_t u_t, Au_t \rangle_{L^2(\partial M)}$ which holds since A is independent of t and self-adjoint. We integrate equation (2.38) over $t \in [0, r]$, use that u is compactly supported

in $Z_{[0,r]}$ and vanishes at $t = 0$, and obtain by $\partial t = \sigma_{\mathcal{D}}(\tau)^{-1}\mathcal{D}_0 - A$

$$\begin{aligned}\|\partial_t u\|_{L^2(Z_{[0,r]})}^2 &\lesssim \|\mathcal{D}_0 u\|_{L^2(Z_{[0,r]})}^2 + \|Au\|_{L^2(Z_{[0,r]})}^2 \lesssim \|\mathcal{D}_0 u\|_{L^2(Z_{[0,r]})}^2 + \|(\partial_t + A)u\|_{L^2(Z_{[0,r]})}^2 \\ &\lesssim \|\mathcal{D}_0 u\|_{L^2(Z_{[0,r]})}^2.\end{aligned}$$

The induction step $\ell \mapsto \ell + 1$ follows as in the proof of Proposition 2.6 (1) and the lemma is proved. $\square$

Proof of Proposition 2.19. Fix some $k \in \mathbb{N}_0$ and some $u \in C_0^\infty(Z_{[0,r]}, E)$. We denote by $\lesssim$ the estimate $\leq$ modulo a constant which is independent of the section u . We obtain by the interior elliptic estimate in Proposition 1.24 for the differential operator $\mathcal{D}_0$ and $K = \partial M \times [-r, r]$, Lemma 2.21 and Proposition 2.6 (2)

$$\begin{aligned}\|u\|_{H^{k+1}(Z_{[0,r]})} &\leq \|\mathcal{M}_{k+1}u\|_{H^{k+1}(Z_{(-r,r]})} \\ &\stackrel{\text{Prop. 1.24}}{\lesssim} \|\mathcal{M}_{k+1}\mathcal{D}_0 u\|_{H^k(Z_{(-r,r]})} + \|\mathcal{M}_{k+1}u\|_{H^k(Z_{(-r,r]})} + \|[\mathcal{D}_0, \mathcal{M}_{k+1}]u\|_{H^k(Z_{(-r,r]})} \\ &\stackrel{\text{Lem. 2.21}}{\lesssim} \|\mathcal{D}_0 u\|_{H^k(Z_{[0,r]})} + \|u\|_{H^k(Z_{[0,r]})} \\ &\stackrel{\text{Prop. 2.6(2)}}{\lesssim} \|\mathcal{D}u\|_{H^k(Z_{[0,r]})} + \|u\|_{H^k(Z_{[0,r]})}.\end{aligned}$$

In the third step it is crucial that the section u vanishes at the boundary of $Z_{[0,r]}$ in order to use equation (2.35) in Lemma 2.21. $\square$

3 Boundary value problems

In this section we define abstract boundary conditions (Definition 3.2) and show that such boundary conditions give rise to a closed unbounded operator between the minimal and the maximal extension of the corresponding differential operator (Theorem 3.3). In Section 3.2 we define right-elliptic boundary conditions (Definition 3.4) and show the elliptic estimates (Theorem 3.6) for smooth sections satisfying the boundary condition as well as the higher elliptic regularity principle (Theorem 3.10). In Section 3.3 we define complementable and self-adjoint boundary conditions and show that the closed unbounded operator induced by such a boundary condition is self-adjoint and regular (Theorem 3.15).

We mainly follow the approach by Bär and Ballmann [BB12] and use the analysis developed in Section 2. The first difference in our setting is that the spectrum of the adapted boundary operator need not be discrete. Consequently, the definition of an elliptic boundary condition in [BB12, Definition 7.5], as well as the proof of the higher elliptic regularity principle, must be modified (Definition 3.4 and Theorem 3.10). The second difference is that an unbounded operator on a Hilbert C^* -module need not be regular, in contrast to the Hilbert space case. This motivates the definition of complementable boundary conditions (Definition 3.13).

Throughout this section we assume the following setup:

Setup 3.1. Let M be a Riemannian manifold with compact boundary ∂M and $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ an $\mathcal{A}$ -linear differential operator as in Setup 2.1. Moreover, suppose that $(M, \mathcal{D})$ is complete and that $\mathcal{D}$ is boundary symmetric with respect to some vector field T along ∂M pointing into the interior of M . We fix a collar of the boundary of M together with the identifications as in Setup 2.5, and fix an adapted boundary operator A (see Definition 2.7 and Proposition 2.6).

3.1 Abstract boundary conditions

In this section we introduce abstract boundary conditions (Definition 3.2) and show that they classify all closed extensions of a boundary symmetric first-order differential operator lying between the minimal and maximal extension (Theorem 3.3).

Definition 3.2 (Boundary condition). An *(abstract) boundary condition* for $\mathcal{D}$ is an $\mathcal{A}$ -submodule B of the hybrid Sobolev space $\check{H}_A(\partial M, E)$ which is closed with respect to the $\check{H}_A$ -norm.

Theorem 3.3. *Every boundary condition $B \subseteq \check{H}_A(\partial M, E)$ for $\mathcal{D}$ yields a closed extension*

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) := \{u \in \text{dom}(\mathcal{D}_{\max}) \mid \mathcal{R}u \in B\} \longrightarrow L^2(M, F)$$

of $\mathcal{D}_{\min}: \text{dom}(\mathcal{D}_{\min}) \rightarrow L^2(M, F)$. Moreover, every closed extension of $\mathcal{D}$ with domain between $\text{dom}(\mathcal{D}_{\min})$ and $\text{dom}(\mathcal{D}_{\max})$ is of this form.

Proof. Let $B \subseteq \check{H}_A(\partial M, E)$ be a boundary condition for $\mathcal{D}$. Using the restriction map $\mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \rightarrow \check{H}_A(\partial M, E)$ given in Theorem 2.3, we define an unbounded operator

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) := \mathcal{R}^{-1}(B) \longrightarrow L^2(M, F), \quad u \longmapsto \mathcal{D}_{\max} u.$$

This operator is closed by the following consideration: Let $(u_j) \subset \text{dom}(\mathcal{D}_B)$ be a sequence such that

$$u_j \xrightarrow{j \rightarrow \infty} u \in L^2(M, E) \quad \text{and} \quad \mathcal{D}u_j \xrightarrow{j \rightarrow \infty} w \in L^2(M, F).$$

We have to show that $u \in \text{dom}(\mathcal{D}_B)$ and $\mathcal{D}_B u = w$. Since $\mathcal{D}_{\max}$ is closed and $\text{dom}(\mathcal{D}_B) \subseteq \text{dom}(\mathcal{D}_{\max})$, we obtain $u \in \text{dom}(\mathcal{D}_{\max})$ and $\mathcal{D}_{\max} u = w$. Moreover, by Theorem 2.3

$$\|\mathcal{R}u_j - \mathcal{R}u\|_{\check{H}_A} \lesssim \|u_j - u\|_{\mathcal{D}} \xrightarrow{j \rightarrow \infty} 0,$$

hence $\mathcal{R}u_j \rightarrow \mathcal{R}u \in \check{H}_A(\partial M, E)$ for $j \rightarrow \infty$. But since $\mathcal{R}u_j \in B$ for all $j \in \mathbb{N}$ and $B \subseteq \check{H}_A(\partial M, E)$ is closed, $\mathcal{R}u \in B$, hence $u \in \text{dom}(\mathcal{D}_B)$. By $\mathcal{D}_B u = \mathcal{D}_{\max} u = w$, the unbounded operator $\mathcal{D}_B$ is closed.

The second statement is similar to [BB12, Proposition 7.2] as the following argumentation shows: Let $\tilde{\mathcal{D}}: \text{dom}(\tilde{\mathcal{D}}) \rightarrow L^2(M, E)$ be some closed extension of $\mathcal{D}: C_c^\infty(M, E) \rightarrow L^2(M, E)$ with $\text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\tilde{\mathcal{D}}) \subseteq \text{dom}(\mathcal{D}_{\max})$. We claim that

$$\text{dom}(\tilde{\mathcal{D}}) = \{u \in \text{dom}(\mathcal{D}_{\max}) \mid \mathcal{R}u \in B\} \quad \text{for} \quad B := \mathcal{R}(\text{dom}(\tilde{\mathcal{D}})).$$

The inclusion $\subseteq$ holds by definition of B . To see the other inclusion, let $u \in \text{dom}(\mathcal{D}_{\max})$ with $\mathcal{R}u \in B$. Then there exists a $v \in \text{dom}(\tilde{\mathcal{D}})$ such that $\mathcal{R}u = \mathcal{R}v$, hence $u - v \in \text{dom}(\mathcal{D}_{\min}) \subseteq \text{dom}(\tilde{\mathcal{D}})$ by Theorem 2.3 and $u \in \text{dom}(\tilde{\mathcal{D}})$.

It remains to show that B indeed defines a boundary condition in the sense of Definition 3.2, hence that $B \subseteq \check{H}_A(\partial M, E)$ is closed with respect to the $\check{H}_A$ -norm. Let $(\varphi_j) \subset B$ be a Cauchy sequence in $\check{H}_A(\partial M, E)$. Then $(\mathcal{E}\varphi_j) \subset \text{dom}(\tilde{\mathcal{D}})$ is also a Cauchy sequence with respect to the graph norm of $\mathcal{D}$ by $\|\mathcal{E}\varphi\|_{\mathcal{D}} \lesssim \|\varphi\|_{\check{H}_A}$ for all $\varphi \in \check{H}_A(\partial M, E)$ (see Proposition 2.15). Let

$$\mathcal{E}\varphi_j \xrightarrow{j \rightarrow \infty} u \in L^2(M, E) \quad \text{and} \quad \mathcal{D}\mathcal{E}\varphi_j \xrightarrow{j \rightarrow \infty} w \in L^2(M, E).$$

Since the unbounded operator $\tilde{\mathcal{D}}$ is closed, $u \in \text{dom}(\tilde{\mathcal{D}})$, in particular $\mathcal{R}u \in B$. We obtain

$$\|\varphi_j - \mathcal{R}u\|_{\check{H}_A} = \|\mathcal{R}(\mathcal{E}\varphi_j - u)\|_{\check{H}_A} \lesssim \|\mathcal{E}\varphi_j - u\|_{\mathcal{D}} \xrightarrow{j \rightarrow \infty} 0,$$

hence $B \subseteq \check{H}_A(\partial M, E)$ is closed with respect to the $\check{H}_A$ -norm. $\square$

3.2 Elliptic boundary conditions and regularity

In general, the higher check spaces $\check{H}_A^k(\partial M, E)$ satisfy for all $k \in \mathbb{N}_0$

$$H^{k-1/2}(\partial M, E) \supseteq \check{H}_A^k(\partial M, E) \supseteq H^{k+1/2}(\partial M, E) \quad (3.1)$$

(see Remark 2.10). A right-elliptic boundary condition will ensure that after intersecting with B the second inclusion in equation (3.1) is an equality. This is exactly what we need to prove the elliptic estimates in Theorem 3.6 and the elliptic regularity principle in Theorem 3.10.

Definition 3.4 (Right-elliptic boundary conditions). A boundary condition $B \subseteq \check{H}_A(\partial M, E)$ is called *right-elliptic* (with respect to $\mathcal{D}$) if there exists for every $k \in \mathbb{N}_0$ a constant $C_{k,B}$ such that

- (1) $\|\varphi\|_{H^{k+1/2}(\partial M)} \leq C_{k,B} \|\varphi\|_{\check{H}_A^k(\partial M)}$ for all $\varphi \in C^\infty(\partial M, E) \cap B$, and
- (2) $C^\infty(\partial M, E) \cap B \subseteq \check{H}_A^k(\partial M, E) \cap B$ is dense with respect to the $\check{H}_A^k$ -norm.

Remark 3.5. (1) We call the boundary condition right-elliptic, instead of elliptic, since we do not require that the adjoint boundary condition (see Definition 3.13 (1)) satisfies the properties (1) and (2) in Definition 3.4 as it is the case for elliptic boundary conditions in [BB12, Section 7]. For the elliptic estimates (Proposition 1.24) as well as the elliptic regularity (Theorem 3.10) it is sufficient to consider right-elliptic boundary conditions as in Definition 3.2.

- (2) Let B be a right-elliptic boundary condition, $k \in \mathbb{N}_0$, and $\varphi \in \check{H}_A^k(\partial M, E) \cap B$. By Definition 3.4 (2) there exists a sequence $(\varphi_j) \subset C^\infty(\partial M, E) \cap B$ approximating φ in the $\check{H}_A^k$ -norm. Then (φ_j) is a Cauchy sequence in $H^{k+1/2}(\partial M, E)$ by Definition 3.4 (1), hence

$\varphi \in H^{k+1/2}(\partial M, E)$. We obtain $\check{H}_A^k(\partial M, E) \cap B = H^{k+1/2}(\partial M, E) \cap B$. In particular, $B \subseteq H^{1/2}(\partial M, E)$.

Theorem 3.6. *Let $k \in \mathbb{N}_0$, B be a right-elliptic boundary condition and suppose $\mathcal{D}$ is elliptic. Then there exists for every compact $K \subseteq M$ a constant C_K such that*

$$\|u\|_{H^{k+1}} \leq C_K (\|\mathcal{D}u\|_{H^k} + \|u\|_{H^k}) \quad [\text{elliptic estimate}]$$

for all $u \in C_c^\infty(M, E)$ with $\text{supp}(u) \subseteq K$ and $u|_{\partial M} \in B$.

Remark 3.7. For the elliptic estimates in Theorem 3.6 it is sufficient to assume that the boundary condition B satisfies (1) in Definition 3.4.

Proof of Theorem 3.6. Fix some $k \in \mathbb{N}_0$, a right-elliptic boundary condition B and some compact subset $K \subseteq M$. Let $\chi: M \rightarrow \mathbb{R}$ be a smooth cutoff function induced by a map as in equation (2.17), and $u \in C_c^\infty(M, E)$ with $\text{supp}(u) \subseteq K$ and $u|_{\partial M} \in B$. Let $C_{k,B}$ be a constant as in Definition 3.4 (1). We write $\lesssim$ for $\leq$ modulo a constant which is independent of the section u . We have

$$\|u\|_{H^{k+1}} \leq \|(1 - \chi)u\|_{H^{k+1}} + \|(1 - \mathcal{E}\mathcal{R})\chi u\|_{H^{k+1}} + \|\mathcal{E}\mathcal{R}\chi u\|_{H^{k+1}}. \quad (3.2)$$

The section $(1 - \chi)u$ is supported in the compact set $K \setminus Z_{[0, r/3]} \subset M \setminus \partial M$. We obtain with the interior elliptic estimates in Proposition 1.24

$$\begin{aligned} \|(1 - \chi)u\|_{H^{k+1}} &\lesssim \|\mathcal{D}(1 - \chi)u\|_{H^k} + \|(1 - \chi)u\|_{H^k} \\ &\leq \|(1 - \chi)\mathcal{D}u\|_{H^k} + \|[\mathcal{D}, (1 - \chi)]u\|_{H^k} + \|(1 - \chi)u\|_{H^k} \\ &\lesssim \|\mathcal{D}u\|_{H^k} + \|u\|_{H^k}. \end{aligned}$$

Here we used in the last step that $[\mathcal{D}, (1 - \chi)]$ and $(1 - \chi)$ are zero-order differential operators.

Since $(1 - \mathcal{E}\mathcal{R})\chi u \in C_0^\infty(Z_{[0, r]}, E)$, we can apply Proposition 2.19 and obtain for the second term in equation (3.2)

$$\begin{aligned} \|(1 - \mathcal{E}\mathcal{R})\chi u\|_{H^{k+1}} &\lesssim \|\mathcal{D}(1 - \mathcal{E}\mathcal{R})\chi u\|_{H^k} + \|(1 - \mathcal{E}\mathcal{R})\chi u\|_{H^k} \\ &\leq \|\mathcal{D}\chi u\|_{H^k} + \|\chi u\|_{H^k} + \|\mathcal{D}\mathcal{E}\mathcal{R}\chi u\|_{H^k} + \|\mathcal{E}\mathcal{R}\chi u\|_{H^k} \\ &\lesssim \|\mathcal{D}u\|_{H^k} + \|u\|_{H^k} + \|\mathcal{E}\mathcal{R}u\|_{H^{k+1}}. \end{aligned} \quad (3.3)$$

In the last step, we used Proposition 1.13 for the compact manifold $Z_{[0, r]}$. For the third term in equation (3.2) and the last term in equation (3.3), we estimate

$$\|\mathcal{E}\mathcal{R}u\|_{H^{k+1}} \lesssim \|\mathcal{R}u\|_{H^{k+1/2}(\partial M)} \leq C_{k,B} \|\mathcal{R}u\|_{\check{H}_A^k(\partial M)} \lesssim \|\mathcal{D}u\|_{H^k} + \|u\|_{H^k}.$$

Here we used Proposition 2.15 (1), that B satisfies (1) in Definition 3.4 and Lemma 2.16. $\square$

The following lemma is a generalization of Lemma 1.21 stating that if the boundary condition

is right-elliptic, then the approximation by compactly supported smooth sections can be chosen to satisfy the boundary condition.

Lemma 3.8. *Let $k \in \mathbb{N}_0$ and B be a boundary condition satisfying (2) in Definition 3.4. Then there exists for every $u \in \text{dom}(\mathcal{D}_B) \cap H_{\text{loc}}^k(M, E)$ with $\mathcal{D}_B u \in H_{\text{loc}}^k(M, F)$, every $g \in C_c^\infty(M, \mathbb{R})$ which is constant at the boundary of M , and every open relatively compact subset $U \subseteq M$ containing the support of gu a sequence of smooth sections supported in $U \cup Z_{[0,r]}$, satisfying the boundary condition and approximating gu in the $\|\cdot\|_{\mathcal{D}, H^k}$ -norm.*

Proof. Let $k \in \mathbb{N}_0$, $u \in \text{dom}(\mathcal{D}_B)$, $g \in C_c^\infty(M, \mathbb{R})$ and $U \subseteq M$ be as in the lemma. Then

$$gu \in \text{dom}(\mathcal{D}_B) \cap H^k(M, E) \quad \text{and} \quad \mathcal{D}_B(gu) = [\mathcal{D}, g]u + g \mathcal{D}_B u \in H^k(M, F).$$

Here we used that $gu \in \text{dom}(\mathcal{D}_{\max})$ by Lemma 1.16, $\mathcal{R}gu = g|_{\partial M} \mathcal{R}u \in B$, $[\mathcal{D}, g]$ extends to $H^k(M, E) \rightarrow H^k(M, F)$ since g is compactly supported, and that $[\mathcal{D}_B, g]u = [\mathcal{D}, g]u$. In order to approximate gu by compactly supported smooth sections satisfying the boundary condition, we use the extension map in Proposition 2.15 and the restriction map in Theorem 2.3

$$\mathcal{E}: \check{H}_A(\partial M, E) \longrightarrow \text{dom}(\mathcal{D}_{\max}), \quad \mathcal{R}: \text{dom}(\mathcal{D}_{\max}) \longrightarrow \check{H}_A(\partial M, E).$$

We write $gu = (1 - \mathcal{E}\mathcal{R})(gu) + \mathcal{E}\mathcal{R}(gu)$ and approximate both terms separately.

Approximation of $(1 - \mathcal{E}\mathcal{R})(gu)$. Since gu is compactly supported and an element of $\text{dom}(\mathcal{D}_{\max}; k)$, there exists by Lemma 1.21 a sequence $(u'_j) \subset C_c^\infty(M, E)$ supported in U and converging to gu in the $\|\cdot\|_{\mathcal{D}, H^k}$ -norm. We define $u_j := (1 - \mathcal{E}\mathcal{R})u'_j \in C_0^\infty(M, E)$ for $j \in \mathbb{N}$ and obtain by Lemma 2.16 and Proposition 2.15 (2)

$$\|u_j - u_\ell\|_{\mathcal{D}, H^k} \leq \|u'_j - u'_\ell\|_{\mathcal{D}, H^k} + \|\mathcal{E}\mathcal{R}(u'_j - u'_\ell)\|_{\mathcal{D}, H^k} \leq C_1 \|u'_j - u'_\ell\|_{\mathcal{D}, H^k}$$

for a suitable constant C_1 and all $j, \ell \in \mathbb{N}$. Since (u'_j) converge in $\text{dom}(\mathcal{D}_{\max}; k)$, the sequence (u_j) is Cauchy in $\text{dom}(\mathcal{D}_{\max}; k)$. It follows that (u_j) converges to some $v \in \text{dom}(\mathcal{D}_{\max}; k)$ since $\text{dom}(\mathcal{D}_{\max}; k)$ is closed by Remark 1.19. Moreover, we have again by Lemma 2.16 and Proposition 2.15 (2) for $k = 0$

$$\|u_j - (1 - \mathcal{E}\mathcal{R})(gu)\|_{\mathcal{D}} \leq \|u'_j - gu\|_{\mathcal{D}} + \|\mathcal{E}\mathcal{R}(u'_j - gu)\|_{\mathcal{D}} \leq C_1 \|u'_j - gu\|_{\mathcal{D}, H^k} \xrightarrow{j \rightarrow \infty} 0,$$

and it follows $v = (1 - \mathcal{E}\mathcal{R})(gu)$.

Approximation of $\mathcal{E}\mathcal{R}(gu)$. Since $\mathcal{R}(gu) \in B$, there exists by Definition 3.4 (2) a sequence $(\varphi_j) \subset C^\infty(\partial M, E) \cap B$ approximating $\mathcal{R}(gu)$ in $\check{H}_A^k(\partial M, E)$. We define $v_j := \mathcal{E}\varphi_j$ for $j \in \mathbb{N}$ and obtain as before

$$\begin{aligned} \|v_j - v_\ell\|_{\mathcal{D}, H^k} &= \|\mathcal{E}(\varphi_j - \varphi_\ell)\|_{\mathcal{D}, H^k} \leq C_2 \|\varphi_j - \varphi_\ell\|_{\mathcal{D}, H^k} \quad \text{and} \\ \|v_j - \mathcal{E}\mathcal{R}(gu)\|_{\mathcal{D}} &= \|\mathcal{E}(\varphi_j - \mathcal{R}(gu))\|_{\mathcal{D}} \leq C_2 \|\varphi_j - \mathcal{R}(gu)\|_{\check{H}_A} \xrightarrow{j \rightarrow \infty} 0, \end{aligned}$$

for all $j, \ell \in \mathbb{N}$ and a suitable constant C_2 . It follows that (v_j) converges to $\mathcal{E}\mathcal{R}(gu)$ in the

$\|\cdot\|_{\mathcal{D}, H^k}$ -norm. Note that the restriction of v_j to the boundary of M satisfies the boundary condition and that v_j is supported in $Z_{[0,r)}$ for all $j \in \mathbb{N}$. $\square$

Lemma 1.17 and Lemma 3.8 for $k = 0$ yield immediately the following corollary:

Corollary 3.9. *Let B be a boundary condition satisfying (2) in Definition 3.4 for $k = 0$. Then the unbounded operator $\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \rightarrow L^2(M, F)$ is the L^2 -closure of*

$$\mathcal{D}: C_c^\infty(M, E; B) := \{u \in C_c^\infty(M, E) \mid u|_{\partial M} \in B\} \longrightarrow L^2(M, F).$$

Theorem 3.10 (Higher elliptic regularity). *Let $\mathcal{D}$ be elliptic, B a right-elliptic boundary condition and $k \in \mathbb{N}_0$. Then the following implication holds:*

$$u \in \text{dom}(\mathcal{D}_B) \cap H_{\text{loc}}^k(M, E) \quad \text{and} \quad \mathcal{D}_B u \in H_{\text{loc}}^k(M, F) \quad \implies \quad u \in H_{\text{loc}}^{k+1}(M, E).$$

Proof. Let $u \in \text{dom}(\mathcal{D}_B) \cap H_{\text{loc}}^k(M, E)$ with $\mathcal{D}_B u \in H_{\text{loc}}^k(M, F)$ and $g \in C_c^\infty(M, \mathbb{R})$ which is constant at the boundary of M . By Lemma 3.8 there exists an open relatively compact subset $U \subseteq M$ and a sequence $(u_j) \subset C_c^\infty(M, E)$ supported within U converging to gu in $\text{dom}(\mathcal{D}_{\max}; k)$ and satisfying $u_j|_{\partial M} \in B$ for all $j \in \mathbb{N}$. Then (u_j) is a Cauchy sequence in $H^{k+1}(M, E)$ by the elliptic estimates in Theorem 3.6 for $K = \overline{U}$, hence (u_j) converges to some $v \in H^{k+1}(M, E)$. Since (u_j) converges in $H^k(M, E)$ to gu , it follows $gu = v \in H^{k+1}(M, E)$, hence $u \in H_{\text{loc}}^{k+1}(M, E)$. $\square$

Corollary 3.11. *Suppose $E = F$ and $\mathcal{D}$ is elliptic. Let B be a right-elliptic boundary condition. Then $\text{dom}(\mathcal{D}_B^k) \subseteq H_{\text{loc}}^k(M, E)$ for all $k \in \mathbb{N}$. Moreover, if $u \in \text{dom}(\mathcal{D}_B^k)$ for all $k \in \mathbb{N}$, then the section u is already smooth. In particular, if the manifold M is compact, we obtain for all $k \in \mathbb{N}$*

$$\text{dom}(\mathcal{D}_B^k) = H^k(M, E; B) := \overline{C_c^\infty(M, E; B)}^{\|\cdot\|_{H^k}}.$$

Here $\mathcal{D}_B^k$ is the k -th composition of the unbounded operator $\mathcal{D}_B$. This means its domain $\text{dom}(\mathcal{D}_B^k)$ is given by

$$\{u \in L^2(M, E) \mid u \in \text{dom}(\mathcal{D}_B), \mathcal{D}_B u \in \text{dom}(\mathcal{D}_B), \dots, \mathcal{D}_B^{k-1} u \in \text{dom}(\mathcal{D}_B)\}. \quad (3.4)$$

Proof. The case $k = 1$ holds by Theorem 3.10. We proceed by an induction on k . By the induction hypothesis and the definition of $\text{dom}(\mathcal{D}_B^{k+1})$ in equation (3.4), every $u \in \text{dom}(\mathcal{D}_B^{k+1})$ satisfies

$$u \in \text{dom}(\mathcal{D}_B^k) \subseteq H_{\text{loc}}^k(M, E) \quad \text{and} \quad \mathcal{D}_B u \in \text{dom}(\mathcal{D}_B^k) \subseteq H_{\text{loc}}^k(M, E),$$

hence $u \in H_{\text{loc}}^{k+1}(M, E)$ by the elliptic regularity in Theorem 3.10. This proves the first assertion. If $u \in \text{dom}(\mathcal{D}_B^k) \subseteq H_{\text{loc}}^k(M, E)$ for all $k \in \mathbb{N}$, we obtain by the Sobolev embedding theorem in Proposition 1.9 that $gu \in C_c^\infty(M, E)$ for all $g \in C_c^\infty(M, \mathbb{R})$, hence the section u is smooth. $\square$

3.3 Complementary and self-adjoint boundary conditions

If the first-order $\mathcal{A}$ -linear differential operator $\mathcal{D}$ is formally self-adjoint and B is a boundary condition for $\mathcal{D}$, then the induced closed unbounded operator given in Theorem 3.3 is not necessarily self-adjoint. To ensure this we introduce the notions of complementary and self-adjoint boundary conditions in Definition 3.13 [cf. BB12, Section 7.2], and show in Theorem 3.15 that the unbounded operator associated to a formally self-adjoint $\mathcal{D}$ together with a complementary self-adjoint boundary condition is self-adjoint and regular. The proof of is based on the localization criterion by Kaad and Lesch [KL11, Theorem 1.1] (see Theorem 1.8) and the localization principle for bundles of Hilbert $\mathcal{A}$ -modules in Section 1.2.4. We orientate towards the proof of the corresponding statement for manifolds without boundary in [Ebe16, Theorem 1.14]. As a corollary we obtain that the unbounded operator $\mathcal{D}_B$ is regular if the boundary condition B is complementary.

Given some state ρ of the C^* -algebra $\mathcal{A}$, we denote by E^ρ , $\mathcal{D}_\rho$ and A_ρ the localizations of E , $\mathcal{D}$ and A , respectively (see Section 1.2.4). Note that $\mathcal{D}_\rho$ is formally self-adjoint and boundary symmetric with respect to the vector field T , and A_ρ is an adapted boundary operator for $\mathcal{D}_\rho$. We start with a generalization of [Ebe16, Lemma 1.18]:

Lemma 3.12. *Let ρ be a state of the C^* -algebra $\mathcal{A}$. Then there exist canonical isometric embeddings $\check{\Phi}: L^2(M, E)^\rho \rightarrow L^2(M, E^\rho)$,*

$$\check{\Phi}: \check{H}_A(\partial M, E)^\rho \longrightarrow \check{H}_{A_\rho}(\partial M, E^\rho) \quad \text{and} \quad \hat{\Phi}: \hat{H}_A(\partial M, E)^\rho \longrightarrow \hat{H}_{A_\rho}(\partial M, E^\rho).$$

Moreover, the subspaces

$$\Phi(C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho) \subseteq \overline{C_c^\infty(M, E^\rho)}^{\|\cdot\|_{\mathcal{D}_\rho}} \quad \text{and} \quad \Phi(C_{cc}^\infty(M, E) \odot_{\mathcal{A}} H_\rho) \subseteq \overline{C_{cc}^\infty(M, E^\rho)}^{\|\cdot\|_{\mathcal{D}_\rho}}$$

are both dense with respect to the graph norm of $\mathcal{D}_\rho$. In particular, Φ is surjective. Here H_ρ is the Hilbert space given in the GNS representation associated to the state ρ (see Section 1.1.2).

Proof. We define

$$\Phi_0: C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho \longrightarrow L^2(M, E^\rho), \quad \Phi_0(u \otimes x)(p) := u(p) \otimes x \in E_p^\rho. \quad (3.5)$$

By definition of the inner product on $L^2(M, E) \otimes_{\mathcal{A}} H_\rho$ and $L^2(M, E^\rho)$, the map Φ_0 preserves the L^2 -inner products as in the proof of [Ebe16, Lemma 1.18], and extends therefore to an isometric embedding $\Phi: L^2(M, E)^\rho \rightarrow L^2(M, E^\rho)$.

We continue with the proof that any $u \in C_c^\infty(M, E^\rho)$ can be approximated in the graph norm of $\mathcal{D}_\rho$ by elements of $\Phi(C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho)$. By a partition of unity argument as well as the fact that the Hilbert $\mathcal{A}$ -module P is finitely generated and projective, we can assume without loss of generality $M = \{x \in \mathbb{R}^n \mid x_1 \geq 0\} =: \mathbb{H}^n$ or $M = \mathbb{R}^n$, and $E = M \times \mathcal{A}$. We focus on $M = \mathbb{H}^n$, and it suffices by the considerations in the proof of [Ebe16, Lemma 1.18] to show the following statement: Any smooth function $u: \mathbb{H}^n \rightarrow H_\rho$ with values in a fixed finite-dimensional subspace $H_0 \subseteq H_\rho$ can be approximated in the C^1 -norm by functions of the form $p \mapsto \pi(v(p))x$ for $v \in C_c^\infty(\mathbb{H}^n, \mathcal{A})$ and $x \in H_\rho$.

We follow the proof in [Ebe16, Lemma 1.18]. Fix a smooth function $u: \mathbb{H}^n \rightarrow H_0 \subseteq H_\rho$, an orthonormal basis $\{x_1, \dots, x_r\}$ of H_0 and some $\varepsilon > 0$. Let $x_\rho \in H_\rho$ be the cyclic unit vector as in Section 1.1.2. Then there exist $a_1, \dots, a_r \in \mathcal{A}$ such that

$$\|\pi_\rho(a_j)x_\rho - x_j\| < (\sqrt{r}\|u\|_{C^1})^{-1}\varepsilon$$

for all $j \in \{1, \dots, r\}$. Let $\sigma: H_0 \rightarrow \mathcal{A}$ be the linear map given by $x_j \mapsto a_j$ for all $j \in \{1, \dots, r\}$. Then the operator norm of $\Psi: H_0 \rightarrow H_\rho$, defined via $x \mapsto \pi_\rho(\sigma(x))x_\rho - x$, is less than $\varepsilon\|u\|_{C^1}^{-1}$. We define $v: \mathbb{H}^n \rightarrow \mathcal{A}$ by $v(p) := \sigma(u(p))$, and obtain

$$\|\pi_\rho(v(\cdot)) - u(\cdot)\|_{C^1} = \|\Psi(u)\|_{C^1} \leq \|\Psi\|_{\text{op}}\|u\|_{C^1} < \varepsilon.$$

This finishes the proof that $\Phi(C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho) \subseteq \overline{C_c^\infty(M, E^\rho)}^{\|\cdot\|_{\mathcal{D}_\rho}}$ is dense with respect to the graph norm of $\mathcal{D}_\rho$. The proof of the corresponding statement about sections with compactly support in the interior of M is similar. We obtain in particular that any $u \in C_c^\infty(M, E^\rho)$ can be approximated by elements in $\Phi(C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho)$ in the L^2 -norm. Therefore, the map Φ has dense range, hence is surjective since it is an isometric embedding.

The maps Φ as in equation (3.5) with M replaced by ∂M preserves the $\check{H}_A$ - and $\hat{H}_A$ -inner products by the same considerations as for the L^2 -inner product, hence the map Φ_0 in equation (3.5) extends to isometric embeddings $\hat{\Phi}$ and $\check{\Phi}$ as in the lemma. $\square$

Let $B \subseteq \check{H}_A(\partial M, E)$ be some boundary condition and ρ a state of the C^* -algebra $\mathcal{A}$. We define the *localized boundary condition* $B_\rho := \check{\Phi}(B \otimes_{\mathcal{A}} H_\rho)$. Since B is a closed $\mathcal{A}$ -submodule of $\check{H}_A(\partial M, E)$ and Φ is an isometric embedding (Lemma 3.12), B_ρ is a closed linear subspace of $\check{H}_{A_\rho}(\partial M, E^\rho)$, hence a boundary condition for $\mathcal{D}_\rho$.

Definition 3.13. Let $B \subseteq \check{H}_A(\partial M, E)$ be a boundary condition for $\mathcal{D}$ and $\tilde{A}$ some boundary operator adapted to the formal adjoint $\mathcal{D}^\dagger$.

- (1) The *adjoint boundary condition* of B is defined as

$$B^{\text{ad}} := \{\psi \in \check{H}_{\tilde{A}}(\partial M, F) \mid \langle \sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle = 0 \forall \varphi \in B\}.$$

- (2) We call B *complementable* if $(B^{\text{ad}})^{\text{ad}} = B$ and $(B_\rho)^{\text{ad}} = (B^{\text{ad}})_\rho$ for every state ρ of the C^* -algebra $\mathcal{A}$.
- (3) Suppose $\mathcal{D}$ is formally self-adjoint (in particular $E = F$). We call B *self-adjoint* if $B = B^{\text{ad}}$ for the adapted operator $\tilde{A} = A$.

In Definition 3.13 (1) $\langle \cdot, \cdot \rangle$ denotes the perfect pairing $\hat{H}_{\tilde{A}}(\partial M, F) \times \check{H}_{\tilde{A}}(\partial M, F) \rightarrow \mathcal{A}$ in Lemma 2.13 which is well-defined since $\sigma_{\mathcal{D}}(\tau)$ extends to a bounded operator $\check{H}_A(\partial M, E) \rightarrow \hat{H}_{\tilde{A}}(\partial M, F)$ by Lemma 2.18 (1). The definition of the adjoint boundary condition is motivated by Green's formula in Theorem 2.3. The adjoint boundary condition defines a boundary condition for $\mathcal{D}^\dagger$, and we will see in Corollary 3.17 that the unbounded operator associated to $\mathcal{D}^\dagger$ with the adjoint boundary condition B^{ad} is indeed the adjoint of $\mathcal{D}_B$.

Remark 3.14. If the C^* -algebra $\mathcal{A}$ is equal to $\mathbb{C}$, then every boundary condition B is complementable. Moreover, $B = \emptyset$ and $B = \check{H}_A(\partial M, E)$ are both complementable boundary conditions for $\mathcal{D}$.

Theorem 3.15. *Let $\mathcal{D}$ be formally self-adjoint (in particular $E = F$) and let $B \subseteq \check{H}_A(\partial M, E)$ be a complementable self-adjoint boundary condition for $\mathcal{D}$. Then the unbounded operator*

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \longrightarrow L^2(M, E)$$

is self-adjoint and regular.

We obtain by the localization criterion by Kaad and Lesch [KL11] and Lemma 3.12:

Lemma 3.16. *Let B be a boundary condition for $\mathcal{D}$. The unbounded operator $\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \rightarrow L^2(M, E)$ is self-adjoint and regular if and only if $(\mathcal{D}_\rho)_{B_\rho}: \text{dom}((\mathcal{D}_\rho)_{B_\rho}) \rightarrow L^2(M, E^\rho)$ is self-adjoint for every state ρ of the C^* -algebra $\mathcal{A}$.*

Proof. We obtain by the criterion of Kaad and Lesch [KL11] in Theorem 1.8 that $\mathcal{D}_B$ is self-adjoint and regular if and only if the closure of

$$\mathcal{D}_B \odot_{\mathcal{A}} \text{id}_{H_\rho}: \text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho \longrightarrow L^2(M, E)^\rho$$

is self-adjoint for every state ρ of the C^* -algebra $\mathcal{A}$. Since $\Phi: L^2(M, E)^\rho \rightarrow L^2(M, E^\rho)$ is a surjective isometric embedding, the closure of $\mathcal{D}_B \odot_{\mathcal{A}} \text{id}_{H_\rho}$ is self-adjoint if and only if the closure of

$$(\mathcal{D}_\rho)_{B_\rho}: \Phi(\text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho) \longrightarrow L^2(M, E^\rho)$$

is self-adjoint. Here we used that $\Phi(\text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho) \subseteq \text{dom}((\mathcal{D}_\rho)_{B_\rho})$ and that $\mathcal{D}_\rho \circ \Phi = \Phi \circ (\mathcal{D} \odot_{\mathcal{A}} \text{id}_{H_\rho})$ on $C_c^\infty(M, E) \odot_{\mathcal{A}} H_\rho$ (look at local trivializations) and therefore by Proposition 1.15 and Lemma 3.12 also on $\text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho \subseteq \text{dom}(\mathcal{D}_{\max}) \odot_{\mathcal{A}} H_\rho$. It remains to show that

$$\Phi(\text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho) \subseteq \text{dom}((\mathcal{D}_\rho)_{B_\rho})$$

is dense with respect to the graph norm of $\mathcal{D}_\rho$.

Let $u \in \text{dom}((\mathcal{D}_\rho)_{B_\rho})$. Then, we obtain by Theorem 2.3 and Lemma 3.12

$$u_0 := \mathcal{E}(\mathcal{R}u) - u \in \ker(\mathcal{R}) = \overline{C_{cc}^\infty(M, E^\rho)}^{\|\cdot\|_{\mathcal{D}_\rho}} = \overline{\Phi(C_{cc}^\infty(M, E^\rho) \odot_{\mathcal{A}} H_\rho)}^{\|\cdot\|_{\mathcal{D}_\rho}},$$

hence u_0 can be approximated by elements in $\Phi(\text{dom}(\mathcal{D}_B) \odot_{\mathcal{A}} H_\rho)$ in the graph norm of $\mathcal{D}_\rho$. It remains to show that $\mathcal{E}(\mathcal{R}u)$ has also such an approximation. By Construction 1.7 (2), we have that $\iota(B) \subseteq B \otimes_{\mathcal{A}} H_\rho$ is dense, and we obtain

$$\mathcal{R}u \in B_\rho = \check{\Phi}(B \otimes_{\mathcal{A}} H_\rho) = \check{\Phi}(\overline{\iota(B)}).$$

Let (φ_j) be a sequence in B such that $\check{\Phi}(\varphi_j \otimes x_\rho)$ approximates $\mathcal{R}u$ in $\check{H}_{A_\rho}(\partial M, E^\rho)$. Here x_ρ is the unique unit vector in H_ρ that is cyclic (see Section 1.1.2). Then there exists by

Proposition 2.15 a constant C such that for all $j \in \mathbb{N}$

$$\|\mathcal{E}(\mathcal{R}u) - \Phi((\mathcal{E}\varphi_j) \otimes x_\rho)\|_{\mathcal{D}_\rho} = \|\mathcal{E}(\mathcal{R}u - \check{\Phi}(\varphi_j \otimes x_\rho))\|_{\mathcal{D}_\rho} \leq C\|\mathcal{R}u - \check{\Phi}(\varphi_j \otimes x_\rho)\|_{\check{H}_{A_\rho}}.$$

This proves that $\Phi(\mathcal{E}\varphi_j \otimes x_\rho) \in \Phi(\text{dom}(\mathcal{D}_B) \odot H_\rho)$ approximates $\mathcal{E}(\mathcal{R}u)$ in the graph norm of $\mathcal{D}_\rho$, and the lemma is proved. $\square$

Proof of Theorem 3.15. Let B be a complementable self-adjoint boundary condition for $\mathcal{D}$, and ρ some state of the C^* -algebra $\mathcal{A}$. By Lemma 3.16 we have to show that $(\mathcal{D}_\rho)_{B_\rho} : \text{dom}((\mathcal{D}_\rho)_{B_\rho}) \rightarrow L^2(M, E^\rho)$ is self-adjoint. Since B is complementable and self-adjoint, we obtain $(B_\rho)^{\text{ad}} = (B^{\text{ad}})_\rho = B_\rho$, hence the localized boundary condition is self-adjoint. From now on we omit the subscript ρ to improve readability. We have to show that the Hilbert spaces

$$\begin{aligned} \text{dom}(\mathcal{D}_B) &\stackrel{B \text{ self-adjoint}}{=} \{v \in \text{dom}(\mathcal{D}_{\max}) \mid \langle \sigma_{\mathcal{D}}(\tau)\varphi, \mathcal{R}v \rangle = 0 \ \forall \varphi \in B\} \quad \text{and} \\ \text{dom}((\mathcal{D}_B)^{\text{ad}}) &\stackrel{\text{def.}}{=} \{v \in L^2(M, E) \mid \exists w \in L^2(M, E) : \langle \mathcal{D}_B u, v \rangle_{L^2} = \langle u, w \rangle_{L^2} \ \forall u \in \text{dom}(\mathcal{D}_B)\} \end{aligned}$$

coincide.

Fix $v \in \text{dom}(\mathcal{D}_B)$. We calculate for $w := \mathcal{D}_B v \in L^2(M, E)$ and $u \in \text{dom}(\mathcal{D}_B)$

$$\langle \mathcal{D}_B u, v \rangle_{L^2} = \langle u, \mathcal{D}_B v \rangle_{L^2} - \langle \sigma_{\mathcal{D}}(\tau)\mathcal{R}u, \mathcal{R}v \rangle = \langle u, w \rangle_{L^2},$$

hence $v \in \text{dom}(\mathcal{D}_B)^{\text{ad}}$. In the first step we used Green's formula (Theorem 2.3), $\mathcal{D} = \mathcal{D}^\dagger$ and $\mathcal{D}_B \subseteq \mathcal{D}_{\max}$, and in the second step $\mathcal{R}u \in B$ and that B is a self-adjoint boundary condition in the sense of Definition 3.13 (3).

Fix $v \in \text{dom}((\mathcal{D}_B)^{\text{ad}})$. Since $\mathcal{D}_{\text{cc}} \subseteq \mathcal{D}_B$ and $\mathcal{D} = \mathcal{D}^\dagger$, we have $(\mathcal{D}_B)^{\text{ad}} \subseteq (\mathcal{D}_{\text{cc}})^{\text{ad}} = \mathcal{D}_{\max}$ and therefore $v \in \text{dom}(\mathcal{D}_{\max})$ with $(\mathcal{D}_B)^{\text{ad}} v = \mathcal{D}_{\max} v$. We obtain by Green's formula in Theorem 2.3, $\mathcal{D} = \mathcal{D}^\dagger$ and the definition of $\text{dom}((\mathcal{D}_B)^{\text{ad}})$

$$\langle \sigma_{\mathcal{D}}(\tau)\mathcal{R}u, \mathcal{R}v \rangle = -\langle \mathcal{D}_B u, v \rangle_{L^2} + \langle u, \mathcal{D}_{\max} v \rangle_{L^2} = \langle u, (-\mathcal{D}_B)^{\text{ad}} + \mathcal{D}_{\max} v \rangle_{L^2} = 0 \quad (3.6)$$

for all $u \in \text{dom}(\mathcal{D}_B) \subseteq \text{dom}(\mathcal{D}_{\max})$. Since we can extend by Proposition 2.15 any $\varphi \in B$ by the extension map to an element in $\text{dom}(\mathcal{D}_B)$, equation (3.6) yields that $v \in \text{dom}(\mathcal{D}_B)$, and the theorem is proved. $\square$

Corollary 3.17. *Let B be a complementable boundary condition for $\mathcal{D}$. Then the unbounded operator $\mathcal{D}_B : \text{dom}(\mathcal{D}_B) \rightarrow L^2(M, F)$ is regular, and its adjoint is given by the unbounded operator $(\mathcal{D}^\dagger)_{B^{\text{ad}}} : \text{dom}((\mathcal{D}^\dagger)_{B^{\text{ad}}}) \rightarrow L^2(M, E)$.*

Proof. Let B be a complementable boundary condition as in the corollary and $\tilde{A}$ a boundary operator adapted to $\mathcal{D}^\dagger$. We consider the formally self-adjoint boundary symmetric first-order differential operator

$$\mathcal{P} := \begin{pmatrix} 0 & \mathcal{D}^\dagger \\ \mathcal{D} & 0 \end{pmatrix} : C^\infty(M, E \oplus F) \longrightarrow C^\infty(M, E \oplus F)$$

with adapted boundary operator $A \oplus \tilde{A}$, and the boundary condition $B \oplus B^{\text{ad}} \subseteq \check{H}_A(\partial M, E) \oplus \check{H}_{\tilde{A}}(\partial M, F) \cong \check{H}_{A \oplus \tilde{A}}(\partial M, E \oplus F)$. Since B is complementable, the boundary condition $B \oplus B^{\text{ad}}$ is complementable and self-adjoint by

$$(B \oplus B^{\text{ad}})^{\text{ad}} = (B^{\text{ad}})^{\text{ad}} \oplus B^{\text{ad}} = B \oplus B^{\text{ad}}, \text{ and}$$

$$((B \oplus B^{\text{ad}})_{\rho})^{\text{ad}} = ((B^{\text{ad}})_{\rho})^{\text{ad}} \oplus (B_{\rho})^{\text{ad}} = B_{\rho} \oplus (B^{\text{ad}})_{\rho} = ((B \oplus B^{\text{ad}})^{\text{ad}})_{\rho}$$

for every state ρ of the C^* -algebra $\mathcal{A}$. We apply Theorem 3.15 and obtain that $\mathcal{P}_{B \oplus B^{\text{ad}}}$ is self-adjoint and regular, hence $\mathcal{D}_B$ is regular with adjoint given by $(\mathcal{D}^{\dagger})_{B^{\text{ad}}}$. $\square$

4 Index theory

In this section we show that a formally self-adjoint boundary symmetric first-order elliptic differential operator which is invertible at infinity (Definition 4.3) equipped with a right-elliptic complementable self-adjoint boundary condition gives rise to a Fredholm operator in the sense of Definition 1.5 (see Theorem 4.5). In the presence of a grading and a Real structure we obtain by Construction 1.6 an associated class in the Real K-theory of the underlying Real C^* -algebra.

Setup 4.1. Let M be a Riemannian manifold with compact boundary ∂M , $\mathcal{A}$ a graded Real unital C^* -algebra, $E \rightarrow M$ a bundle of finitely generated projective graded Real Hilbert $\mathcal{A}$ -modules equipped with an $\mathcal{A}$ -linear metric connection, and $\mathcal{D}: C^{\infty}(M, E) \rightarrow C^{\infty}(M, E)$ an odd Real first-order $\mathcal{A}$ -linear differential operator such that $(M, \mathcal{D})$ is complete. Suppose $\mathcal{D}$ is boundary symmetric with respect to some vector field T along ∂M pointing into the interior of M . Let A be some adapted boundary operator (see Definition 2.7 and Proposition 2.6) and $B \subseteq \check{H}_A(\partial M, E)$ some boundary condition (Definition 3.2) which is invariant under the grading automorphism α and the Real structure κ .

Remark 4.2. In contrast to the previous sections, we assume in Setup 4.1 that E is a bundle of finitely generated projective Hilbert $\mathcal{A}$ -modules. This means in the case $\mathcal{A} = \mathbb{C}$ that E is a vector bundle with finite dimensional fibers and not just a bundle of Hilbert spaces. The reason is that we need Rellich's theorem (Proposition 1.9 (2)) in the proof Theorem 4.5.

Definition 4.3. We call $\mathcal{D}$ *invertible at infinity* if there exists a compact subset $K \subseteq M$ and a constant c such that

$$\langle \mathcal{D}u, \mathcal{D}u \rangle_{L^2} \geq C \langle u, u \rangle_{L^2} \quad (4.1)$$

for all $u \in C_c^{\infty}(M, E)$ with $\text{supp}(u) \cap K = \emptyset$.

Remark 4.4. Suppose $\mathcal{D}$ is invertible at infinity. Since $C_c^{\infty}(M, E) \subseteq \text{dom}(\mathcal{D}_{\text{max}})$ is dense with respect to the graph norm of $\mathcal{D}$ by Proposition 1.15, equation (4.1) holds for all $u \in \text{dom}(\mathcal{D}_{\text{max}})$.

Theorem 4.5. *Let $\mathcal{D}$ be elliptic, formally self-adjoint and invertible at infinity, and let B be right-elliptic complementable and self-adjoint. Then we obtain an unbounded odd Real Fredholm operator*

$$\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \longrightarrow L^2(M, E)$$

in the sense of Definition 1.5, hence it has a well-defined index class $\text{ind}(\mathcal{D}_B) \in \text{KO}_0(\mathcal{A})$.

Proof. Since $\mathcal{D}$ is formally self-adjoint and the boundary condition is complementable and self-adjoint, we obtain by Theorem 3.15 a self-adjoint and regular unbounded operator $\mathcal{D}_B: \text{dom}(\mathcal{D}_B) \rightarrow L^2(M, E)$ together with a functional calculus as in Proposition 1.4. Moreover, the unbounded operator $\mathcal{D}_B$ is odd and Real since the first-order differential operator $\mathcal{D}$ and the boundary condition B preserve the grading and the Real structure. It remains to show that $\mathcal{D}_B$ is Fredholm in the sense of Definition 1.5. Then $\mathcal{D}_B$ gives rise to a well-defined index class in $\text{KO}_0(\mathcal{A})$ by Construction 1.6.

It is sufficient to show that $gf(\mathcal{D}_B)$ and $f(\mathcal{D}_B)g$ are compact for all $f \in C_0(\mathbb{R})$ and all $g \in C_c^\infty(M, \mathbb{R})$. This is a generalization of [Ebe16, Theorem 2.34] to manifolds with boundary. Then the same considerations as in the proof of [Ebe16, Theorem 2.41] yield together with Remark 4.4 that $\mathcal{D}_B$ is a Fredholm operator. Fix some $g \in C_c^\infty(M, \mathbb{R})$. We just show that $gf(\mathcal{D}_B)$ is compact for all $f \in C_0(\mathbb{R})$ since the proof for $f(\mathcal{D}_B)g$ is similar. Consider the closed ideal

$$J := \{f \in C_0(\mathbb{R}) \mid gf(\mathcal{D}_B) \in \text{Kom}_{\mathcal{A}}(L^2(M, E))\} \subseteq C_0(\mathbb{R}).$$

It remains to show that $g(\mathcal{D}_B^2 + 1)^{-1}$ and $g\mathcal{D}_B(\mathcal{D}_B^2 + 1)^{-1}$ are compact operators on $L^2(M, E)$. Then the functions $(x^2 + 1)^{-1}$ and $x(x^2 + 1)^{-1}$ would be in the ideal J , hence by the Stone-Weierstrass theorem the ideal J would be equal to $C_0(\mathbb{R})$.

Let $K \subseteq M$ be a compact submanifold such that $\text{supp}(g) \subseteq K^\circ \subseteq M$. Then $g(\mathcal{D}^2 + 1)^{-1}$ is given by a composition of the adjointable maps

$$L^2(M, E) \xrightarrow{(\mathcal{D}_B - i)^{-1}} L^2(M, E) \xrightarrow{(\mathcal{D}_B + i)^{-1}} \text{dom}(\mathcal{D}_B) \xrightarrow{g} H^1(K, E) \longrightarrow L^2(M, E). \quad (4.2)$$

By the properties (6) and (8) of the functional calculus in Proposition 1.4, the first and second maps in equation (4.2) are well-defined, and the second one is adjointable by $\langle u, v \rangle_{\mathcal{D}} = \langle (\mathcal{D}_B + i)u, (\mathcal{D}_B + i)v \rangle_{L^2}$ which holds for all $u, v \in \text{dom}(\mathcal{D}_B)$. The third map in equation (4.2) is at first defined on the subspace $C_c^\infty(M, E; B) \subseteq \text{dom}(\mathcal{D}_B)$, but it extends to a bounded map $\text{dom}(\mathcal{D}_B) \rightarrow H^1(K, E)$ by the elliptic estimates in Theorem 3.6 and Corollary 3.9. We write the fourth map in equation (4.2) as $H^1(K, E) \rightarrow L^2(K, E) \rightarrow L^2(M, E)$, which is a compact operator since the first map is compact by Rellich's theorem in Proposition 1.9 (2). Finally, we obtain that $g(\mathcal{D}_B^2 + 1)^{-1}: L^2(M, E) \rightarrow L^2(M, E)$ is compact as the composition of adjointable operators and a compact operator. The same argument yields that $g\mathcal{D}_B(\mathcal{D}_B^2 + 1)^{-1}: L^2(M, E) \rightarrow L^2(M, E)$ is compact since $\mathcal{D}_B(\mathcal{D}_B - i)^{-1}$ is an adjointable operator on $L^2(M, E)$, and the theorem is proved. $\square$

5 Examples of right-elliptic complementable self-adjoint boundary conditions

In this section we study the generalized Atiyah-Patodi-Singer boundary condition (Section 5.1) and chiral boundary conditions arising from a suitable involution along the boundary (Section 5.2). We prove in Proposition 5.2 and Proposition 5.5 that these boundary conditions give rise to right-elliptic (complementable self-adjoint) boundary conditions in the sense of Definition 3.2, Definition 3.13 and Definition 3.4.

Setup 5.1. Let M be a Riemannian manifold with compact boundary ∂M , $\mathcal{A}$ a unital C^* -algebra, E and F two bundles of finitely generated projective Hilbert $\mathcal{A}$ -modules over M equipped with an $\mathcal{A}$ -linear connection, and $\mathcal{D}: C^\infty(M, E) \rightarrow C^\infty(M, F)$ a first-order $\mathcal{A}$ -linear differential operator such that $(M, \mathcal{D})$ is complete. Suppose $\mathcal{D}$ is boundary symmetric with respect to a vector field T along ∂M pointing into the interior of M . We denote by τ the co-vector field associated to the vector field T .

5.1 The generalized Atiyah-Patodi-Singer boundary condition

The generalized Atiyah-Patodi-Singer (APS) boundary condition requires that the boundary values of a section lie in a certain part of the spectrum of the adapted boundary operator A . In other words, the projection to a certain part of the spectrum of A vanishes. Since the spectrum of A is not necessarily discrete as in the setting of finite-rank vector bundles, such projections do not always exist. However, if a point $a \in \mathbb{R}$ is not in the spectrum of A , we can use the functional calculus for A (see Theorem 1.26 and Proposition 1.4) to define generalized APS boundary conditions. These boundary conditions are right-elliptic in the sense of Definition 3.4. See [APS75] for the classical APS boundary condition and [BB12, Example 7.27] for the generalized APS boundary condition in the setting of finite-rank vector bundles.

Proposition 5.2. *Let A be an adapted boundary operator for $\mathcal{D}$ and $a \in \mathbb{R} \setminus \text{spec}(A)$. We define*

$$B_{\text{APS}}^a := f_\varepsilon^a(A) H^{1/2}(\partial M, E) \subseteq \check{H}_A(\partial M, E)$$

for some $\varepsilon > 0$ such that $(a - \varepsilon, a + \varepsilon) \cap \text{spec}(A) = \emptyset$, and some function $f_\varepsilon^a \in C^\infty(\mathbb{R}, [0, 1])$ such that $f_\varepsilon^a \equiv 1$ on $(-\infty, a - \varepsilon)$ and $f_\varepsilon^a \equiv 0$ on $(a + \varepsilon, \infty)$. Then the following holds:

- (1) B_{APS}^a is independent of ε and the function f_ε^a .
- (2) B_{APS}^a defines a right-elliptic boundary condition for $\mathcal{D}$.
- (3) Suppose there exist adapted boundary operators A and $\tilde{A}$ of $\mathcal{D}$ and $\mathcal{D}^\dagger$, respectively, and an $\varepsilon > 0$ such that $\sigma_{\mathcal{D}}(\tau) f_\varepsilon^a(A) = f_\varepsilon^a(-\tilde{A}) \sigma_{\mathcal{D}}(\tau)$ and

$$(a - \varepsilon, a + \varepsilon) \cap \text{spec}(A) = \emptyset, \quad (-a - \varepsilon, -a + \varepsilon) \cap \text{spec}(\tilde{A}) = \emptyset.$$

Then B_{APS}^a is a complementable boundary condition with adjoint given by B_{APS}^{-a} .

- (4) If $\mathcal{D}$ is formally self-adjoint (in particular $E = F$) and there exists an adapted boundary operator A which anti-commutes with $\sigma_{\mathcal{D}}(\tau)$, then $B_{\text{APS}} := B_{\text{APS}}^0$ defines a complementable right-elliptic self-adjoint boundary condition for $\mathcal{D}$.

Definition 5.3. Boundary conditions as in Proposition 5.2 are called *generalized Atiyah-Patodi-Singer boundary conditions*.

Proof of Proposition 5.2. We start with part (1). Let $\varepsilon, \varepsilon' > 0$ and $f_\varepsilon^a, f_{\varepsilon'}^a$ be as in the proposition. Since $\text{supp}(f_{\varepsilon'}^a - f_\varepsilon^a) \cap \text{spec}(A) = \emptyset$, we obtain by the properties for the functional

calculus in Proposition 1.4

$$\|(f_{\varepsilon'}^a(A) - f_{\varepsilon}^a(A))\varphi\|_{L^2} \leq \|f_{\varepsilon'}^a(A) - f_{\varepsilon}^a(A)\|_{\text{op}} \|\varphi\|_{L^2} = \sup_{t \in \text{spec}(A)} |f_{\varepsilon'}^a(t) - f_{\varepsilon}^a(t)| \|\varphi\|_{L^2} = 0$$

for all $\varphi \in C^\infty(\partial M, E)$. It follows $f_{\varepsilon}^a(A) C^\infty(\partial M, E) = f_{\varepsilon'}^a(A) C^\infty(\partial M, E)$ and part (1) is proved.

We show next part (2). Fix some $k \in \mathbb{N}_0$, and some $\varepsilon > 0$ and f_{ε}^a as in the proposition. There exists a constant C such that

$$\begin{aligned} \|f_{\varepsilon}^a(A)\varphi\|_{\mathring{H}^{k+1/2}}^2 &= \|\langle (1 + A^2)^{k+1/2} (f_{\varepsilon}^a)^2(A)\varphi, \varphi \rangle_{L^2}\|_{\mathcal{A}} \\ &\leq C \|\langle (1 + A^2)^k \check{f}(A) (f_{\varepsilon}^a)^2(A)\varphi, \varphi \rangle_{L^2}\|_{\mathcal{A}} \\ &= \|f_{\varepsilon}^a(A)\varphi\|_{\check{H}_A^k}^2 \end{aligned}$$

for all $f_{\varepsilon}^a(A)\varphi \in f_{\varepsilon}^a(A) C^\infty(\partial M, E) = C^\infty(\partial M, E) \cap B_{\text{APS}}^a$. Here we used in the second step that $(f_{\varepsilon}^a)^2$ is supported in $(-\infty, a + \varepsilon)$. This shows part (1) in Definition 3.4. In particular, the $\check{H}_A$ -norm is equivalent to the $H^{1/2}$ -norm on $f_{\varepsilon}^a(A) C^\infty(\partial M, E)$, hence B_{APS}^a is closed in $\check{H}_A(\partial M, E)$ and B_{APS}^a defines a boundary condition for $\mathcal{D}$. In order to show that B_{APS}^a is a right-elliptic boundary condition it remains to show part (2) in Definition 3.4, namely that

$$f_{\varepsilon}^a(A) C^\infty(\partial M, E) = C^\infty(\partial M, E) \cap B_{\text{APS}}^a \subseteq \check{H}_A^k(\partial M, E) \cap B_{\text{APS}}^a$$

is dense with respect to the $\check{H}_A^k$ -norm. Fix some $\varphi \in \check{H}_A^k(\partial M, E) \cap B_{\text{APS}}^a$. Let $(\varphi'_j) \subseteq C^\infty(\partial M, E)$ be a sequence converging to φ in the $\check{H}_A^k$ -norm. Since $\varphi \in B_{\text{APS}}^a$ there exists $\varphi' \in H^{1/2}(\partial M, E)$ with $\varphi = f_{\varepsilon}^a(A)\varphi'$, and we obtain $(f_{\varepsilon}^a(A) - 1)\varphi = 0$ by

$$\|(f_{\varepsilon}^a(A) - 1)\varphi\|_{L^2} \leq \max_{t \in \text{spec}(A)} |(f_{\varepsilon}^a(t) - 1)f_{\varepsilon}^a(t)| \|\varphi'\|_{L^2} = 0.$$

We define $\varphi_j := f_{\varepsilon}^a(A)\varphi'_j \in f_{\varepsilon}^a(A) C^\infty(\partial M, E)$ for all $j \in \mathbb{N}_0$ and obtain

$$\|\varphi_j - \varphi\|_{\check{H}_A^k} \leq \|\varphi'_j - \varphi\|_{\check{H}_A^k} + \|(f_{\varepsilon}^a(A) - 1)\varphi'_j\|_{\check{H}_A^k} \xrightarrow{j \rightarrow \infty} 0, \quad (5.1)$$

hence $f_{\varepsilon}^a(A) C^\infty(\partial M, E)$ is dense in $\check{H}_A^k(\partial M, E) \cap B_{\text{APS}}^a$ and B_{APS}^a is right-elliptic. The convergence in equation (5.1) holds since φ'_j converges to φ in $\check{H}_A^k$ and by

$$\begin{aligned} \|(f_{\varepsilon}^a(A) - 1)\varphi'_j\|_{\check{H}_A^k}^2 &= \|\langle (1 + A^2)^k \check{f}(A) (f_{\varepsilon}^a(A) - 1)^2 \varphi'_j, \varphi'_j \rangle_{L^2}\|_{\mathcal{A}} \\ &\leq C \|\langle (1 + A^2)^{k-1/2} (f_{\varepsilon}^a(A) - 1)^2 \varphi'_j, \varphi'_j \rangle_{L^2}\|_{\mathcal{A}} \\ &= C \|(f_{\varepsilon}^a(A) - 1)\varphi'_j\|_{\mathring{H}^{k-1/2}}^2 \xrightarrow{j \rightarrow \infty} C \|(f_{\varepsilon}^a(A) - 1)\varphi\|_{\mathring{H}^{k-1/2}}^2 = 0 \end{aligned}$$

for a suitable constant C independent of n .

We continue with part (3). Fix $\varepsilon > 0$ and $A, \tilde{A}$ as in the proposition. We obtain

$$\begin{aligned} B^{\text{ad}} &= \{\psi \in \check{H}_{\tilde{A}}(\partial M, F) \mid \langle \sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle = 0 \forall \varphi \in B_{\text{APS}}^a\} \\ &= \{\psi \in \check{H}_{\tilde{A}}(\partial M, F) \mid \langle \sigma_{\mathcal{D}}(\tau)\varphi, f_{\varepsilon}^a(-\tilde{A})\psi \rangle = 0 \forall \varphi \in H^{1/2}(\partial M, E)\} \\ &= f_{\varepsilon}^{-a}(\tilde{A}) H^{1/2}(\partial M, F). \end{aligned}$$

It follows $((B_{\text{APS}}^a)^{\text{ad}})^{\text{ad}} = B_{\text{APS}}^a$, and with $(B_{\text{APS}}^a)_{\rho} = f_{\varepsilon}^a(A_{\rho}) H^{1/2}(\partial M, E^{\rho})$ we obtain

$$((B_{\text{APS}}^a)_{\rho})^{\text{ad}} = ((B_{\text{APS}}^a)^{\text{ad}})_{\rho}$$

for all states ρ of the C^* -algebra $\mathcal{A}$. This proves that B_{APS}^a is a complementable boundary condition in the sense of Definition 3.13 (2). The last part of the proposition follows directly from part (2) and (3). $\square$

5.2 Local boundary conditions and chirality

In this section we define chiral boundary conditions. These are boundary conditions induced by a suitable involution χ of E along the boundary of the manifold M [see BB12, Example 7.26]. They play a crucial role in the study of scalar curvature on manifolds with boundary as for instance in [Lot21] and [CZ24]. We show that chiral boundary conditions are right-elliptic complementable self-adjoint and local (see Definition 5.4).

Definition 5.4. A boundary condition $B \subseteq \check{H}_A(\partial M, E)$ for $\mathcal{D}$ is called *local*, if there exists a subbundle E_0 of $E|_{\partial M}$ such that B is equal to the closure of $C^{\infty}(\partial M, E_0)$ in the $\check{H}_A$ -norm.

Proposition 5.5. *Let $\mathcal{D}$ be formally self-adjoint (in particular $E = F$), and χ a formally self-adjoint involution of the bundle $E|_{\partial M}$ such that*

- $\sigma_{\mathcal{D}}(\tau)$ anti-commutes with χ and
- $\sigma_{\mathcal{D}}(\xi)$ commutes with χ for all $\xi \in T\partial M \subseteq TM$.

We denote by $E_1 \subseteq E|_{\partial M}$ the eigenspace corresponding to the eigenvalue one of the involution χ . Then $B_{\chi} := H^{1/2}(\partial M, E_1)$ defines a right-elliptic complementable self-adjoint local boundary condition for $\mathcal{D}$.

Definition 5.6. Boundary conditions as in Proposition 5.5 are called *chiral boundary conditions*.

Proof of Proposition 5.5. Let A_0 be some adapted boundary operator for $\mathcal{D}$. We write A_0 as

$$A_0 = \frac{1}{2}(A_0 - \chi A_0 \chi) + \frac{1}{2}(A_0 + \chi A_0 \chi) =: A + A_1.$$

Note that

$$\sigma_{\chi A_0 \chi}(\xi) = \chi \sigma_{A_0}(\xi) \chi = \chi \sigma_{\mathcal{D}}(\tau)^{-1} \sigma_{\mathcal{D}}(\xi) \chi = -\sigma_{A_0}(\xi)$$

for all $\xi \in T\partial M$ (see proof of Proposition 2.6). We obtain that A is of first order, self-adjoint, elliptic and anti-commutes with χ , and that A_1 is a zero-order differential operator. Therefore, A

is an adapted boundary operator for $\mathcal{D}$ which anti-commutes with χ . We write

$$\check{f} = \frac{1}{2}(\check{f} + \hat{f}) + \frac{1}{2}(\check{f} - \hat{f}) =: f_{\text{ev}} + f_{\text{odd}}$$

and obtain for $k \in \mathbb{N}_0$ and $\varphi \in C^\infty(\partial M, E) \cap B_\chi = C^\infty(\partial M, E_1)$

$$|\varphi|_{\mathbb{H}^{k+1/2}}^2 = \langle (1 + A^2)^{k+1/2} \varphi, \varphi \rangle_{L^2} \leq C_k \langle (1 + A^2)^k f_{\text{ev}}(A) \varphi, \varphi \rangle_{L^2} = C_k |\varphi|_{\check{\mathbb{H}}_A^k}^2 \quad (5.2)$$

for a suitable constant C_k independent of φ . In the second step we used that f_{ev} behaves asymptotically like $|x|$ for $x \rightarrow \pm\infty$, and in the third step

$$\langle (1 + A^2)^k f_{\text{odd}}(A) \varphi, \varphi \rangle_{L^2} = \langle (1 + A^2)^k f_{\text{odd}}(A) \chi \varphi, \chi \varphi \rangle_{L^2} = -\langle (1 + A^2)^k f_{\text{odd}}(A) \varphi, \varphi \rangle_{L^2},$$

which holds since φ satisfies the boundary condition and χ is self-adjoint, anti-commutes with A and $\chi^2 = 1$. Equation 5.2 yields that $B_\chi \subseteq \check{\mathbb{H}}_A^k(\partial M, E)$ satisfies (1) in Definition 3.4 and is closed with respect to $\|\cdot\|_{\check{\mathbb{H}}_A^k}$. In particular, B_χ defines a boundary condition in the sense of Definition 3.2.

We show next that B_χ satisfies part (2) in Definition 3.4, namely that

$$C^\infty(\partial M, E) \cap B \subseteq \check{\mathbb{H}}_A^k(\partial M, E) \cap B$$

is dense with respect to the $\check{\mathbb{H}}_A^k$ -norm for all $k \in \mathbb{N}_0$. Let $k \in \mathbb{N}_0$ and $\varphi \in \check{\mathbb{H}}_A^k(\partial M, E) \cap B$. Note that χ extends to a bounded operator from $\check{\mathbb{H}}_A^k(\partial M, E) \rightarrow \hat{\mathbb{H}}_A^k(\partial M, E)$ by

$$\|\chi\psi\|_{\check{\mathbb{H}}_A^k}^2 = \|\langle (1 + A^2)^k \check{f}(A) \chi \psi, \chi \psi \rangle_{L^2}\|_{\mathcal{A}} = \|\langle (1 + A^2)^k \check{f}(-A) \psi, \psi \rangle_{L^2}\|_{\mathcal{A}} = \|\psi\|_{\hat{\mathbb{H}}_A^k}^2$$

for all $\psi \in C^\infty(\partial M, E)$. Here we used that χ is formally self-adjoint and that $\chi(1 + A^2)^k \check{f}(A) = (1 + A^2)^k \check{f}(-A) \chi$ since A anti-commutes with χ . Since $\varphi \in B$, we obtain

$$\varphi = \chi \varphi \in \check{\mathbb{H}}_A^k(\partial M, E) \cap \hat{\mathbb{H}}_A^k(\partial M, E) = \mathbb{H}^{k+1/2}(\partial M, E).$$

Let $(\varphi'_j) \subset C^\infty(\partial M, E)$ be a sequence converging to φ in the $\mathbb{H}^{k+1/2}$ -norm. We define $\varphi_j := \frac{1}{2}(1 + \chi)\varphi'_j \in C_c^\infty(\partial M, E) \cap B$ for all $j \in \mathbb{N}$, and obtain

$$\begin{aligned} \|\varphi_j - \varphi\|_{\check{\mathbb{H}}_A^k} &\leq \frac{1}{2} \|(1 - \chi)\varphi'_j\|_{\check{\mathbb{H}}_A^k} + \|\varphi'_j - \varphi\|_{\check{\mathbb{H}}_A^k} \\ &\leq C \|(1 - \chi)\varphi'_j\|_{\mathbb{H}^{k+1/2}} + C \|\varphi'_j - \varphi\|_{\mathbb{H}^{k+1/2}} \xrightarrow{j \rightarrow \infty} C \|(1 - \chi)\varphi\|_{\mathbb{H}^{k+1/2}} = 0 \end{aligned}$$

for a suitable constant C independent of j . Here we used that $(1 - \chi)$ is a zero-order differential operator, hence extends to a bounded operator from $\mathbb{H}^{k+1/2}(\partial M, E) \rightarrow \mathbb{H}^{k+1/2}(\partial M, E)$ by Proposition 1.13. This proves part (2) in Definition 3.4 and B_χ is a right-elliptic boundary condition.

It remains to show that B_χ is complementable and self-adjoint in the sense of Defini-

tion 3.13 (2) and (3). We show first that B_χ is self-adjoint, i.e.

$$B_\chi = (B_\chi)^{\text{ad}} \stackrel{\text{def.}}{=} \{\psi \in \check{H}_A(\partial M, E) \mid \langle \sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle = 0 \text{ for all } \varphi \in B_\chi\}. \quad (5.3)$$

For $\varphi, \psi \in B = H^{1/2}(\partial M, E_1)$, we have

$$\langle \sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle_{L^2} = \langle \sigma_{\mathcal{D}}(\tau)\chi\varphi, \chi\psi \rangle_{L^2} = \langle -\sigma_{\mathcal{D}}(\tau)\varphi, \psi \rangle_{L^2}$$

where we used that χ is self-adjoint and that $\sigma_{\mathcal{D}}(\tau)$ anti-commutes with χ . This proves the inclusion $\subseteq$ in equation (5.3). Let $\psi \in \check{H}_A(\partial M, E)$ such that $\langle \sigma_{\mathcal{D}}(\tau), \psi \rangle = 0$ for all $\varphi \in B_\chi$. Fix some sequence $(\psi'_j) \subset C^\infty(\partial M, E)$ converging to ψ in $\check{H}_A(\partial M, E)$. We define $\psi_j := \frac{1}{2}(1 + \chi)\psi'_j \in B_\chi$ for $j \in \mathbb{N}$, and show

$$\|\psi_j - \psi\|_{\check{H}_A} = \frac{1}{2}\|\psi'_j - \psi\|_{\check{H}_A} + \frac{1}{2}\|\chi\psi'_j - \psi\|_{\check{H}_A} \xrightarrow{j \rightarrow \infty} 0. \quad (5.4)$$

Since B_χ is closed in $\check{H}_A(\partial M, E)$, this would imply that $\psi \in B_\chi$. The first term in equation (5.4) converge to zero by choice of the sequence (ψ'_j) . Since $\sigma_{\mathcal{D}}(\tau)$ extends to a bounded operator $\check{H}(\partial M, E) \rightarrow \hat{H}(\partial M, E)$ by Lemma 2.18 (1), we have by Lemma 2.13

$$\|\chi\psi'_j - \psi\|_{\check{H}_A} = \sup_{\varphi \in C^\infty(\partial M, E)} \frac{\|\langle \sigma_{\mathcal{D}}(\tau)\varphi, \chi\psi'_j - \psi \rangle\|_{\mathcal{A}}}{\|\sigma_{\mathcal{D}}(\tau)\varphi\|_{\hat{H}_A}}.$$

We calculate for $\varphi = \frac{1}{2}(\varphi - \chi\varphi) + \frac{1}{2}(\varphi + \chi\varphi) =: \varphi_0 + \varphi_1 \in C^\infty(\partial M, E)$

$$\begin{aligned} \|\langle \sigma_{\mathcal{D}}(\tau)\varphi, \chi\psi'_j - \psi \rangle\|_{\mathcal{A}} &\leq \|\langle \sigma_{\mathcal{D}}(\tau)\varphi_0, \chi\psi'_j - \psi \rangle\|_{\mathcal{A}} + \|\langle \sigma_{\mathcal{D}}(\tau)\varphi_1, \chi\psi'_j \rangle\|_{\mathcal{A}} + \|\langle \sigma_{\mathcal{D}}(\tau)\varphi_1, \psi \rangle\|_{\mathcal{A}} \\ &\leq \|\sigma_{\mathcal{D}}(\tau)\varphi_0\|_{\hat{H}_A} \|\psi'_j - \psi\|_{\check{H}_A} + \|\langle \sigma_{\mathcal{D}}(\tau)\varphi_1, \psi'_j \rangle\|_{\mathcal{A}} \\ &\xrightarrow{j \rightarrow \infty} 0 \end{aligned}$$

Here we used in the second step Lemma 2.13 and that $\langle \sigma_{\mathcal{D}}(\tau)\varphi_1, \psi \rangle = 0$ since $\varphi_1 \in B_\chi$, and in the last step that the sequence (ψ'_j) converges to ψ in $\check{H}_A(\partial M, E)$. This shows equation (5.4), hence B_χ is a self-adjoint boundary condition. Let ρ be a state of the C^* -algebra $\mathcal{A}$. We obtain by the same considerations that $(B_{\chi_\rho})^{\text{ad}} = B_{\chi_\rho}$. Since the localized boundary condition of B_χ is given by B_{χ_ρ} , we obtain

$$((B_\chi)_\rho)^{\text{ad}} = (B_{\chi_\rho})^{\text{ad}} = B_{\chi_\rho} = (B_\chi)_\rho = ((B_\chi)^{\text{ad}})_\rho,$$

hence B_χ is complementable. $\square$

Example 5.7. Let (N, g_N) be a compact connected Riemannian manifold of dimension n with boundary ∂N . Let T be the interior unit normal vector field along the boundary of N and τ the corresponding co-vector field. The indefinite vector bundle $(TN \oplus TN, g_N \oplus (-g_N))$ admits a canonical indefinite spin structure which gives as in [Ton25b, Section 2] rise to a graded Real $\text{Cl}_{n,n}$ -linear Dirac bundle $\mathcal{S} = \mathcal{S}N \hat{\otimes} \mathcal{S}N$. The bundle $\mathcal{S}$ comes canonically with two Clifford

multiplications

$$c: T^*N \longrightarrow \text{End}(\mathcal{S}) \quad \text{and} \quad \bar{c}: T^*N \longrightarrow \text{End}(\mathcal{S})$$

satisfying

$$c(\xi)^2 = -g_N(\xi, \xi), \quad \bar{c}(\xi)^2 = g_N(\xi, \xi), \quad c(\xi)\bar{c}(\eta) = -\bar{c}(\eta)c(\xi), \quad c(\xi)^* = -c(\xi), \quad \bar{c}(\xi)^* = \bar{c}(\xi),$$

and $c(\xi)$ and $\bar{c}(\xi)$ are odd and Real for all $\xi, \eta \in T^*N$. The induced graded Real $\mathbb{Cl}_{n,n}$ -linear Dirac operator $\mathcal{D}_\mathcal{S}$ is a formally self-adjoint elliptic odd Real first-order $\mathbb{Cl}_{n,n}$ -linear differential operator which is of Dirac type, hence by Example 2.8 boundary symmetric with respect to T . Let $\mathcal{A}$ be some adapted boundary operator which anti-commutes with $\sigma_\mathcal{D}(\tau) = c(\tau)$. This choice is for formally self-adjoint Dirac type operator always possible [cf. BB16, Lemma 3.2]. We define a self-adjoint involution

$$\chi := \bar{c}(\tau)c(\tau) \in \text{End}(\mathcal{S}) \tag{5.5}$$

which anti-commutes with $\sigma_\mathcal{D}(\tau) = c(\tau)$ and commutes with $\sigma_\mathcal{D}(\xi) = c(\xi)$ for all $\xi \in T^*\partial N \subseteq T^*N$. The chiral boundary condition $B_\chi := H^{1/2}(\partial M, E_1)$ is by Proposition 5.5 right-elliptic complementable self-adjoint and respects the grading and the Real structure. Here $E_1 \subset \mathcal{S}|_{\partial N}$ is the eigenspace of χ corresponding to the eigenvalue one. We obtain by Theorem 4.5 an unbounded Fredholm operator

$$\mathcal{D}_{\mathcal{S}, B_\chi}: \text{dom}(\mathcal{D}_{\mathcal{S}, B_\chi}) \longrightarrow L^2(N, \mathcal{S}).$$

Its index is equal to the Euler characteristic e of the manifold N

$$\text{ind}(\mathcal{D}_{\mathcal{S}, B_\chi}) = e(N) \in \text{KO}_0(\mathbb{Cl}_{n,n}) \cong \text{KO}_0(\mathbb{C}) \cong \mathbb{Z}. \tag{5.6}$$

So far this is classical and follows already from the theory by Bär and Ballmann [BB12]. See also [CHZ24, Example 2.1].

Let F be a closed connected spin manifold of dimension k . Consider the graded Real $\mathbb{Cl}_k$ -linear spin Dirac bundle $\mathcal{S}^F$ and twist it with the Mishchenko-Fomenko bundle $\mathcal{L}$ of the manifold F . We obtain a graded Real $\mathbb{Cl}_k \otimes C^*\pi_1(F)$ -linear Dirac bundle $\mathcal{S}_\mathcal{L} := \mathcal{S}^F \otimes \mathcal{L}$. The index of the induced graded Real $\mathbb{Cl}_k \otimes C^*\pi_1(F)$ -linear Dirac operator $\mathcal{D}_{\mathcal{S}_\mathcal{L}}$ is the Rosenberg index of F [Ros83]

$$\text{ind}(\mathcal{D}_{\mathcal{S}_\mathcal{L}}) = \alpha(F) \in \text{KO}_k(C^*\pi_1(M)). \tag{5.7}$$

See [Ton25b, Section 2.3] for more details.

Consider the graded Real $\mathbb{Cl}_{n+k,n} \otimes C^*\pi_1(N)$ -linear Dirac bundle $\mathcal{S} \hat{\boxtimes} \mathcal{S}_\mathcal{L} \rightarrow M \times F$ with induced graded Real $\mathbb{Cl}_{n+k,n} \otimes C^*\pi_1(N)$ -linear Dirac operator $\mathcal{D}_{\mathcal{S} \hat{\boxtimes} \mathcal{S}_\mathcal{L}}$. This is a formally self-adjoint elliptic odd Real first-order $\mathbb{Cl}_{n+k,n} \otimes C^*\pi_1(N)$ -linear differential operator which is boundary symmetric with respect to T . We consider the chiral boundary condition B_χ with respect to the self-adjoint involution χ as in equation (5.5), and obtain by Theorem 4.5 an

unbounded Fredholm operator

$$\mathcal{D}_{\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}, B_{\chi}} : \text{dom}(\mathcal{D}_{\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}, B_{\chi}}) \longrightarrow L^2(M \times N, \mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}).$$

Since the metric on $M \times N$ as well as the bundle $\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}$ with all its structures is in product form, and $\text{ind}(\mathcal{D}_{\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}, B_{\chi}})$, $\text{ind}(\mathcal{D}_{\mathcal{S}, B_{\chi}})$ and $\text{ind}(\mathcal{D}_{\mathcal{S}_{\mathcal{L}}})$ are well-defined, we obtain by a product formula for the higher index, equation (5.6) and equation (5.7)

$$\text{ind}(\mathcal{D}_{\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}, B_{\chi}}) = e(N) \cdot \alpha(F) \in \text{KO}_0(\text{Cl}_{n+k, n} \otimes C^*\pi_1(N)) \cong \text{KO}_k(C^*\pi_1(N)).$$

The product formula follows by purely functional analytic considerations for the higher index. Here the boundary of the manifold M does not play a role.

Note that the bundle $\mathcal{S} \hat{\boxtimes} \mathcal{S}_{\mathcal{L}}$ is equal to the graded Real $\text{Cl}_{n+k, n} \otimes C^*\pi_1(N)$ -linear Dirac bundle

$$\mathcal{S}(N \times F) \hat{\otimes} \text{pr}_1^* \mathcal{S}N \otimes \text{pr}_1^* \mathcal{L} \longrightarrow N \times F.$$

This is the bundle used in the proof of the generalization of the rigidity theorem by Goette and Semmelmann [GS02, Theorem 2.4] in [Ton25b, Theorem A] and [RT26, Theorem A]. The boundary condition is the same one used by Lott [Lot21] in the generalization of the Goette-Semmelmann rigidity theorem to manifolds with boundary. This example illustrates how the new developed theory of boundary value problems could be used in order to generalize [Lot21] in the spirit of [Ton25b] and [RT26].

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